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OF ARTHUR CAYLEY, Sc.D., F.R.S.

LATE SADLERIAN PROFESSOR OF PURE MATHEMATICS
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ON A FORM OF QUARTIC SURFACE WITH TWELVE NODES.

[From the British Association Report, (1886), pp. 540, 541.]

USING throughout capital letters to denote homogeneous quadric functions of the coordinates (x, y, z, w) , we have as a form of quartic surface with eight nodes

$$\Omega = (*U, V, W)^2 = 0;$$

viz. the nodes are here the octad of points, or eight points of intersection of the quadric surfaces $U = 0, V = 0, W = 0$; the equation can, by a linear transformation on the functions U, V, W (that is, by substituting for the original functions U, V, W linear functions of these variables), be reduced to the form

$$\Omega = U^2 + V^2 + W^2 = 0.$$

Suppose now that the function Ω can in a second manner be expressed in the like form $\Omega = P^2 + Q^2 + R^2$ (where P, Q, R are not linear functions of U, V, W); that is, suppose that we have identically $U^2 + V^2 + W^2 = P^2 + Q^2 + R^2$, this gives

$$U^2 - P^2 + V^2 - Q^2 + W^2 - R^2 = 0;$$

or, writing $U + P, V + Q, W + R = A, B, C$, and $U - P, V - Q, W - R = F, G, H$, the identity becomes $AF + BG + CH = 0$; and this identity being satisfied, the equation $\Omega = 0$ of the quartic surface may be written in the two forms

$$\Omega = (A + F)^2 + (B + G)^2 + (C + H)^2 = 0,$$

and

$$\Omega = (A - F)^2 + (B - G)^2 + (C - H)^2 = 0,$$

viz. the quartic surface has the nodes which are the intersections of the three quadric surfaces $A + F = 0, B + G = 0, C + H = 0$, and also the nodes which are the intersections of the three quadric surfaces $A - F = 0, B - G = 0, C - H = 0$. We may of course also write the equation of the surface in the form

$$\Omega = A^2 + B^2 + C^2 + F^2 + G^2 + H^2 = 0.$$

An easy way of satisfying the identity $AF + BG + CH = 0$ is to assume

$$A, B, C, F, G, H = ayz, bzx, cxy, fxw, gyw, hzw,$$

where the constants a, b, c, f, g, h satisfy the condition $af + bg + ch = 0$; this being so, the functions A, B, C, F, G, H , and consequently the functions $A + F, B + G, C + H$ and $A - F, B - G, C - H$ each of them vanish for the four points $(y = 0, z = 0, w = 0)$, $(z = 0, x = 0, w = 0)$, $(x = 0, y = 0, w = 0)$, $(x = 0, y = 0, z = 0)$, or say the points $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$. It hence appears that the quartic surface

$$\Omega = a^2 y^2 z^2 + b^2 z^2 x^2 + c^2 x^2 y^2 + f^2 x^2 w^2 + g^2 y^2 w^2 + h^2 z^2 w^2 = 0$$

is a quartic surface with twelve nodes; viz. it has as nodes the last-mentioned four points, the remaining four points of intersection of the surfaces

$$ayz + fxw = 0, \quad bzx + gyw = 0, \quad cxy + hzw = 0,$$

and the remaining four points of intersection of the surfaces

$$ayz - fxw = 0, \quad bzx - gyw = 0, \quad cxy - hzw = 0.$$

The above is the analytical theory of one of the two forms of quartic surface with twelve nodes recently established by Dr K. Rohn in a paper in the *Berichte u. d. Verhandlungen der K. Sächsische Gesellschaft zu Leipzig*, (1884), pp. 52—60.

889.

ON A DIFFERENTIAL EQUATION AND THE CONSTRUCTION OF MILNER'S LAMP.

[From the Proceedings of the Edinburgh Mathematical Society, vol. v. (1887), pp. 99—101.]

WHAT sort of an equation is

$$b^3 \cos(\alpha + \theta) = a \cos \theta \int_{\beta}^{\theta} r^2 d\theta - \frac{1}{2} \left\{ \cos \theta \int_{\beta}^{\theta} r^3 \cos \theta d\theta + \sin \theta \int_{\beta}^{\theta} r^3 \sin \theta d\theta \right\} ? \quad (1)$$

Write

$$X = \int_{\beta}^{\theta} r^2 d\theta, \quad Y = \int_{\beta}^{\theta} r^3 \cos \theta d\theta, \quad Z = \int_{\beta}^{\theta} r^3 \sin \theta d\theta, \quad (2)$$

and start with the equations

$$\frac{dX}{d\theta} = r^2, \quad \frac{dY}{d\theta} = r^3 \cos \theta, \quad \frac{dZ}{d\theta} = r^3 \sin \theta, \quad (3)$$

$$\left(\frac{d}{d\theta} + 1 \right) \left\{ a \cos \theta \cdot X - \frac{1}{2} (Y \cos \theta + Z \sin \theta) \right\} = 0. \quad (4)$$

This last gives

$$(r - a \cos \theta) dr + ar \sin \theta \cdot d\theta = 0, \quad (5)$$

and the system thus is

$$\frac{dX}{d\theta} = r^2, \quad \frac{dY}{d\theta} = r^3 \cos \theta, \quad \frac{dZ}{d\theta} = r^3 \sin \theta, \quad \frac{d\theta}{dr} = -\frac{r - a \cos \theta}{ar \sin \theta};$$

viz. this is a system of ordinary differential equations between the five variables θ, r, X, Y, Z : the system can therefore be integrated with four arbitrary constants, and these may be so determined that for the value β of θ, X, Y, Z shall be each $= 0$; and r shall have the value r_0 .

But this being so, from the assumed equations (3) and (4) we have

$$X = \int_{\beta}^{\theta} r^2 d\theta, \quad Y = \int_{\beta}^{\theta} r^3 \cos \theta d\theta, \quad Z = \int_{\beta}^{\theta} r^3 \sin \theta d\theta,$$

and further, by integration of (4),

$$L \cos \theta + M \sin \theta = a \cos \theta \cdot X - \frac{1}{2} (Y \cos \theta + Z \sin \theta).$$

Here L and M denote properly determined constants: viz. the conclusion is that r, X, Y, Z admit of being determined as functions of θ and of an arbitrary constant r_0 , in such wise that

$$a \cos \theta \cdot X - \frac{1}{2} (Y \cos \theta + Z \sin \theta)$$

shall be a function of θ , of the proper form $L \cos \theta + M \sin \theta$, but not so that it shall be the precise function $b^3 \cos(\alpha + \theta)$. To make it have this value, we must have $L = b^3 \cos \alpha, M = -b^3 \sin \alpha$ (where L, M are given functions of a, b, α, β, r_0), i.e. we must have two given relations between a, b, α, β, r_0 : or treating r_0 as a disposable constant, we must have one given relation between a, b, α, β .

The equation $d\theta = -\frac{ar \sin \theta}{r - a \cos \theta} dr$ gives $r^2 - 2ar \cos \theta = C$, where $C = r_0^2 - 2ar_0 \cos \beta$.

There would be considerable difficulty in working the question out with r_0 arbitrary, but we may do it easily enough for the particular value $r_0 = 0$ or $r_0 = 2a \cos \beta$, giving $C = 0$ and

therefore $r = 2a \cos \theta$: and we ought in this case to be able to satisfy the given equation not in general but with two determinate relations between the constants a, b, α, β .

We have

$$\begin{aligned}\int \cos^2 \theta d\theta &= \frac{1}{2}\theta + \frac{1}{4}\sin 2\theta, \\ \int \cos^4 \theta d\theta &= \frac{3}{8}\theta + \frac{1}{4}\sin 2\theta + \frac{1}{32}\sin 4\theta, \\ \int \cos^3 \theta \sin \theta d\theta &= -\frac{1}{4}\cos^4 \theta,\end{aligned}$$

and thence

$$\begin{aligned}a \cos \theta \cdot X - \frac{1}{2}(Y \cos \theta + Z \sin \theta) \\ &= 4a^3 \cos \theta \left[\frac{1}{2}(\beta - \theta) + \frac{1}{4}(\sin 2\beta - \sin 2\theta) \right] \\ &\quad - \frac{1}{2} \cdot 4a^3 \cos \theta \left[\frac{3}{8}(\beta - \theta) + \frac{1}{4}(\sin 2\beta - \sin 2\theta) + \frac{1}{32}(\sin 4\beta - \sin 4\theta) \right] \\ &\quad - \frac{1}{2} \cdot 4a^3 \sin \theta \left[-\frac{1}{4}(\cos^4 \beta - \cos^4 \theta) \right] \\ &= -\frac{3}{2}a^3 \cos \theta (\sin 2\beta - \sin 2\theta) \\ &\quad - \frac{1}{16}a^3 \cos \theta (\sin 4\beta - \sin 4\theta) + \frac{1}{2}a^3 \sin \theta (\cos^4 \beta - \cos^4 \theta),\end{aligned}$$

where the terms containing β are readily reduced to $\frac{1}{2}a^3 \cos^3 \beta \sin(\theta - \beta)$; hence also the terms without β disappear of themselves: and we have

$$a \cos \theta \cdot X - \frac{1}{2}(Y \cos \theta + Z \sin \theta) = \frac{1}{2}a^3 \cos^3 \beta \cdot \sin(\theta - \beta),$$

which may be put

$$= b^3 \cos(\theta + \alpha) :$$

viz. this will be so if we have the two relations

$$\alpha = \frac{1}{2}\pi - \beta; \quad \text{and} \quad b^3 = -\frac{1}{2}a^3 \cos^3 \beta.$$

I make (see figure) Milner's lamp, with a circular section, β arbitrary, but a segment AM ($\angle SAM = \beta$) made solid. G in the line SG at right angles to AM is the c. G. of the lamp, and G' the c. G. of the oil.

And this seems to be the only form—for the pole of r must, it seems to me, be on the bounding circle—viz. in the equation $r^2 - 2ar \cos \theta = C$, we must have $C = 0$.

890.

NOTE ON THE HYDRODYNAMICAL EQUATIONS.

[From the Proceedings of the Royal Society of Edinburgh, vol. xv. (1889), pp. 342—344.]

WRITING for shortness $D = \frac{d}{dt} + u \frac{d}{dx} + v \frac{d}{dy} + w \frac{d}{dz}$, then if from the hydrodynamical equations

$$Du = \frac{d}{dx} \left(V - \frac{P}{\rho} \right), \quad Dv = \frac{d}{dy} \left(V - \frac{P}{\rho} \right), \quad Dw = \frac{d}{dz} \left(V - \frac{P}{\rho} \right),$$

without the aid of the equation

$$\frac{du}{dx} + \frac{dv}{dy} + \frac{dw}{dz} = 0,$$

we eliminate $V - \frac{P}{\rho}$, we obtain equations not equivalent to those of Helmholtz,

$$\left(D\xi = \xi \frac{du}{dx} + \eta \frac{du}{dy} + \zeta \frac{du}{dz}, \text{ \&c.}, \quad (2\xi, 2\eta, 2\zeta) = \left(\frac{dw}{dy} - \frac{dv}{dz}, \frac{du}{dz} - \frac{dw}{dx}, \frac{dv}{dx} - \frac{du}{dy} \right), \text{ as usual} \right),$$

but which, transforming them by means of the omitted equation, agree as they should do with his equations. But the form of the equations obtained directly by elimination as above, is an interesting one, which it is worth while to give.

We have

$$\begin{aligned} D \left(\frac{dv}{dz} - \frac{dw}{dy} \right) + \frac{dv}{dz} \frac{dw}{dy} - \frac{dv}{dy} \frac{dw}{dz} \\ + \frac{d}{dz} \left(Dv + \frac{dv}{dx} u + \frac{dv}{dy} v + \frac{dv}{dz} w \right) - \frac{d}{dy} \left(Dw + \frac{dw}{dx} u + \frac{dw}{dy} v + \frac{dw}{dz} w \right) \\ = \frac{d}{dz} \left(\frac{dv}{dt} + \frac{dv}{dx} u + \frac{dv}{dy} v + \frac{dv}{dz} w \right) - \frac{d}{dy} \left(\frac{dw}{dt} + \frac{dw}{dx} u + \frac{dw}{dy} v + \frac{dw}{dz} w \right), \end{aligned}$$

where the terms containing second derived functions disappear of themselves, and the expression on the right-hand is thus

$$-\frac{du}{dz} \frac{dv}{dx} - \frac{dv}{dz} \frac{dv}{dy} - \frac{dw}{dz} \frac{dv}{dz} + \frac{du}{dy} \frac{dw}{dx} + \frac{dv}{dy} \frac{dw}{dy} + \frac{dw}{dy} \frac{dw}{dz}.$$

Representing for shortness the Matrix

$$\begin{vmatrix} \frac{du}{dx} & \frac{du}{dy} & \frac{du}{dz} \\ \frac{dv}{dx} & \frac{dv}{dy} & \frac{dv}{dz} \\ \frac{dw}{dx} & \frac{dw}{dy} & \frac{dw}{dz} \end{vmatrix} \text{ by } \begin{vmatrix} a & b & c \\ a' & b' & c' \\ a'' & b'' & c'' \end{vmatrix}, \text{ and its square by } \begin{vmatrix} A & B & C \\ A' & B' & C' \\ A'' & B'' & C'' \end{vmatrix},$$

we have

$$\begin{vmatrix} A & B & C \\ A' & B' & C' \\ A'' & B'' & C'' \end{vmatrix} = \begin{vmatrix} \frac{du}{dx} & \frac{du}{dy} & \frac{du}{dz} \\ \frac{dv}{dx} & \frac{dv}{dy} & \frac{dv}{dz} \\ \frac{dw}{dx} & \frac{dw}{dy} & \frac{dw}{dz} \end{vmatrix} \cdot \left(\left(\frac{du}{dx}, \frac{dv}{dx}, \frac{dw}{dx} \right), \left(\frac{du}{dy}, \frac{dv}{dy}, \frac{dw}{dy} \right), \left(\frac{du}{dz}, \frac{dv}{dz}, \frac{dw}{dz} \right) \right);$$

viz. the combinations which enter into the foregoing formula are

$$C' = \frac{dv}{dx} \frac{du}{dz} + \frac{dv}{dy} \frac{dv}{dz} + \frac{dv}{dz} \frac{dw}{dz},$$

and

$$B'' = \frac{dw}{dx} \frac{du}{dy} + \frac{dw}{dy} \frac{dv}{dy} + \frac{dw}{dz} \frac{dw}{dy},$$

and the equation thus is $D(c' - b'') + C' - B'' = 0$; viz. the three equations are

$$\begin{aligned} D(c' - b'') + C' - B'' &= 0, \\ D(a'' - c) + A'' - C &= 0, \\ D(b - a') + B - A' &= 0, \end{aligned}$$

which are the equations in question.

Observe that we have

$$\begin{aligned} C' - B'' &= (a', b', c')(c, c', c'') - (a'', b'', c'')(b, b', b'') \\ &= a'c + b'c' + c'c'' - a''b - b'b'' - b''c'', \end{aligned}$$

and thence, writing

$$\rho = a(c' - b'') + b(a'' - c) + c(b - a'), \quad = ac' - ab'' + a''b - a'c,$$

we have

$$C' - B'' + \rho = (a + b' + c'')(c' - b'') = 0,$$

if $a + b' + c'' = 0$; viz. this being so, $C' - B'' = -\rho$, or the first equation is

$$D(c' - b'') = \rho, \quad = a(c' - b'') + b(a'' - c) + c(b - a'),$$

that is, $D\xi = \xi \frac{du}{dx} + \eta \frac{du}{dy} + \zeta \frac{du}{dz}$, the first equation of Helmholtz, and we thus have the equations

of Helmholtz, if $a + b' + c'' = 0$, that is, if $\frac{du}{dx} + \frac{dv}{dy} + \frac{dw}{dz} = 0$.

The foregoing three equations $D(c' - b'') + C' - B'' = 0$, &c., are the quaternion equation $\left(\sigma = iu + jv + kw, \nabla = i\frac{d}{dx} + j\frac{d}{dy} + k\frac{d}{dz}, \frac{d}{dt} = D, \text{ denotes a complete differentiation} \right)$,

$$\frac{d}{dt} \nabla \sigma = \nabla \nabla \sigma S \sigma \nabla,$$

of Mr M'Aulay's paper "Some General Theorems in Quaternion Integration," *Messenger of Mathematics*, vol. xiv. (1884), pp. 26—37; see p. 34.

891.

ON THE BINODAL QUARTIC AND THE GRAPHICAL REPRESENTATION OF THE ELLIPTIC FUNCTIONS.

[From the Transactions of the Cambridge Philosophical Society, vol. xiv. (1889), pp. 484—494. Read May 6, 1889.]

I APPROACH the subject from the question of the graphical representation of the elliptic functions: assuming as usual that the modulus is real, positive, and less than unity, and to fix the ideas considering the function sn (but the like considerations are applicable to the functions cn and dn), then the equation $x + iy = \text{sn}(x' + iy')$ establishes a $(1, 1)$ correspondence between the xy infinite quarter plane, and the $x'y'$ rectangle (sides K and K'): viz. to any given point $x + iy$, x and y each positive, there corresponds a single point $x' + iy'$, x' , y' each positive and less than K , K' respectively: and conversely to any such point $x' + iy'$, there corresponds a single point $x + iy$, x and y each positive.

I draw in the $x'y'$ -figure the rectangle $A'B'C'D'$ (sides K and K'), and in the xy -figure, I take on the axis of x , the points B , C where $AB = 1$, $AC = \frac{1}{k}$: and the point D at infinity. We have thus in the $x'y'$ -figure the closed curve or contour $A'B'C'D'A'$: and corresponding hereto we have in the xy -figure the closed curve or contour $ABCD A$, viz. here D is the point at infinity considered as a line always at infinity, extending from the point at infinity on the positive part of the axis of x , to the point at infinity on the positive part of the axis of y , the contour being thus AB , BC , CD (D at infinity on the axis of x); and then D (at infinity on the axis of y) A . And

thus to a point P' describing successively the lines $A'B'$, $B'C'$, $C'D'$, $D'A'$ there corresponds a point P describing successively the lines AB , BC , CD , DA : to P' at D' there corresponds P at D , viz. this is any point at infinity from D on the axis of x to D on the axis of y . There is no real breach of continuity: in further illustration, suppose that P , instead of actually coming to D' , just cuts off the corner, viz. that it passes from a point γ' on $C'D'$ to a point α' on $D'A'$ (γ' , α' each of them very near to D'): then P passes from a point γ very near D on the axis of x (that is, at a great distance from A) to a point α very near D on the axis of y (that is, at a great distance from A): and to the indefinitely small arc $\gamma'\alpha'$ described by P' there corresponds the indefinitely large arc $\gamma\alpha$ described by P .

We thus see that, if P' describe any arc $E'F'$ passing from a point E' of $A'D'$ to a point F' of $B'C'$, then P will describe an arc EF passing from a point E of AD to a point F of BC : and similarly, if P' describe any arc $G'H'$ passing from a point G' of $A'B'$ to a point H' of $C'D'$, then P will describe an arc GH passing from a point G of AB to a point H of CD .

Supposing $E'F'$ is a straight line parallel to $A'x'$, that is, cutting $A'D'$ and $B'C'$ each at right angles, then EF will be an arc cutting AD and BC each at right angles: and so if $G'H'$ is a straight line parallel to $A'y'$, that is, cutting $A'B'$ and $C'D'$ each at right angles, then GH will be an arc cutting AB and CD each at right angles: and moreover, since $E'F'$ and $G'H'$ cut each other at right angles, then also EF and GH cut each other at right angles.

Supposing, as above, that $E'F'$ and $G'H'$ are straight lines, we have EF and GH each of them the arc of a special bicircular quartic: the theory was in fact established in a very elegant manner in a memoir by Siebeck, "Ueber eine Gattung von Curven vierten Grades, welche mit den elliptischen Functionen zusammenhängen," Crelle, t. LVII. (1860), pp. 359—370, and t. LIX. (1861), pp. 173—184.

In particular, if P' describe the line $L'M'$ lying halfway between $A'B'$ and $D'C'$ (that is, if $A'L' = \frac{1}{2}K'$), then P will describe the circular quadrant LM , radius $\frac{1}{\sqrt{k}}$, viz. in this case the bicircular quartic degenerates into a circle twice repeated: and so if P' describe successively the lines $E'F'$ and $E'_1F'_1$ equidistant from $L'M'$ ($A'E' = \frac{1}{2}K' - \eta$, $A'E'_1 = \frac{1}{2}K' + \eta$), then P will describe the arcs EF and E_1F_1 which are the images of each other in regard to the centre A and circular quadrant LM , and which together constitute the quadrant of one and the same bicircular quartic.

A bicircular quartic is of course a binodal quartic with the circular points at infinity for the two nodes: there is no real gain of generality in considering the binodal quartic rather than the bicircular quartic, but I have preferred to do so, and I have accordingly introduced the term *Binodal Quartic* into the title of the present Memoir. I present in a compendious form the properties of the general curve, and I show how the curve is to be particularised so as to obtain from it the special bicircular quartics which present themselves as above in the graphical representation of the elliptic functions.

A binodal quartic has the Plückerian numbers

$$m, n, \delta, \kappa, \tau, \iota = 4, 8, 2, 0, 8, 12.$$

The number of tangents to the curve which can be drawn from either of the nodes is $n - \delta = 4$; and the pencil of tangents from the one node is homographic with the pencil of tangents from the other node. Call the nodes I and J : and let the tangents from I be called (a, b, c, d) and those from J be called (a', b', c', d') , then if the tangents which correspond to (a', b', c', d') respectively are (a, b, c, d) , they may also be taken to be (b, a, d, c) , (c, d, a, b) or (d, c, b, a) : and considering the intersections of corresponding tangents, we have thus four tetrads of points, say the f -points, such that the points of each tetrad lie in a conic through the two nodes: and we have consequently four conics each passing through the two nodes, say these are the f -conics.

Starting as above with the correspondence (a, b, c, d) , (a', b', c', d') , if the intersections of a and a' , b and b' , c and c' , d and d' are called A, B, C, D respectively, then we have A, B, C, D for a tetrad of f -points, lying on the f -conic $(ABCD)$.

Writing AB for the two points, the intersections of IA, JB and of IB, JA respectively, and so in other cases, then the remaining three tetrads of f -points are

$$\begin{aligned} AB, CD & \text{ lying on the } f\text{-conic } (AB, CD), \\ AC, BD & \text{ „ „ „ „ } (AC, BD), \\ AD, BC & \text{ „ „ „ „ } (AD, BC). \end{aligned}$$

The two points AB may be spoken of as the antipoints of A, B : and so in other cases.

Any two of the f -conics have in common the nodes I, J , and they therefore intersect in two points besides: at each of these the tangents to the two conics, and the lines to I, J respectively, form a harmonic pencil.

Consider the two tangents at I and the two tangents at J : we have a conic touching these four lines and passing through the tetrad of f -points, or what is the same thing, intersecting the f -conic in the four f -points: say this is a c -conic. There are thus four c -conics corresponding to the four f -conics respectively.

We may consider a variable conic, passing through the points I and J ; and such that the tangents thereto at these points respectively meet on a point of a c -conic: the variable conic and the corresponding f -conic each pass through the points I, J and they meet in two points besides: and the variable conic may be such that at each of these the tangents to the variable conic and the f -conic form with the lines drawn to the points I and J a harmonic pencil. The variable conic, as thus defined, contains a single variable parameter; and it has for its envelope the binodal quartic: the binodal quartic is thus in four different ways the envelope of a variable conic. This is of course Casey's Theorem for the fourfold generation of the bicircular quartic as the envelope of a variable circle.

One of the f -conics, say $(ABCD)$, may break up into the line IJ and a line, say the axis $(ABCD)$: we have then the four f -points A, B, C, D on this line. The other f -conics say (AB, CD) , (AC, BD) , (AD, BC) are proper conics as before: any one of these meets the line $(ABCD)$ in two points: and at each of these the line and the tangent to the conic form with the lines drawn to the points I and J a harmonic pencil. The binodal quartic is in this case said to have an *axis*.

But a second f -conic, say (AD, BC) , may break up into the line IJ and a line, say the axis (AD, BC) : we have then the four f -points AD, BC on this line. Writing moreover A', D' for the two points AD , or say $(AI, DJ) = A'$, $(AJ, DI) = D'$; and similarly B', C' for the two points BC , or say $(BI, CJ) = B'$, $(BJ, CI) = C'$; then the four f -points are A', B', C', D' , and the axis (AD, BC) may be called $(A'B'C'D')$.

The relation of the f -points A, B, C, D and A', B', C', D' and of the two axes is as shown in the figure: taking X for the intersection of IJ with $(ABCD)$ and Y for that of IJ with $(A'B'C'D')$, also Z for the intersection of the two axes, then X, Z are the sibiconjugate points of the involution AD, BC and Y, Z the sibiconjugate points of the involution $A'D', B'C'$. The two axes intersect in Z , and form with the lines ZI, ZJ a pencil in involution.

The two remaining f -conics (AB, CD) and (AC, BD) , or as they might also be called $(A'B', C'D')$ and $(A'C', B'D')$, are proper conics as before: they touch each other at the points I, J , and have for their common tangents at these points the lines ZI, ZJ respectively. Each conic meets each axis in two points; and at each of these points the axis and the tangent to the conic form with the lines to I, J a harmonic pencil. The binodal quartic is in this case said to be *biaxial*. The point Z , which is the intersection of the two axes, may be called the *centre*.

In the case where an f -conic breaks up into the line IJ , and a line containing four f -points, say an f -line, the corresponding c -conic coincides with the f -conic, viz. it also breaks up into the line IJ and the f -line: the variable conic is a conic through the points I, J such that the tangents thereto at these points respectively meet on the f -line. Moreover, the variable conic must be such that at each of its intersections with the f -conic, that is, the f -line, the tangent to the variable conic and the f -line must be harmonics in regard to the lines drawn from the

point to the points I, J respectively: but this condition is satisfied *ipso facto* for each of the intersections of the variable conic and the f -line. This depends on the theorem that, taking on a conic any three points P, I, J , then the tangent at P and the line drawn from P to the pole of IJ are harmonics in regard to the lines PI, PJ . Thus we have only three conditions for the variable conic, or, as above defined, it would in the case in question (of four f -points in a line) depend upon two variable parameters. There is really another condition—but what this in general is I have not ascertained: and this being so the variable conic in the case in question (of the four f -points in a line) depends upon a single variable parameter, and we have as before the bicircular quartic as the envelope. The foregoing is the axial case; in the biaxial case, the same thing happens in regard to the variable conics belonging to the two axes respectively. Thus in every case we have the fourfold generation of the curve as the envelope of a variable conic: only in the axial case, the variable conics belonging to the axis, and in the biaxial case the variable conics belonging to the two axes respectively, are not by the foregoing definitions completely defined. It will be seen further on how, in the case of the biaxial bicircular quartic, we complete the definition of the variable circles belonging to the two axes respectively.

Taking the points I, J to be the circular points at infinity, we have a bicircular quartic. The f -points are the foci, and the f -conics are circles, viz. we have 16 foci situate in fours upon four focal circles. The harmonic relation of two lines to the lines through I, J means of course that the lines cut at right angles; hence the focal circles cut each other at right angles: this must certainly be a known property, but it is not mentioned in Salmon's Higher Plane Curves, Ed. 3, Dublin, 1879, and I cannot find it in Darboux or Casey: it is given No. 81 in Lachlan's Memoir "On Systems of Circles and Spheres," *Phil. Trans.*, vol. CLXXVII. (1886), and I find it as a question in the Educational Times, March 1, 1889, 10034 (Prof. Morley). "Prove that, of the four focal circles of a circular cubic or bicircular quartic, any two are orthogonal, and the radii are connected by the relation $\sum(R^{-2}) = 0$." The theorem is not as well known as it should be.

The c -conics are confocal conics having for their real foci the so-called double-foci of the quartic (more accurately, the common foci are the four quadruple foci of the quartic); we have thus four conics corresponding to the four focal circles respectively, each conic intersecting the corresponding circle in the four foci upon this circle. And we have then the quartic as the envelope of a variable circle having its centre upon one of these conics and cutting at right angles the corresponding focal circle: the bicircular quartic is thus generated in four different ways.

Instead of one of the focal circles, we may have a line or axis, and the quartic is then said to be *axial*: the foci on the axis may be any four points; and for a real curve they may be all real, or two real and two imaginary, or all four imaginary. The remaining focal circles are real or imaginary circles, cut by the axis at right angles, that is, having their centres on the axis, and cutting each other at right angles.

But instead of another of the focal circles, we may have a line or axis, and the quartic is then said to be *biaxial*: the two axes cut at right angles at a point which may be called the centre of the curve. The foci on each axis form pairs of points situate symmetrically in regard to the centre. If on one of the axes the foci are real, then on the other axis they form two imaginary conjugate pairs; and conversely: but if on one of the axes the foci are two of them real and the other two conjugate imaginaries, then this is so for the other axis also. There are thus only the two cases: 1°, foci on the one axis real, and on the other conjugate imaginaries; 2°, foci on each axis two of them real and the other two conjugate imaginaries: there is however a limiting case where on each axis two foci are united at the centre, the other two foci being real on the one axis and conjugate imaginaries on the other. The remaining two focal circles are real or imaginary circles, cutting each axis at right angles, that is, having their centres at the centre; and cutting each other at right angles, that is, having the sum of the squares of their radii = 0.

The biaxial form of bicircular quartic is, in fact, that which presents itself in the theory of the representation of the elliptic functions.

I consider for a moment the case of a variable circle having its centre upon a given line, and cutting at right angles a given circle. The variable circles pass all of them through two fixed points, the antipoints of the intersections of the given line and circle, and which are thus real or imaginary according as the intersections of the given line and circle are imaginary or real. Hence, considering any one variable circle and the consecutive variable circle, these intersect in two real points, when the given line does not meet the given circle (meets it in two imaginary points); but when the given line meets the given circle in two real points, then the two variable circles intersect in two imaginary points: if however the given line touches the given circle, then the two variable circles touch each other. Taking now the curve of centres to be any given curve whatever, and considering one of the variable circles, and the consecutive variable circle, it at once appears that, if the tangent to the curve of centres at the centre of the variable circle does not meet the given circle, then the two variable circles intersect in two real points (which, if the tangent touch the given circle, unite in a single real point): but if the tangent to the curve of centres meets the given circle, then the two variable circles do not intersect. It hence appears that the real portions of the envelope arise exclusively from those portions of the curve of centres which are such that at any point thereof the tangent to the curve of centres does not meet the given circle. In particular, if the given circle be real, and the curve of centres is a real ellipse enclosing the given circle, then the real portion of the envelope arises from the whole ellipse: but if the curve of centres be a real ellipse cutting the given circle in four real points, then drawing the four common tangents of the ellipse and circle, it is at once seen that there are on the ellipse two detached portions such that, at any point of either portion, the tangent to the ellipse does not meet the circle: and the real portions of the envelope arise exclusively from these portions of the ellipse.

In the case just referred to, there are on the ellipse four portions each lying outside the circle and terminating in the four intersections respectively of the ellipse and circle, such that at a point of any one of these portions the tangent to the ellipse meets the circle in two real points. Starting from the extremity of one of these portions of the ellipse and proceeding to the other extremity on the circle, the corresponding variable circles do not intersect each other, but each of them is a circle lying wholly inside that which immediately precedes it; and the variable circle becomes ultimately a point, viz. this point is a focus of the curve: this agrees with the foregoing statement that the f -conic intersects the circle in the four foci upon this circle. For the two portions of the ellipse which lie inside the circle, the variable circle is of course always imaginary. The like considerations apply to the case where the locus of the centre of the variable circle is a hyperbola or parabola. The foregoing remarks illustrate the actual generation of a bicircular quartic as the envelope of the variable circle.

Starting now from the equation

$$x + iy = \operatorname{sn}(x' + iy'),$$

we have

$$x = \frac{\operatorname{sn} x' \operatorname{cn} iy' \operatorname{dn} iy'}{1 - k^2 \operatorname{sn}^2 x' \operatorname{sn}^2 iy'}, \quad y = \frac{\operatorname{sn} iy' \operatorname{cn} x' \operatorname{dn} x'}{1 - k^2 \operatorname{sn}^2 x' \operatorname{sn}^2 iy'},$$

or putting

$$\operatorname{sn} x' = p, \quad \operatorname{sn} iy' = iq;$$

these equations are

$$x = \frac{p\sqrt{1+q^2}\sqrt{1+k'^2q^2}}{1+k^2p^2q^2}, \quad iy = \frac{iq\sqrt{1-p^2}\sqrt{1-k^2p^2}}{1+k^2p^2q^2},$$

whence also

$$x^2 + y^2 = \frac{p^2 + q^2 + k^2p^2q^2(1-p^2-q^2)}{(1+k^2p^2q^2)^2} = r^2, \quad (\text{if } x^2 + y^2 \text{ be put } = r^2).$$

These equations, considering therein q as a given constant, and p as a variable parameter, determine the curve EF : and considering p as a given constant, and q as a variable parameter, they determine the curve GH . But the eliminations are easily effected; we have

$$p^2(1 - k^2q^2r^2) = r^2 - q^2, \quad q^2(1 - k^2p^2r^2) = r^2 - p^2.$$

Hence, for EF ,

$$p^2 = \frac{r^2 - q^2}{1 - k^2q^2r^2}, \quad 1 - p^2 = \frac{1 + q^2 - (1 + k'^2q^2)r^2}{1 - k^2q^2r^2},$$

$$1 - k^2p^2 = \frac{1 + k'^2q^2 - k^2(1 + q^2)r^2}{1 - k^2q^2r^2}, \quad 1 + k^2p^2 = \frac{1 - k^2q^2}{1 - k^2q^2r^2},$$

and consequently

$$x = \frac{\sqrt{1 + q^2}\sqrt{1 + k'^2q^2}\sqrt{r^2 - q^2}\sqrt{1 - k^2r^2}}{1 - k^2q^2},$$

$$y = \frac{q\sqrt{1 + q^2 - (1 + k'^2q^2)r^2}\sqrt{1 + k'^2q^2 - k^2(1 + q^2)r^2}}{1 - k^2q^2},$$

giving x and y each of them in terms of r . And from the first of these we at once derive

$$(x^2 + y^2)^2 - 2Ax^2 - 2By^2 + \frac{1}{k^2} = 0,$$

where

$$2A = \frac{1 + q^2}{1 + k'^2q^2} + \frac{1}{k^2} \cdot \frac{1 + k'^2q^2}{1 + q^2}, \quad 2B = q^2 + \frac{1}{k^2q^2}.$$

Similarly, for GH ,

$$q^2 = \frac{r^2 - p^2}{1 - k^2p^2r^2}, \quad 1 + q^2 = \frac{(1 - p^2) + r^2(1 - k^2p^2)}{1 - k^2p^2r^2},$$

$$1 + k'^2q^2 = \frac{1 - k^2p^2 + k'^2(1 - p^2)r^2}{1 - k^2p^2r^2}, \quad 1 - k^2q^2 = \frac{1 - k^2p^2r^2 - k^2r^2 + k^2p^2}{1 - k^2p^2r^2},$$

and consequently

$$x = \frac{p\sqrt{(1 - p^2) + r^2(1 - k^2p^2)}\sqrt{1 - k^2p^2 + k'^2(1 - p^2)r^2}}{1 - 2k^2p^2 + k^4p^4},$$

$$y = \frac{\sqrt{1 - p^2}\sqrt{1 - k^2p^2}\sqrt{p^2 - r^2}}{1 - k^2p^2},$$

giving x, y each of them in terms of r . And from the second of them we at once derive

$$(x^2 + y^2)^2 - 2Ax^2 - 2By^2 + \frac{1}{k^2} = 0,$$

where

$$2A = \frac{1 + 2k'^2p^2 + 2k^2p^2 + k^4p^4}{k^2(1 - p^2)}, \quad 2B = \frac{1 - 2k^2p^2 - k'^2(1 - 2k^2)p^2}{k'^2(1 + k^2p^2)}.$$

Consider, in particular, the case where the line $E'F'$ is the midway line $L'M'$: here $y' = \frac{1}{2}K'$, and thence $iq = \text{sn } iy' = \text{sn } \frac{1}{2}iK' = \frac{i}{\sqrt{k}}$, that is, $q = \frac{1}{\sqrt{k}}$: and we thence obtain $A = B = \frac{1}{k}$; the equation of the bicircular quartic is

$$(x^2 + y^2)^2 - \frac{2}{k}(x^2 + y^2) + \frac{1}{k^2} = 0,$$

viz. this is the circle $x^2 + y^2 - \frac{1}{k} = 0$ twice repeated. As a direct verification, observe that we have here

$$x + iy = \operatorname{sn} \left(x' + \frac{1}{2}iK' \right) = \frac{\frac{1+k}{\sqrt{k}} \operatorname{sn} x' + \frac{i}{\sqrt{k}} \operatorname{cn} x' \operatorname{dn} x'}{1 + k \operatorname{sn}^2 x'},$$

and hence

$$x^2 + y^2 = \frac{(1+k)^2 \operatorname{sn}^2 x' + (1 - (1+k')^2 + k'^2)p^2}{k(1 + 2k \operatorname{sn}^2 x' + k^2 \operatorname{sn}^4 x')} = \frac{1}{k},$$

as it should be.

Reverting to the equation

$$x + iy = \operatorname{sn}(x' + iy'),$$

I write successively

$$y' = \frac{1}{2}K' - z', \quad \operatorname{sn} iy' = iq_1 = \operatorname{sn} i(\frac{1}{2}K' - z'),$$

and

$$y' = \frac{1}{2}K' + z', \quad \operatorname{sn} iy' = iq_2 = \operatorname{sn} i(\frac{1}{2}K' + z');$$

we then have

$$iq_1 \cdot iq_2 = \operatorname{sn} i(\frac{1}{2}K' - z') \operatorname{sn} i(\frac{1}{2}K' + z'), = -\frac{1}{k},$$

that is, $q_1 q_2 = \frac{1}{k}$; hence for q writing q_1 or q_2 , we have in each case the same values of A and B ; that is, we have the same bicircular quartic corresponding to the lines $E'F'$ and $E'_1F'_1$ equidistant from the line $L'M'$: but to one of these lines there corresponds the quadrant lying inside, to the other that lying outside, the circular quadrant

$$x^2 + y^2 - \frac{1}{k} = 0.$$

The curve

$$(x^2 + y^2)^2 - 2Ax^2 - 2By^2 + \frac{1}{k^2} = 0$$

is in four different ways the envelope of a variable circle: viz. the circle may have its centre on a conic $\alpha x^2 + \beta y^2 - 1 = 0$, and cut at right angles one of the circles

$$x^2 + y^2 \pm \frac{1}{k} = 0;$$

or it may have its centre on the axis of x , or on the axis of y . The circle, having its centre on either axis, cuts this axis at right angles; but this condition being *ipso facto* satisfied, we do not thereby determine the radius of the circle having for its centre a given point on the axis: the expression for the radius must be sought for independently.

Write for shortness

$$\lambda^2 = \frac{B - \frac{1}{k^2}}{A - B},$$

and consider the circle

$$x^2 + y^2 - 2ax = \lambda \sqrt{A - B - 2a^2} + B,$$

where a is a variable parameter. Differentiating, we have

$$x = \frac{\lambda a}{\sqrt{A - B - 2a^2}}, \quad \text{giving} \quad a = \frac{x \sqrt{A - B}}{\sqrt{2x^2 + \lambda^2}},$$

the equation of the circle then gives

$$x^2 + y^2 - B = \frac{\lambda}{a}(2x^2 + \lambda^2), = \frac{\sqrt{A - B} \sqrt{2x^2 + \lambda^2}}{x};$$

or we have

$$(x^2 + y^2)^2 - 2Ax^2 - 2By^2 + \frac{1}{k^2} = 0$$

as the envelope of the variable circle

$$x^2 + y^2 - 2ax = \lambda\sqrt{A - B - 2a^2} + B,$$

having its centre on the axis of x .

And similarly, the same quartic curve is the envelope of the variable circle

$$x^2 + y^2 - 2by = \frac{\sqrt{A - \frac{1}{k^2}}}{\sqrt{B - A}} \sqrt{B - A - 2b^2} + A,$$

having its centre on the axis of y .

If we have in like manner the equation $x + iy = \text{cn}(x' + iy')$, then writing, as before, $\text{sn } x' = p$, $\text{sn } iy' = iq$, we find

$$r^2 = \frac{(1 - p^2)(1 + q^2)}{(1 + k^2 p^2 q^2)^2}, \quad \text{where } r^2 = x^2 + y^2.$$

For the curve EF corresponding to a line $E'F'$ parallel to the axis of x' , we have to eliminate p from these equations; the expression for r^2 gives

$$p^2 = \frac{-(1 - k^2)q^2 + r^2(1 + k'^2 q^2)}{1 + k'^2 q^2 + k^2 q^2 r^2}, \quad 1 - p^2 = \frac{(1 + q^2)(1 - k^2 r^2)}{1 + k'^2 q^2 + k^2 q^2 r^2},$$

$$1 - k'^2 p^2 = \frac{(1 - k^2)(1 + q^2)}{1 + k'^2 q^2 + k^2 q^2 r^2}, \quad 1 + k'^2 p^2 q^2 = \frac{1 + 2k'^2 q^2 + k'^2 q^2 r^2}{1 + k'^2 q^2 + k^2 q^2 r^2},$$

and thence

$$x = \frac{\sqrt{1 + q^2} \sqrt{-(1 - k^2)q^2 + r^2(1 + k'^2 q^2)} \sqrt{1 + k'^2 q^2 + k^2 q^2 r^2}}{1 + 2k'^2 q^2 + k'^2 q^2},$$

$$y = \frac{q \sqrt{1 + q^2 - r^2} \sqrt{1 + k'^2 q^2} \sqrt{1 - k^2 + k^2 r^2(1 + q^2)}}{1 + 2k'^2 q^2 + k'^2 q^2},$$

from which we deduce

$$(x^2 + y^2)^2 - 2Ax^2 - 2By^2 - \frac{k'^2}{k^2} = 0,$$

where

$$2A = \frac{-1 + 2k^2 + 2k'^2 q^2 + k^2 q^2}{k^2(1 + q^2)}, \quad 2B = \frac{1 - 2k^2 - k'^2(1 - 2k^2)q^2}{k'^2(1 + k^2 q^2)},$$

which is a bicircular quartic of the foregoing form.

Similarly, for the curve GH corresponding to a line $G'H'$ parallel to the axis of y' , we have to eliminate q : the expression for r^2 gives

$$q^2 = \frac{-1 + p^2 + r^2}{1 - k'^2 p^2 - k^2 p^2 r^2}, \quad 1 + q^2 = \frac{(1 - k^2 p^2)(1 - p^2 r^2)}{1 - k'^2 p^2 - k^2 p^2 r^2},$$

$$1 + k'^2 q^2 = \frac{(1 - k^2)(1 - p^2)}{1 - k'^2 p^2 - k^2 p^2 r^2}, \quad 1 - k^2 q^2 = \frac{1 - 2k^2 p^2 + k^2 p^2 r^2}{1 - k'^2 p^2 - k^2 p^2 r^2},$$

Hence

$$x = \frac{p \sqrt{-1 + p^2 + r^2} \sqrt{1 - k^2 p^2} \sqrt{1 - k^2 + k^2 r^2(1 - p^2)}}{1 - 2k^2 p^2 + k^2 p^2},$$

$$y = \frac{\sqrt{1 - p^2} \sqrt{(1 - k^2)p^2 + r^2(1 - k^2 p^2)} \sqrt{1 - k^2 p^2 - k^2 p^2 r^2}}{1 - 2k^2 p^2 + k^2 p^2},$$

from which we deduce

$$(x^2 + y^2)^2 - 2Ax^2 - 2By^2 - \frac{k'^2}{k^2} = 0,$$

where

$$2A = \frac{-1 + 2k^2 + 2k'^2p^2 + k^2p^2}{k^2(1 - p^2)}, \quad 2B = \frac{1 - 2k^2p^2 - k'^2(1 - 2k^2)p^2}{k'^2(1 - k^2p^2)},$$

which is again a bicircular quartic of the foregoing form. And we have the like results for the equation

$$x + iy = \operatorname{dn}(x' + iy');$$

so that, for the sn , cn , and dn , the curve EF or GH is in each case a biaxial bicircular quartic of the form

$$(x^2 + y^2)^2 - 2Ax^2 - 2By^2 + C = 0.$$

892.

NOTE ON THE ORTHOMORPHIC TRANSFORMATION OF A CIRCLE INTO ITSELF.

[From the Proceedings of the Edinburgh Mathematical Society, vol. viii. (1890), pp. 91, 92.]

THE following is, of course, substantially well known, but it strikes me as rather pretty:—to find the orthomorphic transformation of the circle

$$x^2 + y^2 - 1 = 0$$

into itself. Assume this to be

$$x_1 + iy_1 = \frac{A(x + iy) + B}{1 + C(x + iy)}.$$

Then, writing A' , B' , C' for the conjugates of A , B , C , we have

$$x_1 - iy_1 = \frac{A'(x - iy) + B'}{1 + C'(x - iy)},$$

and then

$$x_1^2 + y_1^2 = \frac{AA'(x^2 + y^2) + AB'(x + iy) + A'B(x - iy) + BB'}{1 + C(x + iy) + C'(x - iy) + CC'(x^2 + y^2)},$$

which should be an identity for $x^2 + y^2 = 1$, $x_1^2 + y_1^2 = 1$.

Evidently $C = AB'$, whence $C' = A'B$; and the equation then is

$$1 + AA'BB' = AA' + BB',$$

that is,

$$(1 - AA')(1 - BB') = 0.$$

But $BB' = 1$ gives the illusory result

$$x_1 + iy_1 = B;$$

therefore

$$AA' = 1,$$

and the required solution thus is

$$x_1 + iy_1 = \frac{A(x + iy) + B}{1 + AB'(x + iy)},$$

where A is a unit-vector (say $A = \cos \chi + i \sin \chi$) and B , B' are conjugate vectors. Or, writing $B = b + i\beta$, $B' = b - i\beta$, the constants are χ , b , β ; three constants as it should be.

893.

THE BITANGENTS OF THE QUINTIC.

(LETTER FROM PROFESSOR CAYLEY TO MR HEAL.)

[From the *Annals of Mathematics*, vol. v. (1890), pp. 109, 110.]

DEAR SIR,

I have to thank you very much for your letter concerning the bitangents of the quintic, and am glad you have obtained more simple numerical coefficients than you had at first. I do not see my way to verifying your result, but assuming it to be correct, it is a very interesting and valuable one. The relation to Salmon's *Higher Plane Curves* (p. 351) is rather curious, viz. the result there given is of the deg-order 18-54, with an extraneous factor $(\alpha x + \beta y + \gamma z)^6$, the rejection of which would reduce it to the proper form, deg-order 18-48; whereas yours is of deg-order 24-66, with an extraneous factor H^2 , the rejection of which would reduce it to the same proper form, deg-order 18-48.

But there is another known solution, not by any means a handy one, but which has no extraneous factor*, viz. if

$$\Omega = D^2H - 4DH_1 + 6H_2,$$

then the required equation is

$$D^2\Omega = 0,$$

where the facients of the reciprocant are the first derived functions,

$$\partial_x U, \partial_y U, \partial_z U.$$

Say this is

$$(Ax^4 - \dots)(\partial_x U)^4 = 0;$$

the coefficients A, B, C , &c., are those of Ω , viz. these are of deg-order 3-6, and $\partial_x U$, &c., are of deg-order 1-4; so that the deg-order is

$$4(3-6) + 6(1-4) = (12-24) + (6-24) = (18-48),$$

as it should be. It might be worth while to further consider this form, but I doubt whether, practically, the whole question is not too difficult to be worth working at.

I remain, dear sir, yours very sincerely,

A. CAYLEY.

Cambridge, January 17, 1890.

* See *Phil. Trans.*, vol. CXLIX. pp. 193—212, [260].

894.

THE INVESTIGATION BY WALLIS OF HIS EXPRESSION FOR π .

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xxiii. (1889), pp. 165—169.]

THE following is in effect the investigation by Wallis in the *Arithmetica Infinitorum* (Oxford, 1656) of his well-known expression for π . He obtains the series of equations which in modern

notation are

$$\begin{aligned}\int_0^1 (x - x^2)^0 dx &= 1, \\ \int_0^1 (x - x^2)^1 dx &= \frac{1}{2} \cdot \frac{1}{3}, \\ \int_0^1 (x - x^2)^2 dx &= \frac{1}{3} \cdot \frac{1}{5}, \\ \int_0^1 (x - x^2)^3 dx &= \frac{1}{20} \cdot \frac{1}{7},\end{aligned}$$

or say, in general,

$$\int_0^1 (x - x^2)^n dx = \frac{1}{\phi(n)} \cdot \frac{1}{2n+1}.$$

In the case $n = \frac{1}{2}$, the integral gives the area of a semicircle, diameter = 1, viz. this is $= \frac{1}{8}\pi$, or we have $\frac{1}{8}\pi = \frac{1}{2\phi(\frac{1}{2})}$; or writing with him $\square = \frac{4}{\pi}$, this is $\square = \phi(\frac{1}{2})$, where $\phi(0)$, $\phi(1)$, $\phi(2)$, ... are the series of numbers 1, 2, 6, 20, ..., which are in fact the middle binomial coefficients of the several even powers, and are the diagonal numbers in the table which he subsequently considers. As regards the form in which Wallis exhibits these results, his theorem CXXXIII. is:

Si exponatur series Primanorum mulctata serie secundanorum, Residuorum Quadrata, Cubi, Biquadrata, &c. ad seriem Æqualium rationem habebunt cognitam.

And then, after an algebraical calculation,

$$\begin{aligned}\frac{1}{2} - \frac{1}{3} &= \frac{1}{6}, & \frac{1}{3} - \frac{2}{4} \cdot \frac{1}{5} &= \frac{1}{30}, & \frac{1}{4} - \frac{3}{5} \cdot \frac{3}{6} \cdot \frac{1}{7} &= \frac{1}{140}, \\ \frac{1}{2 \cdot 3} &= \frac{1}{6}, & \frac{1 \cdot 2}{3 \cdot 4 \cdot 5} &= \frac{1}{30}, & \frac{1 \cdot 2 \cdot 3}{4 \cdot 5 \cdot 6 \cdot 7} &= \frac{1}{140},\end{aligned}$$

... et sic deinceps; continue multiplicando Numeratores per numeros quadratos et denominatores per binos continue sequentes arithmetice proportionales—viz. he thus explains the law of his calculated numbers.

He then forms the table

∞	1	$\frac{1}{2}\square$	$\frac{3}{8}$	$\frac{3}{8}\square$	$\frac{5}{16}$	$\frac{15}{48}\square$	$-\frac{1}{2}$
1	1	1	1	1	1	1	0
$\frac{1}{2}\square$	1	$\square$	$\frac{3}{2}$	$\frac{3}{2}\square$	$\frac{15}{8}$	$\frac{105}{48}\square$	$\frac{1}{2}$
$\frac{3}{8}$	$\frac{1}{2}$	$\frac{3}{2}$	2	$\frac{5}{2}$	3	$\frac{7}{2}$	1
$\frac{3}{8}\square$	$\frac{1}{3}$	$\frac{3}{2}\square$	$\frac{5}{2}$	$\frac{15}{4}\square$	$\frac{105}{16}$	$\frac{315}{48}\square$	$\frac{3}{2}$
$\frac{5}{16}$	$\frac{1}{4}$	$\frac{15}{8}$	3	$\frac{105}{16}$	10	$\frac{945}{48}$	2
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

where the figures in the second column are = 1, those in the fourth column are $= \frac{n+1}{1}$, those of the sixth column are $= \frac{(n+1)(n+2)}{1 \cdot 2}$, and so on: n being in each case the rank in the column as shown by the right-hand outside column, viz. these ranks are $-\frac{1}{2}$, 0, $\frac{1}{2}$, 1, ... at intervals of $\frac{1}{2}$. And the several even lines contain the same figures as the several even columns respectively.

Wallis then remarks that, if in any line the first and second terms respectively are A and 1, then the even lines are as follows, viz. the second line is

$$A, 1, A^2_1, 1^3_2, A^{2 \cdot 4}_{1 \cdot 3}, 1^{3 \cdot 5}_{2 \cdot 4}, A^{2 \cdot 4 \cdot 6}_{1 \cdot 3 \cdot 5}, \dots$$

the fourth line is

$$A, 1, A_{\frac{3}{1}}, 1_{\frac{4}{2}}, A_{\frac{3.5}{1.3}}, 1_{\frac{4.6}{2.4}}, A_{\frac{3.5.7}{1.3.5}}, \dots$$

the sixth line is

$$A, 1, A_{\frac{4}{1}}, 1_{\frac{5}{2}}, A_{\frac{4.6}{1.3}}, 1_{\frac{5.7}{2.4}}, A_{\frac{4.6.8}{1.3.5}}, \dots$$

and this being so, he completes the table by inserting a term $\square$ in the diagonal line between the 1 and 2, and calculating the odd lines as follows: viz. if in each case A is the first term and B the second term of the line, then the complete rule is: the first line is

$$A, 1, A_{\frac{0}{1}}, 1_{\frac{2}{1}}, A_{\frac{0.2}{1.3}}, 1_{\frac{2.4}{1.3}}, A_{\frac{0.2.4}{1.3.5}}, \dots$$

the second line is

$$A, 1, A_{\frac{2}{1}}, 1_{\frac{3}{2}}, A_{\frac{2.4}{1.3}}, 1_{\frac{3.5}{2.4}}, A_{\frac{2.4.6}{1.3.5}}, \dots$$

the third line is

$$A, 1, A_{\frac{2}{1}}, 1_{\frac{4}{3}}, A_{\frac{2.4}{1.3}}, 1_{\frac{4.6}{3.5}}, A_{\frac{2.4.6}{1.3.5}}, \dots$$

the fourth line is

$$A, 1, A_{\frac{3}{1}}, 1_{\frac{4}{2}}, A_{\frac{3.5}{1.3}}, 1_{\frac{4.6}{2.4}}, A_{\frac{3.5.7}{1.3.5}}, \dots$$

the fifth line is

$$A, 1, A_{\frac{4}{1}}, 1_{\frac{5}{3}}, A_{\frac{4.6}{1.3}}, 1_{\frac{5.7}{3.5}}, A_{\frac{4.6.8}{1.3.5}}, \dots$$

the sixth line is

$$A, 1, A_{\frac{5}{1}}, 1_{\frac{6}{2}}, A_{\frac{5.7}{1.3}}, 1_{\frac{6.8}{2.4}}, A_{\frac{5.7.9}{1.3.5}}, \dots$$

and so on, the rule for the odd lines being thus an interpolation from that for the even lines. Observe that, in the first line A is $= \infty$, and that, in the several odd terms after the first, A is $= 1$.

The table thus calculated (I have given here the fractions in their least terms) is

∞	1	$\frac{1}{2}\square$	$\frac{3}{8}$	$\frac{3}{8}\square$	$\frac{5}{16}$	$\frac{15}{48}\square$	$\frac{35}{128}$	$-\frac{1}{2}$
1	1	1	1	1	1	1	1	0
$\frac{1}{2}\square$	1	$\square$	$\frac{3}{2}$	$\frac{3}{2}\square$	$\frac{15}{8}$	$\frac{105}{48}\square$	$\frac{315}{128}$	$\frac{1}{2}$
$\frac{3}{8}$	$\frac{1}{2}$	$\frac{3}{2}$	2	$\frac{5}{2}$	3	$\frac{7}{2}$	4	1
$\frac{3}{8}\square$	$\frac{1}{3}$	$\frac{3}{2}\square$	$\frac{5}{2}$	$\frac{15}{4}\square$	$\frac{105}{16}$	$\frac{315}{48}\square$	$\frac{3465}{384}$	$\frac{3}{2}$
$\frac{5}{16}$	$\frac{1}{4}$	$\frac{15}{8}$	3	$\frac{105}{16}$	10	$\frac{945}{48}$	35	2
$\frac{15}{48}\square$	$\frac{1}{5}$	$\frac{105}{48}\square$	7	$\frac{315}{48}\square$	$\frac{945}{48}$	$\frac{9009}{128}\square$	$\frac{19305}{384}$	$\frac{5}{2}$
$\frac{35}{128}$	$\frac{1}{6}$	$\frac{315}{128}$	4	$\frac{3465}{384}$	35	$\frac{19305}{384}$	70	3
$-\frac{1}{2}$	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3	

in which table $\square$, as the term interpolated between the diagonal terms 1, 2, denotes the value $\frac{4}{\pi}$ as before.

The third line of the table is

$$\frac{1}{2}\square, 1, \square, \frac{3}{2}, \frac{3}{2}\square, \frac{15}{8}, \frac{105}{48}\square, \frac{315}{128}, \dots$$

or, in terms involving $\square$:

$$\frac{1}{2}\square, 1, \frac{4}{1.3}\square, \frac{2}{3}, \frac{2.4}{1.3.5}\square, \frac{2.4}{3.5}, \frac{2.4.6}{1.3.5.7}\square, \frac{2.4.6}{3.5.7}, \dots$$

The successive even terms continually increase but tend to equality, and in like manner the successive odd terms continually increase but tend to equality; it seems to have been assumed that, for any three consecutive terms x, y, z , we have $\frac{y}{x} > \frac{z}{y}$, that is, $y^2 > xz$. Taking this to be so, we have

$$1 > \frac{1}{2}\square, \quad \square^2 > \frac{8}{3}, \quad \frac{3}{2^2}\square^2 > \frac{8.16}{3.5}, \quad \frac{3^2.5}{2^2.4^2}\square^2 > \frac{8.16.24}{3.5.7}, \dots$$

and these equations give $\square$

$$\begin{aligned} &\text{less than } \sqrt{2}, \sqrt{\frac{2 \cdot 4}{3}}, \sqrt{\frac{2 \cdot 4 \cdot 4 \cdot 6}{3 \cdot 3 \cdot 5}}, \sqrt{\frac{2 \cdot 4 \cdot 4 \cdot 6 \cdot 6 \cdot 8}{3 \cdot 3 \cdot 5 \cdot 5 \cdot 7}}, \dots \\ &\text{greater than } \sqrt{\frac{8}{3}}, \sqrt{\frac{8 \cdot 16}{3 \cdot 5}}, \sqrt{\frac{8 \cdot 16 \cdot 24}{3 \cdot 5 \cdot 7}}, \dots \end{aligned}$$

limits which tend continually to equality. We thus have

$$\square = \frac{4}{\pi}, = \frac{3 \cdot 3 \cdot 5 \cdot 5 \cdot 7 \cdot 7 \dots}{2 \cdot 4 \cdot 4 \cdot 6 \cdot 6 \cdot 8 \dots},$$

the number of factors in the numerator being always equal to the number in the denominator, and the accuracy of the approximation increasing with the number of factors.

It is to be remarked that for a square; row m and column n ; m or $n = -\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ as before; the term of the square is in general $\Pi(m+n) \div \Pi(m) \Pi(n)$; thus $m = n = \frac{1}{2}$, the term is $\Pi(1) \div \{\Pi(\frac{1}{2})\}^2, = 1 \div (\frac{1}{2}\sqrt{\pi})^2 = \frac{4}{\pi}, = \square$; $m = 3, n = \frac{1}{2}$ it is $\Pi(\frac{7}{2}) \div \Pi(3) \Pi(\frac{1}{2})$, $= \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi} \div 6\sqrt{\pi}, = \frac{35}{32}$; and so in any other case.

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OF ARTHUR CAYLEY, Sc.D., F.R.S.

LATE SADLERIAN PROFESSOR OF PURE MATHEMATICS
IN THE UNIVERSITY OF CAMBRIDGE.

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NOTE ON THE HYDRODYNAMICAL EQUATIONS.

[From the *Proceedings of the Royal Society of Edinburgh*, vol. xv. (1889), pp. 342–344.]

WRITING for shortness $D = \frac{d}{dt} + u \frac{d}{dx} + v \frac{d}{dy} + w \frac{d}{dz}$, then if from the hydrodynamical equations

$$Du = \frac{d}{dx} \left(V - \frac{P}{\rho} \right), \quad Dv = \frac{d}{dy} \left(V - \frac{P}{\rho} \right), \quad Dw = \frac{d}{dz} \left(V - \frac{P}{\rho} \right),$$

without the aid of the equation

$$\frac{du}{dx} + \frac{dv}{dy} + \frac{dw}{dz} = 0,$$

we eliminate $V - \frac{P}{\rho}$, we obtain equations not equivalent to those of Helmholtz,

$$D\xi = \left(\xi \frac{d}{dx} + \eta \frac{d}{dy} + \zeta \frac{d}{dz} \right) u, \quad \&c.,$$

($2\xi, 2\eta, 2\zeta = \frac{dv}{dz} - \frac{dw}{dy}, \frac{dw}{dx} - \frac{du}{dz}, \frac{du}{dy} - \frac{dv}{dx}$, as usual), but which, transforming them by means of the omitted equation, agree as they should do with his equations. But the form of the equations obtained directly by elimination as above, is an interesting one, which it is worth while to give.

We have

$$\begin{aligned} D \left(\frac{dv}{dz} - \frac{dw}{dy} \right) &= D \left(\frac{dv}{dz} - \frac{dw}{dy} \right) - \frac{d}{dz} Dv + \frac{d}{dy} Dw, \\ &= \left(\frac{d}{dt} \left(\frac{dv}{dz} - \frac{dw}{dy} \right) \right) + \left(\frac{dv}{dz} - \frac{dw}{dy} \right) \frac{d}{dx} \left(\frac{dw}{dy} \right) \\ &\quad - \frac{d}{dz} \left(\frac{dv}{dt} + u \frac{dv}{dx} + v \frac{dv}{dy} + w \frac{dv}{dz} \right) \\ &\quad + \frac{d}{dy} \left(\frac{dw}{dt} + u \frac{dw}{dx} + v \frac{dw}{dy} + w \frac{dw}{dz} \right), \end{aligned}$$

where the terms containing second derived functions disappear of themselves, and the expression on the right-hand is thus

$$= -\frac{du}{dz} \frac{dv}{dx} - \frac{dv}{dz} \frac{dv}{dy} - \frac{dw}{dz} \frac{dv}{dz} + \frac{du}{dy} \frac{dw}{dx} + \frac{dv}{dy} \frac{dw}{dy} + \frac{dw}{dy} \frac{dw}{dz}.$$

Representing for shortness the Matrix

$$\begin{vmatrix} \frac{du}{dx} & \frac{du}{dy} & \frac{du}{dz} \\ \frac{dv}{dx} & \frac{dv}{dy} & \frac{dv}{dz} \\ \frac{dw}{dx} & \frac{dw}{dy} & \frac{dw}{dz} \end{vmatrix} \quad \text{by} \quad \begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{vmatrix}, \quad \text{and its square by} \quad \begin{vmatrix} A, & B, & C \\ A', & B', & C' \\ A'', & B'', & C'' \end{vmatrix},$$

we have

$$\begin{vmatrix} A, & B, & C \\ A', & B', & C' \\ A'', & B'', & C'' \end{vmatrix} = \begin{vmatrix} \frac{du}{dx} & \frac{du}{dy} & \frac{du}{dz} \\ \frac{dv}{dx} & \frac{dv}{dy} & \frac{dv}{dz} \\ \frac{dw}{dx} & \frac{dw}{dy} & \frac{dw}{dz} \end{vmatrix} \left(\frac{du}{dx}, \frac{dv}{dx}, \frac{dw}{dx} \right), \left(\frac{du}{dy}, \frac{dv}{dy}, \frac{dw}{dy} \right), \left(\frac{du}{dz}, \frac{dv}{dz}, \frac{dw}{dz} \right),$$

viz. the combinations which enter into the foregoing formula are

$$C' = \frac{dv}{dx} \frac{du}{dz} + \frac{dv}{dy} \frac{dv}{dz} + \frac{dv}{dz} \frac{dw}{dz},$$

and

$$B'' = \frac{dw}{dx} \frac{du}{dy} + \frac{dw}{dy} \frac{dv}{dy} + \frac{dw}{dz} \frac{dw}{dy};$$

and the equation thus is $D(c' - b'') + C' - B'' = 0$; viz. the three equations are

$$\begin{aligned} D(c' - b'') + C' - B'' &= 0, \\ D(a'' - c) + A'' - C &= 0, \\ D(b - a') + B - A' &= 0, \end{aligned}$$

which are the equations in question.

Observe that we have

$$C' - B'' = (a', b', c')(c, c', c'') - (a'', b'', c'')(b, b', b''),$$

and thence, writing

$$\begin{aligned} \rho &= a(c' - b'') + b(a'' - c) + c(b - a'), \\ &= ac' - ab'' + a''b - a'c, \end{aligned}$$

we have

$$C' - B'' + \rho = (a + b' + c'')(c' - b'') = 0,$$

if $a + b' + c'' = 0$; viz. this being so, $C' - B'' = -\rho$, or the first equation is

$$D(c' - b'') = \rho, = a(c' - b'') + b(a'' - c) + c(b - a'),$$

that is, $D\xi = \xi \frac{du}{dx} + \eta \frac{du}{dy} + \zeta \frac{du}{dz}$, the first equation of Helmholtz, and we thus have the equations of Helmholtz, if $a + b' + c'' = 0$, that is, if $\frac{du}{dx} + \frac{dv}{dy} + \frac{dw}{dz} = 0$.

The foregoing three equations $D(c' - b'') + C' - B'' = 0$, &c., are the quaternion equation ($\sigma = i\xi + j\eta + k\zeta$, $\nabla = i\frac{d}{dx} + j\frac{d}{dy} + k\frac{d}{dz}$, $\frac{d}{dt} = D$, denotes a complete differentiation),

$$\frac{d}{dt} V \nabla \sigma = V \nabla \sigma_1 S \sigma_1 \nabla,$$

of Mr M'Aulay's paper "Some General Theorems in Quaternion Integration," *Messenger of Mathematics*, vol. XIV. (1884), pp. 26–37; see p. 34.

891.

**ON THE BINODAL QUARTIC AND THE GRAPHICAL
REPRESENTATION OF THE ELLIPTIC FUNCTIONS.**

[From the *Transactions of the Cambridge Philosophical Society*, vol. XIV. (1889), pp. 484–494. Read May 6, 1889.]

I APPROACH the subject from the question of the graphical representation of the elliptic functions: assuming as usual that the modulus is real, positive, and less than unity, and to fix the ideas considering the function sn (but the like considerations are applicable to the functions cn and dn), then the equation $x + iy = \text{sn}(x' + iy')$ establishes a $(1, 1)$ correspondence between the xy infinite quarter plane, and the $x'y'$ rectangle (sides K and K'): viz. to any given point $x + iy$, x and y each positive, there corresponds a single point $x' + iy'$, x' , y' each positive and less than K , K' respectively: and conversely to any such point $x' + iy'$, there corresponds a single point $x + iy$, x and y each positive.

I draw in the $x'y'$ -figure the rectangle $A'B'C'D'$ (sides K and K'), and in the xy -figure, I take on the axis of x , the points B , C where $AB = 1$, $AC = \frac{1}{k}$: and the point D at infinity. We have thus in the $x'y'$ -figure the closed curve or contour

$A'B'C'D'A'$: and corresponding hereto we have in the xy -figure the closed curve or contour $ABCD$, viz. here D is the point at infinity considered as a line always at infinity, extending from the point at infinity on the positive part of the axis of x , to the point at infinity on the positive part of the axis of y , the contour being thus AB, BC, CD (D at infinity on the axis of x); and then D (at infinity on the axis of y) A . And thus to a point P' describing successively the lines $A'B', B'C', C'D', D'A'$ there corresponds a point P describing successively the lines AB, BC, CD, DA : to P' at D' there corresponds P at D , viz. this is any point at infinity from D on the axis of x to D on the axis of y . There is no real breach of continuity: in further illustration, suppose that P' , instead of actually coming to D' , just cuts off the corner, viz. that it passes from a point γ' on $C'D'$ to a point α' on $D'A'$ (γ', α' each of them very near to D'): then P passes from a point γ very near D on the axis of x (that is, at a great distance from A) to a point α very near D on the axis of y (that is, at a great distance from A): and to the indefinitely small arc $\gamma'\alpha'$ described by P' there corresponds the indefinitely large arc $\gamma\alpha$ described by P .

We thus see that, if P' describe any arc $E'F'$ passing from a point E' of $A'D'$ to a point F' of $B'C'$, then P will describe an arc EF passing from a point E of AD to a point F of BC : and similarly, if P' describe any arc $G'H'$ passing from a point G' of $A'B'$ to a point H' of $C'D'$, then P will describe an arc GH passing from a point G of AB to a point H of CD .

Supposing $E'F'$ is a straight line parallel to $A'x'$, that is, cutting $A'D'$ and $B'C'$ each at right angles, then EF will be an arc cutting AD and BC each at right angles: and so if $G'H'$ is a straight line parallel to $A'y'$, that is, cutting $A'B'$ and $C'D'$ each at right angles, then GH will be an arc cutting AB and CD each at right angles: and moreover, since $E'F'$ and $G'H'$ cut each other at right angles, then also EF and GH cut each other at right angles.

Supposing, as above, that $E'F'$ and $G'H'$ are straight lines, we have EF and GH each of them the arc of a special bicircular quartic: the theory was in fact established in a very elegant manner in a memoir by Siebeck, "Ueber eine Gattung von Curven vierten Grades, welche mit den elliptischen Functionen zusammenhängen," *Crelle*, t. LVII. (1860), pp. 359–370, and t. LIX. (1861), pp. 173–184.

In particular, if P' describe the line $L'M'$ lying halfway between $A'B'$ and $D'C'$ (that is, if $A'L' = \frac{1}{2}K'$), then P will describe the circular quadrant LM , radius $\frac{1}{\sqrt{k}}$, viz. in this case the bicircular quartic degenerates into a circle twice repeated: and so if P' describe successively the lines $E'F'$ and $E'_1F'_1$ equidistant from $L'M'$ ($AE' = \frac{1}{2}K' - \eta$, $AE'_1 = \frac{1}{2}K' + \eta$), then P will describe the arcs EF and E_1F_1 , which are the images of each other in regard to the centre A and circular quadrant LM , and which together constitute the quadrant of one and the same bicircular quartic.

A bicircular quartic is of course a binodal quartic with the circular points at infinity for the two nodes: there is no real gain of generality in considering the

binodal quartic rather than the bicircular quartic, but I have preferred to do so, and I have accordingly introduced the term Binodal Quartic into the title of the present Memoir. I present in a compendious form the properties of the general curve, and I show how the curve is to be particularised so as to obtain from it the special bicircular quartics which present themselves as above in the graphical representation of the elliptic functions.

A binodal quartic has the Plückerian numbers

$$\begin{array}{cccccc} m & n & \delta & \kappa & \tau & \iota \\ = 4 & 8 & 2 & 0 & 8 & 12. \end{array}$$

The number of tangents to the curve which can be drawn from either of the nodes is $n - 4$, $= 4$; and the pencil of tangents from the one node is homographic with the pencil of tangents from the other node. Call the nodes I and J : and let the tangents from I be called (a, b, c, d) and those from J be called (a', b', c', d') , then if the tangents which correspond to (a, b, c, d) respectively are (a', b', c', d') , they may also be taken to be (b, a, d, c) or (c, d, a, b) or (d, c, b, a) : and considering the intersections of corresponding tangents, we have thus four tetrads of points, say the f -points, such that the points of each tetrad lie in a conic through the two nodes: and we have consequently four conics each passing through the two nodes, say these are the f -conics.

Starting as above with the correspondence (a, b, c, d) , (a', b', c', d') , if the intersections of a and a' , b and b' , c and c' , d and d' are called A, B, C, D respectively, then we have A, B, C, D for a tetrad of f -points, lying on the f -conic $(ABCD)$.

Writing AB for the two points, the intersections of IA, JB and of IB, JA respectively, and so in other cases, then the remaining three tetrads of f -points are

$$\begin{array}{llll} AB, CD & \text{lying on the } f\text{-conic } (AB, CD), \\ AC, BD & \text{„ „ } (AC, BD), \\ AD, BC & \text{„ „ } (AD, BC). \end{array}$$

The two points AB may be spoken of as the antipoints of A, B : and so in other cases.

Any two of the f -conics have in common the nodes I, J , and they therefore intersect in two points besides: at each of these the tangents to the two conics, and the lines to I, J respectively, form a harmonic pencil.

Consider the two tangents at I and the two tangents at J : we have a conic touching these four lines and passing through the tetrad of f -points, or what is the same thing, intersecting the f -conic in the four f -points: say this is a c -conic. There are thus four c -conics corresponding to the four f -conics respectively.

We may consider a variable conic, passing through the points I and J ; and such that the tangents thereto at these points respectively meet on a point of a

c-conic: the variable conic and the corresponding *f*-conic each pass through the points *I*, *J* and they meet in two points besides: and the variable conic may be such that at each of these the tangents to the variable conic and the *f*-conic form with the lines drawn to the points *I* and *J* a harmonic pencil. The variable conic, as thus defined, contains a single variable parameter; and it has for its envelope the binodal quartic: the binodal quartic is thus in four different ways the envelope of a variable conic. This is of course Casey's Theorem for the fourfold generation of the bicircular quartic as the envelope of a variable circle.

One of the *f*-conics, say $(ABCD)$, may break up into the line IJ and a line, say the axis $(ABCD)$: we have then the four *f*-points A, B, C, D on this line. The other *f*-conics say (AB, CD) , (AC, BD) , (AD, BC) are proper conics as before: any one of these meets the line $(ABCD)$ in two points: and at each of these the line and the tangent to the conic form with the lines drawn to the points *I* and *J* a harmonic pencil. The binodal quartic is in this case said to have an axis.

But a second *f*-conic, say (AD, BC) , may break up into the line IJ and a line, say the axis (AD, BC) : we have then the four *f*-points AD, BC on this line. Writing moreover A', D' for the two points AD , or say $(AI, DJ) = A'$, $(AJ, DI) = D'$; and similarly B', C' for the two points BC , or say $(BI, CJ) = B'$, $(BJ, CI) = C'$; then the four *f*-points are A', B', C', D' , and the axis (AD, BC) may be called $(A'B'C'D')$.

The relation of the *f*-points A, B, C, D and A', B', C', D' and of the two axes is as shown in the figure: taking X for the intersection of IJ with $(ABCD)$ and Y for that of IJ with $(A'B'C'D')$, also Z for the intersection of the two axes, then X, Z are the sibiconjugate points of the involution AD, BC and Y, Z the sibiconjugate points of the involution $A'D', B'C'$. The two axes intersect in Z , and form with the lines ZI, ZJ a pencil in involution.

The two remaining *f*-conics (AB, CD) and (AC, BD) , or as they might also be called $(A'B', C'D')$ and $(A'C', B'D')$, are proper conics as before: they touch each other at the points *I*, *J*, and have for their common tangents at these points the lines ZI ,

ZJ respectively. Each conic meets each axis in two points; and at each of these points the axis and the tangent to the conic form with the lines to I, J a harmonic pencil. The binodal quartic is in this case said to be biaxial. The point Z , which is the intersection of the two axes, may be called the centre.

In the case where an f -conic breaks up into the line IJ , and a line containing four f -points, say an f -line, the corresponding c -conic coincides with the f -conic, viz. it also breaks up into the line IJ and the f -line: the variable conic is a conic through the points I, J such that the tangents thereto at these points respectively meet on the f -line. Moreover, the variable conic must be such that at each of its intersections with the f -conic, that is, the f -line, the tangent of the variable conic and the f -line must be harmonics in regard to the lines drawn from the point to the points I, J respectively: but this condition is satisfied ipso facto for each of the intersections of the variable conic and the f -line. This depends on the theorem that, taking on a conic any three points P, I, J , then the tangent at P and the line drawn from P to the pole of IJ are harmonics in regard to the lines PI, PJ . Thus we have only three conditions for the variable conic, or, as above defined, it would in the case in question (of four f -points in a line) depend upon two variable parameters. There is really another condition—but what this in general is I have not ascertained: and this being so the variable conic in the case in question (of the four f -points in a line) depends upon a single variable parameter, and we have as before the bicircular quartic as the envelope. The foregoing is the axial case; in the biaxial case, the same thing happens in regard to the variable conics belonging to the two axes respectively. Thus in every case we have the fourfold generation of the curve as the envelope of a variable conic: only in the axial case, the variable conics belonging to the axis, and in the biaxial case the variable conics belonging to the two axes respectively, are not by the foregoing definitions completely defined. It will be seen further on how, in the case of the biaxial bicircular quartic, we complete the definition of the variable circles belonging to the two axes respectively.

Taking the points I, J to be the circular points at infinity, we have a bicircular quartic. The f -points are the foci, and the f -conics are circles, viz. we have 16 foci situate in fours upon four focal circles. The harmonic relation of two lines to the lines through I, J means of course that the lines cut at right angles; hence the focal circles cut each other at right angles: this must certainly be a known property, but it is not mentioned in Salmon's *Higher Plane Curves*, Ed. 3, Dublin, 1879, and I cannot find it in Darboux or Casey: it is given No. 81 in Lachlan's Memoir "On Systems of Circles and Spheres," *Phil. Trans.*, vol. CLXXVII. (1886), and I find it as a question in the *Educational Times*, March 1, 1889, 10034 (Prof. Morley). "Prove that, of the four focal circles of a circular cubic or bicircular quartic, any two are orthogonal, and the radii are connected by the relation $\Sigma(\pm\delta) = 0$." The theorem is not as well known as it should be.

The c -conics are confocal conics having for their real foci the so-called double-foci of the quartic (more accurately, the common foci are the four quadruple foci of the quartic); we have thus four conics corresponding to the four focal circles respectively,

each conic intersecting the corresponding circle in the four foci upon this circle. And we have then the quartic as the envelope of a variable circle having its centre upon one of these conics and cutting at right angles the corresponding focal circle: the bicircular quartic is thus generated in four different ways.

Instead of one of the focal circles, we may have a line or axis, and the quartic is then said to be axial: the foci on the axis may be any four points; and for a real curve they may be all real, or two real and two imaginary, or all four imaginary. The remaining focal circles are real or imaginary circles, cut by the axis at right angles, that is, having their centres on the axis, and cutting each other at right angles.

But instead of another of the focal circles, we may have a line or axis, and the quartic is then said to be biaxial: the two axes cut at right angles at a point which may be called the centre of the curve. The foci on each axis form pairs of points situate symmetrically in regard to the centre. If on one of the axes the foci are real, then on the other axis they form two imaginary conjugate pairs; and conversely: but if on one of the axes the foci are two of them real and the other two conjugate imaginaries, then this is so for the other axis also. There are thus only the two cases: 1°, foci on the one axis real, and on the other conjugate imaginaries; 2°, foci on each axis two of them real and the other two conjugate imaginaries: there is however a limiting case where on each axis two foci are united at the centre, the other two foci being real on the one axis and conjugate imaginaries on the other. The remaining two focal circles are real or imaginary circles, cutting each axis at right angles, that is, having their centres at the centre; and cutting each other at right angles, that is, having the sum of the squares of their radii = 0.

The biaxial form of bicircular quartic is, in fact, that which presents itself in the theory of the representation of the elliptic functions.

I consider for a moment the case of a variable circle having its centre upon a given line, and cutting at right angles a given circle. The variable circles pass all of them through two fixed points, the antipoints of the intersections of the given line and circle, and which are thus real or imaginary according as the intersections of the given line and circle are imaginary or real. Hence, considering any one variable circle and the consecutive variable circle, these intersect in two real points, when the given line does not meet the given circle (meets it in two imaginary points); but when the given line meets the given circle in two real points, then the two variable circles intersect in two imaginary points: if however the given line touches the given circle, then the two variable circles touch each other. Taking now the curve of centres to be any given curve whatever, and considering one of the variable circles, and the consecutive variable circle, it at once appears that, if the tangent to the curve of centres at the centre of the variable circle does not meet the given circle, then the two variable circles intersect in two real points (which, if the tangent touch the given circle, unite in a single real point): but if the tangent to the curve of centres meets the given circle, then the two variable circles do not intersect. It hence appears that the real portions of the envelope arise exclusively

from those portions of the curve of centres which are such that at any point thereof the tangent to the curve of centres does not meet the given circle. In particular, if the given circle be real, and the curve of centres is a real ellipse enclosing the given circle, then the real portion of the envelope arises from the whole ellipse: but if the curve of centres be a real ellipse cutting the given circle in four real points, then drawing the four common tangents of the ellipse and circle, it is at once seen that there are on the ellipse two detached portions such that, at any point of either portion, the tangent to the ellipse does not meet the circle: and the real portions of the envelope arise exclusively from these portions of the ellipse.

In the case just referred to, there are on the ellipse four portions each lying outside the circle and terminating in the four intersections respectively of the ellipse and circle, such that at a point of any one of these portions the tangent to the ellipse meets the circle in two real points. Starting from the extremity of one of these portions of the ellipse and proceeding to the other extremity on the circle, the corresponding variable circles do not intersect each other, but each of them is a circle lying wholly inside that which immediately precedes it; and the variable circle becomes ultimately a point, viz. this point is a focus of the curve: this agrees with the foregoing statement that the f -conic intersects the circle in the four foci upon this circle. For the two portions of the ellipse which lie inside the circle, the variable circle is of course always imaginary. The like considerations apply to the case where the locus of the centre of the variable circle is a hyperbola or parabola. The foregoing remarks illustrate the actual generation of a bicircular quartic as the envelope of the variable circle.

Starting now from the equation $x + iy = \text{sn}(x' + iy')$, we have

$$x = \frac{\text{sn } x' \text{ cn } iy' \text{ dn } iy'}{1 - k^2 \text{sn}^2 x' \text{sn}^2 iy'}, \quad iy = \frac{\text{sn } iy' \text{ cn } x' \text{ dn } x'}{1 - k^2 \text{sn}^2 x' \text{sn}^2 iy'},$$

or putting

$$\text{sn } x' = p, \quad \text{sn } iy' = iq;$$

these equations are

$$x = \frac{p\sqrt{1+q^2}\sqrt{1+k^2q^2}}{1+k^2p^2q^2}, \quad y = \frac{q\sqrt{1-p^2}\sqrt{1-k^2p^2}}{1+k^2p^2q^2},$$

whence also

$$x^2 + y^2 = \frac{p^2 + q^2}{1 + k^2p^2q^2} = r^2, \quad (\text{if } x^2 + y^2 \text{ be put } = r^2).$$

These equations, considering therein q as a given constant, and p as a variable parameter, determine the curve EF : and considering p as given constant, and q as a variable parameter, they determine the curve GH . But the eliminations are easily effected; we have

$$p^2(1 - k^2q^2r^2) = r^2 - q^2, \quad q^2(1 - k^2p^2r^2) = r^2 - p^2.$$

Hence, for EF ,

$$p^2 = \frac{r^2 - q^2}{1 - k^2 q^2 r^2}, \quad 1 - p^2 = \frac{1 + q^2 - (1 + k^2 q^2) r^2}{1 - k^2 q^2 r^2},$$

$$1 - k^2 p^2 = \frac{1 + k^2 q^2 - k^2 (1 + q^2) r^2}{1 - k^2 q^2 r^2}, \quad 1 + k^2 p^2 q^2 = \frac{1 - k^2 q^4}{1 - k^2 q^2 r^2},$$

and consequently

$$x = \frac{\sqrt{1 + q^2} \sqrt{1 + k^2 q^2} \sqrt{r^2 - q^2}}{1 - k^2 q^4},$$

$$y = \frac{q \sqrt{1 + q^2 - (1 + k^2 q^2) r^2} \sqrt{1 + k^2 q^2 - k^2 (1 + q^2) r^2}}{1 - k^2 q^4},$$

giving x and y each of them in terms of r . And from the first of these we at once derive

$$(x^2 + y^2)^2 - 2Ax^2 - 2By^2 + \frac{1}{k^2} = 0,$$

where

$$2A = \frac{1 + q^2}{1 + k^2 q^2} + \frac{1}{k^2} \frac{1 + k^2 q^2}{1 + q^2}, \quad 2B = q^2 + \frac{1}{k^2 q^2}.$$

Similarly, for GH ,

$$q^2 = \frac{r^2 - p^2}{1 - k^2 p^2 r^2}, \quad 1 + q^2 = \frac{(1 - p^2) + r^2(1 - k^2 p^2)}{1 - k^2 p^2 r^2},$$

$$1 + k^2 q^2 = \frac{1 - k^2 p^2 + k^2 r^2(1 - p^2)}{1 - k^2 p^2 r^2}, \quad 1 + k^2 p^2 q^2 = \frac{1 - k^2 p^4}{1 - k^2 p^2 r^2},$$

and consequently

$$x = \frac{p \sqrt{(1 - p^2) + (1 - k^2 p^2) r^2} \sqrt{1 - k^2 p^2 + k^2 (1 - p^2) r^2}}{1 - k^2 p^4},$$

$$y = \frac{\sqrt{1 - p^2} \sqrt{1 - k^2 p^2} \sqrt{r^2 - p^2}}{1 - k^2 p^4},$$

giving x, y each of them in terms of r . And from the second of them we at once derive

$$(x^2 + y^2)^2 - 2Ax^2 - 2By^2 + \frac{1}{k^2} = 0,$$

where

$$2A = \frac{1}{k^2 p^2} + p^2, \quad 2B = \frac{1 - p^2}{1 - k^2 p^2} + \frac{1 - k^2 p^2}{k^2 (1 - p^2)}.$$

Consider, in particular, the case where the line EF is the midway line LM : here $y' = \frac{1}{2}K'$, and thence $iq = \operatorname{sn} iy' = \operatorname{sn} \frac{1}{2}iK' = \frac{i}{\sqrt{k}}$, that is, $q = \frac{1}{\sqrt{k}}$: and we thence obtain $A = B = \frac{1}{k}$; the equation of the bicircular quartic is

$$(x^2 + y^2)^2 - \frac{2}{k}(x^2 + y^2) + \frac{1}{k^2} = 0,$$

viz. this is the circle $x^2 + y^2 - \frac{1}{k} = 0$ twice repeated. As a direct verification, observe that we have here

$$x + iy = \operatorname{sn}(x' + \tfrac{1}{2}iK') = \frac{\frac{1}{\sqrt{k}} \operatorname{sn} x' + \frac{i}{\sqrt{k}} \operatorname{cn} x' \operatorname{dn} x'}{1 + \frac{1}{k} \operatorname{sn}^2 x'} = \frac{(1+k)p + i\sqrt{1-p^2}\sqrt{1-k^2p^2}}{\sqrt{k}(1+kp^2)},$$

and hence

$$x^2 + y^2 = \frac{(1+k)^2p^2 + 1 - (1+k^2)p^2 + k^2p^4}{k(1+2kp^2+k^2p^4)} = \frac{1}{k},$$

as it should be.

Reverting to the equation $x + iy = \operatorname{sn}(x' + iy')$, I write successively

$$y' = \tfrac{1}{2}K' - z', \quad \operatorname{sn} iy' = iq_1 = \operatorname{sn} i(\tfrac{1}{2}K' - z'),$$

and

$$y' = \tfrac{1}{2}K' + z', \quad \operatorname{sn} iy' = iq_2 = \operatorname{sn} i(\tfrac{1}{2}K' + z');$$

we then have

$$iq_1 \cdot iq_2 = \operatorname{sn} i(\tfrac{1}{2}K' - z') \operatorname{sn} i(\tfrac{1}{2}K' + z') = -\frac{1}{k},$$

that is, $q_1q_2 = \frac{1}{k}$; hence for q writing q_1 or q_2 , we have in each case the same values of A and B ; that is, we have the same bicircular quartic corresponding to the lines $E'F'$ and $E'_1F'_1$, equidistant from the line $L'M'$: but to one of these lines there corresponds the quadrant lying inside, to the other that lying outside, the circular quadrant

$$x^2 + y^2 - \frac{1}{k} = 0.$$

The curve

$$(x^2 + y^2)^2 - 2Ax^2 - 2By^2 + \frac{1}{k^2} = 0$$

is in four different ways the envelope of a variable circle: viz. the circle may have its centre on a conic $\alpha x^2 + \beta y^2 - 1 = 0$, and cut at right angles one of the circles

$$x^2 + y^2 \pm \frac{1}{k} = 0;$$

or it may have its centre on the axis of x , or on the axis of y . The circle, having its centre on either axis, cuts this axis at right angles; but this condition being ipso facto satisfied, we do not thereby determine the radius of the circle having for its centre a given point on the axis: the expression for the radius must be sought for independently.

Write for shortness

$$\lambda^2 = \frac{B - \frac{1}{k^2}}{A - B},$$

and consider the circle

$$x^2 + y^2 - 2ax = \lambda\sqrt{A - B - 2a^2} + B,$$

where a is a variable parameter. Differentiating, we have

$$x = \frac{\lambda a}{\sqrt{A - B - 2a^2}}, \quad \text{giving } a = \frac{x\sqrt{A - B}}{\sqrt{2x^2 + \lambda^2}};$$

the equation of the circle then gives

$$x^2 + y^2 - \frac{2}{\lambda}(2a^2 + \lambda^2) = B^2 + \lambda^2(A - B), \quad = -\frac{1}{k^2};$$

that is,

$$(x^2 + y^2)^2 - 2Ax^2 - 2By^2 + \frac{1}{k^2} = 0$$

as the envelope of the variable circle

$$x^2 + y^2 - 2ax = \frac{\sqrt{B - \frac{1}{k^2}}}{\sqrt{A - B}} \sqrt{A - B - 2a^2} + B,$$

having its centre on the axis of x .

And similarly, the same quartic curve is the envelope of the variable circle

$$x^2 + y^2 - 2\beta y = \frac{\sqrt{A - \frac{1}{k^2}}}{\sqrt{B - A}} \sqrt{B - A - 2\beta^2} + A,$$

having its centre on the axis of y .

If we have in like manner the equation $x + iy = \text{cn}(x' + iy')$, then writing, as before,

$$\text{sn } x' = p, \quad \text{sn } iy' = iq,$$

we find

$$x = \frac{(1 - p^2)(1 + q^2)}{(1 + k^2 p^2 q^2)^2}, \quad y = \frac{p^2 q^2 (1 - k^2 p^2)(1 + k^2 q^2)}{(1 + k^2 p^2 q^2)^2},$$

and hence

$$r^2 = \frac{1 - p^2 + q^2 - k^2 p^2 q^2}{1 + k^2 p^2 q^2}, \quad \text{where } r^2 = x^2 + y^2.$$

For the curve EF corresponding to a line $E'F'$ parallel to the axis of x' , we have to eliminate p from these equations; the expression for r^2 gives

$$p^2 = \frac{1 + q^2 - r^2}{1 + k'^2 q^2 + k^2 q^2 r^2}, \quad 1 - p^2 = \frac{-(1 - k^2)q^2 + r^2(1 + k^2 q^2)}{1 + k'^2 q^2 + k^2 q^2 r^2},$$

$$1 - k^2 p^2 = \frac{(1 - k^2) + k^2 r^2(1 + q^2)}{1 + k'^2 q^2 + k^2 q^2 r^2}, \quad 1 + k^2 p^2 q^2 = \frac{1 + 2k^2 q^2 + k^2 q^4}{1 + k'^2 q^2 + k^2 q^2 r^2},$$

and thence

$$x = \frac{\sqrt{1+q^2}\sqrt{-(1-k^2)q^2+r^2(1+k^2q^2)}\sqrt{1+k'^2q^2+k^2q^2r^2}}{1+2k^2q^2+k^2q^4},$$

$$y = \frac{q\sqrt{1+q^2-r^2}\sqrt{1+k^2q^2}\sqrt{1-k^2+k^2r^2(1+q^2)}}{1+2k^2q^2+k^2q^4},$$

from which we deduce

$$(x^2 + y^2)^2 - 2Ax^2 - 2By^2 - \frac{k'^2}{k^2} = 0,$$

where

$$2A = \frac{-1 + 2k^2 + 2k^2q^2 + k^2q^4}{k^2(1+q^2)}, \quad 2B = \frac{-1 - 2k'^2q^2 + k^2(1-2k^2)q^4}{k^2q^2(1+k'^2q^2)},$$

which is a bicircular quartic of the foregoing form.

Similarly, for the curve GH corresponding to a line $G'H'$ parallel to the axis of y' , we have to eliminate q : the expression for r^2 gives

$$q^2 = \frac{-1 + p^2 + r^2}{1 - k^2p^2 - k^2p^2r^2}, \quad 1 + q^2 = \frac{(1 - k^2)p^2 + r^2(1 - k^2p^2)}{1 - k^2p^2 - k^2p^2r^2},$$

$$1 + k^2q^2 = \frac{(1 - k^2) - k^2r^2(1 - p^2)}{1 - k^2p^2 - k^2p^2r^2}, \quad 1 + k^2p^2q^2 = \frac{1 - 2k^2p^2 + k^2p^4}{1 - k^2p^2 - k^2p^2r^2}.$$

Hence

$$x = \frac{\sqrt{1-p^2}\sqrt{(1-k^2)p^2+r^2(1-k^2p^2)}\sqrt{1-k^2p^2-k^2p^2r^2}}{1-2k^2p^2+k^2p^4},$$

$$y = \frac{p\sqrt{-1+p^2+r^2}\sqrt{1-k^2p^2}\sqrt{1-k^2+k^2r^2(1-p^2)}}{1-2k^2p^2+k^2p^4},$$

from which we deduce

$$(x^2 + y^2)^2 - 2Ax^2 - 2By^2 - \frac{k'^2}{k^2} = 0,$$

where

$$2A = \frac{-1 + 2k^2 - 2k^2p^2 + k^2p^4}{k^2(1-p^2)}, \quad 2B = \frac{1 - 2k^2p^2 - k^2(1-2k^2)p^4}{k^2p^2(1-k^2p^2)},$$

which is again a bicircular quartic of the foregoing form. And we have the like results for the equation

$$x + iy = \operatorname{dn}(x' + iy');$$

so that, for the sn , cn , and dn , the curve EF or GH is in each case a biaxial bicircular quartic of the form

$$(x^2 + y^2)^2 - 2Ax^2 - 2By^2 + C = 0.$$

892.

NOTE ON THE ORTHOMORPHIC TRANSFORMATION OF A CIRCLE INTO ITSELF.

[From the *Proceedings of the Edinburgh Mathematical Society*, vol. VIII. (1890), pp. 91, 92.]

THE following is, of course, substantially well known, but it strikes me as rather pretty:—to find the orthomorphic transformation of the circle

$$x^2 + y^2 - 1 = 0$$

into itself. Assume this to be

$$x_1 + iy_1 = \frac{A(x + iy) + B}{1 + C(x + iy)}.$$

Then, writing A' , B' , C' for the conjugates of A , B , C , we have

$$x_1 - iy_1 = \frac{A'(x - iy) + B'}{1 + C'(x - iy)};$$

and then

$$x_1^2 + y_1^2 = \frac{AA'(x^2 + y^2) + AB'(x + iy) + A'B(x - iy) + BB'}{1 + C(x + iy) + C'(x - iy) + CC'(x^2 + y^2)},$$

which should be an identity for $x^2 + y^2 = 1$, $x_1^2 + y_1^2 = 1$.

Evidently $C = AB'$, whence $C' = A'B$; and the equation then is

$$1 + AA'BB' = AA' + BB',$$

that is,

$$(1 - AA')(1 - BB') = 0.$$

But $BB' = 1$ gives the illusory result

$$x_1 + iy_1 = B,$$

therefore

$$AA' = 1,$$

and the required solution thus is

$$x_1 + iy_1 = \frac{A(x + iy) + B}{1 + A'B(x + iy)},$$

where A is a unit-vector (say $A = \cos \lambda + i \sin \lambda$) and B , B' are conjugate vectors. Or, writing $B = b + i\beta$, $B' = b - i\beta$, the constants are λ , b , β ; three constants as it should be.

893.

THE BITANGENTS OF THE QUINTIC.

(LETTER FROM PROFESSOR CAYLEY TO MR HEAL.)

[From the *Annals of Mathematics*, vol. v. (1890), pp. 109, 110.]

DEAR SIR,

I have to thank you very much for your letter concerning the bitangents of the quintic, and am glad you have obtained more simple numerical coefficients than you had at first. I do not see my way to verifying your result, but assuming it to be correct, it is a very interesting and valuable one. The relation to Salmon's *Higher Plane Curves* (p. 351) is rather curious, viz. the result there given is of the deg-order 18-54, with an extraneous factor $(\alpha x + \beta y + \gamma z)^6$, the rejection of which would reduce it to the proper form, deg-order 18-48; whereas yours is of deg-order 24-66, with an extraneous factor H^2 , the rejection of which would reduce it to the same proper form, deg-order 18-48.

But there is another known solution, not by any means a handy one, but which has no extraneous factor¹, viz. if

$$\Omega = D^3H - 4D^2H_1 + 6DH_2,$$

then the required equation is

$$F\Omega = 0,$$

where the facients of the reciprocant are the first derived functions,

$$\partial_x U, \partial_y U, \partial_z U.$$

Say this is

$$(D^4U - \dots)(\partial_x U)^4 = 0;$$

the coefficients A, B, C , &c., are those of Ω , viz. these are of deg-order 3-6, and $\partial_x U$, &c., are of deg-order 1-4; so that the deg-order is

$$4(3-6) + 6(1-4) = (12-24) + (6-24) = (18-48),$$

as it should be. It might be worth while to further consider this form, but I doubt whether, practically, the whole question is not too difficult to be worth working at.

I remain, dear sir, yours very sincerely,

A. CAYLEY.

Cambridge, January 17, 1890.

¹See *Phil. Trans.*, vol. CXLIX. pp. 193-212, [260].

894.

THE INVESTIGATION BY WALLIS OF HIS EXPRESSION FOR π .

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XXIII. (1889), pp. 165–169.]

THE following is in effect the investigation by Wallis in the *Arithmetica Infinitorum* (Oxford, 1656) of his well-known expression for π . He obtains the series of equations which in modern notation are

$$\begin{aligned}\int_0^1 (x - x^2)^0 dx &= 1, \\ \int_0^1 (x - x^2)^1 dx &= \frac{1}{2} \cdot \frac{1}{3}, \\ \int_0^1 (x - x^2)^2 dx &= \frac{2}{3} \cdot \frac{1}{5}, \\ \int_0^1 (x - x^2)^3 dx &= \frac{2 \cdot 3}{4 \cdot 5} \cdot \frac{1}{7},\end{aligned}$$

or say, in general,

$$\int_0^1 (x - x^2)^n dx = \frac{1}{\phi(n)} \cdot \frac{1}{2n+1}.$$

In the case $n = \frac{1}{2}$, the integral gives the area of a semicircle, diameter = 1, viz. this is $= \frac{1}{8}\pi$, or we have $\frac{1}{8}\pi = \frac{1}{2\phi(\frac{1}{2})} \cdot \frac{1}{2}$; or writing with him $\square = \frac{4}{\pi}$, this is $\square = \phi(\frac{1}{2})$, where $\phi(0)$, $\phi(1)$, $\phi(2)$, ... are the series of numbers 1, 2, 6, 20, ..., which are in fact the middle binomial coefficients of the several even powers, and are the diagonal numbers in the table which he subsequently considers. As regards the form in which Wallis exhibits these results, his theorem CXXXIII. is:

Si exponatur series Primanorum mulctata serie secundanorum, Residuorum Quadrata, Cubi, Biquadrata, &c. ad seriem Æqualium rationem habebunt cognitam.

And then, after an algebraical calculation,

$$\frac{1}{2} - \frac{1}{3} = \frac{1}{6}, \quad \frac{1}{3} - \frac{2}{4} \cdot \frac{1}{5} = \frac{1}{30}, \quad \frac{1}{4} - \frac{3}{5} \cdot \frac{3}{6} \cdot \frac{1}{7} = \frac{1}{140},$$

$$\frac{1}{2 \cdot 3} = \frac{1}{6}, \quad \frac{1 \cdot 2}{3 \cdot 4 \cdot 5} = \frac{1}{30}, \quad \frac{1 \cdot 2 \cdot 3}{4 \cdot 5 \cdot 6 \cdot 7} = \frac{1}{140},$$

$$\frac{1}{2 \cdot 3} = \frac{1}{6}, \quad \frac{1 \cdot 4}{2 \cdot 3 \cdot 4 \cdot 5} = \frac{1}{30}, \quad \frac{1 \cdot 4 \cdot 9}{2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7} = \frac{1}{140},$$

... et sic deinceps; continue multiplicando Numeratores per numeros quadratos et denominatores per binos continue sequentes arithmetice proportionates—viz. he thus explains the law of his calculated numbers.

He then forms the table

	1	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{15}{8}$	$\frac{15}{8}$	$\frac{105}{48}$	$\frac{105}{48}$	$\frac{945}{384}$	
1	1	1	1	1	1	1	1	1	0
$\frac{3}{2}$	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3	$\frac{7}{2}$	4	$\frac{9}{2}$	$\frac{1}{2}$
$\frac{3}{2}$	1	2	$\frac{15}{4}$	6	$\frac{35}{4}$	12	$\frac{63}{4}$	20	1
$\frac{15}{8}$	1	$\frac{5}{2}$	6	$\frac{35}{2}$	30	$\frac{99}{2}$	75	$\frac{231}{2}$	$\frac{3}{2}$
$\frac{15}{8}$	1	3	$\frac{35}{4}$	30	$\frac{315}{4}$	165	$\frac{3003}{8}$	715	2
$\frac{105}{48}$	1	$\frac{7}{2}$	12	$\frac{99}{2}$	165	$\frac{1287}{2}$	2002	$\frac{19305}{4}$	$\frac{5}{2}$
$\frac{105}{48}$	1	4	$\frac{63}{4}$	75	$\frac{3003}{8}$	2002	$\frac{19305}{4}$	12870	3
$\frac{945}{384}$	1	$\frac{9}{2}$	20	$\frac{231}{2}$	715	$\frac{19305}{4}$	12870	$\frac{19305}{740}$	$\frac{7}{2}$
	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3	$\frac{7}{2}$	

where the figures in the second column are = 1, those in the fourth column are = $\frac{n+1}{1}$, those of the sixth column are = $\frac{(n+1)(n+2)}{1 \cdot 2}$, and so on: n being in each case the rank in the column as shown by the right-hand outside column, viz. these ranks are $-\frac{1}{2}, 0, \frac{1}{2}, 1, \dots$ at intervals of $\frac{1}{2}$. And the several even lines contain the same figures as the several even columns respectively.

Wallis then remarks that, if in any line the first and second terms respectively are A and 1, then the even lines are as follows, viz. the second line is

$$A, 1, A \cdot \frac{2}{1}, \frac{1 \cdot 2}{1 \cdot 2}, A \cdot \frac{2 \cdot 4}{1 \cdot 3}, \frac{1 \cdot 3}{1 \cdot 3}, A \cdot \frac{2 \cdot 4 \cdot 6}{1 \cdot 3 \cdot 5}, \frac{1 \cdot 3 \cdot 5}{1 \cdot 3 \cdot 5}, A \cdot \frac{2 \cdot 4 \cdot 6 \cdot 8}{1 \cdot 3 \cdot 5 \cdot 7}, \dots$$

the fourth line is

$$A, 1, A \cdot \frac{3}{1}, \frac{1 \cdot 4}{1 \cdot 2}, A \cdot \frac{3 \cdot 5}{1 \cdot 3}, \frac{1 \cdot 4 \cdot 6}{1 \cdot 3 \cdot 5}, A \cdot \frac{3 \cdot 5 \cdot 7}{1 \cdot 3 \cdot 5}, \frac{1 \cdot 4 \cdot 6 \cdot 8}{1 \cdot 3 \cdot 5 \cdot 7}, A \cdot \frac{3 \cdot 5 \cdot 7 \cdot 9}{1 \cdot 3 \cdot 5 \cdot 7}, \dots$$

the sixth line is

$$A, 1, A \cdot \frac{5}{1}, \frac{1 \cdot 6}{1 \cdot 2}, A \cdot \frac{5 \cdot 7}{1 \cdot 3}, \frac{1 \cdot 6 \cdot 8}{1 \cdot 2 \cdot 4}, A \cdot \frac{5 \cdot 7 \cdot 9}{1 \cdot 3 \cdot 5}, \frac{1 \cdot 6 \cdot 8 \cdot 10}{1 \cdot 2 \cdot 4 \cdot 6}, \dots$$

and this being so, he completes the table by inserting a term $\square$ in the diagonal line between the 1 and 2, and calculating the odd lines as follows: viz. if in each case A is the first term and B the second term of the line, then the complete rule is:

the first line is

$$A, 1, A \cdot \frac{0}{1}, \frac{1}{2}, A \cdot \frac{0 \cdot 2}{1 \cdot 3}, \frac{1 \cdot 3}{2 \cdot 4}, A \cdot \frac{0 \cdot 2 \cdot 4}{1 \cdot 3 \cdot 5}, \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}, \dots$$

the second line is

$$A, 1, A \cdot \frac{1}{1}, \frac{2}{2}, A \cdot \frac{1 \cdot 3}{1 \cdot 3}, \frac{2 \cdot 4}{2 \cdot 4}, A \cdot \frac{1 \cdot 3 \cdot 5}{1 \cdot 3 \cdot 5}, \frac{2 \cdot 4 \cdot 6}{2 \cdot 4 \cdot 6}, \dots$$

the third line is

$$A, 1, A \cdot \frac{2}{1}, \frac{3}{2}, A \cdot \frac{2 \cdot 4}{1 \cdot 3}, \frac{3 \cdot 5}{2 \cdot 4}, A \cdot \frac{2 \cdot 4 \cdot 6}{1 \cdot 3 \cdot 5}, \frac{3 \cdot 5 \cdot 7}{2 \cdot 4 \cdot 6}, \dots$$

the fourth line is

$$A, 1, A \cdot \frac{3}{1}, \frac{4}{2}, A \cdot \frac{3 \cdot 5}{1 \cdot 3}, \frac{4 \cdot 6}{2 \cdot 4}, A \cdot \frac{3 \cdot 5 \cdot 7}{1 \cdot 3 \cdot 5}, \frac{4 \cdot 6 \cdot 8}{2 \cdot 4 \cdot 6}, \dots$$

the fifth line is

$$A, 1, A \cdot \frac{4}{1}, \frac{5}{2}, A \cdot \frac{4 \cdot 6}{1 \cdot 3}, \frac{5 \cdot 7}{2 \cdot 4}, A \cdot \frac{4 \cdot 6 \cdot 8}{1 \cdot 3 \cdot 5}, \frac{5 \cdot 7 \cdot 9}{2 \cdot 4 \cdot 6}, \dots$$

and so on, the rule for the odd lines being thus an interpolation from that for the even lines. Observe that, in the first line $A \cdot \frac{0}{1}$ is $= \infty$, and that, in the several odd terms after the first, $A \cdot$ is $= 1$.

The table thus calculated (I have given here the fractions in their least terms) is

∞	1	$\frac{1}{2}\square$	$\frac{1}{2}$	$\frac{3}{8}\square$	$\frac{3}{8}$	$\frac{5}{16}\square$	$\frac{5}{16}$	$\frac{35}{128}\square$	$\frac{35}{128}$	$-\frac{1}{2}$
1	1	1	1	1	1	1	1	1	1	0
$\frac{1}{2}\square$	1	$\square$	$\frac{3}{2}$	$\frac{3}{2}\square$	$\frac{15}{8}$	$\frac{15}{8}\square$	$\frac{35}{16}$	$\frac{35}{16}\square$	$\frac{315}{128}$	$\frac{1}{2}$
$\frac{1}{2}$	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3	$\frac{7}{2}$	4	$\frac{9}{2}$	5	1
$\frac{3}{8}\square$	1	$\frac{3}{2}\square$	$\frac{5}{2}$	$\frac{15}{8}\square$	$\frac{35}{8}$	$\frac{105}{48}\square$	$\frac{63}{8}$	$\frac{315}{128}\square$	$\frac{1155}{128}$	$\frac{3}{2}$
$\frac{3}{8}$	1	$\frac{15}{8}$	3	$\frac{35}{8}$	6	$\frac{99}{2}$	10	$\frac{99}{8}$	15	2
$\frac{5}{16}\square$	1	$\frac{15}{8}\square$	$\frac{7}{2}$	$\frac{105}{48}\square$	$\frac{63}{8}$	$\frac{945}{384}\square$	$\frac{231}{16}$	$\frac{3465}{384}\square$	$\frac{3003}{128}$	$\frac{5}{2}$
$\frac{5}{16}$	1	$\frac{35}{16}$	4	$\frac{63}{8}$	10	$\frac{231}{16}$	20	$\frac{6435}{128}$	35	3
$\frac{35}{128}\square$	1	$\frac{35}{16}\square$	$\frac{9}{2}$	$\frac{315}{128}\square$	$\frac{99}{8}$	$\frac{3465}{384}\square$	$\frac{6435}{128}$	$\frac{45045}{12288}\square$	$\frac{19305}{740}$	$\frac{7}{2}$
$\frac{35}{128}$	1	$\frac{315}{128}$	5	$\frac{1155}{128}$	15	$\frac{3003}{128}$	35	$\frac{19305}{740}$	70	4
	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3	$\frac{7}{2}$	

in which table $\square$, as the term interpolated between the diagonal terms 1, 2, denotes the value $\frac{4}{\pi}$ as before.

The third line of the table is

$$\frac{1}{2}\square, 1, \square, \frac{3}{2}, \frac{4}{2}\square, \frac{3 \cdot 5}{2 \cdot 4}, \frac{4 \cdot 6}{3 \cdot 5}\square, \frac{3 \cdot 5 \cdot 7}{2 \cdot 4 \cdot 6}, \frac{4 \cdot 6 \cdot 8}{3 \cdot 5 \cdot 7}\square, \frac{3 \cdot 5 \cdot 7 \cdot 9}{2 \cdot 4 \cdot 6 \cdot 8}, \dots$$

The successive even terms continually increase but tend to equality, and in like manner the successive odd terms continually increase but tend to equality; it seems to have been assumed that, for any three consecutive terms x, y, z , we have

$$\frac{x}{y} > \frac{y}{z}, \quad \text{that is, } y^2 > xz.$$

Taking this to be so, we have

$$1 > \frac{1}{2}\square^2, \quad \square^2 > \frac{3}{2} \cdot \frac{3}{2}, \quad \frac{3^2}{2^2} > \square^2 \cdot \frac{4}{2} \cdot \frac{4}{2},$$

$$\frac{4^2}{2^2} \cdot \square^2 > \frac{3^2}{2^2} \cdot \frac{3 \cdot 5}{2 \cdot 4}, \quad \frac{3^2 \cdot 5^2}{2^2 \cdot 4^2} > \square^2 \cdot \frac{4}{2} \cdot \frac{4 \cdot 6}{3 \cdot 5},$$

and these equations give $\square$

less than	greater than
$\sqrt{\frac{2}{1}}$	$\sqrt{\frac{3}{2}}$
$\frac{3^2}{2 \cdot 4} \sqrt{\frac{4}{3}}$	$\frac{3^2}{2 \cdot 4} \sqrt{\frac{5}{4}}$
$\frac{3^2 \cdot 5^2}{2 \cdot 4^2 \cdot 6} \sqrt{\frac{6}{5}}$	$\frac{3^2 \cdot 5^2}{2 \cdot 4^2 \cdot 6} \sqrt{\frac{7}{6}}$
$\frac{3^2 \cdot 5^2 \cdot 7^2}{2 \cdot 4^2 \cdot 6^2 \cdot 8} \sqrt{\frac{8}{7}}$	$\frac{3^2 \cdot 5^2 \cdot 7^2}{2 \cdot 4^2 \cdot 6^2 \cdot 8} \sqrt{\frac{9}{8}}$

limits which tend continually to equality. We thus have

$$\square = \frac{4}{\pi} = \frac{3 \cdot 3 \cdot 5 \cdot 5 \cdot 7 \cdot 7 \dots}{2 \cdot 4 \cdot 4 \cdot 6 \cdot 6 \cdot 8 \dots},$$

the number of factors in the numerator being always equal to the number in the denominator, and the accuracy of the approximation increasing with the number of factors.

It is to be remarked that for a square; row m and column n ; m or $n = -\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ as before; the term of the square is in general $\Pi(m+n) \div \Pi(m) \Pi(n)$; thus $m = n = \frac{1}{2}$, the term is $\Pi(1) \div \{\Pi(\frac{1}{2})\}^2 = 1 \div (\frac{1}{2}\pi)^2 = \frac{4}{\pi} = \square$; $m = 3, n = \frac{1}{2}$ it is $\Pi(\frac{7}{2}) \div \Pi(3) \Pi(\frac{1}{2}) = \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \div 6 = \frac{35}{16}$; and so in any other case.

895.

A THEOREM ON TREES.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XXIII. (1889), pp. 376–378.]

THE number of trees which can be formed with $n + 1$ given knots $\alpha, \beta, \gamma, \dots$ is $= (n + 1)^{n-1}$; for instance $n = 3$, the number of trees with the 4 given knots $\alpha, \beta, \gamma, \delta$ is $4^2 = 16$, for in the first form shown in the figure the $\alpha, \beta, \gamma, \delta$ may be arranged

$$12 + 4 = 16$$

in 12 different orders ($\alpha\beta\gamma\delta$ being regarded as equivalent to $\delta\gamma\beta\alpha$), and in the second form any one of the 4 knots $\alpha, \beta, \gamma, \delta$ may be in the place occupied by the α : the whole number is thus $12 + 4 = 16$.

Considering for greater clearness a larger value of n , say $n = 5$, I state the particular case of the theorem as follows:

$$\text{No. of trees } (\alpha, \beta, \gamma, \delta, \epsilon, \zeta) = \text{No. of terms of } (\alpha + \beta + \gamma + \delta + \epsilon + \zeta)^4 \alpha\beta\gamma\delta\epsilon\zeta = 6^4 = 1296,$$

and it will be at once seen that the proof given for this particular case is applicable for any value whatever of n .

I use for any tree whatever the following notation: for instance, in the first of the forms shown in the figure, the branches are $\alpha\beta, \beta\gamma, \gamma\delta$; and the tree is said to be $\alpha\beta^2\gamma^2\delta$ (viz. the knots α, δ occur each once, but β, γ each twice); similarly in the second of the same forms, the branches are $\alpha\beta, \alpha\gamma, \alpha\delta$, and the tree is said

to be $\alpha^3\beta\gamma\delta$ (viz. the knot α occurs three times, and the knots β, γ, δ each once). And so in other cases.

This being so, I write

$$\begin{array}{rcccl}
 & 1 \alpha^4 & 6 & & 6 \\
 & + 4 \alpha^3\beta & 30 & & 120 \\
 (\alpha + \beta + \gamma + \delta + \epsilon + \zeta)^4 \alpha\beta\gamma\delta\epsilon\zeta = & + 6 \alpha^2\beta^2 & 15 & \left. \vphantom{\begin{array}{l} 1 \alpha^4 \\ + 4 \alpha^3\beta \\ + 6 \alpha^2\beta^2 \\ + 12 \alpha^2\beta\gamma \\ + 24 \alpha\beta\gamma\delta \end{array}} \right\} \alpha\beta\gamma\delta\epsilon\zeta; & 90 \\
 & + 12 \alpha^2\beta\gamma & 60 & & 720 \\
 & + 24 \alpha\beta\gamma\delta & 15 & & 360 \\
 & & & & \hline
 & & & & 1296,
 \end{array}$$

where the numbers of the left-hand column are the polynomial coefficients for the index 4; and the numbers of the right-hand column are the numbers of terms of the several types, 6 terms α^4 , 30 terms $\alpha^3\beta$, 15 terms $\alpha^2\beta^2$, &c.: the products of the corresponding terms of the two columns give the outside column 6, 120, 90, &c.; and the sum of these numbers is of course $6^4 = 1296$.

It is to be shown that we have

$$\begin{aligned}
 & 1 \text{ tree } \alpha^4 \cdot \alpha\beta\gamma\delta\epsilon\zeta (= \alpha^5\beta\gamma\delta\epsilon\zeta); \quad 4 \text{ trees } \alpha^3\beta \cdot \alpha\beta\gamma\delta\epsilon\zeta (= \alpha^4\beta^2\gamma\delta\epsilon\zeta), \dots, \\
 & 24 \text{ trees } \alpha\beta\gamma\delta \cdot \alpha\beta\gamma\delta\epsilon\zeta (= \alpha^2\beta^2\gamma^2\delta^2\epsilon\zeta) :
 \end{aligned}$$

for this being so, then by the mere interchange of letters, the numbers 1, 4, 6, ... of the left-hand column have to be multiplied by the numbers 6, 30, 15, ... of the right-hand column, and we have the numbers in the outside column, the sum of which is = 1296 as above.

Start with the last term $\alpha\beta\gamma\delta \cdot \alpha\beta\gamma\delta\epsilon\zeta = \alpha^2\beta^2\gamma^2\delta^2\epsilon\zeta (= \epsilon\alpha \cdot \alpha\beta \cdot \beta\gamma \cdot \gamma\delta \cdot \delta\zeta)$. We have the trees where the $\alpha, \beta, \gamma, \delta$ may be written in any one of the 24 orders, and the number of such trees is thus = 24. If we consider only the 12 orders ($\alpha\beta\gamma\delta$ being regarded as equivalent to $\delta\gamma\beta\alpha$), then the ϵ, ζ may be interchanged; and the number is thus $2 \times 12 = 24$ as before.

Now for the δ of $\alpha\beta\gamma\delta$ substitute α , or consider the form $\alpha\beta\gamma\alpha \cdot \alpha\beta\gamma\delta\epsilon\zeta = \alpha^3\beta^2\gamma^2\delta\epsilon\zeta$. We see at once in the form $\epsilon\alpha \cdot \alpha\beta \cdot \beta\gamma \cdot \gamma\delta \cdot \delta\zeta$, which one it is of the two δ 's that must be changed into α : in fact, changing the first δ , we should have $\epsilon\alpha \cdot \alpha\beta \cdot \beta\gamma \cdot \gamma\alpha \cdot \delta\zeta$ which contains a circuit $\alpha\beta\gamma$, and a detached branch $\delta\zeta$, and is thus not a tree: changing the second δ , we have $\epsilon\alpha \cdot \alpha\beta \cdot \beta\gamma \cdot \gamma\delta \cdot \alpha\zeta$ which is a tree $\alpha^3\beta^2\gamma^2\delta\epsilon\zeta = \alpha\zeta \cdot \alpha\epsilon \cdot \alpha\beta \cdot \beta\gamma \cdot \gamma\delta$. And similarly for any other order of the $\alpha\beta\gamma\delta$, there is in each case only one of the δ 's which can be changed into α ; and thus from each of the 24 forms we obtain a tree $\alpha^3\beta^2\gamma^2\delta\epsilon\zeta$. But dividing the 24 forms into the 12 + 12 forms corresponding to the interchange of the letters ϵ, ζ , then the first 12 forms, and the second 12 forms, give each of them the same trees $\alpha^3\beta^2\gamma^2\delta\epsilon\zeta$; and the number of these trees is thus $\frac{1}{2} \cdot 24 = 12$.

And in like manner reducing the $\alpha\beta\gamma\delta$ to $\alpha^2\beta^2$, $\alpha^3\beta$ or α^4 , we obtain in each case the number of trees equal to the proper sub-multiple of 24, viz. 6, 4, 1 in the three cases respectively (for the last case this is obvious, viz. there is 1 tree $\alpha^5\beta\gamma\delta\epsilon\zeta = \alpha\beta \cdot \alpha\gamma \cdot \alpha\delta \cdot \alpha\epsilon \cdot \alpha\zeta$); and the subsidiary theorem is thus proved. Hence the original theorem is true: as already remarked, it is easy to see that the proof is perfectly general.

The theorem is one of a set as follows:

Let $(\lambda, \alpha, \beta, \gamma, \dots)$ denote as above the trees with the given knots $\lambda, \alpha, \beta, \gamma, \dots$; $(\lambda + \mu, \alpha, \beta, \gamma, \dots)$ the pairs of trees with the given knots $\lambda, \mu, \alpha, \beta, \gamma, \dots$, the knots λ, μ belonging always to different trees; $(\lambda + \mu + \nu, \alpha, \beta, \gamma, \dots)$ the triads of trees with the given knots $\lambda, \mu, \nu, \alpha, \beta, \gamma, \dots$, the knots λ, μ, ν always belonging to different trees; and so on: then if $i + 1$ be the number of the knots $\lambda, \mu, \nu, \dots$, and n the number of the knots $\alpha, \beta, \gamma, \dots$, the number of trees or pairs, or triads, &c., of trees is $= (i + 1)(i + n + 1)^{n-1}$. In particular, if $i = 0$, then n being the number of knots $\alpha, \beta, \gamma, \dots$, and therefore $n + 1$ the whole number of knots $\lambda, \alpha, \beta, \gamma, \dots$, the number of trees is $= (n + 1)^{n-1}$ as before.

As a simple example, consider the pairs $(\lambda + \mu, \alpha, \beta)$: here $i = 1$, $n = 2$, and we have $(i + 1)(i + n + 1)^{n-1} = 2 \cdot 4 = 8$: in fact, the pairs of trees are

$$(\lambda\alpha, \alpha\beta, \mu), (\lambda\beta, \beta\alpha, \mu), (\lambda\alpha, \lambda\beta, \mu),$$

$$(\mu\alpha, \alpha\beta, \lambda), (\mu\beta, \beta\alpha, \lambda), (\mu\alpha, \mu\beta, \lambda); (\lambda\alpha, \mu\beta), (\lambda\beta, \mu\alpha).$$

We may arrange the trees $(\alpha, \beta, \gamma, \delta, \epsilon)$ as follows:

$$\begin{array}{rcl}
 (\alpha, \beta, \gamma, \delta, \epsilon) = \alpha\beta (\beta, \gamma, \delta, \epsilon); & 125 = 4 \times 1 \cdot 4^2 = 64 \\
 + \alpha\beta \cdot \alpha\gamma (\beta + \gamma, \delta, \epsilon) & + 6 \times 2 \cdot 4^1 = 48 \\
 + \alpha\beta \cdot \alpha\gamma \cdot \alpha\delta (\beta + \gamma + \delta, \epsilon) & + 4 \times 3 \cdot 4^0 = 12 \\
 + \alpha\beta \cdot \alpha\gamma \cdot \alpha\delta \cdot \alpha\epsilon & + 1 = 1 \\
 & \hline
 & 125,
 \end{array}$$

viz. to obtain the trees $(\alpha, \beta, \gamma, \delta, \epsilon)$, we join on the branch $\alpha\beta$ to any tree $(\beta, \gamma, \delta, \epsilon)$: the branches $\alpha\beta, \alpha\gamma$ to any pair of trees $(\beta + \gamma, \delta, \epsilon)$; the branches $\alpha\beta, \alpha\gamma, \alpha\delta$ to any triad of trees $(\beta + \gamma + \delta, \epsilon)$; and take lastly the tree $\alpha\beta \cdot \alpha\gamma \cdot \alpha\delta \cdot \alpha\epsilon$: the knots $\beta, \gamma, \delta, \epsilon$ being then interchanged in every possible manner. The whole number of trees 125 is thus obtained as $= 64 + 48 + 12 + 1$; the theorem is of course perfectly general.

The foregoing theory in effect presents itself in a paper by Borchardt, "Ueber eine der Interpolation entsprechende Darstellung der Eliminations-Resultante," *Crelle*, t. LVII. (1860), pp. 111–121, viz. Borchardt there considers a certain determinant, composed of the elements $10, 12, \dots, 1n, 20, 21, 23, \dots, 2n, \dots, n0, n1, \dots, nn-1$, and represented by means of the trees $(0, 1, 2, \dots, n)$; the branches of the tree being the aforesaid elements, and the tree being regarded as equal to the product of the several branches: the number of terms of the determinant is thus $= (n + 1)^{n-1}$ as above.

896.

A TRANSFORMATION IN ELLIPTIC FUNCTIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XXIV. (1890), pp. 259–262.]

THE formula in question is given in Klein's Memoir "Ueber hyperelliptische Sigmafunctionen," *Math. Ann.*, t. XXVII. (1886), pp. 431–464, see p. 454, in the form

$$u = \int_y^x \frac{(x \, dx)}{\sqrt{f(x)}}, \quad \varphi(u) = \frac{\sqrt{f(x)}\sqrt{f(y)} + F(x, y)}{2(x - y)^2},$$

(the discovery of it being ascribed to Weierstrass); and it is also given in Halphen's *Traité des Fonctions Elliptiques*, t. II. (1888), p. 357.

The algebraic foundation of the theorem is as follows: writing

$$z = \frac{1}{2} \left[\sqrt{\varphi(u)} + c \right],$$

and as usual I, J for the two invariants of the quartic function $(a, b, c, d, e \mid x, y)^4$, we have identically, as is easily verified,

$$4z^3 - Iz - J = \{d\sqrt{a} + b\sqrt{e}\}^2,$$

or say

$$\sqrt{4z^3 - Iz - J} = d\sqrt{a} + b\sqrt{e}.$$

Hence putting

$$\begin{aligned} A &= (a, b, c, d, e \mid x_1, x_2)^4, \\ B &= (a, \dots \mid x_1, x_2)^3 (y_1, y_2), \\ &\vdots \\ E &= (a, \dots \mid y_1, y_2)^4, \end{aligned}$$

and

$$\lambda = x_1 y_2 - x_2 y_1,$$

so that

$$\begin{aligned}\lambda^4 I &= AE - 4BD + 3C^2, \\ \lambda^6 J &= ACE - AD^2 - B^2E + 2BCD - C^3,\end{aligned}$$

then writing

$$z = \frac{1}{2\lambda^2} \left[\sqrt{(AE)} + C \right],$$

so that $\lambda^2 z$ is a bipartite irrational quadric function of the variables (x_1, x_2) and (y_1, y_2) , we have

$$\sqrt{4z^3 - Iz - J} = \frac{1}{\lambda^3} \left[D\sqrt{(A)} + B\sqrt{(E)} \right],$$

and we thence infer the existence of a transformation

$$\frac{dz}{\sqrt{4z^3 - Iz - J}} = M \left\{ \frac{(x_1 dx_2 - x_2 dx_1)}{\sqrt{(A)}} + \frac{(y_1 dy_2 - y_2 dy_1)}{\sqrt{(E)}} \right\}.$$

But for the verification hereof and a direct determination of the value of M , observe that we have

$$dz = \frac{1}{2\lambda^2} \left\{ \sqrt{(E)} \frac{dA}{2\sqrt{(A)}} + dC - [\sqrt{(AE)} + C] \frac{2}{\lambda} d\lambda \right\},$$

where, attending only to the terms in dx_1, dx_2 ,

$$\begin{aligned}dA &= 4\{(a, b, c, d \mid x_1, x_2)^3 dx_1 + (b, c, d, e \mid x_1, x_2)^3 dx_2\} \\ &= 4(A_1 dx_1 + A_2 dx_2), \text{ suppose;} \\ C &= (a, b, c \mid x_1, x_2)^2 (y_1, y_2)^2 + 2(b, c, d \mid x_1, x_2) x_1 x_2 (y_1, y_2)^2 + (c, d, e \mid x_1, x_2)^2 (y_1, y_2)^2 \\ &= E_1 x_1^2 + 2E_2 x_1 x_2 + E_3 x_2^2, \text{ suppose;}\end{aligned}$$

and therefore

$$dC = 2\{(E_1 x_1 + E_2 x_2) dx_1 + (E_2 x_1 + E_3 x_2) dx_2\}.$$

Hence

$$\begin{aligned}dz &= \frac{1}{2\lambda^2} \left\{ \frac{\sqrt{(E)}}{\sqrt{(A)}} 4(A_1 dx_1 + A_2 dx_2) + 2(E_1 x_1 + E_2 x_2) dx_1 \right. \\ &\quad \left. + 2(E_2 x_1 + E_3 x_2) dx_2 - [\sqrt{(AE)} + C] \frac{2}{\lambda} (y_2 dx_1 - y_1 dx_2) \right\}.\end{aligned}$$

The whole coefficient herein of dx_1 is

$$= \frac{1}{\lambda^2} \left\{ \frac{\sqrt{(E)}}{\sqrt{(A)}} A_1 + (E_1 x_1 + E_2 x_2) - \frac{\sqrt{(AE)} + C}{\lambda} y_2 \right\},$$

which is

$$\begin{aligned}&= \frac{1}{\lambda^2 \sqrt{(A)}} \left\{ A_1 \sqrt{(E)} + (E_1 x_1 + E_2 x_2) \sqrt{(A)} - \frac{1}{\lambda} [A \sqrt{(E)} + C \sqrt{(A)}] y_2 \right\} \\ &= \frac{1}{\lambda^2 \sqrt{(A)}} \left\{ [\dots] \sqrt{(E)} + [A\lambda - Ay_2] \sqrt{(E)} \right\}.\end{aligned}$$

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which is

$$= \frac{1}{\lambda^2 \sqrt{(A)}} (-x_2) \{D\sqrt{(A)} + B\sqrt{(E)}\};$$

and similarly the whole coefficient of dx_1 is

$$= \frac{1}{\lambda^2 \sqrt{(A)}} (x_1) \{D\sqrt{(A)} + B\sqrt{(E)}\}.$$

Hence the terms in dx_1, dx_2 are together

$$= \frac{1}{\sqrt{(A)}} (x_1 dx_2 - x_2 dx_1) \cdot \frac{D\sqrt{(A)} + B\sqrt{(E)}}{\lambda^2};$$

and since the terms in dy_1, dy_2 are of the like form, we have

$$dz = \left\{ \frac{1}{\sqrt{(A)}} (x_1 dx_2 - x_2 dx_1) + \frac{1}{\sqrt{(E)}} (y_1 dy_2 - y_2 dy_1) \right\} \times \frac{D\sqrt{(A)} + B\sqrt{(E)}}{\lambda^2},$$

and combining herewith the foregoing value

$$\sqrt{(4z^2 - Iz - J)} = \frac{D\sqrt{(A)} + B\sqrt{(E)}}{\lambda^2},$$

we have the required formula

$$\frac{dz}{\sqrt{(4z^3 - Iz - J)}} = \left\{ \frac{x_1 dx_2 - x_2 dx_1}{\sqrt{(A)}} + \frac{y_1 dy_2 - y_2 dy_1}{\sqrt{(E)}} \right\}.$$

As a very simple verification, suppose $a = 1, b = c = d = e = 0$; then

$$(A, B, C, D, E) = (x_1^4, x_1^3 y_1, x_1^2 y_1^2, x_1 y_1^3, y_1^4),$$

and if $\lambda = x_1 y_2 - x_2 y_1$, as before, then

$$z = \frac{x_1^2 y_1^2}{(x_1 y_2 - x_2 y_1)^2} = \frac{1}{\theta^2},$$

where

$$\theta = \frac{-x_1 y_2}{x_1 y_2 - x_2 y_1}, \quad \text{or} \quad \frac{1}{\theta} = \frac{x_2}{x_1} - \frac{y_2}{y_1}.$$

Also $I = 0, J = 0$, and consequently

$$\frac{dz}{\sqrt{(4z^3 - Iz - J)}} = \frac{dz}{2\sqrt{(z^3)}} = \frac{d\theta}{\theta} = \frac{x_1 dx_2 - x_2 dx_1}{x_1^2} - \frac{y_1 dy_2 - y_2 dy_1}{y_1^2},$$

which, in virtue of the foregoing values of A, E , is

$$= \frac{x_1 dx_2 - x_2 dx_1}{\sqrt{(A)}} - \frac{y_1 dy_2 - y_2 dy_1}{\sqrt{(E)}}.$$

I remark that an even more simple transformation from the general quartic radical to the Weierstrassian cubic radical was obtained by Hermite; this is alluded to in my "Note sur les Covariants, &c.," *Crelle*, t. L. (1855), pp. 285–287, [135], and is given with a demonstration in my paper "Sur quelques formules pour la transformation des intégrales elliptiques," *Crelle*, t. LV.

(1858), pp. 15–24, [235], see No. IV.; viz. from the identical relation $JU^3 - IU^2H + 4H^3 = -\Phi^2$, which connects the covariants of a quartic function, it at once follows that if

$$U = (a, b, c, d, e \mid x, 1)^4,$$

and H is the Hessian hereof,

$$H = (ac - b^2, (ad - bc), ae + 2bd - 3c^2, 2(be - cd), ce - d^2 \mid x, 1)^4;$$

then writing

$$z = \frac{H}{U},$$

we have

$$\frac{dz}{\sqrt{(-4z^3 + zI - J)}} = \frac{2dx}{\sqrt{\{(a, b, c, d, e \mid x, 1)^4\}}},$$

which is the transformation in question.

897.

SUR LES RACINES D'UNE ÉQUATION ALGÈBRIQUE.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, t. CX. (Janvier—Juin, 1890), pp. 174–176, 215–218.]

SOIT $f(u)$ une fonction rationnelle et entière avec des coefficients réels ou imaginaires, de l'ordre n ; en supposant que l'équation $f'(u) = 0$, de l'ordre $n - 1$, ait $n - 1$ racines, je démontre que l'équation $f(u) = 0$ aura n racines. Pour cela, soit $f(u) = f(x + iy) = P + iQ$: je suppose que c dénote une quantité positive donnée, et je considère la surface $c - z = P^2 + Q^2$, en attribuant à la coordonnée z des valeurs positives; c'est seulement pour avoir des maxima au lieu de minima, et pour faciliter ainsi l'exposition, que je prends cette surface au lieu de $z = P^2 + Q^2$. On peut donner à c une valeur si grande que la courbe $c = P^2 + Q^2$ soit une courbe fermée qui ne se coupe pas, c'est-à-dire un contour simple: cela étant, on peut se figurer ce contour comme la ligne de rivage d'une île montagneuse; la valeur de z est au plus $= c$, et, en donnant à z une valeur plus petite, $= b$, on a le contour qui correspond à l'altitude b : évidemment, les contours qui correspondent à des altitudes différentes ne se coupent pas. Il s'agit de prouver que l'île a précisément n sommets, chacun de l'altitude c .

J'écris

$$\frac{dP}{dx} = X, \quad \frac{dP}{dy} = Y;$$

donc

$$\frac{dQ}{dx} = -Y, \quad \frac{dQ}{dy} = X;$$

et, de plus,

$$\frac{dX}{dx} = a, \quad \frac{dX}{dy} = \frac{dY}{dx} = b, \quad \frac{dY}{dy} = -a.$$

et de là

$$\begin{aligned} \frac{d^2P}{dx^2} &= a, & \frac{d^2Q}{dx^2} &= -b, \\ \frac{d^2P}{dx dy} &= b, & \frac{d^2Q}{dx dy} &= a, \\ \frac{d^2P}{dy^2} &= -a, & \frac{d^2Q}{dy^2} &= b. \end{aligned}$$

Cela étant, nous avons

$$-\frac{1}{2} \frac{dz}{dx} = PX - QY, \quad -\frac{1}{2} \frac{dz}{dy} = PY + QX;$$

on aura un plan tangent horizontal pour $P = 0$, $Q = 0$, ou pour $X = 0$, $Y = 0$; les valeurs $P = 0$, $Q = 0$, appartiennent à la valeur c de z et correspondent à des sommets de montagne de cette altitude c ; les valeurs $X = 0$, $Y = 0$ correspondent à des sommets de col. En effet, nous avons

$$\begin{aligned} -\frac{1}{2} \frac{d^2z}{dx^2} &= Pa - Qb + X^2 + Y^2, \\ -\frac{1}{2} \frac{d^2z}{dx dy} &= Pb + Qa, \\ -\frac{1}{2} \frac{d^2z}{dy^2} &= -Pa + Qb + X^2 + Y^2; \end{aligned}$$

donc

$$\begin{aligned} \frac{1}{4} \left[\frac{d^2 z}{dx^2} \frac{d^2 z}{dy^2} - \left(\frac{d^2 z}{dx dy} \right)^2 \right] &= (X^2 + Y^2)^2 - (Pa - Qb)^2 - (Pb + Qa)^2 \\ &= (X^2 + Y^2)^2 - (P^2 + Q^2)(a^2 + b^2), \end{aligned}$$

valeur positive pour $P = 0, Q = 0$; négative pour $X = 0, Y = 0$.

À présent, nous avons $f'(x + iy) = X - iY$; donc, en supposant que l'équation $f'(x + iy) = 0$ ait $n - 1$ racines, il y aura $n - 1$ systèmes de valeurs réelles de x, y qui satisfont aux équations $X = 0, Y = 0$; pour chaque système, il y aura des valeurs déterminées de P, Q et de là aussi de z ; ces valeurs de z seront en général différentes. Ainsi il y aura dans l'île $n - 1$ cols, dont les altitudes seront en général différentes: soient $c_1, c_2, \dots, c_{n-1}$ ces altitudes, commençant avec la plus petite.

Pour $z = 0$, nous avons un contour simple, et de même pour une valeur quelconque plus petite que c_1 ; mais, pour $z = c_1$, nous avons un col; le contour est une courbe, figure de 8; en donnant à z une valeur un peu plus grande, le contour se divise en deux courbes fermées, ou bien contours simples, extérieurs l'un à l'autre. Pour $z = c_2$, nous avons encore un col; l'un des contours simples s'est changé en figure de 8, et, pour une valeur un peu plus grande, le contour se divise en trois contours simples, chacun extérieur aux autres. En continuant de cette manière, on a, pour c_{n-1} , le col le plus haut; le contour est composé de $n - 2$ contours simples et d'une figure de 8; et, en donnant à z une valeur un peu plus grande, on obtient un contour composé de n contours simples, chacun extérieur aux autres. Enfin, en faisant croître z , chacun des contours simples doit se réduire à un point, c'est-à-dire qu'il doit y avoir précisément n sommets de montagne; mais il n'y a pas de sommet de montagne, sinon pour la valeur $z = c$; donc il y a précisément n sommets de montagne, chacun de l'altitude c . On suppose toujours que l'équation $f'(x + iy) = 0$ n'ait pas de racines égales, mais il peut bien arriver que deux ou plusieurs des valeurs $c_1, c_2, \dots, c_n$ deviennent égales; la démonstration est très peu changée, en donnant à z une valeur un peu plus grande que celle qui correspond à l'altitude des cols d'altitude égale; le contour se divise toujours en contours simples, extérieurs chacun aux autres.

Il va sans dire que cette démonstration repose sur les mêmes principes que celles de Gauss et de Cauchy.

Je reprends la théorie des racines de l'équation $f(u) = 0$; au lieu de la surface $c - z = P^2 + Q^2$, il convient de considérer la surface $(c - z)^2 = P^2 + Q^2$, en faisant attention seulement aux valeurs de z positives et pas plus grandes que c . La théorie est très peu changée; les contours sont les mêmes qu'auparavant, mais ils appartiennent à des altitudes différentes; et, au lieu de maxima $z = c$ pour $P = 0, Q = 0$, on a des points coniques, c'est-à-dire, dans l'île montagneuse, au lieu d'un sommet arrondi de montagne, on a un cône ou un pic.

Mais avec la nouvelle surface, on construit graphiquement l'approximation de Newton: partant d'une valeur réelle ou imaginaire approximative u , on obtient la nouvelle valeur

$$u_1 = u + h = u - \frac{f(u)}{f'(u)}.$$

Je représente u par le point (x, y, z) de la surface $(c - z)^2 = P^2 + Q^2$, ou le point (x, y) du plan des sommets $z = c$; et de même, u_1 par le point (x_1, y_1, z_1) de la surface, ou (x_1, y_1) du plan des sommets: cela étant, si, par le point (x, y, z) de la surface, on mène la droite de plus grande pente (droite tangente à la surface et perpendiculaire au contour), cette droite rencontrera le plan des sommets en un point (x_1, y_1) , et l'on obtient ainsi le point (x_1, y_1, z_1) de la surface, qui représente la valeur cherchée u_1 . En particulier, si les coefficients de $f(u)$ sont réels, on a

$$Q = 0;$$

l'équation $(c - z)^2 = P^2 + Q^2$ devient

$$(c - z)^2 = P^2,$$

c'est-à-dire

$$c - z = \pm P,$$

ou enfin

$$c - z = \pm f(x),$$

et la section verticale de l'île est formée par des parties de ces deux courbes symétriques: pour la théorie géométrique, on peut évidemment y substituer la seule courbe $c - z = f(x)$.

J'ai proposé, il y a plus de dix ans (*Amer. Math. Journ.*, t. II., 1879, [694]), le problème que je nomme "The Newton-Fourier imaginary Problem," et dans une Note (*Quart. Math. Journ.*, t. XVI., 1879, [736]), "Application of the Newton-Fourier method to an imaginary root of an equation," j'ai considéré le cas d'une équation quadratique. Pour l'équation $u^2 - 1 = 0$, on a

$$u_1 = u - \frac{u^2 - 1}{2u} = \frac{1}{2} \left(u + \frac{1}{u} \right);$$

cela donne

$$u_1 - 1 = \frac{1}{2u}(u - 1)^2, \quad u_1 + 1 = \frac{1}{2u}(u + 1)^2,$$

et de là

$$\frac{u_1 - 1}{u_1 + 1} = \left(\frac{u - 1}{u + 1} \right)^2.$$

Cette dernière équation a rapport aux deux racines $+1$ et -1 , et, quoiqu'elle donne les résultats les plus élégants, cependant, en vue de la théorie générale, il vaut mieux considérer l'équation $u_1 - 1 = \frac{1}{2u}(u - 1)^2$ qui se rapporte à la seule racine $+1$.

Je remarque d'abord que la formule originale $u_1 = \frac{1}{2} \left(u + \frac{1}{u} \right)$ donne

$$x_1 = \frac{1}{2}x \left(1 + \frac{1}{x^2 + y^2} \right), \quad y_1 = \frac{1}{2}y \left(1 - \frac{1}{x^2 + y^2} \right);$$

donc les valeurs de x et x_1 seront à la fois positives ou négatives, et ainsi l'on peut ne faire attention qu'aux valeurs positives. Cela étant, nous avons

$$x_1 + iy_1 - 1 = \frac{(x + iy - 1)^2}{2(x + iy)}.$$

Désignons par A le point $(0, 1)$, par B le point $(0, -1)$, par O le point $(0, 0)$; et aussi par P le point (x, y) , et de même par P_1 le point (x_1, y_1) ; écrivons aussi $x + iy = se^{i\psi}$, $x + iy - 1 = re^{i\theta}$, $x_1 + iy_1 = r_1e^{i\theta_1}$; l'équation est

$$r_1e^{i\theta_1} = \frac{1}{2} \frac{r^2e^{2i\theta}}{se^{i\psi}};$$

donc

$$r_1 = \frac{1}{2} \frac{r^2}{s}, \quad (\text{c'est-à-dire } AP_1 = \frac{AP^2}{2OP}),$$

et

$$\theta_1 = 2\theta - \psi, \quad (\text{c'est-à-dire } \widehat{AP_1x} = 2\widehat{APx} - \widehat{OPx}).$$

Je remarque que, dans la géométrie des vecteurs, la seule équation $AP_1 = \frac{AP^2}{2OP}$ dénote l'équation en $x_1 + iy_1$, $x + iy$, c'est-à-dire les deux équations que je viens de trouver.

Partant d'un point quelconque P , on obtient une suite de points $P_1, P_2, P_3, \dots$; et, si le point P est sur l'axe des y ($x = 0$), tous les autres points seront aussi sur l'axe de y , et l'on n'approche ni du point A ni du point B . Mais, si la coordonnée x a une valeur positive si petite que l'on veut, on arrive enfin infiniment près du point A , et l'on peut même (dans un sens qui sera expliqué plus bas, mais qui n'est pas le sens le plus naturel) dire que l'approximation est régulière. En effet, on n'a pas toujours $AP_1 < AP$, et ainsi, dans le sens le plus naturel, l'approximation n'est pas toujours régulière. Pour étudier cela, j'écris $AP_1 = AP$; cela donne $AP = 2OP$, ou, ce qui est la même chose, $x^2 + y^2 + \frac{2}{3}x = \frac{1}{3}$, c'est-à-dire que le point P sera situé sur le cercle, centre $x = -\frac{1}{3}$ et rayon $= \frac{2}{3}$; en ne faisant attention qu'aux valeurs positives de x , on a un segment compris entre l'axe des y et un arc par les points $\left(x = 0, y = \pm \frac{1}{\sqrt{3}}\right), (x = \frac{1}{3}, y = 0)$. Si le point P est sur l'arc, on aura $AP_1 = AP$; si P est en dedans du segment, alors $AP_1 > AP$; si P est en dehors du segment, $AP_1 < AP$.

Mais, en supposant P en dehors du segment, et ainsi $AP_1 < AP$, il peut bien arriver que P_1 soit en dedans du segment, et, cela étant, on aura $AP_2 > AP_1$, et l'approximation ne sera pas régulière. Mais, en considérant le cercle $x^2 + y^2 - \frac{2}{3}x = \frac{1}{3}$, lequel est le cercle, centre A et rayon $\frac{2}{3}$, qui touche le segment au point $(x = \frac{1}{3}, y = 0)$, alors, en supposant que le point P soit en dedans de ce cercle, on aura $AP_1 < AP$, le point P_1 sera aussi en dedans du cercle, et ainsi en dehors du segment; et les points successifs $P, P_1, P_2, \dots$ approcheront continuellement du point A ; l'approximation sera dans ce cas régulière.

Il y a ainsi trois régions: le segment, le cercle $x^2 + y^2 - \frac{2}{3}x = \frac{1}{3}$ et le résidu du demi-plan; on pourrait les nommer régions *noire*, *blanche* et *grise* respectivement. C'est seulement pour un point P à l'intérieur de la région blanche que l'approximation est certainement régulière.

Nous venons de considérer en effet les cercles qui ont pour centre le point A ; $AP_1 < AP$ veut dire que le point P est situé sur un cercle plus grand, et P_1 sur un cercle plus petit; mais, au lieu de ces cercles concentriques, considérons des cercles quelconques qui entourent le point A , sans se couper les uns les autres; et convenons de dire que c'est un bon pas quand on passe du point P sur un cercle plus grand à un point P_1 sur un cercle plus petit: avec cette convention on aura, en général, trois régions, lesquelles cependant ne seront pas les mêmes comme auparavant. En particulier, si nous considérons les cercles $AP = kBP$ (k une constante quelconque plus petite que l'unité), alors il n'y a pas de région noire, ou, si l'on veut, la région noire se réduit à la seule droite $y = 0$; donc il n'y a pas non plus de région grise, et le demi-plan entier est région blanche, c'est-à-dire, dans le sens que je viens d'expliquer, l'approximation est toujours régulière. En effet, c'est là la théorie à laquelle on est conduit au moyen de l'équation $\frac{u_1 - 1}{u_1 + 1} = \left(\frac{u - 1}{u + 1}\right)^2$ ci-dessus mentionnée.

En parlant de cercles, j'ai fait une restriction qui n'est nullement nécessaire; j'aurais pu parler d'ovales, de forme quelconque, qui entourent le point A sans se couper les uns les autres.

J'espère appliquer cette théorie au cas d'une équation cubique, mais les calculs sont beaucoup plus difficiles.

898.

SUR L'ÉQUATION MODULAIRE POUR LA TRANSFORMATION DE L'ORDRE 11.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, t. CXI. (Juillet—Décembre, 1890), pp. 447–449.]

L'ÉQUATION en u, v , en y écrivant $u = x, v = y$, est

$$\left. \begin{array}{l} y^{12} \\ + y^{11} (32x^3 - 22x^7) \\ + y^{10} \cdot 44x^2 \\ + y^9 \cdot (88x^9 + 22x) \\ + y^8 \cdot 165x^4 \\ + y^7 \cdot 132x^9 \\ + y^6 (-44x^{11} + 44x^3) \\ + y^5 \cdot -132x^8 \\ + y^4 \cdot -165x^3 \\ + y^3 (-22x^{10} - 88x^2) \\ + y^2 \cdot -44x^9 \\ + y (22x^{11} - 22x) \\ + 1 \cdot -x^{12} \end{array} \right\} = 0.$$

Selon un résultat trouvé par H. J. S. Smith, pour la transformation de l'ordre p , la courbe est de l'ordre $2p$, et il y a à l'origine un point double, à l'infini deux points singuliers équivalents chacun à $\frac{1}{2}(p-1)(p-2)$ points doubles, et de plus $(p-1)(p-3)$ points doubles. Au cas $p = 11$, le nombre de ces derniers points doubles est donc $= 80$. Cela s'accorde avec l'expression

$$D = x^{12}(1 - x^8)^{10}(16x^{16} - 31x^8 + 16)^2(x^{64} - 301960x^{56} + \dots + 1)^2,$$

trouvée par M. Hermite pour le discriminant de la fonction; et l'on voit ainsi que les valeurs de x , qui correspondent aux quatre-vingts points doubles, sont données par les équations

$$16x^{16} - 31x^8 + 16 = 0,$$

$$x^{64} - 301960x^{56} + \dots + 1 = 0.$$

Je ne considère que les seize points doubles donnés par la première équation. Cette équation donne

$$x^8 = \frac{1}{32}(31 + 3i\sqrt{7}), \quad x^4 = \frac{1}{4}(i + 3\sqrt{7}), \quad x^2 = \frac{1+i}{4\sqrt{2}}(-3 + i\sqrt{7});$$

il y a ainsi quatre points doubles, pour lesquels les valeurs de x sont

$$x = \pm \sqrt{\frac{1+i}{4\sqrt{2}}}(-3 + i\sqrt{7}), \quad x = \pm i \sqrt{\frac{1+i}{4\sqrt{2}}}(-3 + i\sqrt{7});$$

je trouve que les valeurs correspondantes de y sont $y = \frac{1+i}{\sqrt{2}}x$, savoir que les quatre points sont situés sur la droite $y = \frac{1+i}{\sqrt{2}}x$, et, en changeant successivement les signes de i et $\sqrt{2}$, on voit ainsi que les seize points sont situés, quatre à quatre, sur les droites

$$y = \frac{1+i}{\sqrt{2}}x, \quad y = \frac{1-i}{\sqrt{2}}x, \quad y = -\frac{1+i}{\sqrt{2}}x, \quad y = -\frac{1-i}{\sqrt{2}}x.$$

J'écris, pour abréger,

$$m = \frac{1+i}{\sqrt{2}}, \quad (\text{donc } m^4 = -1),$$

$$p = \frac{1}{4}(-3 + i\sqrt{7}),$$

donc

$$2p^2 + 3p + 2 = 0.$$

En écrivant $y = mx$ dans l'équation et en rejetant le facteur x^2 , puis en écrivant $x^2 = mp$, l'équation se présente sous la forme

$$\left. \begin{array}{l} m^{12}p^6. \quad -32m^{12} \\ + m^{10}p^6. \quad 88m^{11} \\ + m^9p^6. \quad 44m^{10} - 44m^6 \\ + m^8p^5. \quad -22m^{11} + 132m^7 - 22m^3 \\ + m^6p^5. \quad m^{12} + 165m^8 - 165m^4 - 1 \\ + m^5p^5. \quad 22m^9 - 132m^5 + 22m \\ + m^3p^3. \quad 44m^6 - 44m^2 \\ + m^2p^3. \quad -88m^3 \\ + 1. \quad 32m \end{array} \right\} = 0,$$

où les coefficients ne contiennent que les puissances m^{21} , m^{17} , m^{13} , m^9 , m^5 , m^1 de m et se réduisent ainsi à des multiples de m ; il y a aussi un facteur numérique 8, et, en divisant par $-8m$, l'équation devient

$$4p^{10} - 11p^8 - 11p^7 + 22p^6 + 41p^5 + 22p^4 - 11p^3 - 11p^2 + 4 = 0;$$

cette équation est de la forme

$$(2p^2 + 3p + 2)^2(p^6 - 3p^5 + 2p^4 + p^3 + 2p^2 - 3p + 1) = 0.$$

La droite $y = mx$ a donc, avec la courbe, quatre intersections doubles

$$p = \frac{1}{4}(-3 \pm i\sqrt{7}),$$

c'est-à-dire

$$x^2 = \frac{1+i}{4\sqrt{2}}(-3 \pm i\sqrt{7});$$

on démontre sans peine que la droite n'est pas une tangente, et ces valeurs correspondent ainsi à des points doubles de la courbe, c'est-à-dire qu'il y a sur la droite $y = \frac{1+i}{\sqrt{2}}x$ quatre points doubles. Réciproquement, cette valeur de x^2 conduit au facteur $(16x^{16} - 31x^8 + 16)^2$ du déterminant de l'équation modulaire.

899.

SUR LES SURFACES MINIMA.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, t. CXI. (Juillet—Décembre, 1890), pp. 953, 954.]

ON peut généraliser tant la définition que la construction de ces surfaces, en substituant pour le cercle imaginaire à l'infini une conique ou même une surface quadrique quelconque.

Je rappelle que, dans la théorie ordinaire, une surface minima est une surface telle qu'un point quelconque de la surface est situé à mi-chemin entre les deux centres de courbure, et qu'une telle surface est le lieu des points à mi-chemin entre les deux points situés respectivement sur deux courbes de longueur nulle quelconques. Or on peut rattacher la notion d'une courbe de longueur nulle à celle d'une courbe de poursuite. Pour expliquer cela, j'observe que, dans le plan, en supposant comme à l'ordinaire que le lièvre coure selon une droite et que le chien et le lièvre courent avec des vitesses uniformes, la courbe de poursuite est une courbe déterminée; mais si les vitesses varient arbitrairement, alors la définition exprime seulement que la courbe est une courbe plane. Mais, dans l'espace, si au lieu d'une droite on considère une courbe plane ou à double courbure (disons une directrice) quelconque, alors, quelles que soient les vitesses, la définition précise toujours la courbe, savoir: on a toujours pour courbe de poursuite une courbe dont chaque tangente rencontre la courbe directrice. Au lieu d'une courbe, on peut avoir une surface directrice; dans ce cas, le nom est moins applicable, néanmoins je le retiens, et je dis que la courbe de poursuite est une courbe dont chaque tangente touche la surface directrice. Nous avons, de cette manière, la définition d'une courbe de poursuite par rapport à une courbe ou surface directrice quelconque.

À présent, au lieu du cercle imaginaire à l'infini, considérons une conique quelconque, l'absolue: on établit, comme on sait, par rapport à cette conique, la notion de la perpendicularité, et ainsi les notions d'une normale et des centres de courbure ne cessent pas de subsister. On peut donc considérer une surface telle que chaque point de la surface soit l'harmonique par rapport aux deux centres de courbure du point de rencontre de la normale avec le plan de l'absolue: on a ainsi ce que je nomme une surface quasi-minima. Il va sans dire qu'il faut modifier convenablement la notion métrique d'une aire minima pour qu'elle soit applicable à cette nouvelle surface.

Pour construire la surface, on prend par rapport à l'absolue deux courbes de poursuite quelconques, et puis sur la droite, menée par deux points quelconques de ces courbes respectivement, l'harmonique par rapport à ces deux points du point de rencontre de la droite avec le plan de l'absolue: le lieu de ce point harmonical sera la surface quasi-minima.

Il paraît permis de substituer pour la conique absolue une surface quadrique quelconque, que je nomme aussi l'*absolue*: on a, comme on sait, les notions de la normale et des centres de courbure. Pour la surface quasi-minima, le point sur la surface sera l'un des points doubles (foyers) de l'involution formée par les deux centres de courbure et les deux points de rencontre de la normale avec l'absolue; et de même pour la construction de la surface, il faut prendre sur la droite menée par deux points quelconques des deux courbes de poursuite respectivement les points doubles (foyers) de l'involution formée par ces deux points et les deux points de rencontre de la droite avec l'absolue.

900.

JAMES JOSEPH SYLVESTER.

[(*Scientific Worthies in Nature*, xxv.); *Nature*, vol. xxxix. (1889), pp. 217–219.]

JAMES JOSEPH SYLVESTER, born in London on September 3, 1814, is the sixth and youngest son of the late Abraham Joseph Sylvester, formerly of Liverpool¹. He was educated at two private schools in London, and at the Royal Institution, Liverpool, whence he proceeded in due course of time to St John's College, Cambridge. In these early days he manifested considerable aptitude for mathematics, and so it was not matter for surprise that he came out in the Tripos Examination of 1837 as Second Wrangler; being incapacitated, by the fact of his Jewish origin, from taking his degree, he was not able to compete for either of the Smith's Prizes. In more enlightened times (1872), he had the degrees of B.A. and M.A., by accumulation, conferred upon him, and received therewith the honour of a Latin speech from the Public Orator. He himself says: "I am perhaps the only man in England who am a full (voting) Master of Arts for the three Universities of Dublin, Cambridge, and Oxford, having received that degree from these Universities in the order above given: from Dublin, by *ad eundem*; from Cambridge, *ob merita*; from Oxford, by decree." He is now D.C.L. of Oxford; LL.D. of Dublin and Edinburgh, and Hon. Fellow of St John's College, Cambridge. It is still open for him to receive yet higher recognition from his own alma mater².

Prof. Sylvester became a student of the Inner Temple, July 29, 1846, and was called to the Bar on November 22, 1850³. He has been Professor of Natural Philosophy at University College, London; of Mathematics at the University of Virginia, U.S.A.⁴; then ten years later Professor at the Royal Military Academy, Woolwich; and again, after a five years' interval, Professor of Mathematics at the Johns Hopkins University, Baltimore, U.S.A., from its foundation in 1877. Finally, in December 1883, he was elected Savilian Professor of Geometry at Oxford, in succession to Prof. Henry Smith⁵. His first printed paper was on Fresnel's optical theory (in the *Phil. Mag.*, 1837).

We can here only briefly allude to a communication which was accompanied by many important results: we refer to the Friday evening address (January 23, 1874) to the Royal Institution, "On Recent Discoveries in Mechanical Conversion of Motion." He says:—"It would be difficult to quote any other discovery which opens out such vast and varied horizons as this of Peaucellier's,—in one direction, descending to the wants of the workshop, the simplification of the steam-engine, the revolutionising of the mill-wright's trade, the amelioration of garden-pumps, and other domestic conveniences (the sun of science glorifies all it shines upon); and in the other, soaring to the sublimest heights of the most advanced doctrines of modern analysis, lending aid to, and throwing light from a totally unexpected quarter on the researches of such men as Abel, Riemann, Clebsch, Grassmann, and Cayley. Its head towers above the clouds, while its feet plunge into the bowels of the earth."

The only works that Prof. Sylvester has published, we believe, are: (1) "A Probationary Lecture on Geometry, delivered before the Gresham Committee and the Members of the Common Council of the City of London, December 4, 1854," a slight thing which had to be written and

¹Foster's *Hand-book of Men at the Bar*.

²[** The honorary degree Sc.D. was conferred upon him by Cambridge in 1890.]

³Foster, l.c.

⁴The late Prof. Key, of University College and School, was the first occupant of the Chair, founded by Mr Jefferson, once President of the United States, in 1824.

⁵He commences his Oxford lecture (*Nature*, vol. xxxiii. p. 222), of date December 12, 1885, with the words: "It is now two years and seven days since a message by the Atlantic cable containing the single word 'elected' reached me in Baltimore informing me that I had been appointed Savilian Professor of Geometry in Oxford, so that for three weeks I was in the unique position of filling the post and drawing the pay of Professor of Mathematics in each of two Universities."

delivered at a few hours' notice; (2) "Laws of Verse," in 1870; (3) several short poems, sonnets, and translations, which have appeared in our columns and elsewhere.

Our notice would be incomplete without some record of the honours that have been conferred upon Dr Sylvester. He was elected a Fellow of the Royal Society on April 25, 1839; has received a Royal Medal (1860) and the Copley Medal (1880), this latter rarely awarded, we believe, to a pure mathematician. On this last occasion, Mr Spottiswoode accompanied the presentation with the words, "His extensive and profound researches in pure mathematics, especially his contributions to the theory of invariants and covariants, to the theory of numbers and to modern geometry, may be regarded as fully establishing Mr Sylvester's claim to the award." He is a Fellow of New College, Oxford; Foreign Associate of the United States National Academy of Sciences; Foreign Member of the Royal Academy of Sciences, Göttingen, of the Royal Academy of Sciences of Naples, and of the Academy of Sciences of Boston; Corresponding Member of the Institute of France, of the Imperial Academy of Science of St Petersburg, of the Royal Academy of Science of Berlin, of the Lyncei of Rome, of the Istituto Lombardo, and of the Société Philomathique. He has been long connected with the editorial staff of the *Quarterly Journal of Mathematics* (under one or another of its titles), and was the first editor of, and is a considerable contributor to, the *American Journal of Mathematics*; and he was at one time Examiner in Mathematics and Natural Philosophy in the University of London. He was not an original member of the London Mathematical Society (founded January 16, 1865), but was elected a member on June 19, 1865, Vice-President on January 15, 1866, and succeeded Prof. De Morgan as the second President on November 8, 1866. The Society showed its recognition of his great services to them and to mathematical science generally by awarding him its De Morgan Gold Medal in November 1887. Wherever Dr Sylvester goes, there is sure to be mathematical activity; and the latest proof of this is the formation, during the last term at Oxford, of a Mathematical Society, which promises, we hear without surprise, to do much for the advancement of mathematical science there.

The writings of Sylvester date from the year 1837; the number of them in the Royal Society Index up to the year 1863 is 112, in the next ten years 38, and in the volume for the next ten years 81, making 231 for the years 1837 to 1883: the number of more recent papers is also considerable. They relate chiefly to finite analysis, and cover by their subjects a large part of it: algebra, determinants, elimination, the theory of equations, partitions, tactic, the theory of forms, matrices, reciprocants, the Hamiltonian numbers, &c.; analytical and pure geometry occupy a less prominent position; and mechanics, optics, and astronomy are not absent. A leading feature is the power which is shown of originating a theory or of developing it from a small beginning; there is a breadth of treatment and determination to make the most of a subject, an appreciation of its capabilities, and real enjoyment of it. There is not unfrequently an adornment or enthusiasm of language which one admires, or is amused with: we have a motto from Milton, or Shakespeare; a memoir is a trilogy divided into three parts, each of which has its action complete within itself, but the same general cycle of ideas pervades all three, and weaves them into a sort of complex unity; the apology for an unsymmetrical solution is—symmetry, like the grace of an eastern robe, has not unfrequently to be purchased at the expense of some sacrifice of freedom and rapidity of action; and, he remarks, may not music be described as the mathematic of sense, mathematic as the music of the reason? the soul of each the same! &c. It is to be mentioned that there is always a generous and cordial recognition of the merit of others, his fellow-workers in the science.

It would be in the case of any first-rate mathematician—and certainly as much so in this as in any other case—extremely interesting to go carefully through the whole of a long list of memoirs, tracing out as well their connexion with each other, and the several leading ideas on which they depend, as also their influence on the development of the theories to which they relate; but for doing this properly, or at all, space and time, and a great amount of labour, are required. Short of doing so, one can only notice particular theorems—and there are, in the case of Sylvester, many of these, "beautiful exceedingly," which, for their own sakes, one is tempted

to refer to—or one can give titles, which, to those familiar with the memoirs themselves, will recall the rich stores of investigation and theory contained therein.

A considerable number of papers, including some of the earliest ones, relate to the question of the reality of the roots of a numerical equation: in the several connexions thereof with Sturm's theorem, Newton's rule for the number of imaginary roots, and the theory of invariants. Sylvester obtained for the Sturmian functions, divested of square factors, or say for the reduced Sturmian functions, singularly elegant expressions in terms of the roots, viz. these were

$$f_2(x) = \Sigma(a-b)^2(x-c)(x-d)\dots, \quad f_3(x) = \Sigma(a-b)^2(a-c)^2(b-c)^2(x-d)\dots, \text{ \&c.};$$

but not only this: applying the Sturmian process of the greatest common measure (not to $f(x)$, $f'(x)$, but instead) to two independent functions $f(x)$, $\phi(x)$, he obtained for the several resulting functions expressions involving products of differences between the roots of the one and the other equation, $f(x) = 0$, $\phi(x) = 0$; the question then arose, what is the meaning of these functions? The answer is given by his theory of *intercalations*: they are signaletic functions, indicating in what manner (when the real roots of the two equations are arranged in order of magnitude) the roots of the one equation are intercalated among those of the other. The investigations in regard to Newton's rule (not previously demonstrated) are very important and valuable: the principle of Sturm's demonstration is applied to this wholly different question: viz. x is made to vary continuously, and the consequent gain or loss of changes of sign is inquired into. The third question is that of the determination of the character of the roots of a quintic equation by means of invariants. In connexion with it, we have the noteworthy idea of facultative points; viz. treating as the coordinates of a point in n -dimensional space those functions of the coefficients which serve as criteria for the reality of the roots, a point is facultative or non-facultative according as there is, or is not, corresponding thereto any equation with real coefficients: the determination of the characters of the roots depends (and, it would seem, depends only) on the bounding surface or surfaces of the facultative regions, and on a surface depending on the discriminant. Relating to these theories there are two elaborate memoirs, "On the Syzygetic Relations &c." and "Algebraical Researches &c." in the *Philosophical Transactions* for the years 1853 and 1864 respectively; but as regards Newton's rule later papers must also be consulted.

In the years 1851–54, we have various papers on homogeneous functions, the calculus of forms, &c. (*Camb. and Dub. Math. Journal*, vols. VI. to IX.), and the separate work "On Canonical Forms" (London, 1851). These contain crowds of ideas, embodied in the new words, cogredient, contragredient, concomitant, covariant, contravariant, invariant, emanant, combinant, commutant, canonical form, plexus, &c., ranging over and vastly extending the then so-called theories of linear transformations and hyperdeterminants. In particular, we have the introduction into the theory of the very important idea of continuous or infinitesimal variation: say that a function, which (whatever are the values of the parameters on which it depends) is invariant for an infinitesimal change of the parameters, is absolutely invariant.

There is, in 1844, in the *Philosophical Magazine*, a valuable paper, "Elementary Researches in the Analysis of Combinatorial Aggregation," and the titles of two other papers, 1865 and 1866, may be mentioned: "Astronomical Prolusions; commencing with the instantaneous proof of Lambert's and Euler's theorems, and modulating through the construction of the orbit of a heavenly body from two heliocentric distances, the subtended chord, and the periodic time, and the focal theory of Cartesian ovals, into a discussion of motion in a circle and its relation to planetary motion"; and the sequel thereto, "Note on the periodic changes of orbit under certain circumstances of a particle acted upon by a central force, and on vectorial coordinates, &c., together with a new theory of the analogues of the Cartesian ovals in space."

Many of the later papers are published in the *American Mathematical Journal*, founded, in 1878, under the auspices of the Johns Hopkins University, and for the first six volumes of which Sylvester was editor-in-chief. We have, in vol. I., a somewhat speculative paper entitled "An application of the new atomic theory to the graphical representation of the invariants and

covariants of binary quantics,” followed by appendices and notes relating to various special points of the theory; and in the same and subsequent volumes various memoirs on binary and ternary quantics, including papers (by himself, with the aid of Franklin) containing tables of the numerical generating functions for binary quantics of the first ten orders, and for simultaneous binary quantics of the first four orders, &c. The memoir (vols. II. and III.) on “Ternary cubic-form equations” is connected with some early papers relating to the theory of numbers. We have in it the theory of residuation on a cubic curve, and the beautiful chain-rule of rational derivation; viz. from an arbitrary point 1 on the curve it is possible to derive the singly infinite series of points $(1, 2, 4, 5, \dots, 3j \pm 1)$ such that the chord through any two points, m, n , again meets the curve in a point $m + n$, $m \sim n$ (whichever number is not divisible by 3) of the series; moreover, the coordinates of any point m are rational and integral functions of the degree m^2 of those of the point 1.

There is in vol. V. the memoir, “A Constructive Theory of Partitions arranged in three acts, an Interact in two parts, and an Exodion,” and in vol. VI. we have “Lectures on the Principles of Universal Algebra,” (referring to a course of lectures on multinomial quantity, in the year 1881). The memoir is incomplete, but the general theories of nullity and vacuity, and of the corpus formed by two independent matrices of the same order, are sketched out; and there are, in the *Comptes rendus* of the French Academy, later papers containing developments of various points of the theory,—the conception of “nivellators” may be referred to.

The last-mentioned paper in the *American Mathematical Journal* was published subsequently to Sylvester’s return to England on his appointment as Savilian Professor of Mathematics at Oxford. In December 1886, he gave there a public lecture containing an outline of his new theory of reciprocants (reported in *Nature*, January 7, 1887), and the lectures since delivered are published under the title, “Lectures on the Theory of Reciprocants” (reported by J. Hammond), same Journal, vols. VIII. to X.; thirty-three lectures actually delivered, entire or in abstract, in the course of three terms, to a class in the University, with a concluding so-called lecture 34, which is due to Hammond. The subject, as is well known, is that of the functions of a dependent variable, y , and its differential coefficients, y' , y'' , $\dots$, in regard to x (or, rather, the functions of y' , y'' , $\dots$), which remain unaltered by the interchange of the variables x and y : this is a less stringent condition than that imposed by Halphen (“Thèse,” 1878) on his differential invariants, and the theory is accordingly a more extensive one. A passage may be quoted:—“One is surprised to reflect on the change which is come over Algebra in the last quarter of a century. It is now possible to enlarge to an almost unlimited extent on any branch of it. These thirty lectures, embracing only a fragment of the theory of reciprocants, might be compared to an unfinished epic in thirty cantos. Does it not seem as if Algebra had attained to the dignity of a fine art, in which the workman has a free hand to develop his conceptions, as in a musical theme or a subject for painting? Formerly, it consisted in detached theorems, but nowadays it has reached a point in which every properly-developed algebraical composition, like a skilful landscape, is expected to suggest the notion of an infinite distance lying beyond the limits of the canvas.” And, indeed, the theory has already spread itself out far and wide, not only in these lectures by its founder, but in various papers by auditors of them, and others,—Elliott, Hammond, Leudesdorf, Rogers, Macmahon, Berry, Forsyth.

Sylvester’s latest important investigations relate to the Hamiltonian numbers: there is a memoir, *Crelle*, t. C. (1887), and, by Sylvester and Hammond jointly, two memoirs in the *Philosophical Transactions*. The subject is that of the series of numbers 2, 3, 5, 11, 47, 923, calculated thus far by Sir W. R. Hamilton in his well-known Report to the British Association, on Jerrard’s method. A formula for the independent calculation of any term of the series was obtained by Sylvester, but the remarkable law by means of a generating function was discovered by Hammond, viz. $E_0, E_1, E_2, \dots$, being the series 3, 4, 6, $\dots$ of the foregoing numbers, each increased by unity; then these are calculated by the formula

$$(1 - t)^{E_0} \times (1 - t^2)^{E_1} \times (1 - t^4)^{E_2} \times \dots = 1 - 2t,$$

equating the powers of t on the two sides respectively: observe the paradox, $t = \frac{1}{2}$, then the formula gives $0 = \text{sum of a series of positive powers of } \frac{1}{2}$.

Enough has been said to call to mind some of Sylvester's achievements in mathematical science. Nothing further has been attempted in the foregoing very imperfect sketch.

901.

NOTE ON THE SUMS OF TWO SERIES.

[From the *Messenger of Mathematics*, vol. XIX. (1890), pp. 29–31.]

I CONSIDER the two series

$$S_1 = \frac{1}{1 + e^{\pi a}} + \frac{1}{3(1 + e^{3\pi a})} + \frac{1}{5(1 + e^{5\pi a})} + \cdots$$

and

$$S = \frac{1}{1 + e^{\pi a}} + \frac{1}{2(1 + e^{2\pi a})} + \frac{1}{3(1 + e^{3\pi a})} + \cdots,$$

where a is real, positive, and indefinitely small; these would at first sight appear to be equal to each other, but this is not in fact the case.

Taking first the series S_1 , putting therein $\pi a = 2x$, this is

$$2S_1 = \frac{1}{1 + e^{2x}} + \frac{1}{3(1 + e^{6x})} + \frac{1}{5(1 + e^{10x})} + \cdots.$$

Now we have, (Legendre, *Théorie des Fonctions Elliptiques*, t. II. p. 438),

$$\frac{1}{1 + y} + \frac{1}{3(1 + y^3)} + \frac{1}{5(1 + y^5)} + \cdots = \frac{1}{2} \log \Gamma(y + 1),$$

where C is Euler's constant, $= .577\dots$; and if y be real, positive, and very large, then

$$\Gamma(y + 1) = e^{(y + \frac{1}{2}) \log y - y} \sqrt{2\pi} \dots,$$

whence, differentiating the logarithm and neglecting the terms which contain negative powers of y , then the value is $= C + \log y$; hence, writing $y = \frac{1}{x}$, we obtain

$$\frac{1}{1 + x} + \frac{1}{2(1 + 2x)} + \frac{1}{3(1 + 3x)} + \cdots = -C - \log x.$$

Writing herein $2x$ for x , and dividing by 2, we have

$$\frac{1}{2(1 + 2x)} + \frac{1}{4(1 + 4x)} + \cdots = -\frac{1}{2}C - \frac{1}{2} \log 2x = \frac{1}{2}(C - \log 2) - \frac{1}{2} \log x;$$

or, subtracting,

$$\frac{1}{1 + x} + \frac{1}{3(1 + 3x)} + \frac{1}{5(1 + 5x)} + \cdots = -\frac{1}{2}(C + \log 2) - \frac{1}{2} \log x.$$

Hence, writing for x its value, $= \frac{1}{2}\pi a$, we have

$$S_1 = \frac{1}{4}(C + \log 2) - \frac{1}{4} \log \frac{1}{2}\pi a = \frac{1}{4}(C + 2 \log 2 - \log \pi) - \frac{1}{4} \log a.$$

For the series S , we have (*Fundamenta Nova*, p. 103⁶) the formula

$$\frac{2K}{\pi} = \frac{1}{1 + q^{-1}} - \frac{1}{3(1 + q^{-3})} + \frac{1}{5(1 + q^{-5})} - \cdots,$$

⁶[* Jacobi's *Gesammelte Werke*, t. I, p. 159.]

or, putting herein $a = \frac{K'}{K}$, then $q = e^{-\frac{\pi K'}{K}} = e^{-\pi a}$, and thence

$$S = \frac{1}{1 + e^{\pi a}} + \frac{1}{2(1 + e^{2\pi a})} + \frac{1}{3(1 + e^{3\pi a})} + \cdots = \log \sqrt{k'} + \frac{2K}{\pi}.$$

We have $q = e^{-\pi a}$, which is real, positive, and less than but indefinitely near to 1; hence also k is real, positive, and less than but indefinitely near to 1, say the value is $= 1 - \beta$; thence $k' = \sqrt{(2\beta)}$, and $\log \sqrt{k'} = \frac{1}{4} \log 2\beta$; also $K' = \frac{1}{2}\pi$, and $K = \log \frac{4}{k'}$; therefore

$$a = \frac{\frac{1}{2}\pi}{\log \frac{4}{k'}} = \frac{1}{2}\pi \div \log \frac{2\sqrt{2}}{\sqrt{\beta}}, \quad \text{whence } \log \frac{2\sqrt{2}}{\sqrt{\beta}} = \frac{\pi}{2a},$$

which is the relation between a and β ; and we thus have $\frac{2K}{\pi} = \frac{2}{\pi} \log \frac{2\sqrt{2}}{\sqrt{\beta}} = \frac{1}{a}$; and consequently

$$S = \frac{1}{4} \log 2\beta + \frac{1}{a}, = -\frac{1}{4} \log a.$$

The two values thus are

$$S = -\frac{1}{4} \log a, \quad S_1 = \frac{1}{4}(C + 2 \log 2 - \log \pi) - \frac{1}{4} \log a,$$

each depending on $\log a$, and having for this term the same coefficient $-\frac{1}{4}$; but there is in S_1 a constant term $\frac{1}{4}(C + 2 \log 2 - \log \pi)$, where C is the constant $\cdot 577 \dots$

It is easy to see why the series S is not reducible to S_1 ; however small a may be in the general term $\frac{1}{n(1 + e^{n\pi a})}$, then taking n sufficiently large, not only $n\pi a$ is not indefinitely small, but it in fact becomes indefinitely large; the general term of the first series thus approximates to $\frac{1}{ne^{n\pi a}}$, or the terms more rapidly diminish somewhat than in a geometric series with the ratio $e^{-n\pi a}$ (a small positive value less than but very near to 1), whereas in the second series the general term approximates to $\frac{1}{\frac{n}{2} + (n+1)^2}$, or the convergence is ultimately that of the series $\frac{1}{n^2} + \frac{1}{(n+1)^2} + \cdots$.

902.

ON THE FOCALS OF A QUADRIC SURFACE.

[From the *Messenger of Mathematics*, vol. XIX. (1890), pp. 113–117.]

IN plane geometry, the focus of a curve is the node of the circumscribed line-system of the curve and the circular points at infinity; and so, in solid geometry, the focal of a surface or curve is the nodal line of the circumscribed developable of the surface or curve and the circle at infinity. And as in plane geometry the circular points at infinity may be regarded as an indefinitely thin conic, so in solid geometry the circle at infinity may be regarded as an indefinitely thin quadric surface.

In plane geometry, let it be proposed to find the circumscribed line-system (common tangents) of the two quadrics

$$\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} = 0, \quad \frac{x^2}{a'} + \frac{y^2}{b'} + \frac{z^2}{c'} = 0;$$

if a common tangent be

$$\xi x + \eta y + \zeta z = 0,$$

then we have

$$\begin{aligned} a\xi^2 + b\eta^2 + c\zeta^2 &= 0, \\ a'\xi^2 + b'\eta^2 + c'\zeta^2 &= 0, \end{aligned}$$

here writing for shortness

$$f, g, h = bc' - b'c, ca' - c'a, ab' - a'b,$$

we have

$$\xi^2 : \eta^2 : \zeta^2 = f : g : h;$$

and thence the tangent is

$$\sqrt{f}x + \sqrt{g}y + \sqrt{h}z = 0,$$

viz. we have thus four tangents, and the rationalised form is of course

$$f^2x^4 + g^2y^4 + h^2z^4 - 2ghy^2z^2 - 2hfz^2x^2 - 2fgx^2y^2 = 0.$$

In connexion with the corresponding question in solid geometry, I obtain this equation in a different manner. We investigate the envelope of the line $\xi x + \eta y + \zeta z = 0$, considering ξ, η, ζ as parameters connected by the foregoing two equations; by the ordinary process of indeterminate multipliers, we have

$$\begin{aligned} \xi x + \eta y + \zeta z &= 0, \\ a\xi^2 + b\eta^2 + c\zeta^2 &= 0, \\ a'\xi^2 + b'\eta^2 + c'\zeta^2 &= 0, \end{aligned}$$

and thence, eliminating ξ, η, ζ , we obtain

$$\begin{aligned} \frac{x}{\lambda a + \mu a'} &= \frac{y}{\lambda b + \mu b'} = \frac{z}{\lambda c + \mu c'}, \\ \frac{ax^2}{(\lambda a + \mu a')^2} + \frac{by^2}{(\lambda b + \mu b')^2} + \frac{cz^2}{(\lambda c + \mu c')^2} &= 0, \\ \frac{a'x^2}{(\lambda a + \mu a')^2} + \frac{b'y^2}{(\lambda b + \mu b')^2} + \frac{c'z^2}{(\lambda c + \mu c')^2} &= 0, \end{aligned}$$

equations equivalent to two equations, from which λ and μ are to be eliminated. The second and third equations are the derived functions $\frac{d}{d\lambda}, \frac{d}{d\mu}$ of the first equation in regard to λ, μ respectively; and hence, expressing the first equation in an integral form, the result is

$$\text{Disct. } [x^2(\lambda b + \mu b')(\lambda c + \mu c') + \&c.] = 0;$$

viz. this is

$$\{(bc' + b'c)x^2 + (ca' + c'a)y^2 + (ab' + a'b)z^2\}^2 - 4(bc x^2 + ca y^2 + ab z^2)(b'c'x^2 + c'a'y^2 + a'b'z^2) = 0,$$

or developing and reducing, we have the foregoing result, the coefficients entering through the combinations

$$f, g, h = bc' - b'c, ca' - c'a, ab' - a'b.$$

Writing $c = -1$, also $a' = b' = 1, c' = 0$: the equation of the first conic is

$$\frac{x^2}{a} + \frac{y^2}{b} - z^2 = 0,$$

and that of the second may be replaced by the two equations $x^2 + y^2 = 0, z = 0$, viz. these give the circular points at infinity: we have $f, g, h = 1, -1, a - b$, and the equation of the line-system is

$$x^4 + 2x^2y^2 + y^4 - 2h(x^2 - y^2)z^2 + h^2z^4 = 0.$$

If finally $z = 1$, then for the tangents from the circular points at infinity to the quadric

$$\frac{x^2}{a} + \frac{y^2}{b} - 1 = 0,$$

the equation is

$$x^4 + 2x^2y^2 + y^4 - 2h(x^2 - y^2) + h^2 = 0,$$

where, as before, $h = a - b$; the four tangents intersect in pairs in the two circular points at infinity, and in four other points which are the foci of the quadric.

Passing now to the problem in solid geometry, we consider the two quadric surfaces

$$\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} + \frac{w^2}{d} = 0,$$

$$\frac{x^2}{a'} + \frac{y^2}{b'} + \frac{z^2}{c'} + \frac{w^2}{d'} = 0;$$

if a common tangent plane hereof be

$$\xi x + \eta y + \zeta z + \omega w = 0,$$

then we have

$$a\xi^2 + b\eta^2 + c\zeta^2 + d\omega^2 = 0,$$

$$a'\xi^2 + b'\eta^2 + c'\zeta^2 + d'\omega^2 = 0,$$

and the circumscribed developable is the envelope of the plane, considering in the equation thereof ξ, η, ζ, ω as variable parameters connected by the last two equations. By what precedes, it is at once seen that the resulting equation is

$$\text{Disct. } [x^2(b\lambda + b'\mu)(c\lambda + c'\mu)(d\lambda + d'\mu) + \&c.] = 0,$$

viz. this is a quartic equation in (x^2, y^2, z^2, w^2) , the coefficients $a, b, c, d, a', b', c', d'$, entering therein through the combinations

$$\mathfrak{a}, \mathfrak{b}, \mathfrak{c} = bc' - b'c, ca' - c'a, ab' - a'b;$$

$$\mathfrak{f}, \mathfrak{g}, \mathfrak{h} = ad' - a'd, bd' - b'd, cd' - c'd.$$

The developed result is given in my paper "On the developable surfaces which arise from two surfaces of the second order," *Camb. and Dub. Math. Jour.*, t. v. (1850), pp. 45–53, [84]. We require here, for the particular case, $d = -1, a' = b' = c' = 1, d' = 0$, viz. one of the surfaces is taken to be

$$\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} - w^2 = 0,$$

and the other is the circle at infinity $x^2 + y^2 + z^2 = 0, w = 0$; we thus have $\mathfrak{a}, \mathfrak{b}, \mathfrak{c} = 1, 1, 1$; $\mathfrak{f}, \mathfrak{g}, \mathfrak{h} = b - c, c - a, a - b$, or now using the italic letters f, g, h to signify these values, we have $f + g + h = 0$. The equation is

$$\begin{aligned} & f^2 x^8 + g^2 y^8 + h^2 z^8 + f^2 g^2 h^2 w^8 \\ & + 2f(f - h)g^2 x^6 y^2 + 2g(g - h)f^2 x^2 y^6 + 2h(h - f)g^2 y^6 z^2 + 2g^2(h - f)y^2 z^6 \\ & + 2h(h - g)f^2 z^6 x^2 + 2h^2(f - g)x^6 z^2 \\ & - 2f^2 gh(g - h)x^6 w^2 - 2fg^2 h(h - f)y^6 w^2 - 2fgh^2(f - g)z^6 w^2 \\ & + f^2(f^2 - 6gh)x^4 w^4 + g^2(g^2 - 6hf)y^4 w^4 + h^2(h^2 - 6fg)z^4 w^4 \\ & - 2gh(gh - 3f^2)w^4 y^2 z^2 + 2hf(hf - 3g^2)w^4 z^2 x^2 + 2fg(fg - 3h^2)w^4 x^2 y^2 \\ & - 2f(hf - 3g^2)x^4 y^2 w^2 + 2g(fg - 3h^2)x^2 y^4 w^2 \\ & - 2h(hf - 3g^2)z^4 y^2 w^2 - 2f(fg - 3h^2)z^2 x^4 w^2 \\ & - 2(fg - 3h^2)x^4 y^2 z^2 - 2(hf - 3g^2)x^2 y^4 z^2 - 2(fg - 3h^2)x^2 y^2 z^4 \\ & + \dots = 0. \end{aligned}$$

The equation may be written in the following four forms:

$$\begin{aligned} & x^2 \Theta_1 + (gy^2 - hz^2 + ghw^2)^2 \{y^4 + 2y^2 z^2 + z^4 - 2f(y^2 - z^2)w^2 + f^2 w^4\} = 0, \\ & y^2 \Theta_2 + (hz^2 - fx^2 + hfw^2)^2 \{z^4 + 2z^2 x^2 + x^4 - 2g(z^2 - x^2)w^2 + g^2 w^4\} = 0, \\ & z^2 \Theta_3 + (fx^2 - gy^2 + fgw^2)^2 \{x^4 + 2x^2 y^2 + y^4 - 2h(x^2 - y^2)w^2 + h^2 w^4\} = 0, \\ & w^2 \Theta_4 + (x^2 + y^2 + z^2)^2 \times \{f^2 x^4 + g^2 y^4 + h^2 z^4 - 2ghy^2 z^2 - 2hfz^2 x^2 - 2fgx^2 y^2\} = 0; \end{aligned}$$

the last of these shows that the circle at infinity $x^2 + y^2 + z^2 = 0, w = 0$, is a nodal line on the surface; this, however, is not regarded as a focal. The other three show that we have also, as nodal lines, three conics, which are the focals of the given surface; viz. now writing $w = 1$, we have, for the quadric surface

$$\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} - 1 = 0,$$

the three focal conics

$$x = 0, \quad -\frac{y^2}{h} + \frac{z^2}{g} - 1 = 0,$$

$$y = 0, \quad \frac{x^2}{h} - \frac{z^2}{f} - 1 = 0,$$

$$z = 0, \quad -\frac{x^2}{g} + \frac{y^2}{f} - 1 = 0,$$

where, as before, $f, g, h = b - c, c - a, a - b$ respectively. If, as usual, a, b, c are positive and in order of decreasing magnitude, then for the ellipsoid

$$\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} - 1 = 0,$$

we have the focal conics

$$x = 0, \quad -\frac{y^2}{a-b} + \frac{z^2}{a-c} - 1 = 0,$$

$$y = 0, \quad \frac{x^2}{a-b} - \frac{z^2}{b-c} - 1 = 0,$$

$$z = 0, \quad \frac{x^2}{a-c} + \frac{y^2}{b-c} - 1 = 0;$$

viz. these are an imaginary conic, the focal hyperbola, and the focal ellipse, in the coordinate planes $x = 0, y = 0, z = 0$ respectively.

903.

ON LATIN SQUARES.

[From the *Messenger of Mathematics*, vol. XIX. (1890), pp. 135–137.]

IF in each line of a square of n^2 compartments the same n letters $a, b, c, \dots$ are arranged so that no letter occurs twice in the same column, we have what was termed by Euler “a Latin square.” Supposing that in one of the lines the letters are arranged in the natural order $abcde \dots$, then in the remaining lines there must be arrangements beginning with b, c, d, e , &c., respectively, and we may consider the case in which the bottom line has the arrangement $abcde \dots$, and in the other lines, reckoning from the bottom one in order, the arrangements begin with b, c, d, e , &c., respectively: if the number of such squares be $= N$, then, obviously, the whole number of squares which can be formed with the same n arrangements is $= N [n]^n$.

Starting with the bottom line as above, then it is a well-known problem to determine the number of arrangements for the second line; viz. this number is

$$= 1 \cdot \underline{n} \dots \left\{ 1 - \frac{1}{1} + \frac{1}{1 \cdot 2} - \dots \pm \frac{1}{1 \cdot 2 \dots n} \right\},$$

and if we assume, as above, that the second line begins with b , then the whole number of arrangements is this number divided by $(n - 1)$, the quotient being of course integral. For instance, when $n = 5$, the number is $= 120 - 120 + 60 - 20 + 5 - 1, = 44$, which is divisible by 4, and the number of arrangements for the second line is thus $= 11$.

But the number of arrangements for the third line will be different according to the arrangement selected for the second line, and it is not easy to see how in general the whole number of arrangements for the third line is to be calculated, and the difficulty of course increases for the next following lines; it may be remarked that, when all the lines are filled up except the top-line, the top-line is completely determined.

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LATE SADLERIAN PROFESSOR OF PURE MATHEMATICS
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Imagine the square completed: we may write down the substitutions by which we pass from the bottom line to itself (this is of course the substitution 1) and to each of the other lines respectively; we have thus a set of n substitutions, which may form a group; and when this is so, we may conversely from the group construct the Latin square. But it is not every Latin square which is thus connected with a group of n substitutions.

In the cases $n = 2, 3, 4$ there is no difficulty; the squares are

$$\begin{array}{cccccc} ba, & cab, & dcab, & dcba, & dcba, & dabc, \\ ab, & bca, & cdba, & cdab, & cadb, & cdab, \\ & abc, & badc, & badc, & bdac, & bcda, \\ & & abcd, & abcd, & abcd, & abcd; \end{array}$$

viz. when $n = 2$ the number is 1, when $n = 3$ it is 1, when $n = 4$ it is 4; in this last case, the arrangement *badc* for the second line gives two squares, but each of the other arrangements only one square.

In each of the squares of 4, we have a group, viz. for the four squares respectively, these are

$$\begin{array}{l} 1^\circ, \quad 1, (ab)(cd), (acbd), (adb), \\ 2^\circ, \quad 1, (ab)(cd), (ac)(bd), (ad)(bc), \\ 3^\circ, \quad 1, (abdc), (acdb), (ad)(bc), \\ 4^\circ, \quad 1, (abcd), (ac)(bd), (adcb). \end{array}$$

$1^\circ, 3^\circ, 4^\circ$ are the cyclical groups of $(acbd)$, $(abdc)$, and $(abcd)$, respectively: 2° is a different kind of group.

In the case when $n = 5$, the whole number of squares is 56; viz. there are five arrangements of the second line each giving four squares, and six arrangements each giving six squares, $5 \cdot 4 + 6 \cdot 6 = 56$. The five arrangements are

$$\begin{array}{cccccc} baecd, & badec, & bcaed, & bdeac, & bedca, & \\ abcde, & abcde, & abcde, & abcde, & abcde, & \end{array}$$

viz. in these cases, the substitutions for passing to the second line are $(ab)(ced)$, $(ab)(cde)$, $(abc)(de)$, $(abd)(ce)$, $(abe)(cd)$, respectively.

The six arrangements are

$$\begin{array}{cccccc} bdaec, & beacd, & bcead, & bedac, & bcdea, & bdeca, \\ abcde, & abcde, & abcde, & abcde, & abcde, & abcde; \end{array}$$

viz. in these cases, the substitutions for passing to the second line are $(abdec)$, $(abedc)$, $(abced)$, $(abecd)$, $(abcde)$, $(abdce)$, respectively.

A set of four squares is

$$\begin{array}{cccc} ecdab, & edabc, & ecdab, & edbac, \\ debac, & dceba, & deabc, & dcaeb, \\ cdaeb, & cedab, & cdbea, & cedba, \\ baecd, & baecd, & baecd, & baecd, \\ abcde, & abcde, & abcde, & abcde; \end{array}$$

and a set of six squares is

$$\begin{array}{cccccc} ecdab, & eadcb, & ecdab, & eabcd, & eabcd, & ecbad, \\ debca, & dceba, & daecb, & dceba, & dceab, & daecb, \\ caebd, & cebad, & cebad, & cedab, & cedba, & cedba, \\ bdaec, & bdaec, & bdaec, & bdaec, & bdeac, & bdaec, \\ abcde, & abcde, & abcde, & abcde, & abcde, & abcde. \end{array}$$

In a square belonging to a set of four, the substitutions for obtaining from any one line all the other lines are of a form such as 1, $(ab)(ced)$, $(ac)(bde)$, $(ad)(bec)$, $(ae)(bcd)$, which are not a group. In the case of a set of six squares, there is one square of the set (in the foregoing instance the first square) where the substitutions are of a form such as 1, $(abdec)$, $(acedb)$, $(adceb)$, $(aebcd)$, and which thus form a cyclical group of five substitutions.

904.

NOTE ON RECIPROCAL LINES.

[From the *Messenger of Mathematics*, vol. XIX. (1890), pp. 174, 175.]

IF two lines are reciprocal in regard to a quadric surface, then any point on the one line and any point on the other line are harmonics in regard to the surface, viz. the two points and the intersections of their line of junction with the surface form a harmonic range. This is obvious: the polar plane of the first point passes through the second line, and thus the second point is a point in the polar plane of the first point, that is, the two points are harmonics.

But it is worth while to look at the theorem in a different point of view. If the first line meets the surface in the points A, C and the second line meets the surface in the points B, D (in order to the four points being real, the surface must of course be a skew hyperboloid), then AB, BC, CD, DA are lines on the surface, say AB, CD are directrices, and BC, AD are generatrices, the two reciprocal lines being the diagonals AC and BD of the skew quadrilateral. Taking on AC a point P and on BD a point Q , the theorem is that, if PQ meets the surface in the points X, Y , then the four points P, Q, X, Y are harmonics. Let the generatrix through X meet AB, CD in the points S, S' respectively, and the generatrix through Y meet the same lines in the points T, T' respectively. Then the four lines $CB, DA, T'T, S'S$, are all met by an infinite series of lines, and in particular they are met by the lines CD, BA in the points C, D, T', S' and B, A, T, S respectively. Consequently, writing AH for anharmonic ratio, we have

$$\text{AH}(C, D, T', S') = \text{AH}(B, A, T, S).$$

Consider now the four lines $CA, DB, T'T, S'S$: these are each of them met by the lines CD and BA , and also by the line PQ : therefore, by an infinity of lines (that is, they are directrices of a hyperboloid). They are met by the last-mentioned three lines in the points C, D, T', S' ; A, B, T, S ; and P, Q, Y, X respectively; hence

$$\text{AH}(C, D, T', S') = \text{AH}(A, B, T, S) = \text{AH}(P, Q, Y, X).$$

Comparing with the foregoing equation, it appears that

$$\text{AH}(B, A, T, S) = \text{AH}(A, B, T, S),$$

viz. the anharmonic ratio is not altered by the interchange of the two points A, B : this implies that the points A, B, T, S are harmonics, viz. the pairs A, B and S, T are harmonics; and, this being so, the equation

$$\text{AH}(A, B, T, S) = \text{AH}(P, Q, Y, X)$$

shows that P, Q, Y, X are harmonics, viz. that the pairs P, Q and X, Y are harmonics; or say P, Q are harmonics in regard to the quadric surface, which is the theorem in question.

905.

ON THE EQUATION $x^{17} - 1 = 0$.

[From the *Messenger of Mathematics*, vol. XIX. (1890), pp. 184–188.]

WRITING $\rho = \cos \frac{2\pi}{17} + i \sin \frac{2\pi}{17}$, I carry the solution up to the determination of the periods each of two roots, $\rho + \rho^{16} = 2 \cos \frac{2\pi}{17}$, &c. The expressions contain the radicals

$$a = \sqrt{(17)}, \quad b = \sqrt{\{2(17 - a)\}}, \quad c = \sqrt{\{4(17 + 3a) - 2(3 + a)b\}},$$

where a, b, c are taken to be positive ($a = 4.12, b = 5.07, c = 6.72$). Taking for a moment r to be any imaginary seventeenth root, $r = \rho^\theta$, then the algebraical expression for the period P_1 of eight roots is $P_1 = \frac{1}{2}(-1 \pm a)$, but I assume the value to be $P_1 = \frac{1}{2}(-1 + a)$, and thus determine θ to denote some one of the values 1, 2, 4, 8, 9, 13, 15, 16; similarly, I assume the value of Q_1 to be $= \frac{1}{4}(-1 + a + b)$; and thus further determine θ to denote some one of the values 1, 4, 13, 16: and, again, I assume the value of R_1 to be $= \frac{1}{8}(-1 + a + b + c)$, and thus further determine θ to denote one of the values 1 and 16. As regards the values of the periods R , it is obviously indifferent which value is taken, and I assume therefore $\theta = 1$. This comes to saying that the signs of the radicals are determined in suchwise that r shall denote the root $\cos \frac{2\pi}{17} + i \sin \frac{2\pi}{17}$; and it is to be understood that r has this value.

I write now

$$\begin{aligned} P_1 &= r + r^9 + r^{13} + r^{15} + r^{16} + r^8 + r^4 + r^2, \\ P_2 &= r^3 + r^{10} + r^5 + r^{11} + r^{14} + r^7 + r^{12} + r^6, \\ Q_1 &= r + r^{13} + r^{16} + r^4, \\ Q_2 &= r^9 + r^{15} + r^8 + r^2, \\ Q_3 &= r^3 + r^5 + r^{14} + r^{12}, \\ Q_4 &= r^{10} + r^{11} + r^7 + r^6, \\ R_1 &= r + r^{16}, \\ R_2 &= r^{13} + r^4, \\ R_3 &= r^9 + r^8, \end{aligned}$$

$$\begin{aligned} R_4 &= r^{15} + r^2, \\ R_5 &= r^3 + r^{14}, \\ R_6 &= r^5 + r^{12}, \\ R_7 &= r^{10} + r^7, \\ R_8 &= r^{11} + r^6, \end{aligned}$$

and moreover

$$\begin{aligned} a &= P_1 - P_2, \\ b &= 2(Q_1 - Q_2), \\ +b_1 &= 2(Q_3 - Q_4), \\ c &= 4(R_1 - R_2), \\ -c_1 &= 4(R_3 - R_4), \\ +c_2 &= 4(R_5 - R_6), \\ -c_3 &= 4(R_7 - R_8). \end{aligned}$$

It will appear by what follows that a is determined by the quadric equation $a^2 = 17$, but I have assumed that a denotes the positive root $a = \sqrt{(17)}$; similarly, b is determined by the quadric equation $b^2 = 2(17 - a)$, but it is assumed that b denotes the positive root, $b = \sqrt{\{2(17 - a)\}}$; and c is determined by the quadric equation $c^2 = 4(17 + 3a) - 2(3 + a)b$, but it is assumed that c denotes the positive root,

$$c = \sqrt{\{4(17 + 3a) - 2(3 + a)b\}}.$$

If in the equations I had written b_1, c_1, c_2, c_3 , instead of $+b_1, -c_1, +c_2, -c_3$, then b_1 comes out rationally in terms of a, b ; and c_1, c_2, c_3 come out rationally in terms of a, b, c ; the signs were attached to them *à posteriori*, in suchwise that the values of b_1, c_1, c_2, c_3 might be each of them positive; for their independent determination, we have, in fact, for b_1^2 an expression such as that for b^2 ; and for c_1^2, c_2^2, c_3^2 expressions such as that for c^2 ; and taking as above for each of them the positive value of the square root, we have

$$\begin{aligned} a &= \sqrt{(17)}, & (= 4.12), \\ b &= \sqrt{\{2(17 - a)\}}, & (= 5.07), \\ b_1 &= \sqrt{\{2(17 + a)\}}, & (= 6.49), \\ c &= \sqrt{\{4(17 + 3a) - 2(3 + a)b\}}, & (= 6.72), \\ c_1 &= \sqrt{\{4(17 + 3a) + 2(3 + a)b\}}, & (= 13.77), \\ c_2 &= \sqrt{\{4(17 - 3a) + 2(-3 + a)b_1\}}, & (= 5.75), \\ c_3 &= \sqrt{\{4(17 - 3a) - 2(-3 + a)b_1\}}, & (= 2.02). \end{aligned}$$

The relations between the periods P are

$$P_1 + P_2 = -1,$$

P_1	P_2	
P_1	$-P_1 + 4$	-4
P_2		$-P_2 + 4$

a	$\frac{a}{17}$
-----	----------------

that is,

$$P_1^2 = -P_1 + 4, \text{ \&c.; } a^2 = 17.$$

Here for the second square, we have

$$a^2 = (P_1 - P_2)^2 = P_1^2 + P_2^2 - 2P_1P_2 = (-P_1 + 4) + (-P_2 + 4) + 8, = -P_1 - P_2 + 16, = 17;$$

and similarly in the other cases which follow.

For the periods Q , we have

$$Q_1 + Q_2 = P_1, \quad Q_3 + Q_4 = P_2,$$

	Q_1	Q_2	Q_3	Q_4
Q_1	$Q_2 + 2Q_3 + 4$	-1	$2Q_1 + Q_2 + Q_4$	$Q_2 + Q_3 + 2Q_4$
Q_2		$Q_1 + 2Q_4 + 4$	$Q_1 + 2Q_3 + 4$	$Q_1 + 2Q_3 + Q_3$
Q_3			$2Q_2 + Q_4 + 4$	-1
Q_4				$2Q_1 + Q_3 + 4$

	b	b_1
b	$34 - 2a$	$8a$
b_1		$34 + 2a$

so that $bb_1 = 8a$, or b_1 is given rationally in terms of a, b .

And for the periods R , we have

$$R_1 + R_2 = Q_1, \quad R_3 + R_4 = Q_2, \quad R_5 + R_6 = Q_3, \quad R_7 + R_8 = Q_4,$$

	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8
R_1	$R_4 + 2$	$R_3 + R_8$	$R_2 + R_7$	$R_1 + R_5$	$R_2 + R_4$	$R_2 + R_8$	$R_3 + R_8$	$R_8 + R_7$
R_2		$R_5 + 2$	$R_2 + R_8$	$R_4 + R_8$	$R_1 + R_7$	$R_1 + R_3$	$R_2 + R_8$	$R_4 + R_7$
R_3			$R_1 + 2$	$R_7 + R_8$	$R_8 + R_7$	$R_2 + R_8$	$R_1 + R_4$	$R_4 + R_5$
R_4				$R_2 + 2$	$R_1 + R_8$	$R_3 + R_7$	$R_3 + R_6$	$R_2 + R_5$
R_5					$R_6 + 2$	$R_3 + R_4$	$R_2 + R_7$	$R_8 + R_8$
R_6						$R_7 + 2$	$R_4 + R_6$	$R_1 + R_8$
R_7							$R_8 + 2$	$R_1 + R_5$
R_8								$R_3 + 2$

	c	c_1	c_2	c_3
c	$4(17 + 3a) - 2(3 + a)b$	$8(b + b_1)$	$4(2a - b + b_1)$	$4(-2a + b + b_1)$
c_1		$4(17 + 3a) + 2(3 + a)b$	$4(2a + b + b_1)$	$4(2a + b - b_1)$
c_2			$4(17 - 3a) + 2(-3 + a)b_1$	$8(-b + b_1)$
c_3				$4(17 - 3a) - 2(-3 + a)b_1$

where observe that the overlined terms $R_5 + R_6$, $R_7 + R_8$, $R_3 + R_4$, and $R_1 + R_2$, have the values Q_3 , Q_4 , Q_2 , Q_1 , respectively, and in the last table b_1 may be considered as denoting its value $\frac{8a}{b}$, so that c_1, c_2, c_3 are each given rationally in terms of a, b, c .

And from the foregoing results, we have

$$\begin{aligned}
 P_1 &= \frac{1}{2}(-1 + a), & &= 1\cdot56, \\
 P_2 &= \frac{1}{2}(-1 - a), & &= -2\cdot56, \\
 Q_1 &= \frac{1}{4}(-1 + a + b), & &= 2\cdot05, \\
 Q_2 &= \frac{1}{4}(-1 + a - b), & &= -0\cdot49, \\
 Q_3 &= \frac{1}{4}(-1 - a + b_1), & &= 0\cdot34, \\
 Q_4 &= \frac{1}{4}(-1 - a - b_1), & &= -2\cdot90, \\
 R_1 &= \frac{1}{8}(-1 + a + b + c), & &= 1\cdot87, \\
 R_2 &= \frac{1}{8}(-1 + a + b - c), & &= 0\cdot18, \\
 R_3 &= \frac{1}{8}(-1 + a - b - c_1), & &= -1\cdot96, \\
 R_4 &= \frac{1}{8}(-1 + a - b + c_1), & &= 1\cdot47, \\
 R_5 &= \frac{1}{8}(-1 - a + b_1 + c_2), & &= 0\cdot89, \\
 R_6 &= \frac{1}{8}(-1 - a + b_1 - c_2), & &= -0\cdot55, \\
 R_7 &= \frac{1}{8}(-1 - a - b_1 - c_3), & &= -1\cdot70, \\
 R_8 &= \frac{1}{8}(-1 - a - b_1 + c_3), & &= -1\cdot20.
 \end{aligned}$$

The approximate numerical values have been given throughout only for the purpose of showing that the signs of the square roots have been rightly determined.

906.

NOTE ON SCHLAEFLI'S MODULAR EQUATION FOR THE CUBIC TRANSFORMATION; WITH A CORRECTION.

[From the *Messenger of Mathematics*, vol. XX. (1891), pp. 59, 60; 120.]

THE equation in question, *Crelle*, t. LXXII. (1870), p. 369, is

$$S^4 + T^4 + 8ST - S^3T^3 = 0,$$

where

$$S = \frac{\sqrt{2}}{\sqrt[4]{(kk')}} , \quad T = \frac{\sqrt{2}}{\sqrt[4]{(\lambda\lambda')}} ,$$

which must be of course equivalent to the ordinary modular equation

$$u^4 - v^4 + 2uv - 2u^3v^3 = 0,$$

where

$$u = \sqrt[4]{k}, \quad v = \sqrt[4]{\lambda};$$

the resemblance in form between the two equations, with such different meanings of the S and T in the one case, and the u and v in the other, is very noticeable.

Schlaefli's equation is

$$\frac{4}{kk'} + \frac{4}{\lambda\lambda'} + \frac{16}{(kk'\lambda\lambda')^{1/4}} - \frac{8}{(kk'\lambda\lambda')^{3/4}} = 0,$$

that is,

$$kk' + \lambda\lambda' = 2(kk'\lambda\lambda')^{3/4} - 4(kk'\lambda\lambda')^{1/2},$$

or say

$$k^2k'^2 + \lambda^2\lambda'^2 = 4(kk'\lambda\lambda')^{3/2} - 18(kk'\lambda\lambda').$$

To deduce this from the uv -modular equation, we have (Jacobi's *Fund. Nova*, p. 68, *Ges. Werke*, t. I., p. 124),

$$(1 - u^8)(1 - v^8) = (1 - u^2v^2)^4,$$

or, multiplying each side by u^8v^8 , and extracting the fourth root, we have

$$\sqrt{(kk'\lambda\lambda')} = u^2v^2(1 - u^2v^2) = x - x^2,$$

if for shortness we write $x = u^2v^2$.

The equation to be proved thus is

$$u^8(1 - u^8) + v^8(1 - v^8) = 4(x - x^2)^3 - 18(x - x^2)^2 + 16(x - x^2)^3.$$

But from the foregoing equation

$$(1 - u^8)(1 - v^8) = (1 - u^2v^2)^4,$$

we have

$$u^8 + v^8 = 4u^2v^2 - 6u^4v^4 + 4u^6v^6,$$

and thence

$$u^{16} + v^{16} = (4x - 6x^2 + 4x^3)^2 - 2x^4;$$

and the equation to be proved thus becomes

$$(4x - 6x^2 + 4x^3) - (4x - 6x^2 + 4x^3)^2 + 2x^4 = 4(x - x^2)^3 - 18(x - x^2)^2 + 16(x - x^2)^3,$$

which is in fact an identity, each side being

$$= 4x - 22x^2 + 52x^3 - 66x^4 + 48x^5 - 16x^6.$$

CORRECTION, P. 120.

I find that I misquoted Schlaefli's equation, viz. in effect, I took it to be

$$S_1^4 + T_1^4 + 8S_1T_1 - S_1^3T_1^3 = 0, \quad \text{where } S_1 = \frac{\sqrt{2}}{\sqrt[4]{(kk')}}, \quad T_1 = \frac{\sqrt{2}}{\sqrt[4]{(\lambda\lambda')}};$$

whereas his equation really is

$$S^4 + T^4 - 8ST + S^3T^3 = 0, \quad \text{where } S = 2\sqrt[4]{(kk')}, \quad T = 2\sqrt[4]{(\lambda\lambda')}.$$

The change is only a change of form, for writing $S_1 = \frac{2}{S}$ and $T_1 = \frac{2}{T}$, the equation in (S_1, T_1) is converted into that in (S, T) ; but it was quite an unnecessary one, and I cannot account for having made it, as the paper in *Crelle* must have been before me.

907.

NOTE ON THE NINTH ROOTS OF UNITY.

[From the *Messenger of Mathematics*, vol. xx. (1891), p. 63.]

LET θ be a prime ninth root of unity, so that $\theta^6 + \theta^3 + 1 = 0$; and write

$$\begin{aligned} a &= \theta + \theta^8, \\ b &= \theta^2 + \theta^7, \\ c &= \theta^4 + \theta^5; \end{aligned}$$

then

$$a + b + c = \theta + \theta^2 + \cdots + \theta^8 = \frac{1 - \theta^9}{1 - \theta} - 1, = -1,$$

that is,

$$a + b + c = 0.$$

Also

$$\begin{aligned} a^2 &= b + 2, & bc &= b - 1, \\ b^2 &= c + 2, & ca &= c - 1, \\ c^2 &= a + 2, & ab &= a - 1, \end{aligned}$$

whence

$$\begin{aligned} ab + ac + bc &= -3, \\ abc &= -1; \end{aligned}$$

and a, b, c are thus the roots of the equation $x^3 - 3x + 1 = 0$. We have

$$\begin{aligned} a^2b + b^2c + c^2a &= a^2 + b^2 + c^2 = 6, \\ ab^2 + bc^2 + ca^2 &= bc + ca + ab = -3, \end{aligned}$$

and thence

$$-(a^2b + b^2c + c^2a) + (ab^2 + bc^2 + ca^2) = (b - c)(c - a)(a - b) = -6 - 3, = -9.$$

The equation $x^3 - 3x + 1 = 0$ is thus such that $a^2b + b^2c + c^2a$, and consequently any rational function whatever of a, b, c , invariable by the cyclical interchange (abc) of the roots, has a rational value.

908.

ON TWO INVARIANTS OF A QUADRIQUADRIC FUNCTION.

[From the *Messenger of Mathematics*, vol. xx. (1891), pp. 68, 69.]

THE quadriquadric function

$$\begin{aligned} & z^2 (ax^2 + 2hxy + g'y^2) \\ & + 2zw(h'x^2 + 2bxy + fy^2) \\ & + w^2(gx^2 + 2f'xy + cy^2), \end{aligned}$$

considered successively as a function of (z, w) and of (x, y) , has the discriminants U, V , equal to

$$\begin{aligned} & (ax^2 + 2hxy + g'y^2)(gx^2 + 2f'xy + cy^2) - (h'x^2 + 2bxy + fy^2)^2, \\ & (az^2 + 2h'zw + gw^2)(g'z^2 + 2f'zw + cw^2) - (hz^2 + 2bzw + f'w^2)^2, \end{aligned}$$

respectively. As is well known, these quartic functions have each of them the same quadrinvariant and the same cubinvariant; these are the invariants in question of the quadriquadric function.

The quadrinvariant has been calculated in a different notation, but I am not aware that the cubinvariant has been before calculated; the two values are as follows:

Quadrinvariant is

$$\begin{aligned} & a^2c^2 \quad + 3 \\ & ab^2c \quad - 24 \\ & b^4 \quad + 48 \\ & acgg' \quad + 42 \\ & achf' \quad - 12 \\ & ach'f \quad - 12 \\ & abff' \quad + 72 \\ & bchh' \quad + 72 \\ & b^2gg' \quad - 24 \\ & b^2fh' \quad - 96 \\ & b^2f'h \quad - 96 \\ & af^2g \quad - 36 \\ & af'^2g' \quad - 36 \\ & bfg h \quad + 72 \\ & bfg'h' \quad + 72 \\ & cgh^2 \quad - 36 \\ & cg'h'^2 \quad - 36 \\ & g^2g'^2 \quad + 3 \\ & f^2h'^2 \quad + 48 \\ & f'^2h^2 \quad + 48 \\ & gg'hf' \quad - 12 \\ & gg'h'f \quad - 12 \\ & ff'h'h' \quad - 48 \\ & \hline & \pm 480 \end{aligned}$$

Cubinvariant is

The cubinvariant has the value ± 2866 obtained by combining the following groups of terms with the indicated coefficients:

Group 1 (+76, -49):

$$\begin{aligned} a^2c^3 & - 1 \\ a^2b^2c^2 & + 12 \\ ab^4c & - 48 \\ b^6 & + 64 \end{aligned}$$

Group 2 (+381, -624):

$$\begin{aligned} a^2c^2gg' & + 33 \\ ab^2cgg' & - 120 \\ a^2c^2fh' & + 6 \\ a^2c^2f'h & + 6 \\ ab^2cfh' & + 24 \\ ab^2cf'h & + 24 \\ b^4fh' & - 192 \\ b^4f'h & - 192 \\ b^4gg' & - 48 \\ a^2bcff' & - 36 \\ abc^2hh' & - 36 \\ ab^3ff' & + 144 \\ b^3chh' & + 144 \end{aligned}$$

Group 3 (+648, -432):

$$\begin{aligned} acg^2g'^2 & + 33 \\ acfg'h & + 60 \\ acfgh' & + 60 \\ acf^2h'^2 & + 24 \\ acf'^2h^2 & + 24 \\ acff'hh' & + 12 \\ abff'gg' & + 180 \\ abf^2f'h' & - 144 \\ abff'^2h & - 144 \\ bcgg'hh' & + 180 \\ bcfhh'^2 & - 144 \\ bcf'h^2h' & - 144 \\ a^2f^2f'^2 & + 54 \\ c^2h^2h'^2 & + 54 \\ b^2f^2h^2 & + 192 \\ b^2f'^2h'^2 & + 192 \\ b^2ff'hh' & + 96 \\ b^2fgg'h' & + 24 \\ b^2f'gg'h & + 24 \end{aligned}$$

Group 4 (+288, -936):

$$\begin{aligned}
 af^2fgh & - 36 \\
 aff'^2g'h' & - 36 \\
 af^2g^2g' & - 36 \\
 afgg'^2 & - 36 \\
 af^3gh' & + 72 \\
 af'^3g'h & + 72 \\
 cfgh^2h' & - 36 \\
 cf'g'h'h'^2 & - 36 \\
 cg^2g'h^2 & - 36 \\
 cgg'^2h'^2 & - 36 \\
 cfgh^3 & + 72 \\
 cf'g'h'^3 & + 72 \\
 bff'g'h'^2 & - 144 \\
 bf^2g'h'h' & - 144 \\
 bf^2ghh' & - 144 \\
 bf'gh^2 & - 144 \\
 bfg^2g'h & - 36 \\
 bf'gg'^2h' & - 36
 \end{aligned}$$

Group 5 (+1101, -696, +372, -129):

$$\begin{aligned}
 acf'^2g & - 36 \\
 acf^2g' & - 36 \\
 ac^2gh^2 & - 36 \\
 ac^2g'h'^2 & - 36 \\
 ab^2cf'^2g & - 72 \\
 ab^2cf^2g' & - 72 \\
 ab^2f^2g'h' & - 144 \\
 ab^2f'^2gh & - 144 \\
 acf'^2h^2 & + 24 \\
 acf^2h'^2 & + 24 \\
 af^3gh' & + 72 \\
 af'^3g'h & + 72 \\
 f^4h'^4 & - 64 \\
 f'^4h^4 & - 64 \\
 g^3g'^3 & - 1 \\
 g^2g'^2hh' & + 6 \\
 fg^2g'^2h' & + 6 \\
 f'g^2g'^2h & + 6 \\
 fgg'h'^2 & + 24 \\
 f'gg'h^2 & + 24 \\
 ff'gg'h'h' & + 12 \\
 f^3h^3 & - 64 \\
 f'^3h'^3 & - 64 \\
 f^2gh^2 & + 54 \\
 f'^2g'h'^2 & + 54 \\
 f^2f'h'h'^2 & + 96 \\
 ff'^2h^2h' & + 96
 \end{aligned}$$

(The total grand sum is ± 2866 , the cubinvariant of the quadriquadric.)

By writing herein $f', g', h' = f, g, h$, we obtain of course the two invariants of the symmetrical quadriquadric function.

909.

ON A PARTICULAR CASE OF KUMMER'S DIFFERENTIAL EQUATION OF THE THIRD ORDER.

[From the *Messenger of Mathematics*, vol. XX. (1891), pp. 75–79.]

THE general form of equation in question is

$$\frac{x'''}{x'} - \frac{3}{2} \left(\frac{x''}{x'} \right)^2 + x'^2 \left\{ \frac{A}{(x-1)^2} + \frac{B}{x(x-1)} + \frac{C}{x^2} \right\} - \left\{ \frac{A'}{(t-1)^2} + \frac{B'}{t(t-1)} + \frac{C'}{t^2} \right\} = 0,$$

here x is a function of t ; and A, B, C, A', B', C' are numerical constants. For various given values of A, B, C , and values determined thereby of A', B', C' , the equation admits of a solution in the form $x =$ rational function of t ; the theory in reference to the cases considered by Schwarz is considered in my paper "On the Schwarzian Derivative and the Polyhedral Functions," *Camb. Phil. Trans.*, t. XIII. (1883), pp. 5–68, [744]. But the theory is considered in a more general and exhaustive manner in Goursat's memoir, "Recherches sur l'équation de Kummer," *Acta Soc. Sci. Fennicæ*, t. XV. (1888), pp. 47–127. I consider here one of the solutions given by him, viz. writing

$$\begin{aligned} P &= 4t - 5, & X &= t^2 P^3, \\ Q &= 5t - 4, & Y &= Q^3, \\ R &= 8t^2 - 11t + 8, & Z &= -(t-1)^2 R^2, \end{aligned}$$

so that, identically, $X + Y + Z = 0$; then the solution is expressed by either of the equivalent equations

$$\begin{aligned} x &= -\frac{X}{Z} = \frac{t^2 P^3}{(t-1)^2 R^2}, \\ x-1 &= \frac{Y}{Z} = -\frac{Q^3}{(t-1)^2 R^2}. \end{aligned}$$

The values of the constants to which this solution belongs are

$$A = \frac{4}{9}, \quad B = -\frac{32}{63}, \quad C = \frac{4}{9}; \quad A' = \frac{4}{9}, \quad B' = \frac{131}{144}, \quad C' = \frac{5}{9}.$$

But instead of assuming these values in the first instance, I leave the values indeterminate; and starting from the foregoing expression for x , I substitute this in

$$\Omega = \frac{x'''}{x'} - \frac{3}{2} \left(\frac{x''}{x'} \right)^2 + x'^2 \left\{ \frac{A}{(x-1)^2} + \frac{B}{x(x-1)} + \frac{C}{x^2} \right\} - \left\{ \frac{A'}{(t-1)^2} + \frac{B'}{t(t-1)} + \frac{C'}{t^2} \right\},$$

thus obtaining Ω as a function of t which, as will appear, vanishes identically when A, B, C, A', B', C' have the foregoing values.

I remark that this is, in effect, doing in a somewhat different form for the particular case what Goursat does for the general case, viz. starting from

$$\Omega_1 = \frac{x'''}{x'} - \frac{3}{2} \left(\frac{x''}{x'} \right)^2 + x'^2 \left\{ \frac{A}{(x-1)^2} + \frac{B}{x(x-1)} + \frac{C}{x^2} \right\},$$

with values of A, B, C which belong to the solution considered, he shows that this is a function of t having no infinities other than $(0, 1, \infty)$; that ∞ is not an infinity of the function or of the function multiplied into t , and that 0 and 1 are each of them a twofold infinity; that is, that the function is of the form

$$\frac{Lt^2 + Mt + N}{t^2(t-1)^2} \text{ or } \frac{A'}{(t-1)^2} + \frac{B'}{t(t-1)} + \frac{C'}{t^2}.$$

Proceeding to carry out the process, we have

$$\begin{aligned}\frac{x'}{x} &= \frac{2}{t} + \frac{3P'}{P} - \frac{2R'}{R}, \\ \frac{x'}{x-1} &= -\frac{1}{t-1} + \frac{3Q'}{Q} - \frac{2R'}{R}, \\ \frac{x'}{x} &= \frac{2}{t} - \frac{1}{t-1} + \frac{3P'}{P} - \frac{2R'}{R},\end{aligned}$$

and from either of these equations, collecting and reducing,

$$x' = \frac{5t^2P^2Q^2}{(t-1)^3R^3},$$

where observe that, from the values of x and $x-1$ respectively, it appears *à priori* that tP^2 and Q^2 must be factors in the numerator of x' . From this value of x' , we have

$$\frac{x''}{x'} = \frac{1}{t} - \frac{2}{t-1} + \frac{2P'}{P} + \frac{2Q'}{Q} - \frac{3R'}{R};$$

and hence, P' and Q' being mere constants,

$$\frac{x'''}{x'} - \left(\frac{x''}{x'}\right)^2 = -\frac{1}{t^2} + \frac{2}{(t-1)^2} - \frac{3R''}{R} - 2\left(\frac{P'}{P}\right)^2 - 2\left(\frac{Q'}{Q}\right)^2 + 3\left(\frac{R'}{R}\right)^2,$$

and consequently

$$\begin{aligned}\frac{x'''}{x'} - \frac{3}{2}\left(\frac{x''}{x'}\right)^2 &= -\frac{1}{t^2} + \frac{2}{(t-1)^2} - \frac{3R''}{R} - 2\left(\frac{P'}{P}\right)^2 - 2\left(\frac{Q'}{Q}\right)^2 + 3\left(\frac{R'}{R}\right)^2 \\ &\quad - \frac{1}{2}\left(\frac{1}{t} - \frac{2}{t-1} + \frac{2P'}{P} + \frac{2Q'}{Q} - \frac{3R'}{R}\right)^2.\end{aligned}$$

Putting for shortness

$$\frac{1}{t} = \alpha, \quad \frac{1}{t-1} = \beta, \quad \frac{P'}{P} = p, \quad \frac{Q'}{Q} = q, \quad \frac{R'}{R} = r,$$

this equation gives

$$\begin{aligned}\Omega &= -\alpha^2 + 2\beta^2 - \frac{3R''}{R} - 2p^2 - 2q^2 + 3r^2 \\ &\quad - \frac{1}{2}(\alpha - 2\beta + 2p + 2q - 3r)^2 \\ &\quad + A(-\beta + 3q - 2r)^2 \\ &\quad + B(-\beta + 3q - 2r)(2\alpha - \beta + 3p - 2r) \\ &\quad + C(2\alpha - \beta + 3p - 2r)^2 \\ &\quad - C'\alpha^2 - B'\alpha\beta - A'\beta^2,\end{aligned}$$

which is

$$\begin{aligned}
 &= \alpha^2 \left(-\frac{1}{2} + 4C - C' \right) + \frac{Lt^2}{R} \text{ say it is } = L\alpha^2 - \frac{48}{R} \\
 &\quad + \alpha\beta(2 - 2B - 4C - B') \quad \quad \quad + M\alpha\beta \\
 &\quad + \beta^2(A + B + C - A') \quad \quad \quad + N\beta^2 \\
 &\quad + \alpha p(-2 + 12C) \quad \quad \quad + F\alpha p \\
 &\quad + \alpha q(-2 + 6B) \quad \quad \quad + G\alpha q \\
 &\quad + \alpha r(3 - 4B - 8C) \quad \quad \quad + H\alpha r \\
 &\quad + \beta p(4 - 3B - 6C) \quad \quad \quad + F'\beta p \\
 &\quad + \beta q(4 - 6A - 3B) \quad \quad \quad + G'\beta q \\
 &\quad + p^2(-4 + 9C) \quad \quad \quad + A''p^2 \\
 &\quad + q^2(-4 + 9A) \quad \quad \quad + B''q^2 \\
 &\quad + r^2(-1 + 4A + 4B + 4C) \quad \quad \quad + C''r^2 \\
 &\quad + qr(6 - 12A - 6B) \quad \quad \quad + F''qr \\
 &\quad + rp(6 - 6B - 12C) \quad \quad \quad + G''rp
 \end{aligned}$$

By decomposing $\alpha\beta$, αp , &c., into simple fractions, this becomes

$$\begin{aligned}
 \Omega = & Lx - \frac{48}{R} \\
 & + M(-\alpha + \beta) \\
 & + N\beta^2 \\
 & + F\left(-\frac{5}{4}\alpha + \frac{5}{4}p\right) \\
 & + G\left(-\frac{4}{5}\alpha + \frac{4}{5}q\right) \\
 & + H\left(-\frac{11}{8}\alpha + \frac{11t + \frac{5}{8}}{R}\right) \\
 & + F'(-4\beta + 4p) \\
 & + G'(-5\beta - 5q) \\
 & + H'\left(\beta + \frac{-8t + 19}{R}\right) \\
 & + A''p^2 \\
 & + B''q^2 \\
 & + C''r^2 \\
 & + F''\frac{8}{5}\left(q - \frac{8t - 43}{R}\right) \\
 & + G''\frac{8}{5}\left(p - \frac{8t - 13}{R}\right) \\
 & + H''\frac{8t}{5}(p - q).
 \end{aligned}$$

This is

$$\begin{aligned}
 &= \alpha L \\
 &\quad + \alpha(-M - \frac{5}{4}F - \frac{4}{5}G - \frac{11}{8}H) \\
 &\quad + \beta^2 N \\
 &\quad + \beta(M - 4F' + 5G' + H') \\
 &\quad + p^2 A'' \\
 &\quad + p(4F' + \frac{5}{4}G'' + \frac{8t}{5}H'' + \frac{8}{5}F) \\
 &\quad + q^2 B'' \\
 &\quad + q(-5G' - \frac{8}{5}F'' - \frac{8t}{5}H'' + \frac{4}{5}G) \\
 &\quad + r^2 C'' \\
 &\quad + \frac{1}{R}[-48 + H(11t + \frac{5}{8}) + H'(-8t + 19) - \frac{8}{5}F''(8t - 43) - \frac{8}{5}G''(8t - 13)].
 \end{aligned}$$

This should be identically = 0; making $A'' = 0$, $B'' = 0$, $C'' = 0$, we find

$$A = \frac{4}{9}, \quad B = -\frac{32}{63}, \quad C = \frac{4}{9}; \quad (A + B + C = \frac{8}{63}).$$

and thence

$$F = \frac{4.9}{9}, \quad G = -\frac{2.4}{63}, \quad H = \frac{2}{3}; \quad F' = \frac{2.9}{63}, \quad G' = -\frac{1.2}{63}, \quad H' = -\frac{1}{3}; \quad F'' = \frac{1.2}{63}, \quad G'' = \frac{1.2}{63}, \quad H'' = -\frac{2}{3}.$$

These values make the coefficients of p and q to be each = 0; and they make the coefficient of R to be identically = 0, viz. we have

$$0 = -48 + \frac{1}{8}H + 19H' + \frac{344}{5}F'' + \frac{104}{5}G'',$$

and

$$0 = 11H - 8H' - \frac{64}{5}F'' - \frac{64}{5}G''.$$

We have, moreover,

$$L = \frac{1}{2} - C', \quad M = \frac{131}{144} - B', \quad N = \frac{4}{9} - A';$$

and the coefficients of α and β are $= -M + \frac{131}{144}$ and $M - \frac{131}{144}$ respectively; hence the coefficients of α^2 , β^2 , α and β will all vanish if only $L = 0$, $M = \frac{131}{144}$, $N = 0$, that is,

$$A' = \frac{4}{9}, \quad B' = \frac{131}{144}, \quad C' = \frac{5}{9};$$

and we have thus identically $\Omega = 0$, if only A, B, C, A', B', C' have the above-mentioned values.

910.

NOTE ON THE INVOLUTANT OF TWO BINARY MATRICES.

[From the *Messenger of Mathematics*, vol. xx. (1891), pp. 136, 137.]

CONSIDER the two matrices

$$M = \begin{pmatrix} a, & b \\ c, & d \end{pmatrix}, \quad M' = \begin{pmatrix} a', & b' \\ c', & d' \end{pmatrix},$$

and their product in one or the other order

$$MM' = \begin{pmatrix} A, & B \\ C, & D \end{pmatrix}, \quad M'M = \begin{pmatrix} A_1, & B_1 \\ C_1, & D_1 \end{pmatrix}.$$

Then the Involutant is by definition = either of the determinants

$$I = \begin{vmatrix} 1, & a, & a', & A \\ 0, & b, & b', & B \\ 0, & c, & c', & C \\ 1, & d, & d', & D \end{vmatrix}, \quad I_1 = \begin{vmatrix} 1, & a', & a, & A_1 \\ 0, & b', & b, & B_1 \\ 0, & c', & c, & C_1 \\ 1, & d', & d, & D_1 \end{vmatrix};$$

viz. it is to be shown that these two values are in fact equal.

We have

$$MM' = \begin{pmatrix} a, & b \\ c, & d \end{pmatrix} \begin{pmatrix} a', & b' \\ c', & d' \end{pmatrix} = \begin{pmatrix} A, & B \\ C, & D \end{pmatrix},$$

that is,

$$\begin{aligned} A &= aa' + bc', & B &= ab' + bd', \\ C &= ca' + dc', & D &= cb' + dd', \end{aligned}$$

and similarly

$$M'M = \begin{pmatrix} a', & b' \\ c', & d' \end{pmatrix} \begin{pmatrix} a, & b \\ c, & d \end{pmatrix} = \begin{pmatrix} A_1, & B_1 \\ C_1, & D_1 \end{pmatrix},$$

that is,

$$\begin{aligned} A_1 &= aa' + cb', & B_1 &= ba' + db', \\ C_1 &= ac' + cd', & D_1 &= bc' + dd', \end{aligned}$$

viz. A_1, B_1, C_1, D_1 are obtained from A, B, C, D by the interchange of the accented and unaccented letters.

We have then, from the first expression for the Involutant,

$$\begin{aligned} I &= A \begin{vmatrix} 0, & b, & b' \\ 0, & c, & c' \\ 1, & d, & d' \end{vmatrix} - B \begin{vmatrix} 0, & c, & c' \\ 1, & d, & d' \\ 0, & b, & b' \end{vmatrix} + C \begin{vmatrix} 1, & d, & d' \\ 0, & b, & b' \\ 0, & c, & c' \end{vmatrix} - D \begin{vmatrix} 1, & a, & a' \\ 0, & b, & b' \\ 0, & c, & c' \end{vmatrix} \\ &= A(bc' - b'c) - B(ac' - a'c + cd' - c'd) + C(ab' - a'b + bd' - b'd) - D(bc' - b'c); \end{aligned}$$

or substituting for $A - D, B$ and C their values, this is

$$\begin{aligned} &= (aa' - dd' + bc' - b'c)(bc' - b'c) - (ab' + bd')(ac' - a'c + cd' - c'd) \\ &\quad + (ca' + dc')(ab' - a'b + bd' - b'd); \end{aligned}$$

and multiplying out and grouping together the terms in $bc, b'c', bc'$ and $b'c$, this is found to be

$$= -(a' - d')^2 bc + (a - d)(a' - d')(bc' + b'c) - (a - d)^2 b'c' + (bc' - b'c)^2,$$

which is

$$= -\{(a-d)b' - (a'-d')b\}\{(a-d)c - (a'-d')c'\} + (bc' - b'c)^2.$$

Hence, writing

$$a = bc' - b'c, \quad f = ad' - a'd,$$

$$b = ca' - c'a, \quad g = bd' - b'd,$$

$$c = ab' - a'b, \quad h = cd' - c'd,$$

we have

$$I = -(c+g)(-b+h) + a^2,$$

that is,

$$I = bc - ch + bg - gh + a^2.$$

To obtain the value of I_1 , we must interchange the accented and unaccented letters, that is, change the signs of the several quantities a, b, c, f, g, h ; but I , being a quadric function of the six quantities, is not altered by the change; that is, we have $I = I_1$.

911.

ON AN ALGEBRAICAL IDENTITY RELATING TO THE SIX
COORDINATES OF A LINE.

[From the *Messenger of Mathematics*, vol. XX. (1891), pp. 138–140.]

THE following identity may be verified without difficulty; but it is interesting in regard to the analytical theory of the line.

The identity is

$$\begin{aligned}
 0 = & \begin{vmatrix} B, & C, & F \\ b, & c, & f \\ b', & c', & f' \end{vmatrix} \begin{vmatrix} A, & B, & C \\ a, & b, & c \\ a', & b', & c' \end{vmatrix} \\
 & - (bc' - b'c)^2 (AF + BG + CH) \\
 & + (bc' - b'c)(bC - cB)(Af + Bg' + Ch' + Fa' + Gb' + Hc') \\
 & - (bc' - b'c)(b'C - c'B)(Af + Bg + Ch + Fa + Gb + Hc) \\
 & - (bC - cB)^2 (a'f + b'g' + c'h') \\
 & + (bC - cB)(b'C - c'B)(af + a'f' + bg' + b'g + ch' + c'h) \\
 & - (b'C - c'B)^2 (af + bg + ch),
 \end{aligned}$$

where all the letters have arbitrary values.

It follows that, if

$$\begin{aligned}
 af + bg + ch &= 0, \\
 a'f' + b'g' + c'h' &= 0, \\
 af' + a'f + bg' + b'g + ch' + c'h &= 0, \\
 AF + BG + CH &= 0, \\
 Af + Bg + Ch + Fa + Gb + Hc &= 0, \\
 Af' + Bg' + Ch' + Fa' + Gb' + Hc' &= 0,
 \end{aligned}$$

then either

$$\begin{vmatrix} B, & C, & F \\ b, & c, & f \\ b', & c', & f' \end{vmatrix} = 0,$$

or else

$$\begin{vmatrix} A, & B, & C \\ a, & b, & c \\ a', & b', & c' \end{vmatrix} = 0.$$

Supposing the first three of the six equations are satisfied, then (a, b, c, f, g, h) and (a', b', c', f', g', h') are the coordinates of two intersecting lines; and supposing that the last three equations are also satisfied, then (A, B, C, F, G, H) will be the coordinates of a line meeting each of the intersecting lines. The third line is thus either in the plane of the intersecting lines, or else passes through their point of intersection; and in fact, the first of the two determinant equations is the condition in order that the line may be in the plane of the two intersecting lines, and the second determinant equation is the condition in order that it may pass through their point of intersection. Each equation is in fact one out of four equivalent forms, viz. we may have in the first equation (B, C, F) , (C, A, G) , (A, B, H) , or (F, G, H) ; and in the second equation (A, B, C) , (A, H, G) , (H, B, F) , or (G, F, C) .

The analytical theory may be presented in a complete form by means of the formulae of my memoir "On the six coordinates of a line," (1867), *Camb. Phil. Trans.*, t. XI. pp. 290–323, [435]. Considering the two intersecting lines (a, b, c, f, g, h) and (a', b', c', f', g', h') , the coordinates of the plane through these two lines (that is, the coefficients ξ, η, ζ, ω of the equation $\xi x + \eta y + \zeta z + \omega w = 0$ of the plane) are there given (see p. 295 *) in a fourfold form; and if we thence form the condition in order that the line (A, B, C, F, G, H) may lie in this plane, we have

$$\begin{vmatrix} 0, & C, & -B, & F \\ -C, & 0, & A, & G \\ B, & -A, & 0, & H \\ -F, & -G, & -H, & 0 \end{vmatrix}$$

$$\begin{pmatrix} af' + b'g + c'h, & bf - b'f', & cf - c'f', & -(bc' - b'c) \\ ag' - a'g, & a'f + bg' + c'h, & cg' - c'g, & -(ca' - c'a) \\ ah' - a'h, & bh' - b'h, & a'f + b'g + ch', & -(ab' - a'b) \\ gh' - g'h, & hf - h'f', & fg' - f'g, & af + bg' + ch' \end{pmatrix} = 0;$$

viz. the condition is expressible in any one of the 16 forms obtained by combining a line of the first matrix with a line of the second matrix; thus one form is

$$0 \cdot (af' + b'g + c'h) + C(bf - b'f') - B(cf - c'f') - F(bc' - b'c) = 0,$$

that is,

$$\begin{vmatrix} B, & C, & F \\ b, & c, & f \\ b', & c', & f' \end{vmatrix} = 0,$$

the foregoing first determinant equation, which thus belongs to the case where the line lies in the plane of the two intersecting lines.

[* This Collection, vol. VII., p. 71.]

Again we have (see p. 296 *) an expression, in a fourfold form, for the coordinates of the point of intersection of the two intersecting lines; and thence for the condition, in order that the line (A, B, C, F, G, H) may pass through this point, we have

$$\begin{vmatrix} 0, & H, & -G, & A \\ -H, & 0, & F, & B \\ G, & -F, & 0, & C \\ -A, & -B, & -C, & 0 \end{vmatrix}$$

$$\begin{pmatrix} af' + b'g + c'h, & ag' - a'g, & ah' - a'h, & -(bc' - b'c) \\ bf - b'f', & a'f + bg' + c'h, & bh' - b'h, & -(ca' - c'a) \\ cf - c'f', & cg' - c'g, & a'f + b'g + ch', & -(ab' - a'b) \\ gh' - g'h, & hf - h'f', & fg' - f'g, & af + bg' + ch' \end{pmatrix} = 0,$$

viz. the condition is expressible in any one of the 16 forms obtained by combining a line of the first matrix with a line of the second matrix; thus one form is

$$A(bc' - b'c) + B(ca' - c'a) + C(ab' - a'b) + 0 \cdot (af + bg' + ch') = 0,$$

that is,

$$\begin{vmatrix} A, & B, & C \\ a, & b, & c \\ a', & b', & c' \end{vmatrix} = 0,$$

the foregoing second determinant equation, which thus belongs to the case where the line passes through the point of intersection of the two intersecting lines.

I remark that the original identity may be written in the very compendious symbolical form

$$\begin{vmatrix} A, & a, & a' \\ F, & f, & f' \end{vmatrix} \begin{vmatrix} B, & b, & b' \\ C, & c, & c' \end{vmatrix} = \begin{vmatrix} A, & a, & a' \\ B, & b, & b' \\ C, & c, & c' \end{vmatrix} \begin{vmatrix} F, & f, & f' \\ B, & b, & b' \\ C, & c, & c' \end{vmatrix}$$

viz. here on the left-hand side the second factor denotes the three determinants $bc' - b'c$, $b'C - c'B$, $Bc - Cb$: the whole is a quadric function of these, the coefficients being

$$\begin{aligned} &AF + BG + CH, \quad af + bg + ch, \\ &a'f' + b'g' + c'h', \quad af' + bg' + ch' + f'a + g'b + h'c, \\ &a'F + b'G + c'H + f'A + g'B + h'C, \\ &Af + Bg + Ch + Fa + Gb + Hc, \end{aligned}$$

respectively; and the right-hand side is simply the product of two determinants.

[* Loc. cit., p. 72.]

912.

ON THE NOTION OF A PLANE CURVE OF A GIVEN ORDER.

[From the *Messenger of Mathematics*, vol. XX. (1891), pp. 148–150.]

WE have a complete geometrical notion of a curve of a given order, viz. a curve of the order n is a curve which is met by any line whatever in n points and no more; but starting with this definition, how do we know that there exists a curve of the order n ? and, further, how do we know that it depends linearly on $\frac{1}{2}n(n+3)$ parameters, or, what is the same thing, that there is one and only one curve which can be drawn through $\frac{1}{2}n(n+3)$ given points?

The last-mentioned property does not by itself constitute a definition of a curve of the order n ; thus $n = 2$, we cannot define a curve of the second order as a curve which is uniquely determined by the condition of passing through 5 given points; for a cubic passing through 4 given points is a curve uniquely determined by the condition in question; but we may differentiate between these two solutions by adding the further condition that, when 3 of the 5 points are in a line, the curve of the second order shall include as part of itself this line. And we are thus led to the definition: A curve of the order n is a curve which is uniquely determined by the condition of passing through $\frac{1}{2}n(n+3)$ given points; and of being moreover such that, when $n+1$ of these points lie on a line, it includes as part of itself this line.

Starting from the foregoing definition, the first property is, I think, demonstrable, viz. the property that a curve of the order n is met by any line whatever in n points and no more. Thus $n = 2$: start with a line chosen at pleasure, and on it take 2 points which are regarded as indeterminate points: to fix the ideas, let one of these be regarded as depending on a parameter λ , and the other on a parameter μ , (so that when λ has a determinate value assigned to it, the first point becomes a determinate point, and similarly when μ has a determinate value assigned to it, the second point becomes a determinate point; and consequently, when λ and μ have determinate values, the 2 points are determinate points on the line). Take now any other 3 points not on the line; then, for the moment regarding the 2 points on the line as determinate points, we can through the 5 points draw a curve of the second order; this is the general curve of the second order through the 3 points; for it is a curve of the second order through the 3 points, and which, when the parameters λ and μ are regarded as undetermined and arbitrary, or choosable at pleasure, might be made to pass through any other two points whatever. But by hypothesis, the curve meets the line in question, that is, any line, in two points: and the conic cannot meet the line in more than two points; for in like manner, starting with the given line, and upon it 3 points (which may be considered as depending on the parameters λ , μ and ν respectively), and taking any two points not on the line, we have through the 5 points a conic; and this conic, regarding the parameters λ , μ , ν as undetermined and arbitrary, or choosable at pleasure, will be the general conic through the two points; for by a proper determination of the parameters, it might be made to pass through any other 3 points whatever. But, by hypothesis, the conic contains as part of itself the line; that is, the general conic through the 2 points contains as part of itself any line whatever, which is absurd.

So again for the cubic, $n = 3$: here starting with a line taken at pleasure, we take on it 3 points, which may be regarded as depending on the parameters λ , μ , ν respectively, and we take any other 6 points not on the line. We have through the 9 points a cubic; and this is the general cubic through the 6 points, for it depends on the parameters λ , μ , ν , which might be determined so as to make the curve pass through any other 3 points whatever; and by hypothesis the cubic meets the line, that is, any line whatever, in 3 points. And it cannot meet it in more than 3 points: for starting with the same line and upon it 4 points depending on the parameters λ , μ , ν , ρ respectively, and taking any other 5 points not upon the line, we then have through the 9 points a cubic which will be the general cubic through the 5 points (for it depends on the 4 parameters λ , μ , ν , ρ). But by hypothesis, the cubic contains as part of itself the line, viz. the general cubic through the 5 points contains as part of itself any line whatever, which is absurd.

The reasoning is quite general; and applying to a curve of the order n , the conclusion is that such a curve meets any line in n points and no more.

Collected Mathematical Papers of Arthur Cayley
Volume XIII — Pages 81–105
(Papers 913, 914, 915)

913.

ON THE EPITROCHOID.

[From the *Messenger of Mathematics*, vol. xx. (1891), pp. 150–158.]

IF we have a curve C_1 rolling on a fixed curve C , and consider the epitrochoid described by a point P_1 attached to and carried along with the curve C_1 , then there is a known construction for the radius of curvature at any point of the epitrochoid; the construction is probably much older, but I refer to Mannheim, *Géométrie Descriptive*, Paris, 1886, pp. 177 and 194. In fact, if Q be a position of the point of contact, P_1 the corresponding position of the describing point, R_1 and R the radii of curvature at Q of the two curves respectively, ρ the distance QP_1 , and ϕ the inclination of this distance to the common normal at Q , then the radius of curvature x at the point P_1 of the epitrochoid is given by the formula

$$\frac{x}{x - \rho} = \frac{\rho}{\cos \phi} \left(\frac{1}{R} + \frac{1}{R_1} \right).$$

I prove this as follows: take Q, Q' consecutive positions of the point of contact, P_1, P'_1 the corresponding positions of P_1 ; the centre of curvature is the intersection of the lines P_1Q, P'_1Q' ; hence, if for a moment M and N are the perpendicular distances between these two lines at the points P_1 and Q respectively, we have $\frac{x}{x - \rho} = \frac{M}{N}$. Let ds be the element QQ' considered as belonging to each of the two curves respectively, θ and θ_1 the angles which this element subtends at the two centres of curvature respectively; we have $ds = R\theta = R_1\theta_1$, whence

$$\theta + \theta_1 = ds \left(\frac{1}{R} + \frac{1}{R_1} \right).$$

The instantaneous centre is P_1 , and hence

$$M = P_1P'_1 = \rho(\theta + \theta_1); \quad \text{also } N = ds \cos \phi;$$

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we thus have

$$\frac{x}{x - \rho} = \frac{\rho(\theta + \theta_1)}{ds \cos \phi};$$

or, substituting for $\theta + \theta_1$ the foregoing value, we have the required expression

$$\frac{x}{x - \rho} = \frac{\rho}{\cos \phi} \left(\frac{1}{R} + \frac{1}{R_1} \right).$$

If the curve C_1 roll on a straight line, $R = \infty$, and therefore

$$\frac{x}{x - \rho} = \frac{\rho}{R_1 \cos \phi}.$$

Suppose that at any instant we have on the epitrochoid an inflexion, then $x = \infty$, and we have $\rho = R_1 \cos \phi$; this denotes that P_1 is a point of the circle described on the radius of curvature at Q as its diameter, and we have thus the theorem:

If at any instant we consider the circle, which has for its diameter the radius of curvature of the rolling curve at its point of contact with the right line, then each point on this circle is an inflexion on the epitrochoid described by the same point as the curve rolls on the straight line.

The circle in question for any point Q of the rolling curve C_1 is of course a circle, having its centre on the normal at Q and its radius equal to one-half of the radius of curvature; we may call it the *circle of double curvature*. Regarding Q as a given point on the curve C_1 , and supposing that the circle of double curvature intercepts the curve C_1 in a point X , then taking P_1 at X , we have the theorem:

If the curve C_1 roll on a straight line, then the epitrochoid described by the point X has an inflexion corresponding to the position of the rolling curve for which the point Q comes into contact with the straight line.

Consider the curve C_1 referred to axes fixed in the plane of the curve and moveable with it; take x, y as the coordinates of the point Q ; α, β as the coordinates of the corresponding centre of curvature, and X, Y as current coordinates; the equation of the circle of double curvature is

$$(X - x)(X - \alpha) + (Y - y)(Y - \beta) = 0,$$

viz. this is the equation of a circle having for a diameter the two points (x, y) and (α, β) .

Suppose that C_1 is a conic; the circle of double curvature at any point Q meets it in the point Q counting twice, and in two other points, say P_1, P_2 , which are determined as the intersections of the conic by the line P_1P_2 which joins them. I propose to determine the equation of this line. The investigation would be more symmetric for the ellipse, but I nevertheless prefer to consider for the curve C_1 the hyperbola $b^2x^2 - a^2Y^2 - a^2b^2 = 0$.

Representing for a moment the equation of the circle by $U = 0$, and that of the hyperbola by $V = 0$, it must be possible to determine the ratio of the coefficients p, q in such wise that $pU + qV = 0$ shall break up into a pair of lines, one of which is the tangent at Q of the hyperbola; and then the other of them will be the required line P_1P_2 .

The equation of the tangent is $b^2xX - a^2yY - a^2b^2 = 0$; we then see that the equation of the line P_1P_2 must be of the form $b^2xX + a^2yY + \Omega = 0$; and we have to find p, q, Ω so as to verify the identity

$$(p + qb^2)X^2 + (p - qa^2)Y^2 - p(\alpha + x)X - p(\beta + y)Y + p(\alpha x + \beta y) - qa^2b^2 \\ = (b^2xX - a^2yY - a^2b^2)(b^2xX + a^2yY + \Omega);$$

that is, we ought to have

$$\begin{aligned} p + qb^2 &= b^4x^2, \\ p - qa^2 &= a^4y^2, \\ p(x + \alpha) &= (\Omega - a^2b^2)b^2x, \\ qa^2b^2 - p(\alpha x + \beta y) &= \Omega a^2b^2. \end{aligned}$$

It will be recollected that α, β denote the coordinates of the centre of curvature corresponding to the point (x, y) of the hyperbola; their values thus are

$$\alpha = (a^2 + b^2) \frac{x^3}{a^4}, \quad \beta = -(a^2 + b^2) \frac{y^3}{b^4}.$$

There is no difficulty in finding the values

$$p = \frac{a^4b^4}{a^2 + b^2}, \quad q = \frac{b^4x^2 + a^4y^2}{a^2 + b^2};$$

whence also

$$\begin{aligned} q - a^2b^2 &= \frac{a^4b^4}{a^2 + b^2} \left\{ \left(\frac{1}{a^2} + \frac{1}{b^2} \right) y^2 - 1 \right\}, \\ q + a^2b^2 &= \frac{a^4b^4}{a^2 + b^2} \left\{ \left(\frac{1}{a^2} + \frac{1}{b^2} \right) x^2 + 1 \right\}, \\ \Omega &= -a^2b^2 \frac{x^2 + y^2}{a^2 + b^2}; \end{aligned}$$

these satisfy the identities in question.

The equation of the line P_1P_2 is thus

$$b^2xX + a^2yY - \frac{a^2b^2}{a^2 + b^2}(x^2 + y^2) = 0,$$

and combining this with the equation

$$b^2X^2 - a^2Y^2 - a^2b^2 = 0$$

of the hyperbola, we find

$$\left\{ xX - \frac{a^2}{a^2 + b^2}(x^2 + y^2) \right\}^2 = \frac{a^4 y^2 Y^2}{b^4}, = \frac{a^2 y^2}{b^2}(X^2 - a^2),$$

that is,

$$X^2 \left(x^2 - \frac{a^2 y^2}{b^2} \right) - \frac{2a^2 x}{a^2 + b^2}(x^2 + y^2) X + \frac{a^4}{(a^2 + b^2)^2}(x^2 + y^2)^2 + \frac{a^4 y^2}{b^2} = 0,$$

viz.

$$X^2 - 2 \frac{x^2 + y^2}{a^2 + b^2} xX + a^2 \left\{ \left(\frac{x^2 + y^2}{a^2 + b^2} \right)^2 + \frac{y^2}{b^2} \right\} = 0,$$

and we thence obtain

$$X = \frac{x^2 + y^2}{a^2 + b^2} x \pm \frac{ay}{b} \sqrt{\left\{ \left(\frac{x^2 + y^2}{a^2 + b^2} \right)^2 - 1 \right\}},$$

$$Y = -\frac{x^2 + y^2}{a^2 + b^2} y \pm \frac{bx}{a} \sqrt{\left\{ \left(\frac{x^2 + y^2}{a^2 + b^2} \right)^2 - 1 \right\}},$$

values belonging to the points P_1, P_2 respectively: these points are thus real or imaginary, according as $x^2 + y^2$ is greater or less than $a^2 + b^2$. In the limiting case $x^2 + y^2 = a^2 + b^2$, we have $X = x, Y = -y$, viz. here the circle of double curvature has its centre on the axis of x , and has double contact with the hyperbola at the points (x, y) and $(x, -y)$ respectively.

Suppose, to fix the ideas, x and y are both positive, i.e. that Q is on the right-hand upper half branch of the hyperbola; and that $x^2 + y^2 =$ or $> a^2 + b^2$, so that P_1, P_2 are both real. The two values of X are both positive, those of Y are originally both negative, and as Q moves along the half branch of the hyperbola one of these values increases negatively, the other increases positively until it becomes $= 0$, after which it becomes positive and continually increases: say the points P_1, P_2 are originally on the right-hand lower half branch, but one of them, say P_1 , moves up to the vertex, and we afterwards have P_1 on the upper half branch and P_2 on the lower half branch. The limiting case is when P_1 is at the vertex; here

$$\frac{x^2 + y^2}{a^2 + b^2} = \frac{a}{x},$$

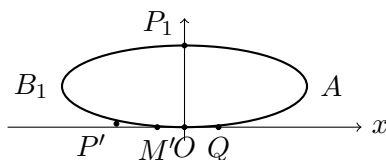
giving for (x, y) a real point, which is easily constructed on the upper half branch. There is thus on the half branch a real position of Q , such that the corresponding circle of double curvature passes through the vertex of the same half branch.

Hence in the epitrochoid described by the vertex of a hyperbola rolling on a right line there is always an inflexion.

Imagine the branch $B'AB$ of a hyperbola rolling on the line $y = 0$, where A is the vertex, B', B the points at infinity, and so that as x increases positively (that is, towards the right) the point of contact Q passes from A towards B ; consider the epitrochoid described by the vertex A . Supposing that the point of contact is at first at A ; the point A is a cusp on the epitrochoid, the motion of A is at first vertically upwards, and towards the right, the curve being concave to the axis; we then come to a position of the point of contact for which there is an inflexion; the motion of A is still upwards and towards the right, but the curve has now

become convex to the axis, and the motion continues in this manner as the point of contact moves along the hyperbola towards B ; when the point of contact is at B , i.e. when the asymptote has come to coincide with the axis, then the motion of A is again vertically upwards; the other half branch of the hyperbola now comes into contact with the axis; the point A still moves upwards but towards the left, the curve being concave to the axis, and this continues until the point of contact arrives at the other vertex (i.e. the vertex on the other branch), when the motion of A is horizontal and towards the left, viz. the coordinate y has here a maximum value, equal transverse axis of the hyperbola; the motion of A is thenceforth towards the left and downwards, until the other asymptote comes to coincide with the curve, after which it is downwards and towards the right (the point of contact being now on the half $B'A$ of the original branch BAB'), and ultimately the point of contact is again at A , and we have $y = 0$, a cusp; the curve is symmetrical in regard to the before-mentioned maximum ordinate: it is to be noticed that the distance between the two cusps is equal to four times the difference between the length of the half asymptote and the half branch of the hyperbola (this difference being a finite value expressible by elliptic functions), and that, as already mentioned, the maximum ordinate is equal to the transverse axis. The above described portion of the curve is of course repeated continually right and left *ad infinitum*; two consecutive portions intersect each other, and there is thus a real node vertically above each cusp of the infinite curve. We have, in what precedes, a complete explanation of the motion of a hyperbola rolling on a straight line, and in particular our explanation of the form of the epitrochoid described by a vertex; the epitrochoid described by any other point of the curve is of a similar character but must be non-symmetrical in regard to the maximum ordinate; and it would not be difficult to explain the form of the curve for other positions of the describing point; but as regards the analytical theory, I prefer to consider the case of the ellipse.

In the case of the ellipse $b^2X^2 + a^2Y^2 - a^2b^2 = 0$ rolling on the line $y = 0$ (see figure), take as origin the point O which comes in contact with the extremity B of



the minor axis; let x_1, y_1 be the coordinates of the describing point P_1 , and for the point Q , taking ϕ as the complement of the eccentric angle, let the coordinates referred to the axes of the ellipse be

$$x_1 = a \sin \phi, \quad y_1 = b \cos \phi;$$

the arc BQ is then

$$= \int_0^\phi \sqrt{(a^2 \cos^2 \phi + b^2 \sin^2 \phi)} d\phi,$$

which is $= aE\phi$, the modulus being the eccentricity e of the ellipse; we then have for the coordinates of P_1

$$\begin{aligned} x &= OP' = OQ - QP', & &= BQ - QP', \\ y &= & &P_1P', \end{aligned}$$

where QP' and P_1P' are the projections of P_1Q in the direction of, and at right angles to, the tangent QO at Q ; we find the values without difficulty, and we thus have

$$\begin{aligned} x &= \int_0^\phi \sqrt{(a^2 \cos^2 \phi + b^2 \sin^2 \phi)} d\phi - \frac{(a^2 - b^2) \sin \phi \cos \phi - ax_1 \cos \phi + by_1 \sin \phi}{\sqrt{(a^2 \cos^2 \phi + b^2 \sin^2 \phi)}}, \\ y &= \frac{ab - bx_1 \sin \phi - ay_1 \cos \phi}{\sqrt{(a^2 \cos^2 \phi + b^2 \sin^2 \phi)}}, \end{aligned}$$

which are the expressions for x, y in terms of a variable parameter ϕ .

In particular, for the point B at the extremity of the minor axis, $x_1 = 0, y_1 = b$, and the equations become

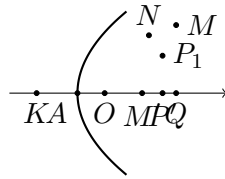
$$\begin{aligned} x &= \int_0^\phi \sqrt{(a^2 \cos^2 \phi + b^2 \sin^2 \phi)} d\phi - \frac{(a^2 - b^2) \sin \phi \cos \phi + b^2 \sin \phi}{\sqrt{(a^2 \cos^2 \phi + b^2 \sin^2 \phi)}}, \\ y &= \frac{ab(1 - \cos \phi)}{\sqrt{(a^2 \cos^2 \phi + b^2 \sin^2 \phi)}}, \end{aligned}$$

which, for $a = b$, become

$$x = a(\phi - \sin \phi), \quad y = a(1 - \cos \phi),$$

the ordinary formulæ for the cycloid.

In the case of the parabola $Y^2 = 4aX$ (see figure), taking x_1, y_1 for the coordinates



of the describing point P_1 , and representing those of the point of contact Q by means of the parameter ϕ , $X = a\phi^2, Y = 2a\phi$, then the arc AQ is

$$= a [\phi \sqrt{(\phi^2 + 1)} + \log\{\phi + \sqrt{(\phi^2 + 1)}\}],$$

so that, taking the origin at the point O , which comes into contact with the vertex A , this is also the value of OQ , and we then have

$$x = OQ - QP', \quad y = P_1P'.$$

Producing the axis of the parabola to meet the line $y = 0$ in k , we have

$$AK = AN = a\phi^2,$$

and thence $KM = a\phi^2 + x_1$; and then, if θ be the inclination AKQ of the axis of the parabola,

$$KM' = (a\phi^2 + x_1) \cos \theta, \quad MM' = (a\phi^2 + x_1) \sin \theta;$$

and thence also

$$KP' = (a\phi^2 + x_1) \cos \theta + y_1 \sin \theta, \quad P_1P' = (a\phi^2 + x_1) \sin \theta - y_1 \cos \theta;$$

also

$$KQ = 2a\phi^2 \sec \theta,$$

and thence

$$QP' = 2a\phi^2 \sec \theta - (a\phi^2 + x_1) \cos \theta - y_1 \sin \theta.$$

We have $\tan \theta = \frac{1}{\phi}$, and thence

$$\sin \theta = \frac{1}{\sqrt{(\phi^2 + 1)}}, \quad \cos \theta = \frac{\phi}{\sqrt{(\phi^2 + 1)}}, \quad \sec \theta = \frac{\sqrt{(\phi^2 + 1)}}{\phi};$$

hence

$$QP' = 2a\phi\sqrt{(\phi^2 + 1)} - \frac{(a\phi^2 + x_1)\phi + y_1}{\sqrt{(\phi^2 + 1)}} = \frac{a(\phi^2 + 2\phi) - x_1\phi - y_1}{\sqrt{(\phi^2 + 1)}},$$

$$P_1P' = \frac{a\phi^2 + x_1 - \phi y_1}{\sqrt{(\phi^2 + 1)}},$$

and the values of x, y thus become

$$x = a[\phi\sqrt{(\phi^2 + 1)} + \log\{\phi + \sqrt{(\phi^2 + 1)}\}] - \frac{a(\phi^2 + 2\phi) - x_1\phi - y_1}{\sqrt{(\phi^2 + 1)}},$$

that is,

$$x = a \log\{\phi + \sqrt{(\phi^2 + 1)}\} + \frac{(x_1 - a)\phi + y_1}{\sqrt{(\phi^2 + 1)}},$$

$$y = \frac{a\phi^2 + x_1 - \phi y_1}{\sqrt{(\phi^2 + 1)}}.$$

In particular, for the focus of the parabola, $x_1 = a, y_1 = 0$, and the equations become

$$x = a \log\{\phi + \sqrt{(\phi^2 + 1)}\},$$

$$y = a\sqrt{(\phi^2 + 1)};$$

that is,

$$\frac{x}{a} = \log\left\{\frac{y}{a} + \sqrt{\left(\frac{y^2}{a^2} - 1\right)}\right\},$$

or say

$$\exp\left(\frac{x}{a}\right) = \frac{y}{a} + \sqrt{\left(\frac{y^2}{a^2} - 1\right)};$$

this gives

$$\exp\left(-\frac{x}{a}\right) = \frac{y}{a} - \sqrt{\left(\frac{y^2}{a^2} - 1\right)};$$

and we have therefore

$$y = \frac{1}{2}a \left\{ \exp\left(\frac{x}{a}\right) + \exp\left(-\frac{x}{a}\right) \right\},$$

viz. as is well known, the epitrochoid described by the focus of the parabola is the catenary.

914.

ON A SOLUBLE QUINTIC EQUATION.

[From the *American Journal of Mathematics*, vol. XIII. (1891), pp. 53–58.]

MR YOUNG, in his paper, “Solvable Quintic Equations with Commensurable Coefficients,” *American Journal of Mathematics*, x. (1888), pp. 99–130, has given, in illustration of his general theory of the solution of soluble quintic equations (founded upon a short note by Abel), no less than twenty instances of the solution of a quintic equation with purely numerical coefficients, having a solution of the form $\sqrt[5]{A} + \sqrt[5]{B} + \sqrt[5]{C} + \sqrt[5]{D}$, where A, B, C, D are numerical expressions involving only square roots. But the solutions are not presented in their most simple form: thus in example 1, $x^5 + 3x^2 + 2x - 1 = 0$, the expression involves a radical

$$\sqrt[5]{\frac{1}{4}(21125 + 9439\sqrt{5})} :$$

here

$$(21125 + 9439)\sqrt{5}, = \sqrt{5}(9439 + 4225\sqrt{5}), = \sqrt{5}\frac{1}{4}(18 + 5\sqrt{5})^2(1 + \sqrt{5})^2(2 + \sqrt{5}),$$

so that, taking out the roots of the squared factors, we have as the proper form of the radical the very much more simple form $\sqrt[5]{47(2 + \sqrt{5})\sqrt{5}}$; where observe that $(2 + \sqrt{5})(2 - \sqrt{5}) = -1$, and thence $(2 + \sqrt{5})\sqrt[5]{-47(2 - \sqrt{5})\sqrt{5}} = \sqrt[5]{47(2 + \sqrt{5})\sqrt{5}}$, viz. the conjugate radicals $\sqrt[5]{-47(2 - \sqrt{5})\sqrt{5}}$ and $\sqrt[5]{47(2 + \sqrt{5})\sqrt{5}}$ differ only by a factor $2 + \sqrt{5}$ which is rational in 1 and $\sqrt{5}$. To avoid fractions I consider the foregoing equation under the form

$$x^5 + 3000x^2 + 20000x - 100000 = 0,$$

and I will presently give the solution; but first I consider the general theory. Writing

$$\begin{aligned} A &= \alpha^5, & A' &= \alpha^2\gamma, & A'' &= \alpha^3\beta, \\ B &= \beta^5, & B' &= \alpha\beta^2, & B'' &= \beta^3\delta, \\ C &= \gamma^5, & C' &= \gamma^2\delta, & C'' &= \alpha\gamma^3, \\ D &= \delta^5, & D' &= \beta\delta^2, & D'' &= \gamma\delta^3, \end{aligned}$$

we have $A'D' = \alpha^2\delta^2\beta\gamma$, $B'C' = \alpha\delta\beta^2\gamma^2$. Also

$$A'' = \frac{A'B'}{\beta\gamma}, \quad B'' = \frac{B'D'}{\alpha\delta}, \quad C'' = \frac{A'C'}{\alpha\delta}, \quad D'' = \frac{C'D'}{\beta\gamma},$$

which determine A'', B'', C'', D'' in terms of $A', B', C', D', \alpha\delta, \beta\gamma$; and then

$$A = \frac{A'A''}{\beta\gamma}, \quad B = \frac{B'B''}{\alpha\delta}, \quad C = \frac{C'C''}{\alpha\delta}, \quad D = \frac{D'D''}{\beta\gamma},$$

which give A, B, C, D .

If now we assume $x = \alpha + \beta + \gamma + \delta$, and regard $A, B, C, D, A', B', C', D', A'', B'', C'', D'', \alpha\delta, \beta\gamma$ each as a rational function, we may express x, x^2, x^3, x^5 each of them by means of rational functions or of rational functions multiplied into $\alpha, \beta, \gamma, \delta$ respectively: thus,

$$\begin{aligned} x &= \alpha + \beta + \gamma + \delta, \\ x^2 &= \alpha^2 + \beta^2 + \gamma^2 + \delta^2 + 2\alpha\beta + 2\alpha\gamma + 2\alpha\delta + 2\beta\gamma + 2\beta\delta + 2\gamma\delta \\ &= \frac{A'\beta}{\beta\gamma} + \frac{B'\delta}{\alpha\delta} + \frac{C'\alpha}{\alpha\delta} + \frac{D'\gamma}{\beta\gamma} + \frac{2B'\gamma}{\beta\gamma} + \frac{2A'\delta}{\alpha\delta} + 2\alpha\delta + 2\beta\gamma + \frac{2D'\alpha}{\alpha\delta} + \frac{2C'\beta}{\beta\gamma}, \end{aligned}$$

&c.; and we thus obtain

$$\begin{aligned} &x^5 + qx^3 + rx^2 + sx + t \\ &= A + B + C + D \\ &\quad + (20\alpha\delta + 30\beta\gamma)(A' + D') + 30(\alpha\delta + 2\beta\gamma)(B' + C') \\ &\quad + 3q(A' + B' + C' + D') + 2r(\alpha\delta + \beta\gamma) + t \\ &\quad + \alpha \left\{ 5A'' + 5\frac{C'^2}{\alpha\delta} + \frac{5B'^2\beta\gamma}{\alpha\delta} + 10C'' + \frac{10D'^2}{\alpha\delta} + 10\alpha^2\delta^2 + 20D'' + 30\beta^2\gamma^2 + 30D''\frac{\beta\gamma}{\alpha\delta} + 60\alpha\beta\gamma\delta \right. \\ &\quad \left. + q\left(\frac{B'}{\alpha\delta} + \frac{3D'}{\alpha\delta} + 3\alpha\delta + 6\beta\gamma\right) + r\left(\frac{C'}{\alpha\delta} + \frac{2D'}{\alpha\delta}\right) + s \right\} \\ &\quad + \beta \left\{ 5B'' + 5\frac{A'^2}{\beta\gamma} + 5\frac{D'^2\alpha\delta}{\beta\gamma} + 10A'' + \frac{10C''^2}{\beta\gamma} \right. \\ &\quad \left. + 10\beta^2\gamma^2 + 20D'' + 20C'' + 30\alpha^2\delta^2 + 30C''\frac{\alpha\delta}{\beta\gamma} + 60\alpha\beta\gamma\delta \right. \\ &\quad \left. + q\left(\frac{D'}{\beta\gamma} + \frac{3C'}{\beta\gamma} + 3\beta\gamma + 6\alpha\delta\right) + r\left(\frac{A'}{\beta\gamma} + \frac{2C'}{\beta\gamma}\right) + s \right\} \\ &\quad + \gamma \left\{ 5C'' + 5\frac{D'^2}{\beta\gamma} + 5\frac{A''\alpha\delta}{\beta\gamma} + 10D'' + \frac{10B'^2}{\beta\gamma} \right. \\ &\quad \left. + 10\beta^2\gamma^2 + 20A'' + 20B'' + 30\alpha^2\delta^2 + 30B''\frac{\alpha\delta}{\beta\gamma} + 60\alpha\beta\gamma\delta \right. \\ &\quad \left. + q\left(\frac{A''}{\beta\gamma} + \frac{3B''}{\beta\gamma} + 3\beta\gamma + 6\alpha\delta\right) + r\left(\frac{D'}{\beta\gamma} + \frac{2B'}{\beta\gamma}\right) + s \right\} \\ &\quad + \delta \left\{ 5D'' + 5\frac{B'^2}{\alpha\delta} + 5\frac{C''\beta\gamma}{\alpha\delta} + 10B'' + \frac{10A'^2}{\alpha\delta} \right. \\ &\quad \left. + 10\alpha^2\delta^2 + 20C'' + 20A'' + 30\beta^2\gamma^2 + 30A''\frac{\beta\gamma}{\alpha\delta} + 60\alpha\beta\gamma\delta \right. \\ &\quad \left. + q\left(\frac{C''}{\alpha\delta} + \frac{3A''}{\alpha\delta} + 3\alpha\delta + 6\beta\gamma\right) + r\left(\frac{B'}{\alpha\delta} + \frac{2A'}{\alpha\delta}\right) + s \right\}. \end{aligned}$$

If, then, $x^5 + qx^3 + rx^2 + sx + t = 0$, we have the rational term $= 0$, and the coefficients of $\alpha, \beta, \gamma, \delta$ each $= 0$; in the class of equations under consideration, these last equations differ only in the signs of the radicals contained therein, so that one of them being satisfied identically, the others will be also satisfied. In particular, if $q = 0$, then $\alpha\delta + \beta\gamma = 0$: the rational term gives

$$A + B + C + D - 10\alpha\delta(A' + D' - B' - C') + t = 0,$$

and the term in α gives

$$5A'' + 15B'' + 10C'' - 10D'' + \frac{5}{\alpha\delta}(C'^2 + 2D'^2) + \frac{r}{\alpha\delta}(C' + 2D') - 20\alpha^2\delta^2 + s = 0.$$

For the equation $x^5 + 3000x^2 + 20000x - 100000 = 0$, the expression for the root is $x = \sqrt[5]{A} + \sqrt[5]{B} + \sqrt[5]{C} + \sqrt[5]{D}$, where

$$\begin{aligned} A &= 39000 + 18200\sqrt{5} + (-1720 + 920\sqrt{5})\sqrt{235 + 94\sqrt{5}}, \\ D &= 39000 + 18200\sqrt{5} + (-1720 - 920\sqrt{5})\sqrt{235 + 94\sqrt{5}}, \\ B &= 39000 - 18200\sqrt{5} + (-1720 + 920\sqrt{5})\sqrt{235 - 94\sqrt{5}}, \\ C &= 39000 - 18200\sqrt{5} + (-1720 - 920\sqrt{5})\sqrt{235 - 94\sqrt{5}}, \end{aligned}$$

and where also

$$\begin{aligned} A' &= -150 - 70\sqrt{5} + (-10 - 2\sqrt{5})\sqrt{235 + 94\sqrt{5}}, \\ D' &= -150 - 70\sqrt{5} + (-10 + 2\sqrt{5})\sqrt{235 + 94\sqrt{5}}, \\ B' &= -150 + 70\sqrt{5} + (-10 - 2\sqrt{5})\sqrt{235 - 94\sqrt{5}}, \\ C' &= -150 + 70\sqrt{5} + (-10 + 2\sqrt{5})\sqrt{235 - 94\sqrt{5}}, \end{aligned}$$

and

$$\begin{aligned} A'' &= -940 - 100\sqrt{5} + (-100 + 20\sqrt{5})\sqrt{235 + 94\sqrt{5}}, \\ D'' &= -940 - 100\sqrt{5} + (-100 - 20\sqrt{5})\sqrt{235 + 94\sqrt{5}}, \\ B'' &= -940 + 100\sqrt{5} + (-100 + 20\sqrt{5})\sqrt{235 - 94\sqrt{5}}, \\ C'' &= -940 + 100\sqrt{5} + (-100 - 20\sqrt{5})\sqrt{235 - 94\sqrt{5}}. \end{aligned}$$

The foregoing forms are in some respects the most convenient; but it is to be observed that we have

$$\begin{aligned} A &= 2600\sqrt{5}(1 + \sqrt{5})(2 + \sqrt{5}) + 40(1 + \sqrt{5})(18 + 5\sqrt{5})\sqrt{47\sqrt{5}(2 + \sqrt{5})}, \text{ \&c.}, \\ A' &= -10\sqrt{5}(1 + \sqrt{5})(2 + \sqrt{5}) - 2\sqrt{5}(1 + \sqrt{5})\sqrt{47\sqrt{5}(2 + \sqrt{5})}, \text{ \&c.}, \\ A'' &= 20(1 - \sqrt{5})(18 + 13\sqrt{5}) + 20\sqrt{5}(1 - \sqrt{5})\sqrt{47\sqrt{5}(2 + \sqrt{5})}, \text{ \&c.}, \end{aligned}$$

or putting for shortness

$$\sqrt{Q} = \sqrt{47\sqrt{5}(2 + \sqrt{5})}, \quad \sqrt{Q_1} = \sqrt{-47\sqrt{5}(2 - \sqrt{5})},$$

(so that, according to a foregoing remark, we have $(2 + \sqrt{5})\sqrt{Q} = \sqrt{Q_1}$), then we have

$$\begin{aligned} A &= 40(1 + \sqrt{5})\{65\sqrt{5}(2 + \sqrt{5}) + (18 + 5\sqrt{5})\sqrt{Q}\}, \text{ \&c.}, \\ A' &= -2\sqrt{5}(1 + \sqrt{5})\{5(2 + \sqrt{5}) + \sqrt{Q}\}, \text{ \&c.}, \\ A'' &= 20(1 - \sqrt{5})\{18 + 13\sqrt{5} + \sqrt{5}\sqrt{Q}\}, \text{ \&c.}, \end{aligned}$$

where observe that the term $2 + \sqrt{5}$ is a factor of Q .

Starting from the values of A' , B' , C' , D' , we have

$$A' = -2\sqrt{5}(1 + \sqrt{5})\{5(2 + \sqrt{5}) + \sqrt{Q}\},$$

and therefore

$$\begin{aligned} D' &= -2\sqrt{5}(1 + \sqrt{5})\{5(2 + \sqrt{5}) - \sqrt{Q}\}, \\ A'D' &= 20(1 + \sqrt{5})^2(2 + \sqrt{5})[25(2 + \sqrt{5}) - 47\sqrt{5}], \end{aligned}$$

where the last factor is

$$= 50 - 22\sqrt{5}, = -2\sqrt{5}(11 - 5\sqrt{5}), = -\sqrt{5}(1 - \sqrt{5})^2(2 - \sqrt{5}).$$

Hence

$$A'D' = -\sqrt{5} \cdot 20(-4)^2(-1) = 320\sqrt{5},$$

that is, $A'D' = (\alpha\delta)^2\beta\gamma = 320\sqrt{5}$, and similarly $B'C' = \alpha\delta(\beta\gamma)^2 = -320\sqrt{5}$, whence $\alpha\delta = -4\sqrt{5}$, $\beta\gamma = 4\sqrt{5}$, and $\alpha\delta + \beta\gamma = 0$, as above.

We have, moreover,

$$\begin{aligned} A' &= -2\sqrt{5}(1 + \sqrt{5})\{5(2 + \sqrt{5}) + \sqrt{Q}\}, \\ B' &= 2\sqrt{5}(1 - \sqrt{5})\{5(2 - \sqrt{5}) - \sqrt{Q_1}\}, \end{aligned}$$

and thence

$$\begin{aligned} A'B' &= 80[-25 - \sqrt{Q}Q_1 + 5(2 - \sqrt{5})\sqrt{Q} - 5(2 + \sqrt{5})\sqrt{Q_1}] \\ &= 80[-25 - 47\sqrt{5} + (5(2 - \sqrt{5}) - 5)\sqrt{Q}], \end{aligned}$$

that is,

$$\begin{aligned} A'B' \div \beta\gamma &= 4\sqrt{5}[-25 - 47\sqrt{5} + 5(1 - \sqrt{5})\sqrt{Q}] \\ &= -20(47 + 5\sqrt{5}) + 20\sqrt{5}(1 - \sqrt{5})\sqrt{Q}, = A''; \end{aligned}$$

and similarly we verify the values of B'' , C'' and D'' .

We have next

$$A'A'' = 160\sqrt{5}\{(10 + 5\sqrt{5} + \sqrt{Q})(18 + 13\sqrt{5} + \sqrt{5}\sqrt{Q})\},$$

or observing that $Q\sqrt{5}$ is $= 235(2 + \sqrt{5})$, the whole term in $\{ \}$ is

$$= (505 + 220\sqrt{5}) + (470 + 235\sqrt{5}) + (18 + 13\sqrt{5} + 25 + 10\sqrt{5})\sqrt{Q},$$

or we have

$$\begin{aligned} A'A'' &= 160\sqrt{5}\{65\sqrt{5}(7 + 3\sqrt{5}) + (43 + 23\sqrt{5})\sqrt{Q}\} \\ &= 160\sqrt{5}(1 + \sqrt{5})\{65\sqrt{5}(2 + \sqrt{5}) + (18 + 5\sqrt{5})\sqrt{Q}\}, \end{aligned}$$

and consequently

$$A'A'' \div \beta\gamma = 40(1 + \sqrt{5})\{65\sqrt{5}(2 + \sqrt{5}) + (18 + 5\sqrt{5})\sqrt{Q}\}, = A;$$

and similarly we verify the values of B , C , D .

In the proposed equation $x^5 + 3000x^2 + 20000x - 100000 = 0$, we have $r = 3000$, $s = 20000$, $t = -100000$, $\alpha\delta = -4\sqrt{5}$; the two equations to be verified thus are

$$A + B + C + D + 40\sqrt{5}(A' + D' - B' - C') - 100000 = 0,$$

and

$$5A'' + 15B'' + 10C'' - 10D'' - \frac{\sqrt{5}}{4}(C'^2 + 2D'^2) - 150\sqrt{5}(C' + 2D') - 1600 + 20000 = 0.$$

As to the first of these, we have $A + B + C + D = 156000$, $A' + D' - B' - C' = -280\sqrt{5}$, and the equation thus is

$$156000 + 40\sqrt{5}(-280\sqrt{5}) - 100000 = 0,$$

which is right.

For the second equation, if in the calculation we keep the radicals in the first instance distinct, we have

$$\begin{aligned} 5A'' + 15B'' + 10C'' - 10D'' &= -18800 + 3000\sqrt{5} + (-1500 + 300\sqrt{5})\sqrt{Q} + (500 + 100\sqrt{5})\sqrt{Q_1}, \\ -150\sqrt{5}(C' + 2D') &= \{-450 - 70\sqrt{5} \\ &\quad + (20 + 4 + \sqrt{5})\sqrt{Q} + (-10 + 2\sqrt{5})\sqrt{Q_1}\}(-150\sqrt{5}) \\ -1600 + 20000 &= 18400, \\ -\frac{\sqrt{5}}{4}(C'^2 + 2D'^2) &= -\frac{\sqrt{5}}{4}\{282000 + 416800\sqrt{5} \\ &\quad + (-8800 - 4000\sqrt{5})\sqrt{Q} + (4400 - 2000\sqrt{5})\sqrt{Q_1}\}. \end{aligned}$$

Substituting in the equation, we ought to have

$$\begin{aligned} 0 &= -18800 + 3000\sqrt{5} + (-1500 + 300\sqrt{5})\sqrt{Q} + (500 + 100\sqrt{5})\sqrt{Q_1} \\ &\quad + 52500 + 67500\sqrt{5} + (-3000 - 3000\sqrt{5})\sqrt{Q} + (-1500 + 1500\sqrt{5})\sqrt{Q_1} \\ &\quad + 18400 \\ &\quad - 52100 - 70500\sqrt{5} + (5000 + 2200\sqrt{5})\sqrt{Q} + (2500 - 1100\sqrt{5})\sqrt{Q_1}, \end{aligned}$$

that is,

$$0 = (500 - 500\sqrt{5})\sqrt{Q} + (1500 + 500\sqrt{5})\sqrt{Q_1},$$

viz.

$$0 = 500\{(1 - \sqrt{5})\sqrt{Q} + (3 + \sqrt{5})\sqrt{Q_1}\},$$

which is satisfied in virtue of $\sqrt{Q} = (2 + \sqrt{5})\sqrt{Q_1}$: this completes the verification.

915.

ON THE PARTITIONS OF A POLYGON.

[From the *Proceedings of the London Mathematical Society*, vol. XXII. (1891), pp. 237–262. Read March 12, 1891.]

1. THE partitions are made by non-intersecting diagonals; the problems which have been successively considered are (1) to find the number of partitions of an r -gon into triangles, (2) to find the number of partitions of an r -gon into k parts, and (3) to find the number of partitions of an r -gon into p -gons, r of the form

$$n(p-2)+2.$$

Problem (1) is a particular case of (2); and it is also a particular case of (3); but the problems (2) and (3) are outside each other; for problem (3) a very elegant solution, which will be here reproduced, is given in the paper, H. M. Taylor and R. C. Rowe, "Note on a Geometrical Theorem," *Proc. Lond. Math. Soc.*, t. XIII. (1882), pp. 102–106, and this same paper gives the history of the solution of (1).

The solution of (2) is given in the memoir, Kirkman "On the k -partitions of the r -gon and r -ace," *Phil. Trans.*, t. CXLVII. (for 1857), p. 225; viz. he there gives for the number of partitions of the r -gon into k parts (or, what is the same thing, by means of $k-1$ non-intersecting diagonals) the expression

$$\frac{[r+k-2]^{k-1} [r-3]^{k-1}}{[k]^{k-1} [k-1]^{k-1}},$$

but there is no complete demonstration of this result.

If $k = r - 2$, we have the solution of the problem (1); viz. the number of partitions of the r -gon into triangles is

$$= [2r-4]^{r-3} \div [r-2]^{r-3}.$$

The present paper relates chiefly to the foregoing problem (2), the determination

of the number of partitions of the r -gon into k parts, or, what is the same thing, by means of $k - 1$ non-intersecting diagonals.

2. Assuming for the moment the foregoing result, then for $k = 1$, the number of partitions is

$$= 1,$$

for $k = 2$ it is

$$= \frac{r.r - 3}{2},$$

for $k = 3$ it is

$$= \frac{r + 1.r.r - 3.r - 4}{12},$$

and so on. As a simple verification, $k = 2$, the number of partitions is equal to the number of diagonals, viz. this is number of pairs of summits less number of sides, that is,

$$\frac{1}{2}r(r - 1) - r, = \frac{1}{2}r(r - 3).$$

For convenience, I give the first Table on the next page, which is a tabulation of the functions

$$\begin{aligned} U_1 &= x^3 + x^4 + x^5 + x^6 + \dots + x^r, \\ U_2 &= 2x^4 + 5x^5 + 9x^6 + \dots + \frac{r.r - 3}{2.1} x^r, \\ U_3 &= 5x^5 + 21x^6 + \dots + \frac{r + 1.r.r - 3.r - 4}{3.2.2.1} x^r, \\ U_4 &= 14x^6 + \dots + \frac{r + 2.r + 1.r.r - 3.r - 4.r - 5}{4.3.2.3.2.1} x^r, \\ U_5 &= \dots + \frac{r + 3.r + 2.r + 1.r.r - 3.r - 4.r - 5.r - 6}{5.4.3.2.4.3.2.1} x^r, \\ &\quad \&c. \end{aligned}$$

3. And in connexion herewith, I give the second Table on the next page, which is a tabulation of the functions

$$\begin{aligned} V_2 &= x^4 + 2x^5 + 3x^6 + 4x^7 + \dots + \frac{r-1}{1} x^{r+2}, \\ V_3 &= 4x^6 + 14x^7 + 32x^8 + \dots + 2 \frac{r+1.r-3.r-4}{3.2.1} x^{r+2}, \\ V_4 &= 14x^8 + 72x^9 + \dots + 3 \frac{r+2.r+1.r-3.r-4.r-5}{4.3.3.2.1} x^{r+2}, \\ V_5 &= 48x^{10} + \dots + 4 \frac{r+3.r+2.r+1.r-3.r-4.r-5.r-6}{5.4.3.3.2.1} x^{r+2}, \\ &\quad \&c. \end{aligned}$$

$r =$	$k=1$	2	3	4	5	6	7	8	9	10	11	12	13
3	1												
4	1	2											
5	1	5	5										
6	1	9	21	14									
7	1	14	56	84	42								
8	1	20	120	300	330	132							
9	1	27	225	825	1485	1287	429						
10	1	35	385	1925	5005	7007	5005	1430					
11	1	44	616	4004	14014	28028	32032	19448	4862				
12	1	54	936	7644	34398	91728	148512	143208	75582	16796			
13	1	65	1365	13650	76440	259896	556920	755820	629850	293930	58786		
14	1	77	1925	23100	157080	659736	1790712	3197700	3730650	2735810	1144066	208012	
15	1	90	2640	37400	302940	1534896	5116320	11511720	17587350	17978180	11767536	4457400	742900

$r =$	$k=2$	3	4	5	6	7	8	9	10	11	12	13
4	1											
5	2	4										
6	3	14	14									
7	4	32	72	48								
8	5	60	225	330	165							
9	6	100	550	1320	1430	572						
10	7	154	1155	4004	7007	6006	2002					
11	8	224	2184	10192	25480	34944	24052	7072				
12	9	312	3822	22932	76440	148512	167076	100776	25194			
13	10	420	6300	47040	199920	514080	813960	775200	396800	90440		
14	11	550	9900	89760	476240	1534996	3197700	4263600	3517470	1634880	326876	
15	12	704	14960	161568	1023264	4093056	10744272	18759840	18573816	15690048	6547520	1188640

4. These functions, U and V , are particular values satisfying the equation

$$(V_2 + V_3y + V_4y^2 + \dots) = (U_1 + U_2y + U_3y^2 + U_4y^3 + \dots)^2;$$

that this is so will appear from the following general investigation.

5. Taking x, y as independent variables, denoting by X an arbitrary function of x , and using accents to denote differentiations in regard to x , we require the following identity:

$$\frac{2}{1.2}X^2 + \frac{4y}{1.2.3}(X^2)' + \frac{6y^2}{1.2.3.4}(X^4)'' + \dots = \left\{ X + \frac{y}{1.2}(X^2)' + \frac{y^2}{1.2.3}(X^3)'' + \dots \right\}^2,$$

which I prove as follows. Writing U to denote the same function of u which X is of x , I start from the equation

$$u = x + yU,$$

which determines u as a function of the independent variables x, y . We have

$$\frac{du}{dy}(1 - yU') = U, \quad \frac{du}{dx}(1 - yU') = 1,$$

where the accent denotes differentiation in regard to u ; hence

$$\frac{du}{dy} = U \frac{du}{dx} = \frac{u - x}{y} \frac{du}{dx},$$

or say

$$y \frac{du}{dy} = (u - x) \frac{du}{dx}.$$

Writing

$$u_1 = \int u \, dx,$$

and therefore

$$\frac{du_1}{dx} = u,$$

this equation may be written

$$y \frac{d^2 u_1}{dx \, dy} = \frac{du_1}{dx} - x - y \frac{du}{dx};$$

or, integrating with respect to x , we have

$$y \frac{du_1}{dy} = u_1 - \frac{1}{2}x^2 - yu,$$

or say

$$\frac{2}{y} \frac{d}{dy} \left(\frac{u_1 - \frac{1}{2}x^2}{y} \right) = \frac{(u - x)^2}{y^2},$$

that is,

$$2 \frac{d}{dy} \left(\frac{u_1 - \frac{1}{2}x^2}{y} \right) = \frac{(u - x)^2}{y^2}.$$

6. But, from the equation $u = x + yU$, we have

$$u = x + yX + \frac{y^2}{1.2}(X^2)' + \frac{6y^3}{1.2.3.4}(X^4)'' + \dots,$$

and thence

$$u_1 = \frac{1}{2}x^2 + yX_1 + \frac{y^2}{1.2}X^2 + \frac{y^3}{1.2.3}(X^3)' + \frac{y^4}{1.2.3.4}(X^4)'' + \dots,$$

if for a moment X_1 is written for $\int X \, dx$. And hence, from the relation obtained above, we have the required identity

$$\frac{2}{1.2}X^2 + \frac{4y}{1.2.3}(X^2)' + \frac{6y^2}{1.2.3.4}(X^4)'' + \dots = \left\{ X + \frac{y}{1.2}(X^2)' + \frac{y^2}{1.2.3}(X^3)'' + \dots \right\}^2.$$

This of course gives the series of identities

$$\frac{2}{1.2}X^2 = X^2,$$

$$\frac{4}{1.2.3}(X^2)' = \frac{2}{1.2}X(X^2)',$$

$$\frac{6}{1.2.3.4}(X^4)'' = \frac{2}{1.2.3}X(X^3)'' + \left\{ \frac{1}{1.2}(X^2)' \right\}^2,$$

⋮

or say

$$X^2 = X^2,$$

$$(X^2)' = \frac{1}{2}X(X^2)',$$

$$(X^4)'' = \frac{1}{2}X(X^3)'' + [(X^2)']^2,$$

⋮

all of which may be easily verified.

7. I multiply each side of the identity by x^2 , and write

$$\begin{aligned} U_1 &= x.X, & V_2 &= x^2 \cdot \frac{2}{1.2}X^2, \\ U_2 &= x \cdot \frac{1}{1.2}(X^2)', & V_3 &= x^2 \cdot \frac{4}{1.2.3}(X^2)', \\ U_3 &= x \cdot \frac{1}{1.2.3}(X^3)'', & V_4 &= x^2 \cdot \frac{6}{1.2.3.4}(X^4)'', \\ U_4 &= x \cdot \frac{1}{1.2.3.4}(X^4)', & V_5 &= x^2 \cdot \frac{8}{1.2.3.4.5}(X^5)', \\ &\vdots & & \end{aligned}$$

We thus obtain two sets of functions U and V , satisfying the before-mentioned equation. We have

$$(V_2 + yV_3 + y^2V_4 + \dots) = (U_1 + yU_2 + y^2U_3 + \dots)^2;$$

and it will be observed that we have, moreover, the relations

$$U_2 = \frac{1}{2}x(x^{-2}V_2)', \quad U_3 = \frac{1}{2}x(x^{-2}V_3)', \quad U_4 = \frac{1}{2}x(x^{-2}V_4)', \quad \dots$$

8. In particular, if

$$X = \frac{x^2}{1-x},$$

then the general term

in X is x^{r-1} ,	the first term occurring when $r = 3$,
in X^2 is $(r-3)x^r$,	„ „ $r = 4$,
in X^3 is $\frac{r-3.r-4}{1.2}x^{r+1}$,	„ „ $r = 5$,
in X^4 is $\frac{r-3.r-4.r-5}{1.2.3}x^{r+2}$,	„ „ $r = 6$,

from which it appears that, for this value of X , U_1, U_2, U_3, U_4 , &c., have the before-mentioned values (No. 2), and further that V_2, V_3, V_4, V_5 , &c., have also the before-mentioned values (No. 3).

9. We do not absolutely require, but it is interesting to obtain, the finite expressions of these functions. We have

$$\begin{aligned}
 (1-x)U_1 &= x^3(1), \\
 (1-x)^3U_2 &= x^4(2-x), \\
 (1-x)^5U_3 &= x^5(5-4x+x^2), \\
 (1-x)^7U_4 &= x^6(14-14x+6x^2-x^3), \\
 (1-x)^9U_5 &= x^7(42-48x+27x^2-8x^3+x^4), \\
 (1-x)^{11}U_6 &= x^8(132-165x+110x^2-44x^3+10x^4-x^5), \\
 &\vdots \\
 (1-x)^2V_2 &= x^4(1), \\
 (1-x)^4V_3 &= x^6(4-2x), \\
 (1-x)^6V_4 &= x^8(14-12x+3x^2), \\
 (1-x)^8V_5 &= x^{10}(48-54x+24x^2-4x^3), \\
 &\vdots
 \end{aligned}$$

and here the factors in () satisfy the series of relations

$$\begin{aligned}
 1 &= 1^2, \\
 4-2x &= 2(2-x), \\
 14-12x+3x^2 &= 2 \cdot 1 \cdot (5-4x+x^2) + (2-x)^2, \\
 48-54x+24x^2-4x^3 &= 2 \cdot 1 \cdot (14-14x+6x^2-x^3) + 2(2-x)(5-4x+x^2), \\
 &\vdots
 \end{aligned}$$

corresponding to

$$V_2 = U_1^2, \quad V_3 = 2U_1U_2, \quad \&c.,$$

given by the before-mentioned equation (No. 7), between the functions V and U .

10. It is to be shown that, taking $U_1, U_2, U_3, \dots$ for the functions which belong to the partitions of the r -gon (assumed to be unknown functions of r and the suffixes), and connecting them with a set of functions $V_2, V_3, V_4, \dots$ by the relations

$$U_2 = \frac{1}{2}x(x^{-2}V_2)', \quad U_3 = \frac{1}{2}x(x^{-2}V_3)', \quad U_4 = \frac{1}{2}x(x^{-2}V_4)', \quad \&c.,$$

then we have the foregoing identical equation

$$(V_2 + V_3y + V_4y^2 + \dots) = (U_1 + U_2y + U_3y^2 + U_4y^3 + \dots)^2.$$

This implies the relations

$$\begin{aligned} V_2 &= U_1^2, \\ V_3 &= 2U_1U_2, \\ V_4 &= 2U_1U_3 + U_2^2, \\ V_5 &= 2U_1U_4 + 2U_2U_3, \\ &\&c. \end{aligned}$$

Thus, if U_1 is known, the equation

$$V_2 = U_1^2$$

determines V_2 , and then the equation

$$U_2 = \frac{1}{2}x(x^{-2}V_2)'$$

determines U_2 , so that U_1, U_2 are known; and we thence in the same way find successively U_3 and V_3, U_4 and V_4 , and so on; that is, assuming only that U_1 has the before-mentioned value,

$$U_1 = x^3 + x^4 + x^5 + \dots + x^r + \dots,$$

it follows that all the remaining functions U and V must have their before-mentioned values. But the function

$$U_1 = x^3 + x^4 + x^5 + \dots,$$

where each coefficient is $= 1$, is evidently the proper expression for the generating function of the number of partitions of the r -gon into a single part; and we thus arrive at the proof that the remaining functions U , which are the generating functions for the number of partitions of the r -gon into 2, 3, 4, $\dots$, k , parts, have their before-mentioned values.

11. Considering, then, the partition problem from the point of view just referred to, I write $A_r, B_r, C_r, \dots$ for the number of partitions of an r -gon into 1 part, 2 parts, 3 parts, &c., and form therewith the generating functions

$$\begin{aligned} U_1 &= A_3x^3 + A_4x^4 + \dots + A_rx^r + \dots, \\ U_2 &= B_4x^4 + \dots + B_rx^r + \dots, \\ U_3 &= C_5x^5 + \dots + C_rx^r + \dots, \\ &\vdots \end{aligned}$$

and also the functions

$$\begin{aligned} V_2 &= \frac{3}{2}B_4x^6 + \dots + \frac{r-1}{r}B_rx^{r+2} + \dots, \\ V_3 &= \frac{4}{3}C_5x^7 + \dots + \frac{4}{r}C_rx^{r+2} + \dots, \\ &\vdots \end{aligned}$$

where observe that the functions U, V are such that

$$U_2 = \frac{1}{2}x(x^{-2}V_2)', \quad U_3 = \frac{1}{2}x(x^{-2}V_3)', \quad U_4 = \frac{1}{2}x(x^{-2}V_4)', \quad \&c.$$

To fix the ideas, consider an r -gon which is to be divided into six parts. Choosing any particular summit, and from this summit drawing a diagonal successively

to each of the non-adjacent $r-3$ summits, we divide the r -gon into two parts in $r-3$ different ways; viz. the two parts are

$$\begin{aligned} & \text{a 3-gon and } (r-1)\text{-gon,} \\ & \text{4-gon and } (r-2)\text{-gon,} \\ & \vdots \\ & (r-1)\text{-gon and 3-gon;} \end{aligned}$$

say any one of these ways is

$$\text{an } a\text{-gon and } \beta\text{-gon, } \alpha + \beta = r + 2.$$

Next, writing

$$a + b = 6,$$

that is,

$$a, b = 1, 5, \quad 2, 4, \quad 3, 3, \quad 4, 2, \quad 5, 1,$$

we divide in every possible way the a -gon into a parts, and the β -gon into b parts (so dividing the r -gon into six parts). Observing that A, B, C, D, E, F are the letters belonging to the numbers 1, 2, 3, 4, 5, 6, respectively, the number of parts which we thus obtain (corresponding to the different values of a, b) are

$$A_\alpha E_\beta + B_\alpha D_\beta + C_\alpha C_\beta + D_\alpha B_\beta + E_\alpha A_\beta,$$

and summing for the different values of α, β ($\alpha + \beta = r + 2$), the whole number of parts is

$$= \text{coeff. } x^{r+2} \text{ in } (U_1 U_5 + U_2 U_4 + U_3 U_3 + U_4 U_2 + U_5 U_1),$$

that is,

$$= \text{coeff. } x^{r+2} \text{ in } (2U_1 U_5 + 2U_2 U_4 + U_3^2).$$

12. To obtain the whole number of the partitions of the r -gon into six parts, we must perform the foregoing process successively with each summit of the r -gon as the summit from which is drawn the diagonal which divides the r -gon into two parts; that is, the number found as above is to be multiplied by r . We thus obtain all the partitions repeated a certain number of times, viz. each partition into six parts is a partition by means of five diagonals, and is thus obtainable by the foregoing process, taking any one of the ten extremities of these diagonals as the point from which is drawn the diagonal which divides the r -gon into two parts; that is, we have to divide the foregoing product by 10. The final result thus is

$$\frac{r}{10} F_r = \text{coeff. } x^{r+2} \text{ in } (2U_1 U_5 + 2U_2 U_4 + U_3^2),$$

where

$$\frac{r}{10} F_r \text{ is } = \text{coeff. } x^{r+2} \text{ in } V_6;$$

we thus have

$$V_6 = 2U_1 U_5 + 2U_2 U_4 + U_3^2.$$

13. The reasoning is perfectly general; and applying it successively to the partitions into two parts, three parts, &c., we have

$$\begin{aligned} V_2 &= U_1^2, \\ V_3 &= 2U_1U_2, \\ V_4 &= 2U_1U_3 + U_2^2, \\ V_5 &= 2U_1U_4 + 2U_2U_3, \\ &\vdots \end{aligned}$$

where any function V is related to the corresponding function U as above. The value of U_1 is obviously

$$U_1 = x^3 + x^4 + x^5 + \dots = \frac{x^3}{1-x};$$

and hence the several functions U and V have the values above written down; the general term of U_k is

$$\frac{[r+k-2]^{k-1} [r-3]^{k-1}}{[k]^{k-1} [k-1]^{k-1}} x^r,$$

and the number of partitions of the r -gon into k parts is equal to the coefficient of x^r in this general term.

14. In the investigations which next follow, I consider, without using the method of generating functions, the problem of the partition of the r -gon into 2, 3, 4 or 5 parts; it will be convenient to state the results as follows:

Number of Partitions.

$$\begin{aligned} 2 \text{ parts, } & \frac{r}{2} A, \\ 3 \text{ parts, } & \frac{r}{4} 2A, \\ 4 \text{ parts, } & \frac{r}{6} (3A + 2B), \\ 5 \text{ parts, } & \frac{r}{8} (4A + 8B + 2C); \end{aligned}$$

where the capital letters refer to different “diagonal-types,” thus:

2 parts, or 1 diagonal:	3 parts, or 2 diagonals:	4 parts, or 3 diagonals:	5 parts, or 4 diagonals:
A	A	$A \quad B$	$A \quad B \quad C$
		$\triangle$	$< \quad \square$

viz. if, in a polygon divided into k parts by means of $k-1$ diagonals, we delete all the sides of the polygon, leaving only the diagonals, then these will present themselves under distinct forms, which are what I call “diagonal-types”; for instance, when

$$k = 4,$$

there are the two types A and B shown in the above diagram for four parts.

15. It is to be observed that we have sub-types corresponding to the coalescence of the terminal points of different diagonals; thus, suppose

$$k = 4.$$

Writing now A° and B° to denote the forms without coalescences, we have the sub-types A° , A' , A'' and B° , B' , B'' , B''' , as follows:

$$\begin{array}{ccccccc} & \text{4 parts, or 3 diagonals:} & & & & & \\ A^\circ & A' & A'' & B^\circ & B' & B'' & B''' \\ ||| & \wedge | & \swarrow \wedge & \triangle & \triangle & \triangle & \triangle \end{array}$$

where observe that under A'' are included two distinct forms, which, nevertheless, by reason that there is in each of them the same number ($= 2$) of coalescences, are reckoned as belonging to the same sub-type.

16. The numbers called A , B , C , &c., have values which may be directly determined. I write down as follows:

$$\text{1 diagonal, } A = \frac{r-3}{1}.$$

$$\text{2 diagonals, } A = \frac{r-3 \cdot r-2 \cdot r-1}{6} - 2 \frac{r-3}{1} = \frac{r-3 \cdot r-4}{6}(r+1);$$

where the calculation is

$$\begin{array}{r} r-2 \cdot r-1 = r^2-3r+2 \\ -6 \quad -6 \\ \hline = r^2-3r-4 \\ = r-4 \cdot r+1. \end{array}$$

$$\begin{aligned} \text{3 diagonals, } A &= \frac{r-3 \cdot r-2 \cdot r-1 \cdot r \cdot r+1}{120} - 2 \frac{r-3 \cdot r-2 \cdot r-1}{6} + 1 \cdot \frac{r-3}{1} \\ &= \frac{r-3 \cdot r-2 \cdot r-1 \cdot r \cdot r+1}{120} - 2 \frac{r-3 \cdot r-2 \cdot r-1}{6} + 1 \cdot \frac{r-3}{1} \\ &= \frac{r-3 \cdot r-4 \cdot r-5}{120}(r^2+7r+2); \end{aligned}$$

where the calculation is

$$\begin{array}{r} r-2 \cdot r-1 \cdot r \cdot r+1 = r^4-2r^3-r^2+2r \\ -40 \cdot r-2 \cdot r-1 \quad -40r^2+120r-80 \\ +1240 \quad +1240 \\ \hline = r^4-2r^3-41r^2+122r+40 \\ = r-4 \cdot r-5 \cdot r^2+\dots \\ \frac{r-5 \cdot r-4 \cdot r-3 \cdot r-2 \cdot r-1}{120}. \end{array}$$

$$\begin{aligned}
4 \text{ diagonals, } A &= \frac{r - 3.r - 2.r - 1.r.r + 1.r + 2.r + 3}{5040} \\
&- 3 \frac{r - 3.r - 2.r - 1.r.r + 1}{120} + 3 \frac{r - 3.r - 2.r - 1}{6} - 1 \frac{r - 3}{1} \\
&= \frac{r - 3.r - 4.r - 5.r - 6}{5040} (r^3 + 18r^2 + 65r);
\end{aligned}$$

where the calculation is

$$\begin{aligned}
r - 2.r - 1.r.r + 1.r + 2.r + 3 &= r^6 + 3r^5 - 5r^4 - 15r^3 + 4r^2 + 12r \\
-126 \cdot r - 2.r - 1.r.r + 1 &\quad -126r^4 + 252r^3 + 126r^2 - 252r \\
+2520 \cdot r - 2.r - 1 &\quad +2520r^2 - 7560r + 5040 \\
-5040 &\quad -5040 \\
&= 2650r^2 - 7800r \\
&= r - 4 \cdot r - 5 \cdot r - 6 \cdot r^3 + 18r^2 + 65r.
\end{aligned}$$

Also

$$\begin{aligned}
B &= \frac{r - 5.r - 4.r - 3.r - 2.r - 1.r.r + 1}{5040} - \frac{r - 5.r - 4.r - 3.r - 2.r - 1}{120} \\
&= \frac{r - 5.r - 4.r - 3.r - 2.r - 1.r.r + 1}{5040} - \frac{r - 5.r - 4.r - 3.r - 2.r - 1 \cdot 42}{5040} \\
&= \frac{r - 3.r - 4.r - 5.r - 6}{5040} (r - 1.r - 2.r + 7), \\
C &= \frac{r - 7.r - 6.r - 5.r - 4.r - 3.r - 2.r - 1}{5040} \\
&= \frac{r - 3.r - 4.r - 5.r - 6}{5040} (r - 1.r - 2.r - 7);
\end{aligned}$$

where the calculation is

$$\begin{aligned}
r.r + 1 &= r^2 + r \\
-42 &\quad -42 \\
&= r^2 + r - 42 \\
&= r - 6 \cdot r + 7.
\end{aligned}$$

17. To explain the formation of these expressions, observe that:

One diagonal.—There must be on each side of the diagonal, or say in each of the two “intervals” formed by the diagonal, two sides; there remain $r - 4$ sides which may be distributed at pleasure between the two intervals, and the number of ways in which this can be done is

$$= \frac{r - 3}{1}.$$

Two diagonals.—There must be on each side of the two diagonals, formed by the two diagonals, two sides; there remain $r - 4$ sides to be distributed between the same four intervals, and the number of ways in which this can be done is

$$\frac{r - 3.r - 2.r - 1}{6}.$$

But we must exclude the distributions where there is no side in the one interval and no side in the other interval between the two diagonals; the number of these is that for the case of the coalescence of the two diagonals into a single diagonal, viz. it is

$$= \frac{r-3}{1};$$

and thus the number required is

$$\frac{r-3.r-2.r-1}{6} - 2 \frac{r-3}{1}.$$

18. *Three diagonals, A.*—There must be on each side of the three diagonals, that is, in two of the six intervals formed by the diagonals, two sides; there remain $r-4$ sides to be distributed between the same six intervals, and the number of ways in which this can be done is

$$= \frac{r-3.r-2.r-1.r.r+1}{120}.$$

But we must exclude distributions which would permit the coalescence of the first and second, or of the second and third, or of all three of the diagonals. For the coalescence of the first and second diagonals (the third diagonal not coalescing), the term to be subtracted is

$$\frac{r-3.r-2.r-1}{6} - \frac{r-3}{1},$$

and the same number for the coalescence of the second and third diagonals (the first diagonal not coalescing); that is, the last-mentioned number is to be multiplied by 2; and for the coalescence of all three diagonals the number to be subtracted is

$$= \frac{r-3}{1};$$

we have thus the foregoing value

$$\frac{r-3.r-2.r-1.r.r+1}{120} - 2 \frac{r-3.r-2.r-1}{6} + 1 \frac{r-3}{1},$$

where it will be observed that we have the binomial coefficients 1, 2, 1 with the signs +, −, +.

Three diagonals, B.—There must be outside each of the three diagonals, that is, in three of the six intervals formed by the diagonals, two sides; and there remain $r-6$ sides to be distributed between the six intervals; the number of ways in which this can be done is

$$= \frac{r-5.r-4.r-3.r-2.r-1}{120},$$

and there is here no coalescence of diagonals, so that this is the number required.

19. *Four diagonals, A.*—There must be on each side of the four diagonals, that is, in two of the eight intervals formed by the diagonals, two sides; there remain

$r - 4$ sides to be distributed between the eight intervals, and the number of ways in which this can be done is

$$\frac{r - 3.r - 2.r - 1.r.r + 1.r + 2.r + 3}{5040}.$$

But this number requires to be corrected for coalescences, as in the case *Three diagonals, A*; and the required number is thus found to be

$$\begin{aligned} & \frac{r - 3.r - 2.r - 1.r.r + 1.r + 2.r + 3}{5040} - 3 \frac{r - 3.r - 2.r - 1.r.r + 1}{120} \\ & + 3 \frac{r - 3.r - 2.r - 1}{6} - 1 \frac{r - 3}{1}. \end{aligned}$$

Four diagonals, B.—There must be outside of three of the diagonals, that is, in each of three of the eight intervals formed by the diagonals, two sides; there remain $r - 6$ sides to be distributed between the eight intervals, and the number of ways in which this can be done is

$$\frac{r - 5.r - 4.r - 3.r - 2.r - 1.r.r + 1}{5040}.$$

There is a correction for the coalescence of two of the diagonals, giving rise to a form such as *Three diagonals, B*; and consequently there is a term

$$- \frac{r - 5.r - 4.r - 3.r - 2.r - 1}{120},$$

which, with the first-mentioned term, gives the required number.

Four diagonals, C.—There must be outside of each of the diagonals, that is, in each of four of the eight intervals formed by the diagonals, two sides; there remain $r - 8$ sides to be distributed between the eight intervals, and the number of ways in which this can be done is

$$\frac{r - 7.r - 6.r - 5.r - 4.r - 3.r - 2.r - 1}{5040},$$

which is the required number.

20. In the expressions of No. 14, A , $2A$, $3A + 2B$, $4A + 8B + 2C$, if we regard the terminals of the diagonals as given points, then (1) we have two summits, which can be joined in one way only, giving rise to the diagonal-type A ; (2) we have four summits, which can be joined in two ways only, so as to give rise to the diagonal-type A ; (3) we have six summits, which can be joined in three ways so as to give rise to a diagonal-type A , and in two ways so as to give rise to a diagonal-type B ; and (4) we have eight summits, which can be joined in four ways so as to give rise to a diagonal-type A , in eight ways so as to give rise to a diagonal-type B , and in two ways so as to give rise to a diagonal-type C ; we have thus the linear forms in question. To obtain the number of partitions, we have in each case to multiply by r . To explain this, after the polygon is drawn, imagine

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the summits to be numbered 1, 2, 3, ..., r in succession (the numbering may begin at any one of the r summits); regarding each of these numberings as giving a different partition, we should have the factor r . But, in fact, the partitions so obtained are not all of them distinct, but we have in each case a system of partitions repeated as many times as there are summits of the diagonals, that is, a number of times equal to twice the number of the diagonals; and we have thus, after the multiplication by r , to divide by the numbers 2, 4, 6, 8, in the four cases respectively.

21. We hence have immediately:—

Two parts, the number of partitions

$$= \frac{r}{2}A = \frac{r \cdot r - 3}{2 \cdot 1};$$

Three parts, the number of partitions

$$= \frac{r}{2}A = \frac{r + 1 \cdot r \cdot r - 3 \cdot r - 4}{3 \cdot 2 \cdot 2 \cdot 1};$$

Four parts, the number of partitions

$$= \frac{r}{6}(3A + 2B) = \frac{r + 2 \cdot r + 1 \cdot r \cdot r - 3 \cdot r - 4 \cdot r - 5}{4 \cdot 3 \cdot 2 \cdot 3 \cdot 2 \cdot 1},$$

the calculation being

$$\begin{aligned} 3(r^2 + 7r + 2) &= 3r^2 + 21r + 6 \\ + 2 \cdot r - 1 \cdot r - 2 &= + 2r^2 - 6r + 4 \\ \hline &5r^2 + 15r + 10 \\ &= 5 \cdot r + 1 \cdot r + 2; \end{aligned}$$

Five parts, the number of partitions

$$= \frac{r}{8}(4A + 8B + 2C) = \frac{r + 3 \cdot r + 2 \cdot r + 1 \cdot r \cdot r - 3 \cdot r - 4 \cdot r - 5 \cdot r - 6}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 4 \cdot 3 \cdot 2 \cdot 1},$$

the calculation being

$$\begin{aligned} 4(r^3 + 18r^2 + 65r) &= 4r^3 + 72r^2 + 260r \\ + 8 \cdot r - 1 \cdot r - 2 \cdot r + 7 &= + 8r^3 + 32r^2 - 152r + 112 \\ + 2 \cdot r - 1 \cdot r - 2 \cdot r - 7 &= + 2r^3 - 20r^2 + 46r - 28 \\ \hline &14r^3 + 84r^2 + 154r + 84 \\ &= 14(r^3 + 6r^2 + 11r + 6) \\ &= 14 \cdot r + 1 \cdot r + 2 \cdot r + 3. \end{aligned}$$

To complete the theory, it would be in the first instance necessary to find for any given number of diagonals, $k - 1$, whatever, the number and form of the diagonal-types, A , B , C , &c.; this is itself an interesting question in the Theory of Partitions, but I have not considered it.

22. Although the foregoing process (which, it will be observed, deals with the diagonal-types, without any consideration of the sub-types) is the most simple for the determination of the numbers A , B , C , &c., yet it is interesting to give a second process. Considering the several cases in order:

One diagonal, A.—The diagonal has two summits; we must have on each side of it one summit, and there remain $r - 4$ summits which may be distributed between the two intervals formed by the diagonals. This can be done in $\frac{r-3}{1}$ ways, or we have, as before,

$$A = \frac{r-3}{1}.$$

Two diagonals, A.—The diagonals have four summits; we must have outside each diagonal one summit, and there remain $r - 6$ summits to be distributed between the four intervals formed by the diagonals; this can be done in $\frac{r-5 \cdot r-4 \cdot r-3}{6}$ ways, or we have this value for A° . But the two top summits of the diagonals, or the two bottom summits, may coalesce; in either case, the diagonals have three summits. We must have outside each diagonal one summit, and there remain $r - 5$ summits to be distributed between the three intervals formed by the diagonals; the number of ways in which this can be done is

$$= \frac{r-4 \cdot r-3}{2},$$

say this is the value of A' . And we then have $A = A^\circ + 2A'$, as before. The calculation is

$$= \frac{r+1 \cdot r-3 \cdot r-4}{6}.$$

$$r-5+6=r+1.$$

23. *Three diagonals, A.*—See No. 15 for the figures of the sub-types. We have

$$A = A^\circ + 4A' + 4A'',$$

where the coefficients, 4 and 4, are the number of ways in which A' and A'' respectively can be derived from A° by coalescences of summits. For A° , the diagonals have six summits, and there must be outside of two diagonals one summit; there remain $r - 8$ summits to be distributed between the six intervals formed by the diagonals, and we have

$$A^\circ = \frac{r-7 \cdot r-6 \cdot r-5 \cdot r-4 \cdot r-3}{120}.$$

For A' , the diagonals have five summits, and we must have outside of each of two diagonals, one summit; there remain $r - 7$ summits to be distributed between the five intervals formed by the diagonals; we thus have

$$A' = \frac{r-6 \cdot r-5 \cdot r-4 \cdot r-3}{24}.$$

For A'' , the diagonals have four summits; there must be outside of each of two diagonals one summit, and there remain $r - 6$ summits to be distributed between the four intervals formed by the diagonals; we thus have

$$A'' = \frac{r - 5 \cdot r - 4 \cdot r - 3}{6}.$$

The foregoing values give

$$A = \frac{r - 3 \cdot r - 4 \cdot r - 5}{120} (r^2 + 7r + 2),$$

as before. The calculation is

$$\begin{array}{rcl} r - 6 \cdot r - 7 & = & r^2 - 13r + 42 \\ + 20 \cdot r - 6 & = & + 20r - 120 \\ + 80 & = & + 80 \\ \hline & & r^2 + 7r + 2. \end{array}$$

Three diagonals, B.—See No. 15 for the figures of the sub-types. We have

$$B = B^\circ + 3B' + 3B'' + B''',$$

For B° , the diagonals have six summits, and there must be outside each of the three diagonals one summit; there remain $r - 9$ summits to be distributed between the six intervals formed by the diagonals. We thus have

$$B^\circ = \frac{r - 8 \cdot r - 7 \cdot r - 6 \cdot r - 5 \cdot r - 4}{120}.$$

Similarly,

$$B' = \frac{r - 7 \cdot r - 6 \cdot r - 5 \cdot r - 4}{24}, \quad B'' = \frac{r - 6 \cdot r - 5 \cdot r - 4}{6}, \quad B''' = \frac{r - 5 \cdot r - 4}{2}.$$

Hence

$$B = \frac{r - 5 \cdot r - 4 \cdot r - 3 \cdot r - 2 \cdot r - 1}{120},$$

as before. The calculation is

$$\begin{array}{rcl} r - 6 \cdot r - 7 \cdot r - 8 & = & r^3 - 21r^2 + 146r - 336 \\ + 15 \cdot r - 6 \cdot r - 7 & = & + 15r^2 - 195r + 630 \\ + 60 \cdot r - 6 & = & + 60r - 360 \\ + 60 & = & + 60 \\ \hline & & r - 1 \cdot r - 2 \cdot r - 3. \end{array}$$

24. *Four diagonals, A.*—The figures of the sub-types of A , B , C can be supplied without difficulty. We have

$$A = A^\circ + 6A' + 12A'' + 8A''',$$

where I remark that the numerical coefficients 1, 6, 12, 8 are the terms of $(1, 2)^3$. We have

$$\begin{aligned} A^\circ &= \frac{r - 9.r - 8.r - 7.r - 6.r - 5.r - 4.r - 3}{5040}, \\ A' &= \frac{r - 8.r - 7.r - 6.r - 5.r - 4.r - 3}{720}, \\ A'' &= \frac{r - 7.r - 6.r - 5.r - 4.r - 3}{120}, \\ A''' &= \frac{r - 6.r - 5.r - 4.r - 3}{24}, \end{aligned}$$

and thence

$$A = \frac{r - 3.r - 4.r - 5.r - 6}{5040} (r^3 + 18r^2 + 65r),$$

as before. The calculation is

$$\begin{array}{rcl} r - 9.r - 8.r - 7 & = & r^3 - 24r^2 + 191r - 504 \\ + 42.r - 8.r - 7 & = & + 42r^2 - 630r + 2352 \\ + 504.r - 7 & = & + 504r - 3528 \\ + 1680 & = & + 1680 \\ \hline & & r^3 + 18r^2 + 65r. \end{array}$$

Four diagonals, B.—We have

$$B = B^\circ + 5B' + 9B'' + 7B''' + 2B^{\text{iv}},$$

where the coefficients, 1, 5, 9, 7, 2, are the terms of $(1, 1)^3(1, 2)$. We have

$$\begin{aligned} B^\circ &= \frac{r - 10.r - 9.r - 8.r - 7.r - 6.r - 5.r - 4}{5040}, \\ B' &= \frac{r - 9.r - 8.r - 7.r - 6.r - 5.r - 4}{720}, \\ B'' &= \frac{r - 8.r - 7.r - 6.r - 5.r - 4}{120}, \\ B''' &= \frac{r - 7.r - 6.r - 5.r - 4}{24}, \\ B^{\text{iv}} &= \frac{r - 6.r - 5.r - 4}{6}, \end{aligned}$$

and thence

$$B = \frac{r - 3.r - 4.r - 5.r - 6}{5040} \frac{r - 1.r - 2.r + 7}{r - 1.r - 2.r + 7},$$

as before. The calculation is

$$\begin{array}{rcl}
 r - 10.r - 9.r - 8.r - 7 = r^4 - & 34r^3 + & 431r^2 - 2414r + 5040 \\
 + 35.r - 9.r - 8.r - 7 = & + 35r^3 - & 840r^2 + 6685r - 17640 \\
 + 378.r - 8.r - 7 = & + 378r^2 - & 5670r + 21168 \\
 + 1470.r - 7 = & + 1470r - & 10290 \\
 + 1680 = & + 1680 & \\
 \hline
 & r^4 + & r^3 - 31r^2 + 71r - 42 \\
 & = r - 3.r - 2.r - 1.r + 7.
 \end{array}$$

Four diagonals, C.—We have

$$C = C^\circ + 4C' + 6C'' + 4C''' + 1C^{\text{iv}},$$

where the coefficients are the terms of $(1, 1)^4$. We have

$$\begin{aligned}
 C^\circ &= \frac{r - 11.r - 10.r - 9.r - 8.r - 7.r - 6.r - 5}{5040}, \\
 C' &= \frac{r - 10.r - 9.r - 8.r - 7.r - 6.r - 5}{720}, \\
 C'' &= \frac{r - 9.r - 8.r - 7.r - 6.r - 5}{120}, \\
 C''' &= \frac{r - 8.r - 7.r - 6.r - 5}{24}, \\
 C^{\text{iv}} &= \frac{r - 7.r - 6.r - 5}{6},
 \end{aligned}$$

and thence

$$C = \frac{r - 7.r - 6.r - 5.r - 4.r - 3.r - 2.r - 1}{5040}.$$

I omit the calculation, as the equation is at once seen to be a particular case of a known factorial formula.

25. We may analyse the partitions of an r -gon into a given number of parts, according to the nature of the parts, that is, the numbers of the sides of the several component polygons. It is for this purpose convenient to introduce the notion of “weight”; say a triangle has the weight 1, then a quadrangle, as divisible into two triangles, has the weight 2, a pentagon, as divisible into three triangles, has the weight 3, . . . , and generally an r -gon, as divisible into $r - 2$ triangles, has the weight $r - 2$. It at once follows that, if

$$W = w + w', \text{ or } = w + w' + w'', \text{ \&c.},$$

then a polygon of weight W is divisible into two polygons of the weights w, w' , or into three polygons of the weights w, w', w'' respectively; and so on. Thus the 2-partitions of an 8-gon (weight = 6) are 15, 24, and 33; the 3-partitions are 114,

123, 222, and so on. Of course the number of the partitions 15, 24, 33, is equal to the whole number of the 2-partitions of the 8-gon, that is, $= 20$; the number of the partitions 114, 123, 222, is equal to the whole number of the 3-partitions of the 8-gon, that is, it is $= 120$; and so in other cases. It is easy to derive in order one from the other the numbers of the partitions of each several kind of the polygons of the several weights 2, 3, 4, 5, 6, &c.; and I write down the accompanying Table (No. 26), facing page 112, the process for the construction being as follows:

27. The first column (2 parts) is at once obtained. For a polygon of an odd number of sides, for instance the 9-gon (weight $= 7$), imagining the summits numbered in order 1, 2, $\dots$, 9, we divide this into a triangle and octagon, or obtain the partitions 16, by drawing the diagonals 13, 24, $\dots$, 81, 92: viz. the number is $= 9$.

In the Table this is written, $16 = 9$; and so in other cases. Similarly we divide it into a quadrangle and heptagon, or obtain the partitions 25, by drawing the diagonals 14, 25, $\dots$, 82, 93: viz. the number is again $= 9$; and we divide it into a pentagon and a hexagon, or obtain the partitions 34, by drawing the diagonals 15, 26, $\dots$, 83, 94: viz. the number is $= 9$, and here

$$9 + 9 + 9 = 27,$$

the whole number of 2-partitions of the 9-gon. For a polygon of an even number of sides, for instance the 10-gon (weight $= 8$), the process is a similar one, the only difference being that for the division into two hexagons, (that is, for the partitions 44), each partition is thus obtained twice, or the number of such partitions is $\frac{1}{2} \cdot 10, = 5$; the numbers for the partitions 17, 26, 35, 44, thus are 10, 10, 10, 5; and we have

$$10 + 10 + 10 + 5 = 35,$$

the whole number of the 2-partitions of the 10-gon.

28. To obtain the second column (3 parts)—suppose, for instance, the 3-partitions of the 9-gon; these are 115, 124, 133, 223. We obtain the number of the partitions 115 from the terms

$$16 = 9 \quad \text{and} \quad 25 = 9$$

of the first column: viz. in 16, changing the 6 into 15, that is, dividing the polygon of weight 6 into two parts of weights 1 and 5 respectively: this can be done in eight ways (see, higher up, $15 = 8$ in the first column); and we thus obtain the number of partitions

$$9 \times 8 = 72;$$

and again, in 25, changing the 2 into 11, that is, dividing the polygon of weight 2 into two parts each of weight 1: this can be done in two ways (see, higher up, $11 = 2$ in the first column); and we thus obtain the number of partitions

$$9 \times 2 = 18;$$

we should thus have, for the number of partitions 115, the sum

$$72 + 18 = 90,$$

only, as it is easy to see, each partition is obtained twice, and the number of the

partitions 115 is the half of this, = 45. And by the like process it is found that the numbers of the partitions 124, 133, 223 are equal to 90, 45, 45 respectively; and then, as a verification, we have

$$45 + 90 + 45 + 45 = 225,$$

the whole number of the 3-partitions of the 9-gon.

29. The third column (4 parts) is derived in like manner from the second column by aid of the first column; and so in general, each column is derived in like manner from the column which immediately precedes it, by aid of the first column. And we have for the numbers in each compartment of any column the verification that the sum of these numbers is equal to the whole number (for the proper values of k and r) of the k -partitions of the r -gon.

It might be possible, by an application of the method of generating functions, to find a law for the numbers in any compartment of a column of the table; but I have not attempted to make this investigation.

30. In the table in No. 2, the numbers 1, 2, 5, 14, 42, &c., of the diagonal line show the number of partitions of the triangle, the quadrangle, the 5-gon, ..., r -gon into triangles: viz. these numbers show the number of partitions of the r -gon into $r - 2$ parts, that is, into triangles; and, for the r -gon, writing $k = r - 2$, the number is

$$= \frac{[2r - 4]^{r-3}}{[r - 2]^{r-3}}.$$

If, as above, taking the weight of the triangle to be 1, we write

$$r - 2 = w,$$

then the number is

$$= \frac{[2w]^{w-1}}{[w]^{w-1}},$$

viz. this is the expression for the number of partitions of the polygon of weight w , or $(w + 2)$ -gon, into triangles.

31. The question considered by Taylor and Rowe, in the paper referred to in No. 1, is that of the partition of the r -gon into p -gons, for p , a given number > 3 ; this implies a restriction on the form of r , viz. we must have $r - 2$ divisible by $p - 2$. In fact, generalizing the definition of w , if we attribute to a p -gon the weight 1, and accordingly to a polygon divisible into w p -gons the weight w , then, r being the number of summits, we must have

$$r = (p - 2)w + 2.$$

In particular, if $p = 4$, so that the r -gon is to be divided into quadrangles, then r is necessarily even, and for the values

$$w = 1, 2, 3, \dots,$$

we have

$$r = 4, 6, 8, \dots$$

[To face page 112.]

r	W	7 PARTS	8 PARTS
3	1		
4	2		
5	3		
6	4		
7	5		
8	6		
9	7 57	1111111 $1287 \times 2 = 2574$ 6) $\overline{2574} = 429$	
10	8 12 05 07	1111112 $2002 \times 5 = 10010$ $5005 \times 4 = 20020$ 6) $\overline{30030} = 5005$	11111111 $5005 \times 2 = 10010$ 7) $\overline{10010} = 1430$

[Note: this is the surviving fragment of the original folding-table facing page 112; the remaining compartments are blank in the printed volume.]

32. To fix the ideas, I assume $p = 4$, and thus consider the problem of the division of the $(2w + 2)$ -gon into quadrangles. Writing

$$w - 1 = a + b + c,$$

we take at pleasure any one side of the $(2w + 2)$ -gon, making this the first side of a quadrangle, the second, third, and fourth sides being diagonals of the polygon such that outside the second side we have a polygon of weight a , outside the third side a polygon of weight b , and outside the fourth side a polygon of weight c . Any one or more of the numbers a, b, c may be $= 0$; they cannot be each of them $= 0$ except in the case $w = 1$. The meaning is that the corresponding side of the quadrangle, instead of being a diagonal, is a side of the $(2w + 2)$ -gon, viz. there is no polygon outside such side. Suppose, in general, that P_w is the number of ways in which a polygon of weight w can be divided in quadrangles, and let each of the polygons of weights a, b, c respectively, be divided into quadrangles: the number of ways in which this can be done is $P_a P_b P_c$; and it is to be noticed that, if for instance $a = 0$, then the number is $= P_b P_c$, viz. the formula remains true if only we assume $P_0 = 1$. The number of partitions thus obtained is $\Sigma P_a P_b P_c$, where the summation extends to all the partitions of $w - 1$ into the parts a, b, c (zeros admissible and the order of the parts being attended to). And we thus obtain all the partitions of the $(2w + 2)$ -gon into w parts; for first, the partitions so obtained are all distinct from each other, and next every partition of the $(2w + 2)$ -gon into w parts is a partition in which the selected side of the $(2w + 2)$ -gon is a side of some one of the quadrangles. That is, we have

$$P_w = \Sigma P_a P_b P_c \text{ (} a, b, c \text{ as above);}$$

and it hence appears that, considering the generating function

$$f = 1 + P_1 x + P_2 x^2 + P_3 x^3 + \dots,$$

we have

$$f = 1 + x f^3.$$

The reasoning is precisely the same if, instead of a division into quadrangles, we have a division into p -gons; the only difference is that instead of the three parts a, b, c , we have the $p - 1$ parts $a, b, c, \dots$, and the equation for f thus is

$$f = 1 + x f^{p-1}.$$

33. Writing for a moment

$$f = u + x f^{p-1},$$

and expanding by Lagrange's theorem, we have

$$f = u + \frac{x}{1} (u^{p-1}) + \frac{x^2}{1 \cdot 2} (u^{2(p-1)})' + \dots + \frac{x^w}{1 \cdot 2 \dots w} (u^{w(p-1)})^{(w-1)} + \dots,$$

viz. after the differentiation, writing $u = 1$, we have

$$P_w = \frac{[(p-1)w]^{w-1}}{[w]^{w-1}},$$

where it will be recollected that, for the number of sides of the polygon, we have

$$r = (p-2)w + 2.$$

In the case of the partition into triangles, $p = 3$, and we have the before-mentioned value

$$[2w]^{w-1} \div [w]^{w-1}, \quad w = r - 2.$$

916.

[NOTE ON A THEOREM IN MATRICES.]

[From the *Proceedings of the London Mathematical Society*, vol. XXII. (1891), p. 458.]

PROF. CAYLEY remarks that a “simple instance [of the theorem] is that, if the real symmetric matrix

$$\begin{pmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{pmatrix}$$

has two latent roots each = 0, and therefore a vacuity = 2, then it has also a nullity = 2 [which may be shown as follows], viz. the conditions for a vacuity = 2 are

$$\begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix} = 0, \quad bc - f^2 + ca - g^2 + ab - h^2 = 0,$$

or, if as usual the determinant is called K , and if

$$(A, B, C, F, G, H) = (bc - f^2, ac - g^2, \dots),$$

then, if

$$K = 0, \quad A + B + C = 0,$$

these equations give

$$BC = F^2, \quad AC = G^2, \quad AB = H^2,$$

i.e.

$$BC = F^2, \quad AC = G^2, \quad AB = H^2;$$

and therefore

$$\begin{aligned} A(A + B + C) &= A^2 + H^2 + G^2, \\ B(A + B + C) &= H^2 + B^2 + F^2, \\ C(A + B + C) &= G^2 + F^2 + C^2; \end{aligned}$$

or, if $A + B + C = 0$, then for real values

$$A = B = C = F = G = H = 0,$$

i.e. nullity = 2.”

917.

[NOTE ON THE THEORY OF RATIONAL TRANSFORMATION.]

[From the *Proceedings of the London Mathematical Society*, vol. XXII. (1891),
pp. 475, 476.]

IN my paper, "Note on the Theory of the Rational Transformation between Two Planes, and on Special Systems of Points," *Proc. Lond. Math. Soc.* t. III. (1870), pp. 196–198, [450], I notice a difficulty which presents itself in the theory. The transformation is given by the equations

$$x' : y' : z' = X : Y : Z,$$

where X, Y, Z are functions $(x, y, z)^n$, such that $X = 0, Y = 0, Z = 0$ are curves in the first plane passing through α_1 points each once, α_2 points each twice (that is, having each of the α_2 points for a double point), α_3 points each 3 times, and so on. We have as the condition of a single variable point of intersection,

$$\alpha_1 + 4\alpha_2 + 9\alpha_3 + \dots = n^2 - 1,$$

and as the condition in order that each of the curves $X = 0, Y = 0, Z = 0$, or say the curve $aX + bY + cZ = 0$, may be unicursal,

$$\alpha_2 + 3\alpha_3 + \dots = \frac{1}{2}(n-1)(n-2);$$

and we thence deduce

$$\alpha_1 + 3\alpha_2 + 6\alpha_3 + \dots = \frac{1}{2}n(n+3) - 2;$$

viz. the postulation of the fixed points quoad a curve of the order n is less by 2 than the postulandum (or, as I prefer to call it, the capacity) $\frac{1}{2}n(n+3)$ of the curve of the order n ; that is, there are precisely the three aszygetic curves $X = 0, Y = 0, Z = 0$. This is as it should be, assuming that the $(\alpha_1, \alpha_2, \alpha_3, \dots)$ points are an ordinary system of points: but what if they form a special system having a postulation less

15–2

than $\alpha_1 + 3\alpha_2 + 6\alpha_3 + \dots$? If, for instance, the postulation is $= \alpha_1 + 3\alpha_2 + 6\alpha_3 + \dots - 1$, then this would be $= \frac{1}{2}n(n+3) - 3$, and there would be four aszygetic curves $X = 0$, $Y = 0$, $Z = 0$, $W = 0$. I believe this to be impossible; but the only proof which I can offer rests upon a remark in regard to the form of the tables¹, pp. 148, 149, of my paper "On the Rational Transformation between Two Spaces," *Proc. Lond. Math. Soc.*, t. III. (1870), pp. 127–180, [447]. I recall that the Jacobian curve $J(X, Y, Z) = 0$ consists of α'_1 lines, α'_2 conics, α'_3 cubics, $\dots$, &c., each passing a certain number of times through the $(\alpha_1, \alpha_2, \alpha_3, \dots)$ points, and that the number of times of passage is shown by these tables; thus, *loc. cit.*, $n = 5$, $\alpha_1 = 8$, $\alpha_4 = 1$: the Jacobian consists of eight lines and a quartic, and we have the table ($\alpha'_1 = 8$, $\alpha'_4 = 1$),

	α_1	α_2	α_3	α_4
	8	0	0	1
$\alpha'_1 = 8$	1			1
$\alpha'_2 = 0$				
$\alpha'_3 = 0$				
$\alpha'_4 = 1$	8			1^3

showing that the quartic passes through the eight points α_1 , and through the point α_4 three times (has α_4 for a triple point). Imagine a new function W . Then in like manner $J(X, Y, W) = 0$ consists of eight lines and a quartic, and this quartic passes through the eight points α_1 and the point α_4 three times; that is, the two quartics intersect in $8 + 3 \cdot 3 = 17$ points; and thus the two quartics must be one and the same curve; this implies a syzygy between X, Y, Z, W , viz. W is a mere linear function of X, Y, Z . The general remark is that, if in the tables mp is reckoned as mp^2 , then in the table for the several lines (exclusive of those for which the outside accented letter is $= 0$, and therefore the tabular numbers of the line are each $= 0$), i.e. for the lines which correspond to a line, a conic, a cubic, a quartic, &c., respectively, the sums of the tabular numbers are $1^2 + 1$, $2^2 + 1$, $3^2 + 1$, $4^2 + 1$, &c., respectively. This is, in fact, the case for each of the eleven tables (*loc. cit.*).

¹[* This Collection, vol. VII., pp. 208, 209.]

918.

**ON THE SUBSTITUTION GROUPS FOR TWO, THREE, FOUR,
FIVE, SIX, SEVEN, AND EIGHT LETTERS.**

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xxv. (1891),
pp. 71–88, 137–155.]

THE substitution groups for two, three, four, and five letters were obtained by Serret: those for six, seven and eight letters have recently been obtained by Mr Askwith. I wish to reproduce these results in a condensed form.

The following table shows for the several cases respectively, the orders of the several groups, and for any order the number of distinct groups. As regards the case of eight letters, the numbers mentioned do not exactly agree with Mr Askwith: he gives a few non-existent groups, and omits some which I have supplied (see *post*, the list of the groups for eight letters); and it is possible that there are other omissions: the several numbers in the column and the sum total of 155 are given subject to correction.

2	3	4	5	6	7	8
1 2	1 3 1 6	1 2 3 4 1 8 1 12 1 24	1 5 2 6 1 10 1 12 1 20 1 60 1 120	1 2 1 3 4 4 3 6 6 8 1 9 1 12 1 16 3 18 4 24 2 36 2 48 1 60 1 72 1 120 1 360 1 720	2 6 1 7 2 10 9 12 1 14 2 20 1 21 7 24 1 36 1 40 1 42 1 48 3 72 2 120 1 144 1 240 1 2520 1 5040	1 2 8 4 4 6 25 8 6 12 1 15 22 16 3 18 4 24 3 30 12 32 5 36 14 48 3 60 3 64 3 72 10 96 3 120 2 144 1 168 1 180 3 192 1 240 3 288 1 336 3 360 1 384 2 576 3 720 1 1152 1 1440 1 20160 1 40320
1	2	7	8	34	38	155

Here the top line shows the numbers of letters; each second column shows the order of the groups, and each first column the number of groups of the several orders: the sums at the foot of the first columns show therefore the whole number of groups, viz.

No. of letters = 2, 3, 4, 5, 6, 7, 8

No. of groups = 1, 2, 7, 8, 34, 38, 155.

In the enumeration of the groups, I use some notations which must be explained. For greater simplicity I omit parentheses, and write ab , abc , $ab.cd$, $abc.def$, &c., to denote substitutions, viz. ab is the interchange of a and b ; abc the cyclical change a into b , b into c , c into a ; $ab.cd$ the combined interchange of a and b and of c and d ; $abc.def$ the combined cyclical changes a into b , b into c , c into a , and d into e , e into f , f into d ; and so in other cases.

Again, (abc) all, means the complete group of all the substitutions

$$(1, abc, acb, bc, ca, ab)$$

upon the three letters; and so in other cases. In the case, however, of two letters, I write simply (ab) to denote the complete group $(1, ab)$ of substitutions, and so also for any substitution such as $ab.cd$, where the complete group is $(1, ab.cd)$, I denote the group by $(ab.cd)$. Moreover, (abc) cyc., denotes the group of cyclical substitutions $(1, abc, acb)$ upon the three letters; and so in other cases.

A group will in general contain positive and negative substitutions, and, when this is so, the positive substitutions will form a group which is denoted by the symbol, POS.; the negative substitutions (which of course do not form a group) are denoted in like manner by the symbol, NEG. Thus, (abc) all, pos., or for shortness, (abc) pos., will denote the group formed by the positive substitutions of (abc) all; (abc) pos. is thus the same thing as (abc) cyc., but obviously $(abcd)$ pos. and $(abcd)$ cyc. have quite different meanings. It is to be noticed that, for any odd number of letters, the substitutions of a group (abc) cyc. are all positive, and thus (abc) cyc. pos. is the original group: for any even number of letters, the group $(abcd)$ cyc. or $(abcdef)$ cyc. contains positive and negative substitutions, but the positive substitutions thereof form a cyclical group, thus

$$(abcd) \text{ cyc. pos.} = (ab.cd) \text{ cyc.}, \quad (abcdef) \text{ cyc. pos.} = (ace.bdf) \text{ cyc.},$$

and the notation, $()$ cyc. pos., is thus unnecessary, and it will not be used.

Substitutions or groups which have no letter in common are said to be independent. The product of two independent groups is of course a group, and the components may be called independent factors of the resultant group: we use for such a product the notation $A.B$, and call the group a composite group. If each of the groups A and B contain positive and negative substitutions, we thence derive a new group, $(A.B)$ pos., viz. the substitutions hereof are the products of a positive substitution of A and a positive substitution of B , and the products of a negative substitution of A and a negative substitution of B , say

$$(A.B) \text{ pos.} = (A \text{ pos. } B \text{ pos.}) + (A \text{ neg.})(B \text{ neg.}) :$$

obviously the number of substitutions or order of the group $(A.B)$ pos. is one half of that of the group $A.B$.

A more general, but not perfectly definite, notation is that of $(A.B)$ dim., the dimidiate of the group $A.B$. Suppose, for instance, that A is a group of substitutions of the letters (a, b, c, d) ; and that B is the group (ef) . Here if A is composed of two equal sets $1, P, Q, \dots; R, S, T, \dots$, where the first set $1, P, Q, \dots$, is a group, then we have a group $1, P, Q, \dots, ef.R, ef.S, ef.T, \dots$, which is a group of the form in question, $(A.B)$ dim. But in some cases the group A can be in more than one way divided into two equal sets the first of which forms a group, and a further explanation of the notation is required. I do not give at present a more complete explanation, nor explain the analogous notions trisection (tris.), &c.

Substitutions or groups having a letter or letters in common are non-independent. If A, B are such groups, then the substitutions of A are not commutable with

those of B , and we have not in general such a group as $B.A$ or $A.B$. It may happen that, although the individual substitutions of A are not commutable with those of B , yet that the groups are commutable, say we have $A.B = B.A$, viz. here the substitutions of $A.B$ are in a different order identical with those of $B.A$. We have in this case a group $A.B$; this is not a composite group; and the notation will never be employed without explanation.

I consider, in particular, the two groups $(abc.def)$ cyc. and (abc) cyc. (def) cyc.; in each of them, the six letters (a, b, c, d, e, f) are divided into two sets (a, b, c) , (d, e, f) , and the substitutions are the product of an abc -substitution into a def -substitution: viz. in the first group the substitutions are

$$1, abc.def, acb.dfe,$$

and in the second group they are

$$\begin{aligned} &1, \quad abc, \quad abc.def, \\ &\quad \quad \quad acb, \quad acb.def, \\ &def, \quad abc.dfe, \\ &dfe, \quad acb.dfe. \end{aligned}$$

We can from each of the groups, introducing substitutions which interchange (a, b, c) with (d, e, f) , or say by combination with the group $(ad.be.cf)$, derive a group of double the order; and it is worth while to consider the two cases in detail. First, for the group

$$(ad.be.cf)(abc.def) \text{ cyc.}$$

We have

Arrangements.	Substitutions.
$abcdef$,	1,
$bcaefd$,	$abc.def$,
$cabfde$,	$acb.dfe$,
$defabc$,	$ad.be.cf$,
$efdbca$,	$aecdbf$,
$fdecab$,	$afbdce$,

say

$$(ad.be.cf)(abc.def) \text{ cyc.} = (abcdef)_6.$$

This group of the order 6 is, as it happens, the group $(aecdbf)$ cyc.

It may be remarked that if, instead of $(abc.def)$ cyc., we consider the precisely similar group $(abc.dfe)$ cyc., then operating upon it with the same group $(ad.be.cf)$, we obtain quite a different form of group $(abcdef)_6$. In fact, for $(ad.be.cf)(abc.dfe)$ cyc., we have

Arrangements.	Substitutions.
$abcdef,$	$1,$
$bca fde,$	$abc . dfe,$
$cabefd,$	$acb . def,$
$defabc,$	$ad . be . cf,$
$efdcab,$	$ae . bf . cd,$
$fdebca;$	$af . bd . ce;$

which is not the cyclical group. See *post*, six letters, ord. 6. 3.

Next, for the group $(ad . be . cf)(abc) \text{ cyc. } (def) \text{ cyc.};$ we have here

Arrangements.	Substitutions.
$abcdef,$	$1,$
$bcadef,$	$abc,$
$cabdef,$	$acb,$
$abcefd,$	$def,$
$bcaefd,$	$abc . def,$
$cabefd,$	$acb . def,$
$abc fde,$	$dfe,$
$bca fde,$	$abc . dfe,$
$cab fde,$	$acb . dfe,$
$defabc,$	$ad . be . cf,$
$efdabc,$	$aeb fcd,$
$fdeabc,$	$afcebd,$
$defbca,$	$adbecf,$
$efdbca,$	$aecdbf,$
$fdebca,$	$af . bd . ce,$
$defcab,$	$adc fbe,$
$efdcab,$	$ae . bf . cd,$

say

$$fdecab; \quad afbdce;$$

$$(ad . be . cf)\{(abc) \text{ cyc. } (def) \text{ cyc.}\} = (abcdef)_{18}.$$

See *post*, six letters, ord. 18.

The groups thus obtained, with substitutions which interchange the two sets of letters, are said to be “woven” groups.

By means of the foregoing notations a large number, but by no means all, of the groups belonging to the several cases of 2, 3, 4, 5, 6, 7, and 8 letters can be

represented in a very compendious form. It is, even in the case of 4 letters, proper to introduce special notations: thus for the order 4, we have $(ab)(cd)$, (or as I generally write it $(ac)(bd)$), and $(abcd)$ cyc. (which it is convenient thus to represent), and another group $(1, ab.cd, ac.bd, ad.bc)$: this is a symmetrical character of the woven group $(ac.bd)(ab)(cd)$, but in thus representing it, we fail to exhibit the symmetrical character of the group, and I prefer to represent it as $(abcd)_4$. Again for the order 8, there is a single group, which I write $(abcd)_8$: this can be, in regard to dimidiation, divided in three different ways into two sets of four, and to distinguish them I write

$$\begin{aligned}(abcd)_8 \text{ com.} &= (1, ac, bd, ac.bd; abcd, adbc, ab.cd, ad.bc), \\ (abcd)_8 \text{ cyc.} &= (1, abcd, ac.bd, adbc; ac, bd, ab.cd, ad.bc), \\ (abcd)_8 \text{ pos.} &= (1, ab.cd, ac.bd, ad.bc; ac, bd, abcd, adcb);\end{aligned}$$

viz. com. denotes that the first set of four is the composite group $(ac)(bd)$; cyc. that it is the cyclical group $(abcd)$ cyc.; and pos. that it is the positive group $(abcd)_4$. We have thus, in the case of 6 letters, the three dimidiation forms,

$$\begin{aligned}\{(abcd)_8 \text{ com.}(ef)\} \text{ dim.} &= (ac)(bd) + ef(abcd, adbc, ab.cd, ad.bc), \\ \{(abcd)_8 \text{ cyc.}(ef)\} \text{ dim.} &= (abcd) \text{ cyc.} + ef(ac, bd, ab.cd, ad.bc), \\ \{(abcd)_8 \text{ pos.}(ef)\} \text{ dim.} &= (abcd)_4 + ef(ac, bd, abcd, adcb),\end{aligned}$$

the last of which may be more simply written as $\{(abcd)_8(ef)\} \text{ pos.}$

In the cases of 2, 3, 4, and 5 letters, the groups are as follows:

2 letters ord.	3 letters ord.	4 letters ord.	5 letters ord.
2 (ab)	3 (abc) cyc. 6 (abc) all	2 $(ab.cd)$ 4 $(ab)(cd)$ 4 $(abcd)$ cyc. 4 $(abcd)_4$ 8 $(abcd)_8$ 12 $(abcd)$ pos. 24 $(abcd)$ all	5 $(abcde)$ cyc. 6 (abc) cyc. (de) 10 $\{(abcd)_8(de)\}$ pos. 10 $(abcde)_{10}$ 12 (abc) all (de) 20 $(abcde)_{20}$ 60 $(abcde)$ pos. 120 $(abcde)$ all

The nature of the group is in most cases at once intelligible from the foregoing explanations: thus (ab) denotes the group $(1, ab)$; (abc) cyc., the cyclical group $(1, abc, acb)$; (abc) all, the group of the six substitutions $(1, abc, acb, ab, ac, bc)$, and so in other cases: but, when a further explanation is required, the nature of the group is merely indicated by writing down the order as a suffix: thus $(abcd)_4$ means a group of four substitutions, $(abcd)_8$ a group of 8 substitutions, and so in other cases. In the case of four letters, the necessary explanations have already been

given: it may be remarked that the group $(abcd)_8$ is that of the substitutions which leave unaltered the three-valued function $ac + bd$.

Five letters. Explanations.

Five letters.

ord. 10. $(abcde)_{10}$. This may be written $\{(abcde)_{20}\}$ pos., viz. it consists of the 10 positive substitutions out of the next-mentioned group of 20 substitutions. Referring to that group, and writing $\sigma = S^2 = ac \cdot bd$, and $T = aecdb$, the present group is $(1, \sigma)(1, T, T^2, T^3, T^4)$, where $\sigma^2 = 1$, $T^5 = 1$, $\sigma T = T^4 \sigma$, $\sigma T^4 = T \sigma$, $\sigma T^3 = T^2 \sigma$, $\sigma T^2 = T^3 \sigma$: or, retaining S^2 instead of σ , say the group is $(1, S^2)(1, T, T^2, T^3, T^4)$.

ord. 20. $(abcde)_{20}$. The substitutions, distinguishing the positive and the negative ones, are

+	+	+	-
1,	$ab \cdot de$,	$abdce$,	$abcd$,
	$ac \cdot bd$,	$achbed$,	$abec$,
	$ad \cdot ce$,	$adebc$,	$acde$,
	$ae \cdot bc$,	$aecdb$,	$aceb$,
	$bc \cdot cd$,		$adcb$,
			$aebd$,
			$aedc$,
			$bced$,
			$bdec$;

and here, if $S = abcd$, $T = aecdb$, then the group is $(1, S, S^2, S^3)(1, T, T^2, T^3, T^4)$, where $S^4 = 1$, $T^5 = 1$, and generally $S^i T^j = T^k \cdot S^l$; that is,

	1	T	T^2	T^3	T^4
1	1	T	T^2	T^3	T^4
S	S	$T^2 S$	$T^4 S$	TS	$T^3 S$
S^2	S^2	$T^4 S^2$	$T^3 S^2$	$T^2 S^2$	TS^2
S^3	S^3	$T^3 S^3$	TS^3	$T^4 S^3$	$T^2 S^3$

read

$$ST = T^2 S, \quad S^2 T^2 = T^3 S^2, \text{ \&c.}$$

In the cases of 6, 7, and 8 letters, where the number of groups is larger, I introduce current numbers for the groups of the same order: thus, *infra*, 6 letters order 4, we have the groups 4. 1, 4. 2, 4. 3, 4. 4; and so in other cases, where there is more than one group for the same order.

Six letters. Six letters: the groups are

ord.

2. $(ab.cd.ef),$
3. $(abc.def) \text{ cyc.}$
4. 1 $(ab.cd)(ef),$
 - 2 $\{(ab)(cd)(ef)\} \text{ pos.}$
 - 3 $\{(abcd) \text{ cyc.}(ef)\} \text{ pos.}$
 - 4 $\{(abcd)_4(ef)\} \text{ dim.}$
6. 1 $(abcdef) \text{ cyc.}$
 - 2 $(abc.def) \text{ all},$
 - 3 $(ad.bf.ce)(abc.def) \text{ cyc.}$
8. 1 $(ab)(cd)(ef),$
 - 2 $(abcd) \text{ cyc.}(ef),$
 - 3 $(abcd)_4(ef),$
 - 4 $\{(abcd)_8 \text{ com.}(ef)\} \text{ dim.}$
 - 5 $\{(abcd)_8 \text{ cyc.}(ef)\} \text{ dim.}$
 - 6 $\{(abcd)_8 \text{ pos.}(ef)\} \text{ dim.} = \{(abcd)_8(ef)\} \text{ pos.}$
9. $(abc) \text{ cyc.}(def) \text{ cyc.}$
12. $(abcdef)_{12},$
16. $(abcd)_8(ef),$
18. 1 $(abc) \text{ all}(def) \text{ cyc.},$
 - 2 $\{(abc) \text{ all}(def) \text{ all}\} \text{ pos.},$
 - 3 $(ad.be.cf)\{(abc) \text{ cyc.}(def) \text{ cyc.}\},$
24. 1 $(abcd) \text{ pos.}(ef),$
 - 2 $\{(abcd) \text{ all}(ef)\} \text{ pos.},$

The $\pm$ and $+$ are used for distinction, the $\pm$ showing that the group is one with positive and negative substitutions, the $+$ that it is a group with positive substitutions only.
 - 3 $(\pm abcdef)_{24}$
 - 4 $(+ abcdef)_{24},$
36. 1 $(abc) \text{ all}(def) \text{ all},$
 - 2 $(abcdef)_{36},$
48. 1 $(abcd) \text{ all}(ef),$
 - 2 $(abcdef)_{48},$
60. $(abcdef)_{60},$
72. $(abcdef)_{72},$
120. $(abcdef)_{120},$

360. $(abcdef)$ pos.,

720. $(abcdef)$ all.

*Six letters. Explanations.**Six letters.*

ord. 6. 2. $(abc.def)$ all; the substitutions are those of (abc) all each, compounded with the corresponding substitution of (def) all; viz. they are

$$\begin{aligned} 1, & \ ab.def, \ abc.def, \\ & \ ac.df, \ acb.dfe, \\ & \ bc.ef. \end{aligned}$$

ord. 6. 3. $(ad.bf.ce)(abc.def)$ cyc. This is, in fact, the group $(ad.be.cf)(abc.dfe)$ cyc. explained in the introductory paragraphs; only for convenience the letters e, f have been interchanged. Making this interchange, the substitutions of the group $(ad.bf.ce)(abc.def)$ cyc. are

$$\begin{aligned} 1, & \ abc.def, \ ad.bf.ce, \\ & \ acb.dfe; \ ae.bd.cf, \\ & \ af.be.cd. \end{aligned}$$

It may be added that, writing $\beta = ad.bf.ce$ and $\theta = abc.def$, the group is $(1, \beta)(1, \theta, \theta^2)$, where $\beta^2 = 1$, $\theta^3 = 1$, and $\beta\theta = \theta^2\beta$, $\beta\theta^2 = \theta\beta$. If $U = adbecf$, then $U^2 = abc.def = \theta$, and the form is $(1, \beta)(1, U^2, U^4)$: see $(abcdef)_{12}$, *infra*.

ord. 8. 4, 5, 6; the nature of these groups is explained in the introductory paragraphs.
ord. 12. $(abcdef)_{12}$. The substitutions are

$$\begin{aligned} 1, & \ ab.ef, \ ad.bf.ce, \ abc.def, \ adbecf, \\ & \ ac.de, \ af.be.cd, \ acb.dfe, \ afcebd, \\ & \ bc.df, \ ae.bd.cf, \\ & \ ae.bf.cd, \end{aligned}$$

writing $U = adbecf$ and $\beta = ad.bf.ce$, this is

$$(1, \beta)(1, U, U^2, U^3, U^4, U^5),$$

where $U^6 = 1$, $\beta^2 = 1$, $\beta U = U^5\beta$,

$$\beta(1, U, U^2, U^3, U^4, U^5) = (1, U^5, U^4, U^3, U^2, U)\beta.$$

ord. 18. $(ad.be.cf)\{(abc) \text{ cyc. } (def) \text{ cyc.}\}$; this has been explained in the introductory paragraphs. Arranging them in a more convenient order, the substitutions are

$$\begin{aligned} 1, & \ abc, \ abc.def, \ ad.be.cf, \ adcfbe, \\ & \ acb, \ abc.dfe, \ ae.bf.cd, \ adbecf, \\ & \ def, \ acb.def, \ af.bd.ce, \ aebfcd, \\ & \ dfe; \ acb.dfe; \ aecdbf, \\ & \ afbdce, \\ & \ afcebd. \end{aligned}$$

Six letters. Ord. 24. 3. $(\pm abcdef)_{24}$: the substitutions are

$$\begin{array}{ccccc}
 + & + & + & - & - \\
 1, & ac.bd, & abe.cdf, & abcd, & ab.cd.ef, \\
 & ac.ef, & abf.cde, & adcb, & ac.be.df, \\
 & bd.ef, & ade.bfc, & aecf, & ac.bf.de, \\
 & & adf.bec, & afce, & ad.be.cf, \\
 & & aeb.cfd, & bcdf, & ae.bd.cf, \\
 & & aed.bcf, & bfde, & af.bd.ce, \\
 & & afb.ced, & & \\
 & & afd.bce, & &
 \end{array}$$

This includes, as part of itself, the group

$$\{(abcd)_8 \text{ cyc. } (ef)\} \text{ dim.};$$

and it may be written

$$(ade.bcf) \text{ cyc. } \{(abcd)_8 \text{ cyc. } (ef)\} \text{ dim.}$$

24. 4. $(+ abcdef)_{24}$; the substitutions are

$$\begin{array}{llll}
 1, & ab.cd, & abe.cdf, & abcd.ef, \\
 & ac.bd, & abf.cde, & adcb.ef, \\
 & ac.ef, & ade.bfc, & aecf.bd, \\
 & ad.bc, & adf.bec, & afce.bd, \\
 & ae.cf, & aeb.cfd, & bcdf.ac, \\
 & af.ce, & aed.bcf, & bfde.ac, \\
 & bd.ef, & afb.ced, & \\
 & be.df, & afd.bce, & \\
 & bf.de, & &
 \end{array}$$

This includes, as part of itself, the group $\{(abcd)_8(ef)\} \text{ pos.}$; and it may be written $(ade.bfc) \text{ cyc. } \{(abcd)_8(ef)\} \text{ pos.}$ It consists of the positive terms of $(abcdef)_{48}$, and might thus be written $(abcdef)_{48} \text{ pos.}$

ord. 36. $(abcdef)_{36}$; this consists of the positive terms of $(abcdef)_{72}$. The group consists of the substitutions which leave unaltered $b - c.c - a.a - b.e - f.f - d.d - e.$

ord. 48. $(abcdef)_{48}$. The substitutions are*Six letters.*

+	+	+	+	—	—
1,	$ab.cd,$	$abe.cdf,$	$abcd.ef,$	$ac,$	$abcd,$
	$ac.bd,$	$abf.cde,$	$adcb.ef,$	$bd,$	$adcb,$
	$ac.ef,$	$ade.cbf,$	$aecf.bd,$	$ef;$	$aecf,$
	$ad.bc,$	$adf.cbe,$	$afce.bd,$		$afce,$
	$ae.cf,$	$aeb.cfd,$	$bcd.f.ac,$		$bcd.f,$
	$af.ce,$	$aed.cfb,$	$bfde.ac;$		$bfde;$
	$bd.ef,$	$afb.ced,$			$af.ce.bd,$
	$be.df,$	$afd.ceb;$			$afdceb;$
	$bf.de.$				

—	—
$ab.cd.ef,$	$abecfd,$
$ac.bd.ef,$	$abecdf,$
$ac.be.df,$	$abfcde,$
$ac.bf.de,$	$adecbf,$
$ad.be.cf,$	$adfcbe,$
$ae.cf.bd,$	$aedcfb,$
$af.ce.bd;$	$afbced;$

ord. 60. $(abcdef)_{60}$; this consists of the positive terms of $(abcdef)_{120}$.ord. 72. $(abcdef)_{72}$. The substitutions are

+	+	+	+	+
1,	$abc,$	$abc.def,$	$ab.de,$	$ad.bccf,$
	$acb,$	$abc.dfe,$	$ab.df,$	$ad.bfce,$
	$def,$	$acb.def,$	$ab.ef,$	$ae.bdcf,$
	$dfe;$	$acb.dfe;$	$ac.de,$	$ae.bfcd,$
			$ac.df,$	$af.bdce,$
			$ae.cf,$	$af.becd,$
			$bc.de,$	$bd.aecf,$
			$bc.df,$	$bd.afce,$
			$bc.ef;$	$be.adcf,$
				$be.afcd,$
				$bf.adce,$
				$bf.aecd,$
				$cd.aebf,$
				$cd.afbe,$
				$ce.adbf,$
				$ce.afbd,$
				$cf.adbe,$
				$cf.aebd.$

—	—	—	—	—
$ab,$	$ab.def,$	$ad.be.cf,$	$adbecf,$	
$ac,$	$ab.dfe,$	$ad.bf.ce,$	$adbfce,$	
$bc,$	$ac.def,$	$ae.bd.cf,$	$adcebf,$	
$de,$	$ac.dfe,$	$ae.bf.cd,$	$adcfbe,$	
$df,$	$bc.def,$	$af.bd.ce,$	$aebdcf,$	
$ef;$	$bc.dfe,$	$af.be.cd;$	$aebfcd,$	
	$de.abc,$		$aecdbf,$	
	$de.acb,$		$aecfbd,$	
	$df.abc,$		$afbdce,$	
	$df.acb,$		$afbecd,$	
	$ef.abc,$		$afcdbce,$	
	$ef.acb;$		$afcebd;$	

This contains, as part of itself, the group (abc) all (def) all; and it may be written $(ad.be.cf)\{(abc)$ all (def)
It leaves unaltered the function $abc + def$.

Six letters.

ord. 120. $(abcdef)_{120}$. The substitutions are

+	+	+	+	—	—	—
1,	$ab.cf,$	$abcef,$	$abc.dfe,$	$abcd,$	$abcfd,$	$ab.cd.ef,$
	$ab.de,$	$abdce,$	$abd.cfe,$	$abdf,$	$abdefc,$	$ab.ce.df,$
	$ac.bd,$	$abefd,$	$abe.cdf,$	$abec,$	$abedcf,$	$ac.be.df,$
	$ac.ef,$	$abfdc,$	$abf.cde,$	$abfe,$	$abfced,$	$ac.bf.de,$
	$ad.bf,$	$acbed,$	$acb.def,$	$acbf,$	$acbdfe,$	$ad.bc.ef,$
	$ad.ce,$	$acdfe,$	$acd.bfe,$	$acde,$	$acdbef,$	$ad.be.cf,$
	$ae.bc,$	$acedf,$	$ace.bfd,$	$aceb,$	$acefbd,$	$ae.bd.cf,$
	$ae.df,$	$acfbe,$	$acf.bde,$	$acfd,$	$acfedb,$	$ae.bf.cd,$
	$af.be,$	$adbcf,$	$adb.cef,$	$adbe,$	$adbfec,$	$af.bc.de,$
	$af.cd,$	$adbef,$	$adc.bef,$	$adcb,$	$adcebf,$	$af.bd.ce;$
	$bc.df,$	$adcfe,$	$ade.bfc,$	$adef,$	$adecfb,$	
	$bd.ef,$	$adfeb,$	$adf.bec,$	$adfc,$	$adfbce,$	
	$bc.cd,$	$aebdf,$	$aeb.cfd,$	$aebd,$	$aebcdf,$	
	$bf.ce,$	$aebfc,$	$aec.bdf,$	$aecd,$	$aecbfd,$	
	$cf.de;$	$aefcd,$	$aed.bcf,$	$aecf,$	$aedfcb,$	
		$aecdb,$	$aef.bdc,$	$aefb,$	$aefdbc,$	
		$afbde,$	$afb.ced,$	$afbc,$	$afbecd,$	
		$afcdb,$	$afc.bed,$	$afce,$	$afcdeb,$	
		$afdec,$	$afd.bce,$	$afdb,$	$afdcbe,$	
		$afecb,$	$afe.bcd;$	$afed,$	$afebdc;$	
		$bcdef,$		$bcfd,$		
		$bdfce,$		$bcde,$		
		$becfd,$		$bdfe,$		
		$bfedc;$		$becf,$		
				$bfcd,$		
				$bfde,$		
				$cdfe,$		
				$cefd.$		

ON ORTHOMORPHOSIS (continued)

[920, pp. 179–190]

and the two equations give $x^2 + y^2 - 1 = 0$; the two circumferences thus correspond to each other. It is to be noticed that, if the numerator function contain a factor $(x + iy)^m$, say if $\phi(x + iy) = (x + iy)^m \psi(x + iy)$, then the denominator will be

$$(x + iy)^{-m} \bar{\psi} \left(\frac{1}{x + iy} \right),$$

and the formula thus becomes

$$x_1 + iy_1 = \frac{(x + iy)^m \psi(x + iy)}{\frac{1}{(x + iy)^m} \bar{\psi} \left(\frac{1}{x + iy} \right)},$$

which is thus not really more general than the original form. I recur further on to this transformation between the two circumferences.

10. The curve S may be taken to be a closed curve enclosing a singly connected area F (say such a curve is a Contour), and the curve S' a like curve lying wholly inside S , and such that the normal distance between the two curves is everywhere infinitesimal. It is not in general the case, but it may happen that the successive curves S'' , S''' , &c. will be all of them like curves (each enclosed in the preceding and enclosing the succeeding one), of continually diminishing area, and continually approximating to a point: the curves T will then all of them pass through this point and may be called Radials. We may say that the area F is then squarewise contractible, or contracted, into a point.

11. In particular, a circle is squarewise contractible into a point, viz. if the point be the centre, then the curves S are concentric circles, and the curves T are radii; as is known and will appear, the circle is also contractible into any interior eccentric point whatever. It may be remarked that (the squares being infinitesimal) the numbers of the contours and radials are each of them infinite, but further the number of contours is infinitely great in comparison with that of the radials. Thus in the case just referred to, if to construct a figure we imagine the circumference of the circle divided into any large number n of equal parts, the radials will be the n radii drawn to the points of division: taking the radius to be $= 1$, and writing for shortness $\alpha = \frac{2\pi}{n}$, the sides of the successive squares along any radius will be $\alpha, \alpha(1 - \alpha), \alpha(1 - \alpha)^2, \dots$, and we require an infinite number of these to make up the entire radius; $1 = \alpha\{1 + (1 - \alpha) + (1 - \alpha)^2 + \dots\}$; viz. the number of radials being any large number n whatever, the number of contours will be actually infinite.

12. We may compare the circle as thus contracted into its centre, or rather the quadrant of a circle, with an infinite strip of finite breadth divided into squares by two sets of parallel orthotomic lines: here if x_1, y_1 refer to the positive quadrant of the circle $x_1^2 + y_1^2 = 1$, and x, y to the infinite strip $x = 0$ to $-\infty$, $y = 0$ to $\frac{1}{2}\pi$, we have $x + iy = \log(x_1 + iy_1)$, that is,

$$x = \log \sqrt{(x_1^2 + y_1^2)}, \quad y = \tan^{-1} \frac{y_1}{x_1};$$

the successive concentric quadrants correspond to equal lines parallel to the axis of y , and the successive radii to infinite lines parallel to the axis of x .

13. Considering the contour S as given, then for a consecutive interior contour S' assumed at pleasure, the area F does not contract into a point: but we have the theorem that the consecutive contour S' can be found, and that in one way only, such that the area F shall contract into a given interior point. To do this is, in effect, to make the given area to correspond say to the circle $x_1^2 + y_1^2 - 1 = 0$, the contour S and the successive interior contours to the circumference

of the circle and to the circumferences of the successive interior concentric circles, and finally the given interior point of F to the centre $x_1 = 0, y_1 = 0$ of the circle: and thus the theorem is identical with Riemann's theorem "It is possible and that in one way only to make a given singly connected area F correspond to a circle, in such wise that a given interior point of F shall correspond to the centre of the circle, and that a given boundary point of F shall correspond to a given point on the circumference of the circle." The last clause as to the given boundary point of F is necessary in order to make the correspondence a completely definite one, for without the clause it would obviously be allowable to give to the circle an arbitrary rotation about its centre.

14. The proof depends on the following lemma¹.

For the given area F , it is possible to find, and that in one way only, a real function ξ of the coordinates (x, y) satisfying the following conditions:

(1) ξ is throughout the area finite and continuous, except only that in the neighbourhood of a given point thereof, taken to be the point $x = 0, y = 0$, it is $= \log \sqrt{(x^2 + y^2)}$.

(2) At the boundary of the area $\xi = 0$.

(3) Throughout the area ξ satisfies the partial differential equation

$$\frac{d^2\xi}{dx^2} + \frac{d^2\xi}{dy^2} = 0.$$

In fact, if ξ be thus determined, it follows from (3) that we have

$$-\frac{d\xi}{dy} dx + \frac{d\xi}{dx} dy = d\eta,$$

an exact differential, whence putting this $= d\eta$, or determining η by the quadrature

$$\eta = \int \left(-\frac{d\xi}{dy} dx + \frac{d\xi}{dx} dy \right),$$

we have

$$\frac{d\eta}{dx} = -\frac{d\xi}{dy}, \quad \frac{d\eta}{dy} = \frac{d\xi}{dx},$$

and thence $\xi + i\eta$ is a function of $x + iy$.

Writing then $x_1 + iy_1 = e^{\xi + i\eta} = \psi(x + iy)$ suppose, we have an orthomorphosis of the area into the circle $x_1^2 + y_1^2 - 1 = 0$, viz. for any point (x, y) of the boundary $\xi = 0$, and consequently $x_1 + iy_1 = e^{i\eta}$, thence $x_1 - iy_1 = e^{-i\eta}$, and therefore $x_1^2 + y_1^2 - 1 = 0$, or the point corresponds to a point on the circumference of the circle. Also, for a point x, y in the neighbourhood of $x = 0, y = 0$, we have

$$x_1 + iy_1 = e^{\log \sqrt{(x^2 + y^2)} + i\eta} = \sqrt{(x^2 + y^2)} e^{i\eta},$$

that is, for $x = 0, y = 0$ we have $x_1 = 0, y_1 = 0$, or the given point of F corresponds to the centre of the circle.

Some further considerations as to the continuity of η would be requisite in order to show that to the series of circles $x_1^2 + y_1^2 = c^2$, as c continuously increases from 0 to 1, there correspond closed curves surrounding the assumed origin and each other, and passing continuously to the boundary of F , and thus to complete the proof that we thus obtain an orthomorphosis of F into the circle $x_1^2 + y_1^2 = 1$; but I abstain from a further discussion of the question.

15. Reverting to the lemma, this in the first place asserts the existence of a function ξ satisfying the prescribed conditions, and next that the function is completely determinate, or say that there is but one such function. As to the latter point, suppose that there is a second function ξ_1 satisfying the same conditions, then throughout the area $\xi - \xi_1$ is finite and continuous; by general principles in the theory of functions, this implies that the function has throughout

¹This proof is taken from the paper, Christoffel, "Sul problema delle temperature stazionarie et la rappresentazione di una data superficie," *Annali di Matem.*, t. i. (1867), pp. 89—103.

the area a constant value, and since this value at the boundary is $= 0$, it must be always $= 0$, that is, $\xi - \xi_1 = 0$, or $\xi_1 = \xi$. As to the former point, it is to be remarked that we obtain

$$\frac{d^2\xi}{dx^2} + \frac{d^2\xi}{dy^2} = 0$$

as the general condition for a minimum value of the double integral

$$\iint \left\{ \left(\frac{d\xi}{dx} \right)^2 + \left(\frac{d\xi}{dy} \right)^2 \right\} dx dy;$$

if then we assume that there exists a function ξ , $= 0$ at the boundary of the area, and finite and continuous throughout the area, except only that in the neighbourhood of a given point thereof, say $x = 0$, $y = 0$, it becomes $= \log \sqrt{(x^2 + y^2)}$, and that for all the values which satisfy these conditions it is such that the above-mentioned double integral taken over the given area shall be a minimum, it follows that there exists a function ξ satisfying the conditions of the lemma.

16. As a simple illustration, suppose that the area F is that of the circle $x^2 + y^2 = 1$, and that the excepted point is the point $x = 0$, $y = 0$, the centre of the circle. We have here a function ξ satisfying the conditions of the lemma, viz. this is $\xi = \log \sqrt{(x^2 + y^2)}$; the resulting value of η is $\eta = \tan^{-1} \frac{y}{x}$, and we have then $\xi + i\eta$ a function of $x + iy$, viz. this is $= \log(x + iy)$. Hence $\xi + i\eta = x + iy$, and the resulting orthomorphosis of the two circles is the identical one

$$x_1 + iy_1 = x + iy,$$

viz. we have

$$x_1 = x, \quad y_1 = y.$$

17. The contraction of a circle to a given eccentric point has, in fact, been exhibited in the foregoing example in No. 5, but I resume the question from a different point of view. Writing for shortness $z_1 = x_1 + iy_1$, $z = x + iy$; using also a bar for the conjugate function, $\bar{z} = x - iy$, and so in other cases $\alpha = a + a'i$, $\bar{\alpha} = a - a'i$, $\beta = b + b'i$, &c., the general formula of (7) may be written in the form

$$z_1 = \frac{z - \alpha}{1 - \bar{\alpha}z} \cdot \frac{z - \beta}{1 - \bar{\beta}z} \cdots,$$

viz. we thus have a transformation between the circumferences $x^2 + y^2 - 1 = 0$, $x_1^2 + y_1^2 - 1 = 0$. In fact, repeating the verification, the change of sign of i gives

$$\bar{z}_1 = \frac{\bar{z} - \bar{\alpha}}{1 - \alpha\bar{z}} \cdot \frac{\bar{z} - \bar{\beta}}{1 - \beta\bar{z}} \cdots$$

Here the product of the α -factors is

$$\frac{z\bar{z} - (\alpha\bar{z} + \bar{\alpha}z) + \alpha\bar{\alpha}}{1 - (\alpha z + \bar{\alpha}\bar{z}) + \alpha\bar{\alpha}z\bar{z}},$$

which, if $z\bar{z} = 1$, becomes $= 1$; the same property exists as to the β -factors, &c., and hence putting $z\bar{z} = 1$, we have $z_1\bar{z}_1 = 1$, that is, the two circumferences correspond to each other. The correspondence would have subsisted if a factor $e^{i\kappa}$ had been introduced into the expression of z_1 , but the effect is merely to rotate the circle $x_1^2 + y_1^2 - 1 = 0$ about its centre, and there is no real gain of generality.

18. The most simple case is

$$z_1 = \frac{z - \alpha}{1 - \bar{\alpha}z}.$$

I assume here that $\alpha = a + a'i$ is an interior point of the circle $x^2 + y^2 - 1 = 0$ (viz. $a^2 + a'^2 < 1$); hence $z = \alpha$ gives $z_1 = 0$, viz. to the point $x = a$, $y = a'$, within the circle $x^2 + y^2 - 1 = 0$, there

corresponds the centre of the circle $x_1^2 + y_1^2 - 1 = 0$; and we have in this equation the theory of the squarewise contraction of the circle to the eccentric point $x = a, y = a'$. There is no loss of generality in assuming $a' = 0$; doing this the formula becomes

$$z_1 = \frac{z - a}{1 - az},$$

where $a^2 < 1$, or if to fix the ideas a be taken to be positive, then $a < 1$.

19. To compare with a former investigation, writing $\xi + i\eta = \log(x_1 + iy_1)$, we have

$$\xi + i\eta = \log \sqrt{((x - a)^2 + y^2)} + i \tan^{-1} \frac{y - ay}{x - a + a(x^2 + y^2)},$$

and consequently

$$\xi = \frac{1}{2} \log \frac{(x - a)^2 + y^2}{(1 - ax)^2 + a^2 y^2},$$

which is $= 0$ at the boundary $x^2 + y^2 - 1 = 0$, and is finite and continuous throughout the area except in the neighbourhood of the point $x = a, y = 0$, for which it is

$= \log \sqrt{((x - a)^2 + y^2)}$; moreover, ξ satisfies the partial differential $\frac{d^2 \xi}{dx^2} + \frac{d^2 \xi}{dy^2} = 0$, and starting from the given value of ξ we should obtain the foregoing value of η , and thence $x_1 + iy_1 = e^{\xi + i\eta}$, that is,

$$x_1 + iy_1 = \frac{z - a}{1 - az}, \quad \text{or} \quad z_1 = \frac{z - a}{1 - az},$$

as the equation for the orthomorphosis of the circle $x^2 + y^2 - 1 = 0$ into the circle $x_1^2 + y_1^2 - 1 = 0$, the centre $x_1 = 0, y_1 = 0$ corresponding to the eccentric point $x = a, y = 0$.

20. In further development of the solution, writing

$$z_1 \bar{z}_1 = \frac{z \bar{z} - a(z + \bar{z}) + a^2}{1 - a(z + \bar{z}) + a^2 z \bar{z}},$$

that is,

$$x_1^2 + y_1^2 = \frac{x^2 + y^2 - 2ax + a^2}{a^2(x^2 + y^2) - 2ax + 1},$$

and then assuming $x_1^2 + y_1^2 = \lambda^2$, we have as the contour corresponding thereto

$$x^2 + y^2 + a^2 - 2ax = \lambda^2 \{a^2(x^2 + y^2) - 2ax + 1\},$$

that is,

$$(x^2 + y^2)(1 - a^2 \lambda^2) - 2a(1 - \lambda^2)x + a^2 - \lambda^2 = 0,$$

or say

$$x^2 + y^2 - \frac{2a(1 - \lambda^2)}{1 - a^2 \lambda^2} x + \frac{a^2 - \lambda^2}{1 - a^2 \lambda^2} = 0,$$

or, what is the same thing,

$$\left(x - \frac{a(1 - \lambda^2)}{1 - a^2 \lambda^2}\right)^2 + y^2 = \frac{\lambda^2(1 - a^2)^2}{(1 - a^2 \lambda^2)^2}.$$

The equation may also be written

$$\left\{x - \frac{a(1 - \lambda^2)}{1 - a^2 \lambda^2}\right\}^2 + y^2 = \frac{\lambda^2(1 - a^2)^2}{(1 - a^2 \lambda^2)^2},$$

showing that the contour circles all pass through two imaginary points

$$x = \frac{1}{a} \left(1 - \frac{1}{a}\right), \quad y = \pm \frac{i}{a} \left(\frac{1}{a} - a\right).$$

21. Writing next

$$z_1 = \frac{(z-a)(1-a\bar{z})}{(1-az)(1-a\bar{z})} = \frac{(z-a)(1-a\bar{z})}{|1-az|^2},$$

that is,

$$x_1 = \frac{-a + x(1+a^2) - a(x^2+y^2)}{(1-ax)^2 + a^2y^2}, \quad y_1 = \frac{y(1-a^2)}{(1-ax)^2 + a^2y^2},$$

then to the radius $x_1 - \theta y_1 = 0$ corresponds the radial

$$-a + x(1+a^2) - a(x^2+y^2) - \theta y(1-a^2) = 0,$$

that is,

$$(x-a) \left(x - \frac{1}{a}\right) + y^2 + \theta y \left(\frac{1}{a} - a\right) = 0,$$

which is a circle passing through the two real points $y = 0$, $x = a$, or $\frac{1}{a}$, being the antipoints of the before-mentioned pair of points.

22. Hence for the contraction of the circle $x^2 + y^2 - 1 = 0$ to the interior point $x = a$, $y = 0$, calling this point A , and taking A' the image hereof (coordinates $x = \frac{1}{a}$, $y = 0$), we see that the contour circles are the circles all passing through the antipoints of A , A' , and having thus for their common chord the line bisecting AA' at right angles, and that the radial circles are the circles all passing through the two points A , A' .

In what precedes, the distance AA' is $= \frac{1}{a} - a$, or say the half distance is $= \frac{1}{2} \left(\frac{1}{a} - a\right)$; but altering the scale so as to make this half distance $= 1$, and moreover taking the origin at the middle point of AA' , the equations of the contour circles and the radial circles assume the forms

$$x^2 + y^2 - 2\alpha x + 1 = 0,$$

and

$$x^2 + y^2 - 2\beta y - 1 = 0,$$

respectively.

23. It is better, interchanging x , y and altering the constants, to write these in the forms

$$x^2 + y^2 - \left(b + \frac{1}{b}\right)y + 1 = 0,$$

$$x^2 + y^2 - \left(a - \frac{1}{a}\right)x - 1 = 0,$$

viz. as in effect appearing in No. 5, these are the equations derived from $x + iy = \tan(u + iv)$ where $\tan u = a$, $\tan iv = ib$.

24. Passing now to the form

$$z_1 = \frac{z - \alpha}{1 - \bar{\alpha}z} \cdot \frac{z - \beta}{1 - \bar{\beta}z},$$

to obtain the most simple instance, I write $\alpha = 0$, $\beta = \sqrt{2}$, so that the form is

$$z_1 = z \cdot \frac{z - \sqrt{2}}{1 - \sqrt{2}z};$$

we thus obtain

that is,

$$x_1 + iy_1 = \frac{(x^2 + y^2)(x^2 + y^2 - 2\sqrt{2}x + 2)}{2(x^2 + y^2) - 2x\sqrt{2} + 1},$$

and thence

$$x_1^2 + y_1^2 = \frac{(x^2 + y^2)\{x^2 + y^2 - 2\sqrt{2}x + 2\}}{2(x^2 + y^2) - 2x\sqrt{2} + 1},$$

so that the circumference $x_1^2 + y_1^2 - 1 = 0$ corresponds to the two circumferences $x^2 + y^2 - 1 = 0$, $x^2 + y^2 - 2\sqrt{2}x + 1 = 0$. The second of these is $(x - \sqrt{2})^2 + y^2 - 1 = 0$; hence, putting $x + \frac{1}{\sqrt{2}}$ instead of x , the two circles are

$$\left(x + \frac{1}{\sqrt{2}}\right)^2 + y^2 - 1 = 0, \quad \left(x - \frac{1}{\sqrt{2}}\right)^2 + y^2 - 1 = 0,$$

(the former of these being the circle originally represented by $x^2 + y^2 - 1 = 0$), and the last equation becomes

$$x_1^2 + y_1^2 - 1 = \frac{\left\{\left(x + \frac{1}{\sqrt{2}}\right)^2 + y^2 - 1\right\} \left\{\left(x - \frac{1}{\sqrt{2}}\right)^2 + y^2 - 1\right\}}{2(x^2 + y^2)}.$$

25. Writing here $x_1^2 + y_1^2 - c^2 = 0$, we have

$$\left\{\left(x + \frac{1}{\sqrt{2}}\right)^2 + y^2 - 1\right\} \left\{\left(x - \frac{1}{\sqrt{2}}\right)^2 + y^2 - 1\right\} = 2c^2(x^2 + y^2)(1 - c^2) = 0,$$

that is,

$$(x^2 + y^2)^2 - (2c^2 + 1)x^2 - (2c^2 - 1)y^2 + \frac{1}{4} = 0,$$

a bicircular quartic which (or rather a branch whereof) is the contour corresponding to the circle $x_1^2 + y_1^2 - c^2 = 0$. For $c = 1$, the curve breaks up into the two circles

$$\left(x + \frac{1}{\sqrt{2}}\right)^2 + y^2 - 1 = 0, \quad \left(x - \frac{1}{\sqrt{2}}\right)^2 + y^2 - 1 = 0,$$

and for $c = 0$, it breaks up into

$$\left(x + \frac{1}{\sqrt{2}}\right)^2 + y^2 = 0, \quad \left(x - \frac{1}{\sqrt{2}}\right)^2 + y^2 = 0,$$

that is, into the two points $y = 0$, $x = \pm \frac{1}{\sqrt{2}}$. We have to consider the curves for values of c between these two values $c = 1$ and $c = 0$.

26. For any such value of c , the curve consists of two symmetrical ovals, situate within the two circles respectively: we are concerned only with the oval lying within the circle $\left(x + \frac{1}{\sqrt{2}}\right)^2 + y^2 - 1 = 0$; this is shown in the figure for a value of c slightly inferior to 1, viz. it is an indented oval close approximating to the unshaded portion of the circle, say this is the contour S' consecutive to the circle which is the contour S .

As c diminishes and becomes ultimately $= 0$, this oval continually diminishes,

after a time losing the indentation, and approximating more and more to the form of a circle having for its centre the point $x = -\frac{1}{\sqrt{2}}$, $y = 0$, which is the centre of the circle, and ultimately reducing itself to this point.

It will be noticed that the consecutive contour S' as shown in the figure is not at an infinitesimal distance from every part of the circle which is the contour S ; for a part of the circle, the consecutive contour or contour at an infinitesimal distance is a part of the symmetrically situated oval, lying outside of the circle. And thus in the present case we do not obtain a contraction of the circle into its centre: what we do obtain is a contraction of the unshaded portion of the circle into the centre; this is as it should be, for the only contraction of the circle into its centre is by the series of concentric circles.

27. For the radials, writing as before $x + \frac{1}{\sqrt{2}}$ in place of x , that is, $z + \frac{1}{\sqrt{2}}$ in place of z , we have

$$z_1 = z + \frac{1}{\sqrt{2}} \cdot \frac{\left(z + \frac{1}{\sqrt{2}}\right)\left(z - \frac{1}{\sqrt{2}}\right)}{-z\sqrt{2}} = -\frac{\left(z + \frac{1}{\sqrt{2}}\right)\left(z^2 - \frac{1}{2}\right)}{z\sqrt{2}}.$$

Hence

$$x_1 = -x - \frac{\frac{1}{2}x}{x^2 + y^2} \cdot \frac{x(x^2 + y^2 - \frac{1}{2})}{x^2 + y^2}, \quad y_1 = -y - \frac{\frac{1}{2}y}{x^2 + y^2} \cdot \frac{y(x^2 + y^2 + \frac{1}{2})}{x^2 + y^2},$$

and thus to the radius $x_1 - \theta y_1 = 0$ there corresponds the curve

$$x(x^2 + y^2 - \frac{1}{2}) - \theta y(x^2 + y^2 + \frac{1}{2}) = 0,$$

or say

$$(x - \theta y)(x^2 + y^2) - \frac{1}{2}(x + \theta y) = 0,$$

a circular cubic. In particular, for the line $x_1 = 0$, we have $\theta = 0$, and the curve is $x(x^2 + y^2 - \frac{1}{2}) = 0$, or say $x^2 + y^2 - \frac{1}{2} = 0$, the half circumference of which is shown in the figure: for the line $y_1 = 0$, we have $\theta = \infty$, and the curve is $y(x^2 + y^2 + \frac{1}{2}) = 0$, that is, $y = 0$.

The curve passes through the origin $x = 0, y = 0$, and the equation of the

tangent there is $x + \theta y = 0$. Moreover it passes through the centre of the circle, $x = -\frac{1}{\sqrt{2}}, y = 0$, and the equation of the tangent there is easily found to be

$$x + \frac{1}{\sqrt{2}} - \theta y = 0.$$

We may find where the curve cuts the circle $\left(x + \frac{1}{\sqrt{2}}\right)^2 + y^2 - 1 = 0$, viz. this is $x^2 + y^2 = -x\sqrt{2} + \frac{1}{2}$, and substituting in the equation of the curve, we find

$$x^2 - \theta xy + \frac{\theta}{\sqrt{2}}y = 0,$$

or say

$$\left(x - \frac{1}{\sqrt{2}}\theta\right)\left(x - \theta y + \frac{1}{\sqrt{2}}\right) + \frac{1}{2} = 0,$$

a hyperbola which by its intersections with the circle determines the points in question. There are two real intersections, but of these only one is in the arc bounding the unshaded portion of the circle: the other intersection serves to determine by symmetry the intersection of the circular cubic with the other boundary of the unshaded portion: see the figure which exhibits the path of the circular cubic through the unshaded portion, and into and beyond the shaded portion or lens. I remark that the two systems of curves are considered by Meyer and shown in his Plate XIII.

28. The formula

$$z_1 = \frac{z - \alpha}{1 - \bar{\alpha}z}$$

has been considered for the case of an interior point α ; the case of an exterior point α might be considered in like manner, but it is obvious that we pass from one to the other, by a transformation $\frac{1}{z_1}$ for z_1 ; and it thus easily appears that, if α be an exterior point, we obtain a transformation between the area of the circle $x_1^2 + y_1^2 - 1 = 0$ and the infinite area exterior to the circle $x^2 + y^2 - 1 = 0$.

Similarly, in the formula

$$z_1 = \frac{z - \alpha}{1 - \bar{\alpha}z} \cdot \frac{z - \beta}{1 - \bar{\beta}z},$$

the case considered ($\alpha = a = 0$, $\beta = b = \sqrt{2}$) has been that of an interior point α , and an exterior point β . I omit the case of two exterior points, since this can be by the transformation $\frac{1}{z_1}$ for z_1 reduced to that of two interior points, but it is proper to consider this last case.

29. Considering then the case

$$z_1 = \frac{z - \alpha}{1 - \bar{\alpha}z} \cdot \frac{z - \beta}{1 - \bar{\beta}z},$$

where α, β are interior points, the most simple case is when $\alpha = a$, $\beta = -a$, positive and < 1 . Here

$$z_1 = \frac{z^2 - a^2}{1 - a^2 z^2},$$

and we have

$$x_1 + iy_1 = \frac{(x^2 - y^2)(x^2 + y^2) - 2a^2(x^2 - y^2) + a^4}{1 - 2a^2(x^2 - y^2) + a^4(x^2 + y^2)^2},$$

or say

$$x_1^2 + y_1^2 - 1 = \frac{(1 - a^4)\{(x^2 + y^2)^2 - 1\}}{1 - 2a^2(x^2 - y^2) + a^4(x^2 + y^2)^2},$$

which last form shows that to the circumference $x^2 + y^2 - 1 = 0$, there corresponds the circumference $x_1^2 + y_1^2 - 1 = 0$ (and besides this, only the imaginary curve $x^2 + y^2 + 1 = 0$). Writing $x_1^2 + y_1^2 = c^2$, we have the contour

$$(c^2 a^4 - 1)(x^2 + y^2)^2 - 2a^2(c^2 - 1)(x^2 - y^2) + c^2 - a^4 = 0,$$

a bicircular quartic. For $c = 1$, we have as already mentioned the circle $x^2 + y^2 - 1 = 0$; for c less than 1, a curve lying within this circle, diminishing with c , and after a time acquiring on each side of the axis of x an indentation or assuming an hour-glass form; for the value $c^2 = a^4$, the equation becomes

$$(a^4 + 1)(x^2 + y^2)^2 - 2a^2(x^2 - y^2) = 0,$$

and the curve is a figure of eight, the two loops enclosing the points $y = 0$, $x = a$ and $x = -a$ respectively; and as c^2 diminishes to zero, the curve consists of two detached ovals lying within the two loops of the figure of eight, and ultimately reducing themselves to the two points respectively. There is no difficulty in finding the equation of the curves corresponding to a radius $x_1 - \theta y_1 = 0$, but the configuration of these curves is at once seen from that of the former system. We may in the present case say that the circle is squarewise contracted to the figure of eight; and then further that each loop of the figure of eight is squarewise contracted to a point; but we do not have a squarewise contraction of the circle to a point.

30. A closed curve or contour may be squarewise contracted not into a point but into a finite line: we see this in the case of a system of confocal ellipses, which gives the contraction of an ellipse into the thin ellipse which is the finite line joining the two foci. There is a like contraction of the circle; this is, in fact, given by the formula due to Schwarz, "Ueber einige

Abbildungsaufgaben," *Crelle*, t. LXX. (1869), pp. 105—120 (see p. 115) for the orthomorphosis of the ellipse into a circle: this is

$$x_1 + iy_1 = \operatorname{sn} \frac{2K}{\pi}(X + iY),$$

where, if $a^2 - b^2 = 1$ and $a = \cos \frac{i\pi K'}{4K}$, or, what is the same thing, $ib = \sin \frac{i\pi K'}{4K}$, then the ellipse $\frac{x^2}{a^2} + \frac{y^2}{a^2 - 1} = 1$ is transformed into the circle $x_1^2 + y_1^2 - \frac{1}{k} = 0$. The contours and trajectories for the ellipse are the confocal ellipses and hyperbolas respectively;

and for the circle, they are two sets of bicircular quartics, such that the portions within the circle have a configuration resembling that of the confocal ellipses and hyperbolas within the ellipse. To investigate the formulae, it is convenient to introduce the function

$$X + iY = \sin^{-1}(x + iy);$$

we then have

$$x + iy = \sin(X + iY),$$

or say

$$x = \sin X \cos iY, \quad y = \cos X \sin iY,$$

so that to any given value of Y there corresponds the ellipse

$$\frac{x^2}{\cos^2 iY} + \frac{y^2}{-\sin^2 iY} = 1,$$

and then

$$x_1 + iy_1 = \operatorname{sn} \frac{2K}{\pi}(X + iY),$$

or if

$$\begin{aligned} s &= \operatorname{sn} \frac{2KX}{\pi}, & s_1 &= \operatorname{sn} \frac{2KiY}{\pi} = \operatorname{sn} \frac{2KX}{\pi} \sqrt{(1 - s^2)}, \\ c &= \operatorname{cn} \frac{2KX}{\pi} = \sqrt{(1 - s^2)}, & c_1 &= \operatorname{cn} \frac{2KiY}{\pi} = \sqrt{(1 + s_1^2)}, \\ d &= \operatorname{dn} \frac{2KX}{\pi} = \sqrt{(1 - k^2 s^2)}, & d_1 &= \operatorname{dn} \frac{2KiY}{\pi} = \sqrt{(1 + k^2 s_1^2)}, \end{aligned}$$

then

$$x_1 = \frac{sc_1 d_1}{1 + k^2 s^2 s_1^2}, \quad y_1 = \frac{s_1 c d}{1 + k^2 s^2 s_1^2},$$

giving

$$x_1^2 + y_1^2 = \frac{s^2 + s_1^2}{1 + k^2 s^2 s_1^2},$$

and therefore $x_1^2 + y_1^2 - \frac{1}{k} = 0$ if $1 - ks_1^2 = 0$, that is, if $s_1 = \frac{1}{\sqrt{k}}$, or since $s_1 = \operatorname{sn} \frac{2KiY}{\pi}$, $\frac{2KiY}{\pi} = \frac{1}{2}iK'$, or $Y = \frac{\pi K'}{4K}$; hence defining a as above, it appears that the elliptic periphery $\frac{x^2}{a^2} + \frac{y^2}{a^2 - 1} = 1$ corresponds to the circumference $x_1^2 + y_1^2 - \frac{1}{k} = 0$. By the introduction of $X + iY$ as above, the circle and the ellipse are each compared with a rectangle; the reduction of the circle to the rectangle, as given by the foregoing equation $x_1 + iy_1 = \operatorname{sn} \frac{2K}{\pi}(X + iY)$, or what is substantially the same thing by an equation $x_1 + iy_1 = \operatorname{sn}(X + iY)$, is more fully discussed in my paper, Cayley: "On the Binodal Quartic and the graphical representation of the

Elliptic Functions," *Camb. Phil. Trans.*, t. XIV. (1889), pp. 484—494, [891], and in a paper "On some problems of orthomorphosis," *Crelle*, t. CVII. (1891), pp. 262—277, [921].

31. The whole theory, and in particular Riemann's theorem before referred to, are intimately connected with Cauchy's theorem, "If a function $f(z)$ is holomorphic over a simply connected plane area, and if t denote any point within the area, then

$$f(t) = \frac{1}{2\pi i} \int \frac{f(z)}{z - t} dz,$$

where z denotes $x + iy$, and the integral is taken in the positive sense along the boundary of the area." See Briot and Bouquet, *Théorie des fonctions elliptiques* (Paris, 1875), p. 136.

Here in order to obtain by means of the theorem the value of the function $f(z)$ for a given point $t (= a + ib)$ within the area, we require to know the values of $f(z)$ for the several points of the boundary: viz. if z refers to a point P on the boundary, and if we represent the value $f(z)$ by a point P_1 in a second figure, then these points P_1 form a closed curve or boundary in this second figure, and we require to know not only the form of this boundary, but also the several points P_1 thereof which correspond to the several points P of the first-mentioned boundary, or say we require to know the correspondence of the two boundaries: this being known, we have by the theorem the value of $f(t)$, that is, the point $a_1 + ib_1$ within the second area, which corresponds to the point $t = a + ib$ within the first area. The (1, 1) correspondence of the two areas is of course implied in the assertion that $f(t)$ has a determinate value, determined by means of the given values of $f(z)$ along the boundary.

ON SOME PROBLEMS OF ORTHOMORPHOSIS

[921]

From Crelle's Journal der Mathem., t. cvii. (1891), pp. 262—277.

In the interesting Memoir, Schwarz, "Ueber einige Abbildungsaufgaben," *Crelle*, t. LXX. (1869), pp. 105—120, the author considers the orthomorphic transformation (or, as I call it, the orthomorphosis) of a square into the infinite half-plane, or into a circle, and of a rectangle into the infinite half-plane. It is of course easy to deduce the orthomorphosis of the rectangle into a circle; and then, by giving a proper value to the modulus of the elliptic function involved in the formula, we obtain the orthomorphosis of the square into a circle, this solution (although equivalent thereto) being under a different form from that previously given for the square. But as appears from my paper "On the Binodal Quartic and the Graphical Representation of the Elliptic Functions," *Camb. Phil. Trans.*, vol. XIV. (1889), pp. 484—494, [891], there is for the rectangle (and consequently also for the square) an orthomorphosis wherein the boundary of the rectangle or square corresponds, not to the circumference, but to the circumference together with two twice-repeated portions of a diameter. I propose to consider in I. these several transformations: II. relates to the orthomorphosis of a circle into a circle.

I.

1. We are concerned with the elliptic function sn for which $K' = 2K$, and also with the elliptic function snl of the lemniscate form. The modulus in the former case is

$$k^2 = \frac{\sqrt{2}-1}{\sqrt{2}+1} = (\sqrt{2}-1)^2 = 3-2\sqrt{2}, \quad \text{or say } \sqrt{k} = \sqrt{2}-1, \quad \frac{1}{\sqrt{k}} = \sqrt{2}+1.$$

For the lemniscate function snl , the modulus is $= i$, and if (with Gauss) we write $\frac{1}{2}\varpi$ for the value of the complete function K or F_1 , then we have $\frac{1}{2}\varpi = K(\sqrt{2}-1)$,

$= K\sqrt{k}$, where k has the foregoing special value: I notice the numerical values $\frac{1}{2}\varpi = 1.311028$, $k = 3-2\sqrt{2} = \sin 9^\circ 52'$, $\frac{1}{\sqrt{k}} = 1.553774$. The relation between the lemniscate function snl , and the sn with the foregoing value of k , is easily shown to be

$$\sqrt{k} \text{sn} U = \frac{(i+1) \text{snl} u}{(i-1) \text{snl} u + \sqrt{2}}, \quad \text{where } U = \frac{1}{2}(K' - iK) + i\sqrt{2} \frac{u}{\sqrt{k}}, \quad u = \frac{1+i}{\sqrt{2}} U,$$

and it may be added that we have

$$\begin{aligned} \text{cn} U &= \frac{\sqrt{2}}{(i-1) \text{snl} u + \sqrt{2}} \sqrt{\frac{1+k}{k}} \sqrt{(1+\text{snl} u)(1-i \text{snl} u)}, \\ \text{dn} U &= \frac{\sqrt{2}}{(i-1) \text{snl} u + \sqrt{2}} \sqrt{1+k} \sqrt{(1-\text{snl} u)(1+i \text{snl} u)}. \end{aligned}$$

2. The Schwarzian orthomorphosis of the rectangle into the infinite half-plane is given (Memoir, p. 113) by the formula $X_1 + iY_1 = \text{sn}(X + iY)$, where the modulus is real, positive, and less than unity. Here, see the figures 1 (XY) and 2 (X_1Y_1),

Fig. 1 (XY).

Fig. 2 (X_1Y_1).

the rectangle $ABCD$, the sides of which are $AB = 2K$ and $BC = K'$, is transformed into the upper infinite half-plane ($Y_1 = +$), the four corners of the rectangle corresponding to the points A, B, C, D on the axis of X_1 , where $OB(=OA) = 1$, $OC(=OD) = \frac{1}{k}$.

3. We can, by a properly determined quasi-inversion (as will be explained), transform the X_1Y_1 -figure into a new figure see figure 3 (X_2Y_2), the infinite X_1 -axis being transformed into the circumference of a circle (the radius of which may be taken to be $= 1$) and the infinite half-plane into the area within the circle. The four points A, B, C, D are thus transformed into points on the circle, which if the quasi-inversion be a symmetrical one, will be situate, A and B symmetrically, and

also C and D symmetrically, in regard to the axis OY_2 : and in the case of the before-mentioned modulus $k = 3 - 2\sqrt{2}$, for which the rectangle $ABCD$ becomes a

Fig. 3 (X_2Y_2).*

square, the quasi-inversion may be so determined that the points A, B, C, D shall be situate midway (that is, at inclinations $\pm 45^\circ$, $\pm 135^\circ$) in the four quadrants of the circle.

4. But if in figure 1 we take $EL = \frac{1}{2}K'$ and draw FL parallel to OX , then as shown in my memoir above referred to, the foregoing transformation $X_1 + iY_1 = \text{sn}(X + iY)$ changes the rectangle $OBLF$, the sides of which are $OB = K$ and $BL = \frac{1}{2}K'$, into the quadrant $OBLF$ of figure 2, $OB = 1$ (as already mentioned) and radius $OL = \frac{1}{\sqrt{k}}$. Hence completing the rectangle $RSLM$, the sides of which are $RS = 2K$ and $SL = K'$, (viz. this rectangle differs only in position from the first-mentioned rectangle $ABCD$), we have an orthomorphosis of the rectangle $RSLM$ into the circle of figure 2: but so nevertheless that to the boundary of the rectangle there corresponds not the circumference of the circle but the boundary $RGSBLFMAR$ composed of the circumference and the portions SB, BL and MA, AR (that is, BL, AM each twice) of a diameter of the circle. See post No. 12.

5. The Schwarzian orthomorphosis of the square into a circle is given (Memoir, pp. 111—113) by the formula $x_1 + iy_1 = \text{snl}(x + iy)$; viz. here—see figures 4 (xy) and 5 (x_1y_1)—the square $ABCD$, the half-diagonals whereof are each $= \frac{1}{2}\varpi$, corresponds to the circle radius $= 1$ of figure 5, the four points A, B, C, D of the circle being the quadrantal points as shown in the figure. Figure 5 is, in fact, figure 3 turned through an angle of 45° ; and it thus appears that Schwarz's lemniscate solution for the square into a circle is the quasi-inversion of his solution for the rectangle into the infinite half-plane, when by putting $k = 3 - 2\sqrt{2}$ the rectangle is made to be a square. See post Nos. 13 and 14.

6. The general formula of quasi-inversion whereby the infinite X_1 -axis is changed

Fig. 4 (xy). Fig. 5 (x_1y_1).*

into a circumference², radius unity, is

$$X_2 + iY_2 = \frac{1 + Mi(X_1 + iY_1)}{M(X_1 + iY_1) + i},$$

where M is real. In fact, writing this equation in the form

$$X_2 + iY_2 = \frac{1 + Mi(X_1 + iY_1)}{i\{1 + Mi(X_1 - iY_1)\}^{-1} \cdot M(X_1 + iY_1) + i},$$

we have

$$X_2 - iY_2 = \frac{1 - Mi(X_1 - iY_1)}{-i\{1 + Mi(X_1 - iY_1)\}},$$

¹* The scale of this figure is double that of figures 1, 2, 4, 6, 7 in this paper.

²I use here and elsewhere the term circumference rather than circle, to mark more clearly the distinction between the curve and the included area.

and thence $X_2^2 + Y_2^2 - 1 = 0$, if only $Y_1 = 0$; that is, the infinite X_1 -axis is transformed into the circumference $X_2^2 + Y_2^2 - 1 = 0$.

Writing $Y_1 = 0$, we have

$$X_2 + iY_2 = \frac{1 + MiX_1}{MX_1 + i} = \frac{2MX_1 + i(M^2X_1^2 - 1)}{M^2X_1^2 + 1},$$

so that to a pair of points $(\pm X_1, 0)$ there corresponds a pair of points $(\mp X_2, Y_2)$ situate symmetrically in regard to the axis OY_2 .

The equation gives

$$M(X_1 + iY_1) = \frac{1 - i(X_2 + iY_2)}{-i\{1 + i(X_2 - iY_2)\}^{-1}},$$

Hence, writing this in the form

$$M(X_1 + iY_1) = \frac{1 - i(X_2 + iY_2)}{i\{1 + i(X_2 + iY_2)\}},$$

we have

$$M(X_1 - iY_1) = \frac{1 + i(X_2 - iY_2)}{i\{1 - i(X_2 - iY_2)\}},$$

and consequently $M^2(X_1^2 + Y_1^2) - \frac{1}{M^2} = 0$, if only $Y_2 = 0$; that is, the circumference $X_1^2 + Y_1^2 - \frac{1}{M^2} = 0$ is transformed into the infinite X_2 -axis.

Although for the purposes of the present memoir we require the coefficient M , yet there is no real loss of generality in assuming $M = 1$, and the transformation is best studied under the form

$$X_2 + iY_2 = \frac{1 + i(X_1 + iY_1)}{(X_1 + iY_1) + i};$$

see post No. 11.

7. As already mentioned, the coordinates (X_1, Y_1) and (X_2, Y_2) are connected by an equation of the foregoing form, and in the case of the square ($k = 3 - 2\sqrt{2}$ as before), the corresponding values for the points A, B, C, D should be

$$\begin{aligned} A, \quad X_1 + iY_1 &= -1, & X_2 + iY_2 &= \frac{-1 - i}{\sqrt{2}}, \\ B, \quad X_1 + iY_1 &= 1, & X_2 + iY_2 &= \frac{1 - i}{\sqrt{2}}, \\ C, \quad X_1 + iY_1 &= \frac{1}{k}, & X_2 + iY_2 &= \frac{1 + i}{\sqrt{2}}, \\ D, \quad X_1 + iY_1 &= -\frac{1}{k}, & X_2 + iY_2 &= \frac{-1 + i}{\sqrt{2}}. \end{aligned}$$

The proper value of M is $M = \sqrt{2} - 1 = \sqrt{k}$; the required formula thus is

$$\sqrt{k}(X_1 + iY_1) = \frac{1 + i\sqrt{k}(X_2 + iY_2)}{\sqrt{k}(X_2 + iY_2) + i},$$

or say

$$\sqrt{k}(X_1 + iY_1) = \frac{1 - i(X_2 + iY_2)}{X_2 + iY_2 - i}.$$

^{2*} The scale of this figure is double that of figures 1, 2, 4, 6, 7 in this paper.

Thus for the point B , we should have

$$\sqrt{k} = \frac{1-i \cdot \frac{1-i}{\sqrt{2}}}{\frac{1-i}{\sqrt{2}} - i}, \quad \text{that is,} \quad \sqrt{k} = \frac{\sqrt{2}-i-1}{1-i\sqrt{2}-i} = \frac{\sqrt{2}-1-i}{1-i-i\sqrt{2}},$$

which is right. And similarly for the point C , we should have

$$\frac{1}{\sqrt{k}} = \frac{1-i \cdot \frac{1+i}{\sqrt{2}}}{\frac{1+i}{\sqrt{2}} - i}, \quad \text{that is,} \quad \frac{1}{\sqrt{k}} = \frac{\sqrt{2}-i+1}{1+i-i\sqrt{2}} = \sqrt{2}+1,$$

which is right. And similarly for the points A and D .

8. We connect the square $ABCD$ of figure 1 with that of figure 4 by the equation

$$X + iY - \frac{1}{2}iK' = \frac{1+i}{2\sqrt{k}}(x + iy);$$

in fact, recollecting that $\frac{1}{2}K' = K = \frac{\varpi}{4\sqrt{k}}$, we have for the four points respectively

$$\begin{aligned} A, \quad X + iY &= -K, & x + iy &= -\frac{1}{2}\varpi, \\ B, \quad X + iY &= K, & x + iy &= -\frac{1}{2}\varpi i, \\ C, \quad X + iY &= K + iK', & x + iy &= \frac{1}{2}\varpi, \\ D, \quad X + iY &= -K + iK', & x + iy &= \frac{1}{2}\varpi i, \end{aligned}$$

values which satisfy the relation in question.

9. Hence writing

$$X + iY = U, \quad x + iy = u,$$

we have

$$U - \frac{1}{2}iK' = \frac{1+i}{2\sqrt{k}}u,$$

which is the relation in No. 1 between the arguments U , u of the elliptic functions sn , snl . We have $X_1 + iY_1 = \text{sn } U$, $x_1 + iy_1 = \text{snl } u$; and consequently

$$\sqrt{k}(X_1 + iY_1) = \frac{(i+1)(x_1 + iy_1) + i\sqrt{2}}{(i-1)(x_1 + iy_1) + \sqrt{2}}.$$

Hence, substituting for $\sqrt{k}(X_1 + iY_1)$ its value in terms of $X_2 + iY_2$, we find

$$\frac{1-i(X_2 + iY_2)}{X_2 + iY_2 - i} = \frac{(i+1)(x_1 + iy_1) + i\sqrt{2}}{(i-1)(x_1 + iy_1) + \sqrt{2}},$$

an equation which (multiplying on the left-hand side the numerator and the denominator each by $\frac{-i}{\sqrt{2}}$) may be written

$$\frac{1-i + \frac{1+i}{\sqrt{2}}(x_1 + iy_1)}{\frac{1+i}{\sqrt{2}}(x_1 + iy_1) - \frac{1-i}{\sqrt{2}}} = \frac{1-i(X_2 + iY_2)}{X_2 + iY_2 - i},$$

that is, we have

$$X_2 + iY_2 = \frac{1+i}{\sqrt{2}}(x_1 + iy_1),$$

an equation which shows that the figure 5 is in fact the figure 3 turned through an angle of 45° . We have thus proved the conclusion stated in No. 5 as to the connexion between the lemniscate solution for the square and the solution for the rectangle.

10. It is convenient to collect here the several equations relating to the orthomorphosis of the square. We have

$$X_1 + iY_1 = \text{sn}(X + iY), \quad k = 3 - 2\sqrt{2};$$

$$x_1 + iy_1 = \text{snl}(x + iy);$$

$$\sqrt{k}(X_1 + iY_1) = \frac{1 - i(X_2 + iY_2)}{X_2 + iY_2 - i};$$

$$X + iY - \frac{1}{2}iK' = \frac{1 + i}{2\sqrt{k}}(x + iy);$$

$$\sqrt{k}(X_1 + iY_1) = \frac{(i + 1)(x_1 + iy_1) + i\sqrt{2}}{(i - 1)(x_1 + iy_1) + \sqrt{2}};$$

$$X_2 + iY_2 = \frac{1 + i}{\sqrt{2}}(x_1 + iy_1),$$

which are the equations connecting together the coordinates of the five figures.

11. I examine more in detail the above-mentioned transformation

$$X_2 + iY_2 = \frac{1 + i(X_1 + iY_1)}{(X_1 + iY_1) + i};$$

see the foregoing figures 1 (XY) and 2 (X_1Y_1), in which we now regard the two circles as having each of them the radius unity; changing the sign of i , the equation gives

$$X_2 - iY_2 = \frac{1 - i(X_1 - iY_1)}{(X_1 - iY_1) - i},$$

and we hence find

$$X_2^2 + Y_2^2 + 1 = \frac{2(X_1^2 + Y_1^2 + 1)}{X_1^2 + Y_1^2 + 2Y_1 + 1};$$

consequently if $X_1^2 + Y_1^2 + 1 = 0$, then also $X_2^2 + Y_2^2 + 1 = 0$, or the transformation changes the first of these imaginary circles into the second of them: or say it changes the concentric orthotomic of the circle $X_1^2 + Y_1^2 - 1 = 0$ into the concentric orthotomic of the circle $X_2^2 + Y_2^2 - 1 = 0$.

We have moreover

$$X_2^2 + Y_2^2 - 1 = \frac{X_1^2 + Y_1^2 - 1 - 2Y_1}{X_1^2 + Y_1^2 + 2Y_1 + 1}, \quad Y_2 = \frac{Y_1 - 1 \cdot \frac{1}{2}(X_1^2 + Y_1^2 - 1)}{X_1^2 + Y_1^2 + 2Y_1 + 1},$$

values which give the foregoing expression for $X_2^2 + Y_2^2$. But we further obtain

$$X_2^2 + Y_2^2 - 1 - 2\mu Y_2 = \frac{X_1^2 + Y_1^2 - 1 + \frac{2}{\mu}Y_1}{X_1^2 + Y_1^2 + 2Y_1 + 1} \cdot (1 - \mu),$$

and it thus appears that the circumferences

$$X_1^2 + Y_1^2 - 1 + 2\mu Y_1 = 0, \quad X_2^2 + Y_2^2 - 1 - \frac{2}{\mu}Y_2 = 0,$$

correspond to each other. These are circles passing through the pairs of points ($Y_1 = 0, X_1 = \pm 1$), ($Y_2 = 0, X_2 = \pm 1$) respectively; or imagining them in the same figure, say they are circles which belong each to the series of circles $x^2 + y^2 - 1 + 2\beta y = 0$, and which moreover cut at

right angles at the points $y = 0, x = \pm 1$. But attending more carefully to the nature of the correspondence, it is to be observed that taking Y_1 positive, and giving to X_1 any positive value from 0 to ∞ , we have in figure 2 an arc LJM lying wholly within the upper semicircle LFM ; and that, corresponding hereto in figure 3, we have an arc LJM lying wholly within the lower semicircle LOM ; and that as in figure 2, the arc LJM lies nearer to the semicircumference LFM or to the diameter LOM , so in figure 3 the arc LJM lies nearer to the diameter LFM or to the semicircumference LOM . Thus the upper semicircle LFM of figure 2 corresponds to the lower semicircle LOM of figure 3; but so that the semicircumference LFM of the first figure corresponds to the diameter LFM of the second figure; and the diameter LOM of the first figure to the semicircumference LOM of the second figure. And further supposing that Y_1 is still positive, but that μ has any negative value from $-\infty$ to 0, we have in figure 2 an arc in the upper half-plane lying wholly outside the semicircle; and corresponding thereto in figure 3, an arc lying wholly inside the upper semicircle LHM ; that is, in figure 2 the infinite space in the upper half-plane outside the semicircle corresponds in figure 3 to the space within the upper half-circle LHM ; the infinity of figure 2 corresponding to the semicircumference LHM of figure 3, and the semicircumference LFM of figure 2 to the diameter LFM of figure 3. And thus in figure 2, the upper half-plane inside and outside the semicircle corresponds in figure 3 to the lower and upper half-circles, that is, to the whole circular area $OLHM$ of figure 3.

It is to be observed, moreover, that we have

$$X_2^2 + Y_2^2 + 1 + 2\lambda X_2 = \frac{2(X_1^2 + Y_1^2 + 1 + 2\lambda X_1)}{X_1^2 + Y_1^2 + 2Y_1 + 1},$$

that is, the circles $X_1^2 + Y_1^2 + 1 + 2\lambda X_1 = 0$ and $X_2^2 + Y_2^2 + 1 + 2\lambda X_2 = 0$ correspond to each other. Imagining the circles as belonging to the same figure, these are one and the same circle of the series $x^2 + y^2 + 1 - 2\alpha x = 0$ each passing through the pair of points $(x = 0, y = \pm i)$ which are the antipoints of the pair $(y = 0, x = \pm 1)$; these circles thus cut at right angles those of the series $x^2 + y^2 - 1 + 2\beta y = 0$. We have, by means of the two series of circles, an easy construction for the correspondence between the figures 2 and 3.

12. In explanation of No. 4, observe that, starting from the equation

$$X_1 + iY_1 = \text{sn}(X + iY),$$

and writing $\text{sn } X = P$, $\text{sn } iY = iQ$, we have

$$X_1 + iY_1 = \frac{P\sqrt{1+Q^2}\sqrt{1+k^2Q^2} + iQ\sqrt{1-P^2}\sqrt{1-k^2P^2}}{1+k^2P^2Q^2},$$

that is,

$$X_1 = \frac{P\sqrt{1+Q^2}\sqrt{1+k^2Q^2}}{1+k^2P^2Q^2}, \quad Y_1 = \frac{Q\sqrt{1-P^2}\sqrt{1-k^2P^2}}{1+k^2P^2Q^2},$$

and thence

$$X_1^2 + Y_1^2 = \frac{P^2 + Q^2}{1+k^2P^2Q^2} = r^2 \quad (\text{if } X_1^2 + Y_1^2 \text{ be put } = r^2);$$

hence

$$P^2(1 - k^2Q^2r^2) = r^2 - Q^2, \quad Q^2(1 - k^2P^2r^2) = r^2 - P^2.$$

Now considering in figure 1 any line in the rectangle parallel to the axis of X , that is, taking Y constant and therefore also Q constant, and proceeding to eliminate P , we have

$$1 - P^2 = \frac{1 + Q^2 - (1 + k^2Q^2)r^2}{1 - k^2Q^2r^2}, \quad 1 - k^2P^2 = \frac{1 + k^2Q^2 - k^2(1 + Q^2)r^2}{1 - k^2Q^2r^2},$$

and consequently

$$Y_1^2 = \frac{Q^2\{1 + Q^2 - (1 + k^2Q^2)r^2\}\{1 + k^2Q^2 - k^2(1 + Q^2)r^2\}}{(1 - k^2Q^2r^2)^2},$$

giving X_1, Y_1 each of them in terms of Q^2 and r^2 , $= X_1^2 + Y_1^2$. The former of these equations, replacing therein r^2 by its value, gives easily

$$(X_1^2 + Y_1^2)^2 - 2AX_1^2 - 2BY_1^2 + \frac{1}{k^2} = 0,$$

where

$$2A = \frac{1 + Q^2}{1 + k^2Q^2} + \frac{1}{k^2} \cdot \frac{1 + k^2Q^2}{1 + Q^2}, \quad 2B = Q^2 + \frac{1}{k^2Q^2};$$

viz. we have thus the equation of the curve, a bicircular quartic which in figure 2 corresponds to the line parallel to the axis of X_1 in figure 1.

In particular, for the line LM of figure 1 we have $Y = \frac{1}{2}K'$ and thence $iQ = \operatorname{sn} \frac{1}{2}iK' = \frac{i}{\sqrt{k}}$, that is, $Q = \frac{1}{\sqrt{k}}$, and thence $A = B = \frac{1}{k}$; the equation of the bicircular quartic is

$$(X_1^2 + Y_1^2)^2 - \frac{2}{k}(X_1^2 + Y_1^2) + \frac{1}{k^2} = 0,$$

viz. this is the circle $X_1^2 + Y_1^2 - \frac{1}{k} = 0$ twice repeated, and we have thus this circle, or rather the half-circumference LFM of figure 2, corresponding to the line LM of figure 1. More simply,

$$X_1 + iY_1 = \frac{1}{\sqrt{k}} \operatorname{sn} X + i \frac{i}{\sqrt{k}} \cdot \operatorname{cn} X \operatorname{dn} X,$$

and thence

$$X_1^2 + Y_1^2 = \frac{\operatorname{sn}^2 X + (\operatorname{cn} X \operatorname{dn} X)^2/k^2}{1 + k \operatorname{sn}^2 X \cdot (1/k)} = \frac{1}{k},$$

that is, $X_1^2 + Y_1^2 - \frac{1}{k} = 0$ as before. It is easy to see that the points A, O, B of figure 1 correspond to the points A, O, B of figure 2, and hence that the area of the rectangle $AOBLFMA$ of figure 1 corresponds to that of the semicircle $AOBLFMA$ of figure 2.

Returning to the equation $X_1 + iY_1 = \operatorname{sn}(X + iY)$, if we write herein successively

$$Y_1 = \frac{1}{2}K' + \beta, \quad \operatorname{sn} iY_1 = iQ_1 = \operatorname{sn} i(\frac{1}{2}K' + \beta),$$

and

$$Y_2 = \frac{1}{2}K' - \beta, \quad \operatorname{sn} iY_2 = iQ_2 = \operatorname{sn} i(\frac{1}{2}K' - \beta),$$

then we have

$$Q_1Q_2 = \frac{1}{k};$$

hence for q writing Q_1 or Q_2 , we have in each case the same values of A and B , that is, we have the same bicircular quartic for two lines parallel to and equidistant from the line LM , but to one of these (viz. the line between LM and BA) there corresponds the half-perimeter lying within the semicircle LFM , and to the other of them (viz. the line between LM and CD) there corresponds the half-perimeter lying without the semicircle LFM .

It may be shown in like manner that, to any line in figure 1 parallel to the axis OY , there corresponds in figure 2 a bicircular quartic of the like form

$$(X_1^2 + Y_1^2)^2 - 2AX_1^2 - 2BY_1^2 + \frac{1}{k^2} = 0.$$

13. Similarly in explanation of No. 5, observe that, starting from the equation $x_1 + iy_1 = \text{snl}(x + iy)$ and writing $\text{snl } x = p$, $\text{snl } iy = i \text{snl } y = iq$, we have

$$x_1 + iy_1 = \frac{p\sqrt{1-q^4} + iq\sqrt{1-p^4}}{1-p^2q^2},$$

that is,

$$x_1 = \frac{p\sqrt{1-q^4}}{1-p^2q^2}, \quad y_1 = \frac{q\sqrt{1-p^4}}{1-p^2q^2},$$

and thence

$$x_1^2 + y_1^2 = \frac{p^2 + q^2}{1-p^2q^2};$$

writing $x + y = \frac{1}{2}\varpi$, we have $\text{snl } y = \text{cnl } x$, that is, $q = \frac{\sqrt{1-p^2}}{\sqrt{1+p^2}}$, or $q^2 = \frac{1-p^2}{1+p^2}$, that is,

$\frac{p^2}{1+p^2} + \frac{q^2}{1-p^2} = 1$; and thus to the line $x + y = \frac{1}{2}\varpi$, there corresponds the circle $x_1^2 + y_1^2 - 1 = 0$.

More precisely, to the line CD of figure 4, there corresponds the quarter-circumference CD

of figure 5: and similarly to DA , AB and BC of figure 4 the remaining quarter-circumferences DA , AB , BC of figure 5; that is, to the whole boundary of the square in figure 4 there corresponds the whole circumference of the circle in figure 5.

14. To any line in figure 4, parallel to the axis Ox or to the axis Oy , there corresponds in figure 5 a bicircular quartic of the form $x_1^4 + y_1^4 - 2Ax_1^2 - 2By_1^2 - 1 = 0$. The investigation is substantially the same as that contained in No. 12, and need not be here given. But it is remarkable that also, to any line of figure 4 parallel to a side of the square (that is, to any line $x \pm y = c$), there corresponds in figure 5 a bicircular quartic of the like form (for the sides of the square, or lines $x \pm y = \pm \frac{1}{2}\varpi$ of figure 4, this bicircular quartic becomes the twice repeated circle $x_1^2 + y_1^2 - 1 = 0$ of figure 5, which is the result just obtained in No. 13). I investigate this result as follows. Writing, as in No. 13,

$$\text{snl } x = p, \quad \text{snl } iy = i \text{snl } y = iq,$$

we have, as above,

$$x_1 = \frac{p\sqrt{1-q^4}}{1-p^2q^2}, \quad y_1 = \frac{q\sqrt{1-p^4}}{1-p^2q^2}, \quad \text{and thence} \quad x_1^2 + y_1^2 = \frac{p^2 + q^2}{1-p^2q^2}.$$

Now assuming between x and y the relation $x + y = C$, and writing $\text{snl } C = c$, this gives

$$c = \frac{p\sqrt{1-q^4} + q\sqrt{1-p^4}}{1-p^2q^2};$$

to obtain the required curve, we must between these equations (three independent equations) eliminate p and q . We have

$$x_1 + y_1 = \frac{p\sqrt{1-q^4} + q\sqrt{1-p^4}}{1-p^2q^2},$$

and consequently

$$(1 + p^2q^2)c = (1 - p^2q^2)(x_1 + y_1),$$

or, writing for convenience $\Omega = 1 - p^2q^2$, this equation gives

$$\Omega = \frac{2c}{x_1 + y_1 + c}.$$

Hence $\Omega x_1 = p\sqrt{1-q^4}$, $\Omega y_1 = q\sqrt{1-p^4}$; these equations may be written

$$\Omega^2 x_1^2 = p^2(1-q^2)(1+q^2), \quad \Omega^2 y_1^2 = q^2(1-p^2)(1+p^2),$$

and from these equations obtaining the expressions for p^2 and q^2 , and thence the expression for $p^2 q^2 = 1 - \Omega$, we find, after some easy reductions,

$$(x_1^2 + y_1^2)^2 - \frac{4c^2 x_1^2 y_1^2}{(x_1 + y_1 + c)^2} = (1 - c^2) \Omega.$$

But we have

$$\Omega = \frac{2c}{x_1 + y_1 + c}, \quad \text{or} \quad \Omega = \frac{2c}{P + 2Q} \quad (P = x_1^2 + y_1^2, \quad Q = x_1 y_1);$$

or substituting this value, the equation becomes

$$(x_1^2 + y_1^2)^2 - \frac{4c^2 x_1^2 y_1^2}{(x_1 + y_1 + c)^2} = \frac{2c(1 - c^2)}{x_1 + y_1 + c},$$

that is,

$$\{(x_1^2 + y_1^2)^2(x_1 + y_1 + c)^2 - 4c^2 x_1^2 y_1^2\}(x_1 + y_1 + c) - 4c(x_1^2 + y_1^2)(1 - c^2)(x_1 + y_1 + c)^2 = 0.$$

Writing for a moment $x_1^2 + y_1^2 = P$, $x_1 y_1 = Q$, this is

$$\left\{P - \frac{1}{c^2}(P + 2Q)\right\}\{P + 2Q - c^2\} + 4c^2 Q^2 = 0,$$

an equation which contains the factor $P + 2Q$; throwing this out, the equation becomes, after an easy reduction,

$$(P - 1)^2 + (1 - c^2) \left\{2P_1^2 - \frac{1}{c^2}(P + 2Q)\right\} = 0,$$

that is,

$$(x_1^2 + y_1^2 - 1)^2 + (1 - c^2) \left\{(x_1 - y_1)^2 - \frac{1}{c^2}(x_1 + y_1)^2\right\} = 0,$$

the required equation. Transforming through an angle of 45° by writing

$$x_1 = \frac{x_2 + y_2}{\sqrt{2}}, \quad y_1 = \frac{x_2 - y_2}{\sqrt{2}},$$

(where observe that the axis of x_2 is the line FH of figure 5), the equation becomes

$$(x_2^2 + y_2^2 - 1)^2 + (1 - c^2) 2y_2^2 - \frac{2}{c^2} x_2^2 = 0,$$

or writing $c = \cos \gamma$ and therefore $1 - c^2 = \sin^2 \gamma$, this equation becomes

$$(x_2^2 + y_2^2)^2 - \frac{2}{\cos^2 \gamma} x_2^2 - 2 \cos^2 \gamma \cdot y_2^2 + 1 = 0,$$

a curve consisting of two indented ovals situate symmetrically in regard to the axis Fy_2 of figure 5. In fact, writing in the equation $y_2 = 0$, we have for x_2^2 two real positive values; but, writing $x_2 = 0$, we have for y_2^2 two imaginary values. For $\gamma = 0$, the equation becomes

$$(x_2^2 + y_2^2)^2 - 2(x_2^2 + y_2^2) + 1 = 0,$$

that is, we have the circle $x_2^2 + y_2^2 - 1 = 0$ twice repeated. One of the ovals is shown in figure 5; the portion of it lying within the circle agrees with Schwarz's figure, p. 113, turning this round through an angle of 45° .

For the lines $x - y = C$ of figure 4, we have in figure 5 the same system of bicircular quartics turned round through an angle of 90° .

II.

15. I consider the general problem of the orthomorphosis of a circle into a circle: we can, for the transformation of the circumference of the circle $x^2 + y^2 - 1 = 0$ into that of the circle $x_1^2 + y_1^2 - 1 = 0$, find a formula involving an arbitrary function

or (what is the same thing) an indefinite number of arbitrary constants. In fact, writing for shortness $z = x + iy$, $z_1 = x_1 + iy_1$, and $\bar{z}$, $\bar{z}_1$ for the conjugate functions $x - iy$, $x_1 - iy_1$; also $\phi(z)$ for a function of z involving in general imaginary coefficients $a + ib$, &c., and $\bar{\phi}(z)$ for the like function with the conjugate coefficients $a - ib$, &c.; then if we assume

$$z_1 = \frac{\phi(z)}{z^m \bar{\phi}\left(\frac{1}{z}\right)},$$

where m is any positive or negative integer, this implies

$$\bar{z}_1 = \frac{\bar{\phi}(\bar{z})}{\bar{z}^m \phi\left(\frac{1}{\bar{z}}\right)};$$

consequently, if $x^2 + y^2 - 1 = 0$, that is, $z\bar{z} = 1$, or $\bar{z} = \frac{1}{z}$, we have

$$\bar{z}_1 = \frac{\bar{\phi}\left(\frac{1}{z}\right)}{\left(\frac{1}{z}\right)^m \phi(z)} = \frac{z^m \bar{\phi}\left(\frac{1}{z}\right)}{\phi(z)} = \frac{1}{z_1},$$

or $z_1 \bar{z}_1 = 1$, that is, $x_1^2 + y_1^2 - 1 = 0$.

In a slightly different form, taking α , β , &c., to denote any imaginary quantities, and $\bar{\alpha}$, $\bar{\beta}$, ... the conjugate quantities; assuming

$$\phi(z) = (z - \alpha)(z - \beta) \dots,$$

and taking m for the number of factors, we have

$$z_1 = \frac{(z - \alpha)(z - \beta) \dots}{(1 - \bar{\alpha}z)(1 - \bar{\beta}z) \dots},$$

and then (repeating the demonstration) we have

$$\bar{z}_1 = \frac{(\bar{z} - \bar{\alpha})(\bar{z} - \bar{\beta}) \dots}{(1 - \alpha\bar{z})(1 - \beta\bar{z}) \dots},$$

which, writing therein $\bar{z} = \frac{1}{z}$, becomes

$$\bar{z}_1 = \frac{\left(\frac{1}{z} - \bar{\alpha}\right)\left(\frac{1}{z} - \bar{\beta}\right) \dots}{\left(1 - \frac{\alpha}{z}\right)\left(1 - \frac{\beta}{z}\right) \dots} = \frac{1}{z_1},$$

and consequently, if $\bar{z} = \frac{1}{z}$, then also $\bar{z}_1 = \frac{1}{z_1}$ as before.

We may in the expression for z_1 introduce a factor $\frac{A}{A}$, or, what is the same thing, a factor A which is such that $A\bar{A} = 1$. In particular, we thus have the solution

$$z_1 = \frac{A(z - \alpha)}{(1 - \bar{\alpha}z)}.$$

ON SOME PROBLEMS OF ORTHOMORPHOSIS (continued)

[921, pp. 204–205]

giving

$$z_1 = \frac{A(z-a)}{1-az},$$

which, for $z\bar{z} = 1$, becomes

$$z_1\bar{z}_1 = A\bar{A} \frac{z-a}{1-az} \cdot \frac{\bar{z}-a}{1-a\bar{z}},$$

so that ($A\bar{A}$ being = 1) this gives $z_1\bar{z}_1 = 1$.

16. This is a solution with three arbitrary constants, viz. A (which may be put $= \cos \lambda + i \sin \lambda$) is a single arbitrary constant, and $\alpha, = a + ib$, is two arbitrary constants; and these constants may be so determined that a given point in the interior of the one circle, and a given point on the circumference thereof, shall correspond respectively to a given point in the interior of the other circle and to a given point on the circumference thereof. According to a well-known theorem of Riemann's, any two simply connected areas included within given closed curves respectively may be made to correspond to each other, and that in one way only, under the foregoing conditions as to a pair of interior points and a pair of boundary points; and we have, in what just precedes, the solution of the problem in the case of two equal circles.

17. In the case of any other solution, we thus know that the correspondence between the two circumferences cannot and does not imply a (1, 1) correspondence between the areas of the two circles: but it is interesting to enquire what happens. I take a very particular case

$$z_1 = \frac{z(z-2)}{1-2z},$$

and therefore

$$\bar{z}_1 = \frac{\bar{z}(\bar{z}-2)}{1-2\bar{z}},$$

and consequently

$$z_1\bar{z}_1 = \frac{z\bar{z}\{z\bar{z}-2(z+\bar{z})+4\}}{1-2(z+\bar{z})+4z\bar{z}},$$

that is,

$$x_1^2 + y_1^2 = \frac{(x^2 + y^2)\{x^2 + y^2 - 4x + 4\}}{4(x^2 + y^2) - 4x + 1},$$

so that, writing $x_1^2 + y_1^2 - 1 = 0$, we have $x^2 + y^2 - 1 = 0$, a correspondence of the two circumferences. But to the circumference $x_1^2 + y_1^2 - 1 = 0$ there corresponds not only the circumference $x^2 + y^2 - 1 = 0$, but another circumference. In fact, writing $x_1^2 + y_1^2 = 1$, we have

$$4(x^2 + y^2) - 4x + 1 = (x^2 + y^2)(x^2 + y^2 - 4x + 4),$$

that is,

$$(x^2 + y^2)^2 - 4x(x^2 + y^2) + 4x - 1 = 0,$$

or

$$(x^2 + y^2 - 1)(x^2 + y^2 - 4x + 1) = 0,$$

and there is thus the other circle

$$x^2 + y^2 - 4x + 1 = 0, \quad \text{or say } (x-2)^2 + y^2 - 3 = 0,$$

viz. this is a circle, coordinates of centre (2, 0) and radius $= \sqrt{3}$, cutting the circle $x^2 + y^2 - 1 = 0$ in two real points. Referring to the figures 6 (xy) and 7 (x_1y_1), and observing that $x_1 = 0$,

$y_1 = 0$, that is, $z_1 = 0$, gives $z = 0$, or $z = 2$, that is, the points $x = 0, y = 0$, and $x = 2, y = 0$, we see that to the centre O in figure 7, there correspond in figure 6 the points O, M which are the centres of the two circles. To any small closed curve, or say any small circle surrounding the point O of figure 7, there correspond in figure 6 small closed curves surrounding the points O, M respectively; and if in figure 7 the radius of the circle continually increases and becomes nearly equal to unity, the closed curves of figure 6 continually increase, changing at the same time their forms, and assume the forms shown by the dotted lines of figure 6. It thus appears that, to the whole area of the circle $x_1^2 + y_1^2 - 1 = 0$ of figure 7, there correspond the two lunes ACB and ABD of figure 6; or if we attend only to the area included within the circle $x^2 + y^2 - 1 = 0$ of this figure, then there corresponds not the whole area of this circle, but only the area of the lune ACB : and thus that the assumed relation $z_1 = \frac{z(z-2)}{1-2z}$ establishes, in fact, an orthomorphosis of the circle $x_1^2 + y_1^2 - 1 = 0$ into the lune ACB which lies inside the circle $x^2 + y^2 - 1 = 0$ and outside the circle $(x-2)^2 + y^2 - 3 = 0$. It may be added that, to the infinite area outside the circle $x_1^2 + y_1^2 - 1 = 0$ of figure 7, there correspond in figure 6 first the area of the lens AB common to the two circles, and secondly the area outside the two circles: we have thus an orthomorphosis of the area outside the circle $x_1^2 + y_1^2 - 1 = 0$ into these two areas respectively.

A somewhat more elegant example would have been that of the correspondence

$$z_1 = \frac{z(z - \sqrt{2})}{1 - \sqrt{2}z};$$

here, corresponding to the circumference $x_1^2 + y_1^2 - 1 = 0$, we have the two equal circumferences $x^2 + y^2 - 1 = 0$, and $(x - \sqrt{2})^2 + y^2 - 1 = 0$: and to the whole area of the circle $x_1^2 + y_1^2 - 1 = 0$, there correspond two equal lunes ACB and ABD .

NOTE ON THE LUNAR THEORY.

[922]

From the *Monthly Notices of the Royal Astronomical Society*, vol. LII. (1892), pp. 2–5.

In the Lunar Theory, in whatever way worked out, the values ultimately obtained for the coordinates r , v , y should of course satisfy identically the equations of motion; and that they do so, is the ultimate verification of the correctness of the results obtained. It can hardly be hoped for that such a verification will ever be made for Delaunay's results; and yet it would seem generally that the labour of such a verification of the results to any extent, while exceeding (and possibly greatly exceeding) that of obtaining these results by any method employed for that purpose, ought still to be, so to speak, a labour of the same order. And one can, moreover, imagine the process of verification so arranged as to be a process of mere routine which could be carried out by ordinary computers. But, however this may be, I think it is not without interest to exhibit the verification to a very small extent, viz. to e , m^4 .

I think there is an advantage in using capital letters for the arguments, and I accordingly write G (instead of Delaunay's g), to denote the mean anomaly.

The equations of motion are:

$$\frac{d}{dt} \left(\frac{dr}{dt} \right) - r \left\{ \cos^2 y \left(\frac{dv}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 \right\} + \frac{m^2 r}{r^2} = \frac{d\Omega}{dr},$$

$$\frac{d}{dt} \left(r^2 \cos^2 y \frac{dv}{dt} \right) = \frac{d\Omega}{dv},$$

$$\frac{d}{dt} \left(r^2 \frac{dy}{dt} \right) + r^2 \sin y \cos y \left(\frac{dv}{dt} \right)^2 = \frac{d\Omega}{dy},$$

$$\Omega = \frac{m' r^2}{r'^3} \left\{ \frac{3}{2} \cos^2 y \cos^2(v - v') - \frac{1}{2} \right\};$$

or say

$$\Omega = \frac{m' r^2}{r'^3} \left\{ \frac{3}{2} \cos^2 y \cos^2(v - v') - \frac{1}{2} \right\};$$

and thus the equations become

$$\frac{d^2 r}{dt^2} - r \left\{ \cos^2 y \left(\frac{dv}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 \right\} + \frac{m^2}{r^2} = \frac{m'^2 r}{r'^3} \left\{ 3 \cos^2 y \cos^2(v - v') - 1 \right\},$$

$$\frac{d}{dt} \left(r^2 \cos^2 y \frac{dv}{dt} \right) = - \frac{3m'^2 r^2}{r'^3} \cos^2 y \sin(v - v') \cos(v - v'),$$

$$\frac{d}{dt} \left(r^2 \frac{dy}{dt} \right) + r^2 \sin y \cos y \left(\frac{dv}{dt} \right)^2 = - \frac{3m'^2 r^2}{r'^3} \sin y \cos y \cos^2(v - v').$$

To simplify as much as possible, take the Sun's orbit to be circular, i.e. $r' = a'$, $v' = mnt$; also neglect y^2 : the first and second equations are

$$\frac{d^2 r}{dt^2} - r \left(\frac{dv}{dt} \right)^2 + \frac{m^2}{r^2} = m'^2 r \left\{ \frac{1}{2} + \frac{3}{2} \cos(2v - 2v') \right\},$$

$$\frac{d}{dt} \left(r^2 \frac{dv}{dt} \right) = - \frac{3}{2} m'^2 r^2 \sin(2v - 2v'),$$

and if for convenience of working we write $a = 1$, $n = 1$, then the first equation may be written

$$\frac{d^2 r}{dt^2} - r \left(\frac{dv}{dt} \right)^2 + \frac{1}{r^2} = m^2 r \left\{ \frac{1}{2} + \frac{3}{2} \cos(2v - 2v') \right\},$$

and similarly the second equation is

$$\frac{d}{dt} \left(r^2 \frac{dv}{dt} \right) = -\frac{3}{2} m^2 r^2 \sin(2v - 2v'),$$

and the third equation may be disregarded.

The two equations should be satisfied by Delaunay's values, putting therein $e' = 0$, $y = 0$; say by the values

$$\begin{aligned} \frac{1}{r} &= 1 + \frac{1}{2} m^2 + \frac{1}{16} m^4 \\ &+ \left(m^2 + \frac{11}{8} m^3 + \frac{59}{18} m^4 \right) \cos 2D \\ &+ e \cos G \\ &+ e \left(\frac{15}{8} m - \frac{19}{16} m^2 \right) \cos(2D + G) \\ &+ e \left(\frac{1}{2} m + \frac{17}{16} m^2 \right) \cos(2D - G) \\ r \frac{dv}{dt} &= 1 + 2e \cos G + e \left(\frac{15}{8} m - \dots \right) \sin G + \dots \end{aligned}$$

where

$$D = (1 - m)nt, \quad G = cnt.$$

The verification for the first equation is

$$-\frac{1}{r^2} \frac{d^2 r}{dt^2} - \left(\frac{dv}{dt} \right)^2 + \frac{1}{r^2} \cdot \frac{1}{r} - m^2 r \left\{ \frac{1}{2} + \frac{3}{2} \cos(2v - 2v') \right\} = 0$$

	A	B	C	D	total
Const.	-1	+1			= 0
	$+\frac{1}{2}m^2$	$-\frac{1}{2}m^2$			= 0
	$+\frac{1}{16}m^4$	$-\frac{1}{16}m^4$			= 0
	$+2m^4$	$-\frac{1}{4}m^4$	$-\frac{15}{16}m^4$	$-\frac{3}{8}m^4$	= 0
$\cos 2D$	$+4m^2$	$-\frac{9}{2}m^2$	$+3m^2$	$-\frac{3}{2}m^2$	= 0
$\cos 4D$				$-\frac{15}{16}m^4$	= 0
$\cos G$	e	$-\frac{1}{2}e$	$\frac{3}{2}e$		= 0
	$-\frac{3}{4}em^2$	$+3em^2$	$-\frac{3}{2}em^2$		= 0
$\cos(2D - G)$	$em(\frac{1}{2})$	$em(-\frac{1}{4})$	$em(-\frac{1}{4})$		= 0

The verification for the second equation is

$$\frac{2}{r} \frac{dr}{dt} r \frac{dv}{dt} + r^2 \frac{d^2 v}{dt^2} + \frac{3}{2} m^2 \sin(2v - 2v') = 0$$

$\sin 2D$	$+4m^2$	$-\frac{11}{2}m^2$	$+\frac{3}{2}m^2$	= 0
$\sin 4D$	$+\frac{15}{8}m^4$	$-\frac{59}{16}m^4$	$+\frac{53}{32}m^4$	= 0
$\sin G$	$e \cdot 2$	$+e \cdot (-2)$		= 0
	$e \cdot 3m^2$	$+e \cdot 3m^2$		= 0
$\sin(2D + G)$	$e \cdot \frac{3}{4}m^2$	$e \cdot (-\frac{19}{4})m^2$	$e \cdot 3m^2$	= 0
$\sin(2D - G)$	$e \cdot \frac{1}{2}m$	$e \cdot (-\frac{1}{2})m$		= 0

and the verification is thus completed.

The following intermediate results may be recorded; $\frac{1}{r} = 1 + x$,

$$y^2 = \dots, \quad -\log r = \dots, \quad \frac{dv}{dt} = \dots,$$

$\cos 2D$	$\frac{1}{2}m^2$	$\frac{1}{2}m^2 - \frac{15}{16}m^4$
$\cos 4D$	$+\frac{1}{8}m^4$	$\frac{1}{4}m^4$
$\cos G$	$+\frac{1}{2}em^2$	$e(1 - \frac{1}{2}m^2)$
$\cos(2D - G)$		
$\sin 2D$	$-2m^2 + \dots$	
$\sin 4D$		
$\sin G$		
$\sin(2D - G)$	e	

These were, in fact, made use of for finding the foregoing values of r , $\frac{dv}{dt}$, &c.

NOTE ON A HYPERDETERMINANT IDENTITY.

[923]

From the *Messenger of Mathematics*, vol. XXI. (1892), pp. 131, 132.

The following is in effect a well-known theorem; but I am not sure whether it has been stated in a form at once so general and so precise.

If

$$\Omega = \phi(x_1, y_1)^A (x_2, y_2)^B (x_3, y_3)^C (x_4, y_4)^D \dots$$

be a function separately homogeneous, and of the degrees $A, B, C, D, \dots$ in the sets of variables $(x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4), \dots$ respectively; and if

$$12 = \xi_1 \eta_2 - \xi_2 \eta_1 = \partial_{x_1} \partial_{y_2} - \partial_{x_2} \partial_{y_1}, \quad \&c.,$$

then

$$(A \cdot 23 + B \cdot 31 + C \cdot 12)\Omega = 0,$$

when the variables $(x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4), \dots$, or only the variables $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ are therein severally replaced by (x, y) .

In fact, we have

$$A\Omega = (x_1 \xi_1 + y_1 \eta_1)\Omega, \quad B\Omega = (x_2 \xi_2 + y_2 \eta_2)\Omega, \quad C\Omega = (x_3 \xi_3 + y_3 \eta_3)\Omega;$$

thus the expression is

$$\{(x_1 \xi_1 + y_1 \eta_1) \cdot 23 + (x_2 \xi_2 + y_2 \eta_2) \cdot 31 + (x_3 \xi_3 + y_3 \eta_3) \cdot 12\}\Omega,$$

and if we herein replace the variables $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ in so far as they appear explicitly by (x, y) , the expression becomes

$$= \{(x \xi_1 + y \eta_1) \cdot 23 + (x \xi_2 + y \eta_2) \cdot 31 + (x \xi_3 + y \eta_3) \cdot 12\}\Omega,$$

where the factor in $\{ \}$, substituting for 23, 31, 12 their values $\xi_2 \eta_3 - \xi_3 \eta_2, \xi_3 \eta_1 - \xi_1 \eta_3, \xi_1 \eta_2 - \xi_2 \eta_1$, becomes identically $= 0$. The value of the expression is thus $= 0$; and of

course it remains $= 0$ when, consequently upon the foregoing change $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ each into (x, y) , we also change $(\xi_1, \eta_1), (\xi_2, \eta_2), (\xi_3, \eta_3)$ each into (ξ, η) ; and if we also change $(x_4, y_4), \&c.$, into (x, y) .

It is clear that Ω may denote the covariant symbol

$$\Omega = 23^\alpha 31^\beta 12^\gamma 14^\delta 24^\epsilon 34^\zeta \dots U_1 V_2 W_3 T_4 \dots,$$

where $U, V, W, T, \dots$ denote quantics

$$(a, \dots | x, y)^m, \quad (a', \dots | x, y)^n, \quad \dots$$

of the degrees $m, n, p, q, \dots$ respectively, and $U_1, V_2, \&c.$, are the corresponding functions $(a, \dots | x_1, y_1)^m, (a', \dots | x_2, y_2)^n, \&c.$, the values of A, B, C being here

$$A = m - \beta - \gamma - \delta - \dots,$$

$$B = n - \gamma - \alpha - \epsilon - \dots,$$

$$C = p - \alpha - \beta - \zeta - \dots;$$

the theorem expresses the property that the covariants $12\Omega, 23\Omega, 31\Omega$, are linearly connected together; or, writing it in the form $(A \cdot 12 - C \cdot 13 + B \cdot 23)\Omega = 0$, we have the proper linear combination $A \cdot 12\Omega - C \cdot 13\Omega$ of the two covariants 12Ω and 13Ω , equal to $-B \cdot 23\Omega$, a determinate multiple of 23Ω . Speaking roughly, we say that the difference of the covariants 12Ω and 13Ω is equal to 23Ω .

ON THE NON-EXISTENCE OF A SPECIAL GROUP OF POINTS.

[924]

From the *Messenger of Mathematics*, vol. XXI. (1892), pp. 132, 133.

It is well known that, taking in a plane any eight points, every cubic through these passes through a determinate ninth point. It is interesting to show that there is no system of seven points such that every cubic through these passes through a determinate eighth point.

Assuming such a system: first, no three of the points can be in a line: for, if they were, then among the cubics through the seven points we have the line through the three points and an arbitrary conic through the remaining four points, and these composite cubics have no common eighth point of intersection.

Secondly, no six of the points can be on a conic: for, if they were, then among the cubics through the seven points we have the conic through the six points and an arbitrary line through the remaining point, and these composite cubics have no common eighth point of intersection.

Taking now the points to be 1, 2, 3, 4, 5, 6, 7; among the cubics through these, we have the composite cubics (A, P) , (B, Q) , (C, R) , where A, B, C are the lines 67, 75, 56, and P, Q, R the conics 12345, 12346, 12347 respectively; by what precedes, the points 5, 6, 7 do not lie on a line, and the points (6, 7), (7, 5) and (5, 6) neither of them lie on the conics P, Q, R respectively.

The common eighth point of intersection, if it exists, must be

$$(A, B, C); \quad (A, Q, R), (B, R, P), (C, P, Q);$$

$$(P, B, C), (Q, C, A), (R, A, B); \text{ or } (P, Q, R).$$

There is no point (A, B, C) .

There is no point (A, Q, R) : for Q, R intersect only in the points 1, 2, 3, 4, no one of which lies on A ; and similarly, there is no point (B, R, P) or (C, P, Q) .

B, C intersect in 5, which is a point on P ; and thus 5 is the only point (P, B, C) . Similarly, 6 is the only point (Q, C, A) and 7 is the only point (R, A, B) .

P, Q intersect in 1, 2, 3, 4, which are each of them on R ; hence 1, 2, 3, 4 are the only points (P, Q, R) .

Hence the points 1, 2, 3, 4, 5, 6, 7 present themselves each once, and only once, among the intersections of the three cubics, and there is no common eighth point of intersection.

ON WARING'S FORMULA FOR THE SUM OF THE m TH POWERS OF THE ROOTS OF AN EQUATION.

[925]

From the *Messenger of Mathematics*, vol. XXI. (1892), pp. 133–137.

The formula in question, Prob. I. of Waring's *Meditationes Algebraicae*, Cambridge, 1782, making therein a slight change of notation, is as follows: viz. the equation being

$$x^m + bx^{m-1} + cx^{m-2} + dx^{m-3} + \dots = 0,$$

then we have

$$(-)^m S_{\alpha^m} =$$

$$\begin{aligned} & b^m + \cdot 1 \\ & - mc \cdot b^{m-2} + \frac{1}{2}m \cdot m - 3 \cdot c^2 \cdot b^{m-4} \\ & - md \cdot b^{m-3} + m \cdot m - 4 \cdot cd \cdot b^{m-5} \\ & - me \cdot b^{m-4} + \frac{1}{2}m \cdot m - 5 \cdot (d^2 + 2ce) \cdot b^{m-6} \\ & - mf \cdot b^{m-5} + m \cdot m - 6 \cdot (de + cf) \cdot b^{m-6} \\ & - mg \cdot b^{m-6} + m \cdot m - 7 \cdot df \cdot b^{m-7} \\ & \dots \\ & - \frac{1}{6}m \cdot m - 4 \cdot m - 5 \cdot c^3 \cdot b^{m-6} \\ & + \frac{1}{6}m \cdot m - 5 \cdot m - 6 \cdot c^2 d \cdot b^{m-7} \\ & - \frac{1}{6}m \cdot m - 6 \cdot m - 7 \cdot c^2 e \cdot b^{m-8} \\ & - \frac{1}{6}m \cdot m - 6 \cdot m - 7 \cdot cd^2 \cdot b^{m-8} \\ & + \frac{1}{24}m \cdot m - 5 \cdot m - 6 \cdot m - 7 \cdot c^4 \cdot b^{m-8} \\ & + \&c., \end{aligned}$$

where, reckoning the weights of $b, c, d, e, \dots$ as 1, 2, 3, 4, $\dots$, respectively, the several terms are all the terms of the weight m , or (what is the same thing) in the

coefficient of $b^{m-\theta}$ we have all the combinations of $c, d, e, \dots$, (or say all the non-unitary combinations) of the weight θ , and where the numerical coefficient of

$$b^{m-\theta} c^\gamma d^\delta e^\varepsilon \dots \quad (\gamma + \delta + \varepsilon + \dots = \theta),$$

is

$$(-)^{\gamma+\delta+\varepsilon+\dots} \frac{m \cdot m - (\theta - \Sigma + 1) \cdot m - (\theta - \Sigma + 2) \dots m - (\theta - 1)}{\Pi \gamma \cdot \Pi \delta \cdot \Pi \varepsilon \dots}$$

(Π being the factorial). Thus for the term $b^{m-8} c^2 e^1$, $\theta = 8$; $\gamma, \delta, \varepsilon = 2, 4, 1$ respectively (the other exponents each vanishing), and the coefficient is

$$(-)^3 \frac{m \cdot m - 6 \cdot m - 7}{1 \cdot 2 \cdot 1} = -\frac{1}{2}m \cdot m - 6 \cdot m - 7,$$

as above; and so in other cases.

For the MacMahon form

$$1 + bx + \frac{c}{1 \cdot 2} x^2 + \frac{d}{1 \cdot 2 \cdot 3} x^3 + \dots = (1 - ax)(1 - \beta x)(1 - \gamma x) \dots,$$

or say

$$1 + bx + \frac{c}{1 \cdot 2} x^2 + \dots = (1 - ax)(1 - \beta x) \dots$$

we must for $b, c, d, \dots$, write $b, \frac{c}{1 \cdot 2}, \frac{d}{1 \cdot 2 \cdot 3}, \frac{e}{1 \cdot 2 \cdot 3 \cdot 4}, \dots$ respectively: we thus have

$$\begin{aligned} (-)^m S_{\alpha^m} &= b^m \\ &+ b^{m-1} \cdot (-mc) \\ &+ b^{m-2} \cdot \left(-\frac{m}{1 \cdot 2} d + \frac{1}{2} m \cdot m - 3 \cdot \left(\frac{c}{1 \cdot 2} \right)^2 \right) \\ &+ b^{m-3} \cdot \left(-\frac{m}{1 \cdot 2 \cdot 3} e + m \cdot m - 4 \cdot \frac{c}{1 \cdot 2} \cdot \frac{d}{1 \cdot 2 \cdot 3} \right) \\ &+ \&c., \end{aligned}$$

or say

$$\begin{aligned} (-)^m \Pi(m-1) S_{\alpha^m} &= \Pi(m-1) \left[b^m - \Pi m \frac{c}{1 \cdot 2} b^{m-2} + \dots \right] \\ &+ \Pi m \frac{d}{1 \cdot 2 \cdot 3} b^{m-3} \\ &+ \Pi m \frac{e}{1 \cdot 2 \cdot 3 \cdot 4} b^{m-4} \\ &+ \&c., \end{aligned}$$

the numerical coefficient of

$$b^{m-\theta} c^\gamma d^\delta e^\varepsilon \dots \quad (\gamma + \delta + \varepsilon + \dots = \theta)$$

being

$$(-)^{\gamma+\delta+\varepsilon+\dots} \frac{\Pi m \cdot m - (\theta - \Sigma + 1) \cdot m - (\theta - \Sigma + 2) \cdots m - (\theta - 1)}{\Pi \gamma \cdot \Pi \delta \cdot \Pi \varepsilon \cdots (\Pi 2)^\gamma (\Pi 3)^\delta (\Pi 4)^\varepsilon \cdots}.$$

It is convenient to write down the literal terms in alphabetical order (AO), calculating and affixing to each term the proper numerical coefficient; thus taking

$$1 + bx + \frac{c}{1 \cdot 2} x^2 + \frac{d}{1 \cdot 2 \cdot 3} x^3 + \dots = (1 - ax)(1 - \beta x)(1 - \gamma x) \cdots,$$

we find

$$\begin{aligned} -120 S_6 &= g \cdot 1 \\ &bf \cdot -6 \\ &ce \cdot -15 \\ &d^2 \cdot -10 \\ &b^2 e \cdot +30 \\ &bcd \cdot +120 \\ &c^3 \cdot +30 \\ &b^3 d \cdot -120 \\ &b^2 c^2 \cdot -270 \\ &b^4 c \cdot +360 \\ &b^6 \cdot -120 \\ &\pm 541 \end{aligned}$$

this expression, as representing the value of the non-unitary function S_6 , being in fact a semi-invariant.

It is to be remarked that the foregoing expression for the sum of the m th powers of the roots of the equation

$$x^m + bx^{m-1} + cx^{m-2} + \dots = 0$$

is, in fact, the series for x^m continued so far only as the exponent of b is not negative: see as to this Note XI. of Lagrange's *Equations Numeriques*. For the *a posteriori* verification, observe that we have

$$x = -b + \frac{c}{x} - \frac{d}{x^2} + \frac{e}{x^3} - \cdots,$$

or writing for a moment $u = -b$, say this

$$x = u + fx,$$

where

$$fx = \frac{c}{x} - \frac{d}{x^2} + \frac{e}{x^3} - \&c.$$

Hence, by Lagrange's theorem,

$$\begin{aligned} x^m &= u^m + \frac{m u^m}{u} \left(\frac{c}{u} + \frac{-d}{u^2} + \frac{e}{u^3} + \cdots \right) \\ &+ \frac{1}{1 \cdot 2} \left(\frac{c}{u} + \frac{-d}{u^2} + \frac{e}{u^3} + \cdots \right)' \cdot \frac{m u^{m-1}}{1 \cdot 2} \\ &+ \frac{1}{1 \cdot 2 \cdot 3} (\cdots)'' \cdot \frac{m u^{m-2}}{1 \cdot 2 \cdot 3} \\ &+ \&c., \end{aligned}$$

where the accents denote differentiations in regard to u . This is

$$\begin{aligned} &u^m + m \{ c u^{m-2} + d u^{m-3} + e u^{m-4} + f u^{m-5} + \cdots \} \\ &+ \frac{1}{2} m \{ (m-3) c^2 u^{m-4} + (m-4) \cdot 2cd \cdot u^{m-5} + (m-5)(d^2 + 2ce) u^{m-6} + \cdots \} \\ &- \frac{1}{6} m \{ (m-4)(m-5) c^3 u^{m-6} + \cdots \} \\ &+ \&c. \\ &= u^m \\ &+ u^{m-1} \cdot (-mc) \\ &+ u^{m-2} \cdot (-md) \\ &+ u^{m-3} \cdot (-me) + \frac{1}{2} m \cdot m-3 \cdot c^2 \\ &+ u^{m-4} \cdot (-mf) + \frac{1}{2} m \cdot m-4 \cdot 2cd \\ &+ u^{m-5} \cdot (-mg) + \frac{1}{2} m \cdot m-5 \cdot (d^2 + 2ce) - \frac{1}{6} m \cdot m-4 \cdot m-5 \cdot c^3 \\ &+ \&c., \end{aligned}$$

which, putting therein $u = -b$ and multiplying each side by $(-)^m$, is the before-mentioned formula for $(-)^m S_{\alpha^m}$: in that formula the series being continued only so far as the exponent of b is not negative.

I notice also that we cannot easily, by means of the known formula

$$S_{\alpha^m \beta^p} = S_{\alpha^m} S_{\alpha^p} - S_{\alpha^{m+p}},$$

deduce an expression for $S_{\alpha^m \beta^p}$: in fact, forming the product of the series for S_{α^m} , S_{α^p} respectively, this product is identically equal to the series for $S_{\alpha^{m+p}}$, or we seem to obtain $0 = S_{\alpha^m} S_{\alpha^p} - S_{\alpha^{m+p}}$: to obtain the correct formula, we have to take each of the three series only so far as the exponent of b therein respectively is not negative: and it is not easy to see how the resulting formula is to be expressed.

CORRECTED SEMINVARIANT TABLES FOR THE WEIGHTS 11 AND 12.

[926]

From the *American Journal of Mathematics*, vol. XIV. (1892), pp. 195–200.

The tables in my paper, “Seminvariant Tables.” *American Journal of Mathematics*, vol. VII. (1885), pp. 59–73, [831], are not in the best form, but the deviations present themselves only in a few columns of the tables for the weights 11 and 12, viz. in the former of these two columns, and in the latter a single column, ought to be replaced by linear combinations of other columns; there are, besides, columns which should be new named; in regard hereto, there is a point of theory which requires to be made clear. I remark that in each table the literal terms are in alphabetical order (AO); this is the proper order for the final terms, and although (as about to be explained) the proper order for the initial terms is the counter order (CO), yet as the tables cannot be at the same time arranged in the one and the other order, I adhere to the AO as the proper arrangement for the terms of the tables; we have, however, to introduce the notion that, in general, it is not the top term of a column which is to be regarded as the initial term, and in connexion herewith to consider how the columns are to be named. An instance first presents itself for the weight 11: we have, see column 5 of the table for that weight here given, a seminvariant

$$\begin{aligned} fg + 1 \\ bf^2 \\ bcg \\ bdh \\ beg - 5 \\ bf^2 - 3 \\ c^2h - 16 \end{aligned}$$

$$b^3e^2 + 70;$$

this is to be regarded, not as a seminvariant $fg - b^3e^2$ (it is hardly necessary to remark that, here and elsewhere the $-$ is not a minus sign, but a mere stroke), but as a seminvariant $c^2h - b^3e^2$, viz. c^2h is a term entering into the seminvariant, and which, although it is in AO subsequent, it is in CO precedent, to the terms fg , bcg and bf^2 . The seminvariant contains the term $-16c^2h$ and other terms with the letter h , and it is a misnomer to call it $fg - b^3e^2$, a name implying that the highest letter thereof is g . Instead of the stroke, it would perhaps be better to write ∞ , for instance $c^2h\infty b^3e^2$, where of course ∞ would be used as a mere conventional symbol.

For greater clearness, I give here the express definition of counter order, (CO), viz. whereas in AO we begin with the lowest letters, in CO we begin contrariwise with the highest letters. A term containing a higher letter or higher power of such letter precedes a term containing a lower letter or lower power of the same letter — or in the easiest form, the counter order is the alphabetical order corresponding to the reversed arrangement $z, y, \dots, f, e, d, c, b$ of the letters.

A symbol, as $c^2h - b^3e^2$ above, may be regarded as referring to a set of terms c^2h , b^3e^2 and all the terms which are in CO subsequent to c^2h and in AO precedent to b^3e^2 : as by supposition the terms are arranged in AO, the set includes no term lower than b^3e^2 , or say the bottom term b^3e^2 is also the final term of the set, but it does include terms fg , bcg and bf^2 higher than c^2h , and thus the top term fg is not, but c^2h is, the initial term of the set. It should be remarked that a seminvariant $c^2h - b^3e^2$ need not include all the terms of the set as just defined: there may very well be terms with a coefficient zero, or say accidental zeros; an instance presents itself, weight 10, where in the column $eg - bd^3$ we have $0ce^2$, no term in ce^2 .

The changes actually required are very slight, viz.

Weight 11, instead of

$fg - b^3e^2$, we require $c^2h - b^3e^2$, old $fg - b^3e^2$, new named,
 $c^2h - b^3d^2$, „ $fg - b^3d^2$, linear combination $(fg - b^3e^2) + 8(c^2h - b^3d^2)$,
 $de^2 - b^3c^4$, „ $c^3f - b^3d^2$, old $de^2 - b^3d^2$, new named,
 $c^3f - b^3c^4$, „ $de^2 - b^3c^4$, linear combination $(de^2 - b^3d^2) + 6(c^3f - b^3c^4)$.

Weight 12, instead of

$cf^2 - bcd^3$, we require $d^2g - bcd^3$, old $cf^2 - bcd^3$, new named,
 $d^2g - c^3d^2$, „ $cf^2 - c^3d^2$, linear combination $(cf^2 - bcd^3) - 5(d^2g - c^3d^2)$;

but I have thought it desirable to give the complete tables for the weights in question, 11 and 12; and I have also rearranged the entire columns of the two tables so as to present in each of them the finals in AO. This is the case with the existing tables, except that there is a single transposition in the table weight 10.

Instead of the columns $cdf - b^4d^2$, $ce^2 - c^5$, we ought to have $ce^2 - c^5$, $cdf - b^4d^2$. The complete list up to the weight 12 is

$w = 2:$ $c - b^2$
 $w = 3:$ $d - b^3$
 $w = 4:$ $e - c^2, e - b^4$
 $w = 5:$ $f - be^2c^2, cd - b^5$
 $w = 6:$ $g - d^2, ce - c^3, \dots - b^6$
 $w = 7:$ $h - cf, \dots - bc^3, \dots - b^2c^2, de, c^2d - b^7$
 $w = 8:$ $i, df - e^2, cg - cd^2, \dots - b^2d^2, c^2e - b^4c^2, \dots - b^8$
 $w = 9:$ $j - bc^2, ch - d^3, dg - bcd^2, \dots, ef, c^2f - bc^4, cde - b^2c^3, d^3 - b^3c^3, c^3d - b^9$
 $w = 10:$ $bf^2, ci - ce^2, dh - b^2c^2, eg - bd^3, f^2 - c^2d^2, ce^2 - c^5, cdf - b^4d^2, \dots - b^{10}$
 $w = 11:$ $bk, di - b^4d^2e^2, eh - b^4, ef - b^5, c^2h - b^3e^2, \dots, cdg - b^3cd^2, d^2f - bc^5, de^2 - b^3d^2, \dots, c^3f - b^3c^4, c^{11}$
 $w = 12:$ $m - g^2, \dots - cf^2, ei - c^3, \dots - b^2f^2, dj, fh - bde^2, g^2 - c^2e^2, c^2i - b^2ce^2, ceg - d^4, d^2g - bcd^3, cf^2 - c^3d^2, cdh - b^4e^2, \dots, def - b^3d^3, \dots, c^2de - b^6e^2, d^4 - b^4d^2, \dots, c^2df - b^6d^2, c^4e - b^8c^2, c^3d^2 - b^8c^2, c^6 - b^{12}$

The two new tables are as follows: some accidental numerical errors have been corrected.

TABLE, WEIGHT 11. (Complete table with 14 columns and 56 rows, double-page; cf. PDF pp. 242–243. The columns are headed (in CO/initial-term order): $fg, bf^2, di, eh, c^2h, bcd, bdh, cdg, c^2f, cef, bf^2, b^3i, b^2dg, b^2ch, \dots$; the column totals are $\pm 638, \pm 188, \pm 70, \pm 132, \pm 664, \pm 1640, \pm 57, \pm 170, \pm 43, \pm 278, \pm 192, \pm 35, \pm 98, \pm 48$. The detailed entries reproduce those of the original printed tables.)

TABLE, WEIGHT 12. (Complete table with 21 columns and 77 rows, double-page; cf. PDF pp. 244–245. The column totals are $\pm 1024, \pm 212, \pm 82, \pm 1740, \pm 2854, \pm 946, \pm 12, \pm 46, \pm 325, \pm 190, \pm 298, \pm 1664, \pm 442, \pm 18, \pm 52, \pm 51, \pm 258, \pm 156, \pm 48, \pm 68, \pm 32$. The detailed entries reproduce those of the original printed tables.)

ON CLIFFORD'S PAPER "ON SYZYGETIC RELATIONS AMONG THE POWERS OF LINEAR QUANTICS."

[927]

From the *Proceedings of the London Mathematical Society*, vol. XXIII. (1892), pp. 99–104.

The paper in question, originally printed, *Proc. Lond. Math. Soc.*, t. III. (1869), pp. 9–12, is reproduced No. XIV., pp. 119–122, of the *Mathematical Papers* (8vo. Lond. 1882), where it is immediately followed by the paper No. XV., "On Syzygetic Relations connecting the Powers of Linear Quantics," pp. 123–129. The author, after referring to theorems in M. Paul Serret's *Geometrie de Direction* (Paris, 1869), proceeds as follows:—"By the use of Professor Sylvester's method of contravariant differentiation, I have arrived at certain extensions of these theorems, which I now proceed to explain," and he then states his Theorem I.: In order that a system of N points in a plane should all lie on a curve of the order n , it is sufficient that the p th powers of their distances from an arbitrary line should satisfy a linear homogeneous relation, the number N being given by the formula

$$N = \frac{1}{2}\alpha(n+3) + \frac{1}{2}(\beta+1)(\beta+2),$$

where α is the quotient, and β the remainder of the division of p by n , so that $p = \alpha n + \beta$, and $\beta < n$. And he then gives Theorem II., a like theorem as regards points in space; and, further, two Tables, A and B, for the values of N corresponding to given values of n and p in the two cases respectively.

Theorem I. is incorrect for the first value of N in Table A, viz. if $n = 1$, $p = 2$, then $N = 5$; the theorem here is: In order that a system of five points in a plane may lie in a line, the sufficient condition is that the squares of their distances from an arbitrary line shall satisfy a linear homogeneous relation; or, what is the same thing, if the squares of the distances of the five points satisfy a linear homogeneous relation, then the five points will lie in a line. The right conclusion is that four of the five points will lie in a line.

Before proceeding further, I slightly modify the form of the enunciation by defining (in the case of the plane figure) the power of a point to be its distance from an arbitrary line; or, what is the same thing, to be the nilfactum of the line-equation of the point; that is, for the point (x_1, y_1, z_1) , the power is

$$\alpha x_1 + \beta y_1 + \gamma z_1,$$

where α, β, γ are arbitrary coefficients, or, if we please, line-coordinates. I say that $\alpha x_1 + \beta y_1 + \gamma z_1$ is the first power, $(\alpha x_1 + \beta y_1 + \gamma z_1)^2$ the second power, and so on. Clifford's Theorem I. thus is: If the p th powers of the N points satisfy a homogeneous linear relation, the N points are on a curve of the n th order.

In the case $n = 1$, $p = 2$, we have five points whose second powers satisfy a homogeneous linear relation; that is, if $(x_1, y_1, z_1), \dots, (x_5, y_5, z_5)$ are the coordinates of the five points respectively, we have

$$\lambda_1(\alpha x_1 + \beta y_1 + \gamma z_1)^2 + \dots + \lambda_5(\alpha x_5 + \beta y_5 + \gamma z_5)^2 = 0.$$

It is implicitly assumed that the points are distinct points, and we thus exclude such solutions as

$$\lambda_1 = \lambda_2 = \lambda_3 = 0, \quad (x_4, y_4, z_4) = k(x_5, y_5, z_5), \quad k^2\lambda_4 + \lambda_5 = 0.$$

Here α, β, γ are arbitrary, and the relation is equivalent to the six equations

$$\lambda_1 x_1^2 + \dots + \lambda_5 x_5^2 = 0, \quad \lambda_1 y_1^2 + \dots + \lambda_5 y_5^2 = 0, \quad \lambda_1 z_1^2 + \dots + \lambda_5 z_5^2 = 0,$$

$$\lambda_1 y_1 z_1 + \dots = 0, \quad \lambda_1 z_1 x_1 + \dots = 0, \quad \lambda_1 x_1 y_1 + \dots = 0;$$

and hence, eliminating the λ 's, we have the relation

$$\begin{vmatrix} x_1^2 & y_1^2 & z_1^2 & y_1 z_1 & z_1 x_1 & x_1 y_1 \\ x_2^2 & y_2^2 & z_2^2 & y_2 z_2 & z_2 x_2 & x_2 y_2 \\ x_3^2 & y_3^2 & z_3^2 & y_3 z_3 & z_3 x_3 & x_3 y_3 \\ x_4^2 & y_4^2 & z_4^2 & y_4 z_4 & z_4 x_4 & x_4 y_4 \\ x_5^2 & y_5^2 & z_5^2 & y_5 z_5 & z_5 x_5 & x_5 y_5 \end{vmatrix} = 0,$$

viz. this means that each of the determinants, obtained by selecting in any manner five out of the six columns, is $= 0$. This is a twofold relation, and thus it cannot be equivalent to the threefold relation

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ \vdots & \vdots & \vdots \\ x_5 & y_5 & z_5 \end{vmatrix} [\text{all minors} = 0]$$

which expresses that the five points are in a line.

But we see further that the twofold relation expresses that the five points are such that we can, through them and an arbitrary sixth point (x_6, y_6, z_6) , draw a determinate conic; and this is the case only if four of the five points are in a line; viz. the conic is then the line-pair composed of this line and of the line joining the remaining two points. The right conclusion therefore is that, if the above linear relation is satisfied, then four of the five points lie in a line.

Clifford's proof is rather indicated than carried out, but, from the reference to contravariant differentiation, and from the second paper mentioned above, it must have been as follows:

Starting from the linear relation considered as an identity in (α, β, γ) , and operating upon it with $p\partial_\alpha + q\partial_\beta + r\partial_\gamma$, p, q, r denoting arbitrary coefficients, we obtain

$$\lambda_1(px_1 + qy_1 + rz_1)(\alpha x_1 + \beta y_1 + \gamma z_1) + \cdots + \lambda_5(px_5 + qy_5 + rz_5)(\alpha x_5 + \beta y_5 + \gamma z_5) = 0;$$

hence, determining the ratios of p, q, r , say by the equations

$$px_4 + qy_4 + rz_4 = 0, \quad px_5 + qy_5 + rz_5 = 0,$$

and, instead of $\lambda_1(px_1 + qy_1 + rz_1)$, &c., writing A_1 , &c., we have

$$A_1(\alpha x_1 + \beta y_1 + \gamma z_1) + A_2(\alpha x_2 + \beta y_2 + \gamma z_2) + A_3(\alpha x_3 + \beta y_3 + \gamma z_3) = 0,$$

a homogeneous linear relation between the first powers of the three points (x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) . We thus see that these points, say the points 1, 2, 3, are in a line; and, similarly, by different determinations of p, q, r , that the points 1, 2, 4 are in a line; and that the points 1, 2, 5 are in a line; that is, the points 3, 4 and 5 are each of them in the line 12 joining the points 1 and 2; that is, the points 1, 2, 3, 4, 5 are in a line.

But, if we examine the reasoning more closely, it appears that the first conclusion, 1, 2, 3 are in a line, depends on the assumption that neither 1, 2, nor 3 is in the line 45. Suppose, for instance, that 3 was in the line 45, the equations

$$px_4 + qy_4 + rz_4 = 0, \quad px_5 + qy_5 + rz_5 = 0,$$

imply $px_3 + qy_3 + rz_3 = 0$, and we have

$$A_1(\alpha x_1 + \beta y_1 + \gamma z_1) + A_2(\alpha x_2 + \beta y_2 + \gamma z_2) = 0,$$

satisfied by making the points 1 and 2 coincident. Thus, if 3 be on the line 45 (and similarly if 1 or if 2 be on the line 45), we cannot infer that 1, 2, 3 are in a line.

I assume, therefore, that neither 1, 2, nor 3 is in the line 45; we here conclude, as above, that 1, 2, 3 are in a line. Suppose for a moment that 5 is not on this line; 4 cannot be on each of the lines 15, 25, 35, and I assume, in the first instance,

that it is not on any one of these lines; thus the points 1, 2, 4 are no one of them on the line 35, and hence, by the like reasoning, 1, 2, 4 are in a line; that is, 4 is on the line 12, or, the points 1, 2, 3, 4 are in a line. If, however, 4 is on one of the lines 15, 25, 35, say it is on 35; then it is not on 25, and no one of the points 1, 3, 4 is on 25; hence, by the like reasoning, 1, 2, 4 are in a line. In this case, however, 4 being, by supposition, on the line 35, can only be the point 3; that is, 3 and 4 would be coincident, a case which need not be considered. The correct conclusion from the reasoning thus is, not that 1, 2, 3, 4, 5 are in a line, but that some four of these five points, say 1, 2, 3, 4, are in a line.

It is easy to see that, if we have on a line four points, then there exists between the second powers of these points a linear homogeneous relation. For let the distances of the points from a fixed point of the line be a, b, c, d respectively; and let the arbitrary line meet the line in a point O at a distance r from the fixed point. The distances of the four points from the point O are thus equal to $r + a, r + b, r + c, r + d$ respectively; and hence, writing k for the squared cosine of the inclination of the two lines, the second powers of the four points are $= k(r + a)^2, k(r + b)^2, k(r + c)^2, k(r + d)^2$ respectively; and we have thus, between the second powers p_1, p_2, p_3, p_4 of the four points, the homogeneous linear relation

$$\begin{vmatrix} p_1 & 1 & a & a^2 \\ p_2 & 1 & b & b^2 \\ p_3 & 1 & c & c^2 \\ p_4 & 1 & d & d^2 \end{vmatrix} = 0.$$

Moreover we can, in an infinite number of ways, form a linear combination

$$A_1p_1 + A_2p_2 + A_3p_3 + A_4p_4$$

of these second powers, which shall be independent of r , and have any given value whatever. We can therefore make this sum to be equal to the second power of any point 5 whatever (not on the line containing the four points) in relation to the arbitrary line; that is, given the four points in a line, and any other fifth point, we can establish between the second powers of these five points a linear homogeneous relation

$$A_1p_1 + A_2p_2 + A_3p_3 + A_4p_4 + A_5p_5 = 0,$$

with non-evanescent values for each of the coefficients A . We thus see how, to the linear homogeneous relation between the second powers of the five points, there corresponds properly the non-symmetric relation: four of the five points are in a line.

The most simple cases are when $p = n$, viz. $p = n = 2, N = 6$; $p = n = 3, N = 10$, and generally $N = \frac{1}{2}(n + 1)(n + 2)$; and for these Clifford's theorem is easily verified. I have not examined the other cases, but it is probable that in each of them a correction is required.

ON THE ANALYTICAL THEORY OF THE CONGRUENCY.

[928]

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If the lines of a congruency are considered as issuing from the several points of a surface, or say as the quasi-normals of a surface, then the fundamental geometrical theory is established by an analysis closely similar to that for the theory of the curvature of a surface; viz. it is shown that each quasi-normal is intersected by two consecutive quasi-normals in two points respectively (corresponding to the centres of curvature), or say in two foci; we have on the surface two series of curves of quasi-curvature—only these do not in general intersect at right angles; the intersecting quasi-normals form two series of developable surfaces, each touching the surface of centres (or focal surface), along its cuspidal edge, &c.; and, in particular, each quasi-normal is a bitangent of the focal surface, touching it at the two foci respectively.

But the analysis assumes a very different form if we consider the congruency by itself, without thus connecting it with a surface. Regarding the congruency as determined by means of two equations,

$$U(a, b, c, f, g, h) = 0, \quad V(a, b, c, f, g, h) = 0,$$

between the six coordinates of a line ($af + bg + ch = 0$, as usual), I take (a, b, c, f, g, h) for the coordinates of a particular line of the congruency, $(a_1, b_1, c_1, f_1, g_1, h_1)$ for those of a consecutive line. Denoting, for shortness, the derived functions

$$(\partial_a, \partial_b, \partial_c, \partial_f, \partial_g, \partial_h)U$$

by F, G, H, A, B, C , and similarly those of V by F', G', H', A', B', C' , we have

$$a_1F + b_1G + c_1H + f_1A + g_1B + h_1C = 0,$$

$$a_1F' + b_1G' + c_1H' + f_1A' + g_1B' + h_1C' = 0;$$